ASMITA

NEW edition

A Textbook of PROBABILITY AND STATISTICS

For Engineers

For all Engineering programs of Pokhara University

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Dr. Himalal Gautam • Madhav Prasad Poudel

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SYLLABUS

Pokhara University
Faculty of Science and Technology

Course Code: MTH 216 (3 – 2 - 0 Credit)
Course Title: Probability and Statistics
Nature of the Course: Theory
Level: Bachelor

Full Marks: 100
Pass Mark: 45
Total Lectures: 30 hours
Program: BE

1. Course Description

This course is designed to familiarize students with various statistical models and techniques for analyzing data. The contents include descriptive statistics, probability, probability distributions, sampling and estimation, hypothesis testing, simple correlation and regression analysis with emphasis on engineering field.

2. General Objectives

The general objectives of this course are;

- To familiarize students with various statistical models and techniques for analyzing data.
- To impart analytical skills in the students required for the application of statistical methods for analyzing data in the field of engineering.
- To enable students with the skills to use of real data in the practical engineering-based applications.

3. Methods of Instruction

Lecture, Tutorial, Discussion and Readings

4. Contents in Detail

Specific Objectives	Contents
▪ Identify concepts of statistics and its application in the field of engineering ▪ Summarize, present and compute various descriptive statistics	**Unit I: Introduction and Descriptive Statistics (3 hrs)** 1.1 Introduction of statistics and its applications in engineering 1.2 Collection and presentation of data (Diagrammatic as well as graphical presentation) 1.3 Measure of central tendency, location and Measures of variability
▪ Identify basic probability concepts ▪ Define conditional probability and use Bayes' theorem to revise probabilities ▪ Define random variable and compute expected value and variance of a probability distribution	**Unit II: Probability (5 hrs)** 2.1 Basic probability, additive law, multiplicative law and Bayes' theorem 2.2 Random variables and probability distribution function, 2.3 Mathematical expectation of random variables
▪ Explain and apply discrete probability distributions (Binomial, Poisson distribution, Negative Binomial and Hyper geometric distribution)	**Unit III: Discrete Probability Distributions (3 hrs)** 3.1 Binomial distribution, 3.2 Poisson distribution 3.3 Negative Binomial distribution 3.4 Hyper geometric distribution
▪ Explain and apply the Normal distribution and other continuous probability distributions (uniform distribution, Gamma and Beta distributions, and Exponential distribution)	**Unit IV: Continuous Probability Distributions (4 hrs)** 4.1 Rectangular or uniform distribution 4.2 Normal distribution 4.3 Gamma and Beta distributions 4.4 Exponential distribution
▪ Define the concept of bivariate random variables and joint probability distribution ▪ Explain and calculate joint probability mass and density function	**Unit V: Bivariate Random Variables and Joint Probability Distribution (2 hrs)** 5.1 Joint probability mass function, Marginal probability mass function, 5.2 Joint probability density function, marginal probability density function
▪ Define and apply sampling and sampling distribution and central limit theorem ▪ Construct and interpret confidence interval estimates for the means and proportion	**Unit VI: Sampling Distribution and Estimation (5 hrs)** 6.1 Review of terms used in sampling 6.2 Probability and non-probability sampling 6.3 Sampling distribution of mean and standard error 6.4 Central limit theorem 6.5 Concept of point and interval estimation 6.6 Sample size determination 6.7 Confidence interval for single mean and difference of two population means and population proportions
▪ Describe and apply the procedures of hypothesis testing of various tests	**Unit VII: Hypothesis Testing (5 hrs)** 7.1 Basic concept in hypothesis testing 7.2 One sample test for mean and proportion 7.3 Two sample test for mean and proportions 7.4 Paired t – test 7.5 Chi-square test of independence
▪ Define and apply correlation and regression in the field of engineering	**Unit VIII: Correlation and Regression (3hrs)** 8.1 Simple correlation and its properties 8.2 Simple linear regression

Note: The figures in the parentheses indicate the approximate periods for the respective units.

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5. List of Tutorials (30 Hours)

6. Numerical problems as demanded by the theory of each chapter will be assigned for the students and they are encouraged to solve the problems.

Unit No.	Unit Name	List of Tutorials	Hours
I	Introduction and Descriptive Statistics	1.1 Collection and presentation of data (Diagrammatic as well as Graphical presentation)	1 hr
		1.2 Measure of central tendency, Location and Measure of variability	1 hr
II	Probability	2.1 Basic probability, additive law, multiplicative law and Baye's theorem	2 hrs
		2.2 Random variables (Discrete and Continuous) and probability distribution function	1 hr
		2.3 Mathematical expectation and random variables	1 hr
III	Discrete Probability Distributions	3.1 Binomial distribution	1 hr
		3.2 Poisson distribution	1 hr
		3.3 Negative Binomial distribution	1 hr
		3.4 Hypergeometric distribution	1 hr
IV	Continuous Probability Distributions	4.1 Rectangular or Uniform distribution	1 hr
		4.2 Normal distribution	2 hrs
		4.3 Gamma and Beta distribution	2 hrs
		4.4 Exponential distribution	2 hrs
V	Bivariate Random Variables and Joint Probability Distribution	6.1 Joint probability mass function, Marginal probability mass function	1 hr
		6.2 Joint probability density function, Marginal probability density function	2 hrs
VI	Sampling Distribution and Estimation	6.1 Sampling distribution of mean and standard error	1 hr
			1 hr
		6.2 Central limit theorem	1 hr
		6.3 Concept of point and interval estimation and sample size determination	1 hr
		6.4 Confidence interval for single mean and difference of two population means and population proportion	
VII	Hypothesis Testing	7.1 One sample test of mean and proportion	1 hr
		7.2 Two sample test for mean and proportions	1 hr
		7.3 Paired t-test	1 hr
		7.4 Chi-square test of independence	1 hr
VIII	Correlation and Regression	8.1 Simple correlation and its properties	1 hr
		8.2 Simple linear regression	1 hr

7. Evaluation system and Students' Responsibilities

Evaluation System

In addition to the formal exam(s), the internal evaluation of a student may consist of quizzes, assignments, project work, class participation, etc. The tabular presentation of the internal evaluation is as follows.

Internal Evaluation	Weight	Marks	External Evaluation	Marks
Attendance & Class Participation	10%	50	Semester-End Examination	50
Assignments	20%			
Presentations/Quizzes	10%			
Term Exam	60%			
Total Internal				
Full Marks: 50 + 50 = 100				

Student's Responsibilities

Each student must secure at least 45% marks separately in internal assessment and practical evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such score will be given NOT QUALIFIED (NQ) to appear the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

8. Prescribed Books and References

Prescribed Books

1. Johnson, R. A. (2018). *Probability and Statistics for Engineers*. New Delhi: Pearson Education Limited.

Reference Books

1. Devore, J. L.(2010). Probability and Statistics for Engineering and Sciences. New Delhi: Cengage learning.
2. Sheldom, M. R. (2014). Probability *and Statistics for Engineers and Scientist*. New Delhi: Cengage learning.
3. Gupta, S.C & V.K. Kapoor. (2000). *Fundamentals of Mathematical Statistics: A Modern Approach*. Sultan Chand & Sons Educational Publishers.

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CONTENTS

Unit 1 Introduction and Descriptive Statistics

1.1 Introduction of statistics 1
1.1.1 Importance of statistics 2
1.1.2 Functions of statistics 3
1.1.3 Limitations of statistics 3
1.1.4 Data collection 4
1.1.5 Types of data according to sources 4
1.1.6 Common statistical Terminology 6
Exercise 1.1 9
1.2 Classification and presentation of data 9
1.2.1 Objectives of classification 9
1.2.2 Different types of classification 9
1.2.3 Construction of continuous frequency distribution/ series 11
1.2.4 Displaying frequency distributions 14
1.2.5 Diagrammatic representation of data 17
1.2.6 Difference between diagrams and graphs 18
1.2.7 Multiple bar diagram 19
1.2.8 Sub divided bar diagram 19
1.2.9 Pie-diagram or pie chart 22
Exercise 1.2 23
1.3 Measures of central tendency, location and variability 26
1.3.1 Measures of central tendency 26
1.3.2 Partition values 35
Exercise 1.3 38
1.3.3 Measure of dispersion 41
1.3.4 Various measures of dispersion 41
1.3.5 Box and Whisker plot 49
Exercise 1.4 49

Unit 2 Probability

2.1 Probability theory 53
2.1.1 Basic terminologies used in probability 53
2.1.2 Counting rules 55
2.1.3 Approaches to probability 56
2.1.4 Subjective probability 57
2.1.5 Laws of probability 59
2.1.6 Multiplicative law of probability 64
2.1.7 Baye's theorem 70
Exercise 2.1 81
2.2 Random variables and probability distribution 85
2.2.1 Random variable 85
2.2.2 Probability distribution 85
2.2.3 Discrete random variable 86
2.2.4 Probability mass function 86
2.2.5 Probability distribution function 86
2.2.6 Properties of probability distribution function 86
2.2.7 Continuous random variable 89
2.2.8 Probability density function (p.d.f.) 89
2.2.9 Continuous probability distribution function 90
2.2.10 Properties of F(x) 90
Exercise 2.2 94
2.3 Mathematical expectation 96
2.3.1 Expectation of discrete random variables 96
2.3.2 Expectation of continuous random variable 96
2.3.3 Covariance 97
Exercise 2.3 104

notes.arpanadhikari7.com.np

Unit 3 Theoretical Discrete Probability Distribution 107

3.1 Introduction 107
3.1.1 Bernoulli trial 107
3.1.2 Binomial distribution 108
3.2 Poisson distribution 113
3.3 Hypergeometric distribution 119
3.4 Negative binomial distribution 122
Exercise 3.1 125

Available on notes.arpanadhikari7.com.np

Unit 4 Theoretical Continuous Probability Distribution

4.1 Normal distribution 129
4.2 Curve of the normal distribution 130
4.2.1 Properties of normal distribution 130
4.2.2 Importance of normal distribution 130
4.3 Standard normal distribution 131
4.3.1 Area property of normal / standard normal distribution 131
4.4 The normal / Z table 132
4.5 Relation between binomial and normal distribution 139
4.5.1 Relation between Poisson and normal distribution 139
4.5.2 Continuity correction 140
Exercise 4.1 143
4.6 Rectangular or uniform distribution 145
4.7 Gamma distribution 147
4.8 Beta distribution 149
4.9 Exponential distribution 152
4.9.1 Memory loss property of exponential distribution 152
4.10 Log-normal distribution 155
Exercise 4.2 157

Unit 5 Joint Probability Distribution

5.1 Introduction 159
5.1.1 Joint probability mass function 160
5.1.2 Marginal probability mass function 160
5.1.3 Conditional probability 160
5.1.4 Independence 161
5.2 Joint probability distribution function 161
5.3 Continuous random variables 161
5.3.1 Joint probability density function 161
5.3.2 Marginal probability density function 161
5.3.3 Conditional probability density function 162
5.3.4 Joint probability distribution function 162
Exercise 5.1 174

Unit 6 Sampling and Estimation

6.1 Introduction 179
6.2 Techniques in sampling 180
6.2.1 Probability sampling 180
6.2.2 Non-probability sampling 182
6.3 Errors in sampling 183
6.3.1 Sampling errors 183
6.3.2 Non-sampling errors 183
6.4 Sampling distribution of mean 183
6.4.1 Standard error 185
6.4.2 Application of standard error 186
6.5 Central limit theorem 186
6.6 Estimation 188
6.6.1 Point estimation 189
6.6.2 Interval estimation 190
6.6.3 Confidence interval for mean (μ) 190
6.7 Confidence limits for population proportion 194
6.8 Determination of sample size by using mean 196
6.9 Determination of sample size using proportion 196
6.10 Confidence interval for difference between means of two normal populations 198
6.10.1 Large sample case 198
6.10.2 Small sample case ($n_1 < 30$ and $n_2 < 30$) 199
Exercise 6.1 200

Unit 7 Hypothesis Testing

7.1 Introduction 205
7.2 Systematic procedure for testing of hypothesis 205
Exercise 7.1 228
7.3 Chi-square (χ^2) Test 234
Exercise 7.2 241

notes.arpanadhikari7.com.np
Available on notes.arpanadhikari7.com.np

Unit 8 Correlation and Regression Analysis

8.1 Simple correlation 243
8.1.1 Introduction 243
8.1.2 Karl Pearson's correlation coefficient 245
8.1.3 Spearman's rank correlation coefficients 248
8.2 Regression analysis 249
8.3 Regression equation of Y on X 249
8.4 Regression equation of X on Y 250
8.5 Measure of variation 250
8.6 Coefficient of determination 251
8.7 Standard error of estimate 251
8.8 Properties of regression coefficient 257
8.8.1 Difference between correlation and regression analysis 259
Exercise 8.1 261

References 265
Appendix 266
Exam Question 270

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Unit 1

Introduction and Descriptive Statistics

1.1 Introduction of statistics

The term statistics has been originated from the Latin word 'status' or the Italian word 'statista' or the German word 'statistik' each of which means political state. Some statistician defines: statistics is the science of counting, some other says statistics is the science of averages, some says science of measurement of social phenomenon; so different author define statistics differently. A more exhaustive definition is given by Professor Horace Secrist as "By statistics we mean aggregates of facts to a marked extent by multiplicity of causes numerically expressed, enumerated or expressed or estimated according to the reasonable standard of accuracy, collected in a systematic manner for pre-determined purpose and placed in relation to each other". Lovit defines "statistics is the science which deals with collection, classification, and tabulation of numerical facts as the basis for explanation, description and comparison of phenomenon".

The best definition seems to be given by Croxton and Cowden and according to them "Statistics is the science which deals with collection, presentation, analysis and interpretation of numerical facts" Statistics may broadly categorize into two headings. They are:

a. **Descriptive statistics**: The reductions of voluminous data into readily comprehend form, which helps analyzing and describing the main features of the data without making conclusion is called descriptive statistics. Descriptive statistics are brief informational coefficients that summarize the given data set, which can be representation of the whole population. Measures of central tendency which include mean, median, mode; and measures of dispersion which include range, quartile deviation, mean deviation, standard deviation and frequency distribution includes descriptive statistics. It summarizes the features of data.

b. **Inferential statistics**: Statistical technique which provides some information about population parameter, based on the sample drawn from the population is called inferential statistics. It helps to understanding of the population parameter by analyzing the samples obtained from it. It helps in making generalizations about the population parameter by using various analytical tools and tests. Estimation (Point or interval estimation), hypothesis testing, etc. includes in inferential statistics.

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1.1.1 Importance of statistics

Now a day's statistics become indispensible tool almost in all areas of human activities of knowledge, so the subject becoming more importance of its wide applications. It is used in social sciences, medical sciences, pharmacy, agriculture, economics, psychology, astronomy, business, management, information technology, insurance all fields of engineering and others.

Statistics is used in planning

To be successful in any part of the research planning is more important, and it is the first phase in any works to be done. It deals with factors associated with work to be done and statistical tools come too used for it. So statistics plays significant role in planning of any project or research works.

Statistics is important in mathematics and economics

Initially statistics was thought to be part of mathematics. But now statistics is treated as a separated subject because it has made its identification as a separate discipline. The uses of statistics are beyond the use of mathematics. Further it is broadly used in economics; every economic activity of nation, organization or community can be analyzed by using statistics.

Statistics is equally important in business and management

Statistics has a close relation with business. Before launching any product or to produce any schemes, the research should be done about its implication. For that correlation may be find the relation between the variables, time series and regression model can be used to predict the future trends of the sales and future values. These are only the examples, simply speaking mean, median or mode are the simplest and basic tools to measure the level of production or sales.

Biological sciences and statistics

In biological sciences statistics presents facts and figures, simplifies complexity and facilities to comparison between the variables. It helps formulation of policies, formulation of hypothesis, to study the relationship between variables, to study trend behavior, to study the biological variation, etc. statistical tools are useful. More and more things are measured quantitatively in medicine and public health science, for planning to conduct and interpretation of much medical and health research based works are increasingly depends on statistical technology.

Statistics in social sciences

Statistical tools are widely useful in social sciences such as political science, history, anthropology, sociology, demography, population studies etc. Every social phenomenon is affected by many remarkable factors; the impacts of these factors can be analyzed by using statistics. Different sampling techniques, estimation, hypothesis testing, correlation regression analysis etc are applied to study the various impacts of social phenomenon.

Statistics in science and technology

Different field's of science such as physics, chemistry, astronomy, meteorology, and engineering uses large number of different types of data and instruments, the variations in data and different measurements can be analyzed by statistics.

In different fields of engineering statistical tools are widely used, design of experiments is a methodology for formulating scientific and engineering problems using statistical methods. Process and quality control for manufacturing process and products, probabilistic design used in product and system design and system identification uses statistical methods to build mathematical models. Statistics is a useful tool for test data analysis, system error analysis, and probabilistic risk assessment. To design new computers, network devices and other hardware statistical tools are useful. Data collection and modeling also determine the best method for manufacturing hardware, estimating the mean time between failures for components contributes to informed decision method.

1.1.2 Functions of statistics

The functions of the statistics cab be summarised as follows.

- To represent facts from numerical figures in a definite form.
- To condense the widely and voluminous data.
- To help classification of data.
- To provide methods for making comparison.
- To help in formulating policies.
- To determine relationship between different phenomenon.
- To help in predicting future trends.
- To formulate and test the hypothesis.
- To have the idea about occurrence or non occurrence of certain events.
- To draw valid inferences or conclusions.

1.1.3 Limitations of statistics

Even though it is mentioning that it has many uses and functions it does have certain limitations as well. The limitations of Statistics can be described as follows.

Statistics does not study qualitative phenomenon: All the quantitative variables as marks, weights heights, sales, production level, costs etc can be studied through statistics. There are qualitative variables such as intelligence, beauty, ugly, honesty which cannot be measured through statistics directly because they do not have exactly numerical value.

- **Statistics does not study individuals:** Statistical tool is used to measure collective data. Single value in statistical methods is almost useless.
- **Statistical laws are not exact:** It should be noted that the statistical measures are in terms of the averages. The statistical laws based on approximation but not exact.

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- **Statistics is liable can be misused:** Since statistics are confined to certain variables it is liable to be misused it is also dependent on the biasness of the surveyor. Many liars can misuse statistics, by miss interpretation of statistical facts and figures.
- Statistics may lead to fallacious conclusion.

1.1.4 Data collection

Variable: Characteristics which describes the place, thing, person, opinion, idea, velocity, temperature, acceleration, volume, area etc is called variable in statistics. Generally variables are denoted by X, Y, Z etc. The value of the variable varies from one entity to another. The marks obtained by the students of a class, their weight, height, their interest in mathematics, physics, English etc are example of variables. In statistics the data relating to different variable under study can be collected and analyzed. The collection of raw material or facts is called data. There are two types of data according to nature of variables.

a. **Qualitative Variable:** That fact which cannot be expressed in numerical value is called qualitative data. Qualitative data derived from qualitative variables. Beauty, honesty, intelligence, attributes, nationalities, occupations, smoking/alcoholic habit etc are examples of qualitative data.

b. **Quantitative Variable:** Those facts which can be collected in numerical value are called quantitative data; these types of data derived from quantitative variables. Height, weight, ages, body temperature, area volume, acceleration, velocity, strength of material, cost of commodity etc are the examples. Quantitative variables can further classified into two groups.

   i. **Discrete variable:** The variable which takes fixed or integer values i.e. whole number values is called discrete variable. The numbers of parasites, number of person, number of animal, number of accidents, number of cars, number of buildings, number of plants etc are example of discrete variables.

   ii. **Continuous variable:** The variable which takes all possible values i.e. whole number as well as fractional values within certain range is called continuous variable. Temperature, weights, heights, lengths of certain object, time, age, area, volume, acceleration, velocity etc are examples of continuous variables.

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1.1.5 Types of data according to sources

According to sources the data can be categorized into two headings.

a. **Primary data:** Those data, which are collected by the investigator himself for the purpose of a specific inquiry or study is called primary data. Such data are original in character and are mostly generated by surveys conducted by individuals or research institutions. The first hand information obtained by the investigator is more reliable and accurate since the investigator can extract the correct information by removing doubts, if any, in the minds of the respondents regarding certain questions. High response rates might be obtained since the answers to various questions are obtained on the spot. It permits explanation of questions concerning difficult subject matter. Before collection of primary data following points should be considered.

- **Objective and scope of the inquiry:** It refers to the aim if the survey.
- **Types of inquiry:** It refers to the method of study; it may be official or unofficial, initial or final, conditional or non conditional etc.
- **Statistical units should be used:** It refers to the size of the unit to be covered in the study.
- **Sources of data:** It refers to the person or place or institution or any formal or informal means that you obtain the data.
- **Method adopted for data collection:** It refers to the way that the data is collected. For the smaller field or with smaller number of variables census can be conducted otherwise samples can be taken.
- **Degree of accuracy aimed at the final results:** It depends upon the data that is used for the analysis. To get the highest degree of accuracy whatever may be the drawbacks the primary data are taken into consideration under the condition that the data are taken from right place from the right person at the right time.

Methods of collecting primary data

1. **Direct interview method:** In this method the data are collected directly from the field in a face to face interview with the concerned person. In this method the means of communication can be adjusted so that higher accuracy can be obtained in the result with greater degree of accuracy. The supplementary information can also be obtained from this method. The disadvantage of this method is:

   It is expensive as the person has to approach to each and every individual, there is liability of having biasness in the correspondent, the time of survey may extend because a lot of time takes during the communication process.

2. **Indirect oral interview:** This method is used if a third party is able to communicate all the information regarding the second party. In this case the third party should be totally aware of the second one. Especially in the study of negative characteristics this method is most suitable. This is because if the data is intended to collect through direct method the respond will not provide the correct information. This method can be adopted to find out the habits of alcoholics or the druggists in some particular community.

3. **Information received through local agencies:** In this method, the investigator appoints certain persons as a correspondent for the availability of the information. The correspondent may be reporter or any governmental or nongovernmental agency as well. The data so obtained may not be perfectly correct but somehow the result is obtained within the certain degree of accuracy.

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4. **Mailed Questionnaire method:** Under this method, the investigator constructs the sets of questions that are required in the investigation. These set of questions are mailed to the correspondents or the informants through the post. After the informant receives the questions then he/she replies and again mails back to the investigator. This can be a cheap method and is possible to carry out the survey for a large range of geographic location. This method is limited in the sense that the respondents may not be literate and no one can assure the question will be back after being filled up.
5. **Schedules sent through enumerators:** In this method the information is obtained by investigator by sending the schedules through its enumerators. Required information from respondents is filling up in the questionnaire of schedule and data can be collected by enumerators. It needs to train enumerators.

b. **Secondary data:** When an investigator uses data, which have already been collected by others, such data are called "Secondary Data". Such data are primary data for the agency that collected them, and become secondary for someone else who uses these data for his own purposes. The secondary data can be obtained from journals, reports, government publications, publications of professionals and research organizations. Secondary data are less expensive to collect both in money and time. These data can also be better utilized and sometimes the quality of such data may be better because these might have been collected by persons who were specially trained for that purpose. On the other hand, such data must be used with great care, because such data may also be **full of errors** due to the fact that the purpose of the collection of the data by the primary agency may have been different from the purpose of the user of these secondary data. Secondly, there may have been bias introduced, the size of the sample may have been inadequate, or there may have been arithmetic or definition errors, hence, it is necessary to critically investigate the validity of the secondary data

For any research to be conducted, data collection is the most important part. Before the data is collected, the aim of the data collection should be well defined. The data so collected should meet the requirement of the objective of the research. This is because it guides the method of data collection i.e. to use primary data or to use secondary data, the area to be studied, etc. If the aim is not specific then the aim of the survey may not meet. If the data to be studied are too large, or if the operating cost for the survey exceeds the cost limit then the data obtained from the earlier source can be used. These data are called the secondary data.

1.1.6 Common statistical Terminology

Population: Total number of things, objects, people or items under specific study region is called population. A population is the largest collection of entities (elements or individuals) in which we are interested at a particular time and about which we want to draw some conclusions. Example: If we are interested in the weights of students enrolled in the college of engineering at PU, then our population consists of the weights of all of these students, and our variable of interest is the weight.

Population Size (N): The total number of elements/objects in the population is called the population size and is denoted by N

Census: The complete enumeration or counting of all population under study area is called census. It requires more cost, man power and time than sampling. In modern usage, the term "census" refers to a nation-wide counting of population. It is obtained by a direct canvass of each person or household, which is a large and complicated undertaking.

Sample: Part of the population is called sample. From the population, we select various elements on which we collect our data. This part of the population on which we collect data is called the sample. Example: Suppose that we are interested in studying the characteristics of the weights of the students enrolled in the college of engineering at PU. If we randomly select 50 students among the students of the college of engineering at PU and measure their weights, then the weights of these 50 students are sample.

Sample Size (n): The number of elements in the sample is called the sample size and is denoted by n.

Sampling: Process of selecting samples from the population is called sampling. There are two types of sampling; they are probability sampling and non probability sampling.

Probability Sampling

In the probalility sampling all the observations have equal chances of being selected as a sample. It does not depend upon the type of information obtained for its effectiveness,the parameters obtained from probability sampling are non biased and have a measurable precision. It requires skilled and efficient manpower for its use, more time and more cost to collect data from population. The frequently used probability sampling are simple random sampling, startified random sampling, systematic sampling, cluster sampling and multistage sampling.

Non probability sampling

In the probalility sampling all the observations have equal chances of being selected as a sample instead the investigator is bias in making a sample. Where as the non probability sampling method does not obey any probabilistic rule when collecting samples from the population, so each unit of the population don not have eqaul chance being included in the sample. Purposive or judgement sampling, convinience sampling, quota sampling and snow ball sampling are all non probability sampling.

Measurement scale of data

Data are obtained through measurements. Measurement means the assigning of numbers to observations. Analysis of data means manipulating or operating on these numbers. There are four types of measurement scale.

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i. **Nominal scale**: Just giving the name of the new event/phenomenon or goods and number assigned to non quantitative answers is called nominal scale and it is lowest measurement scale. Such a number only represents the name of the results: they have no real numerical value. Value assigned is completely arbitrary. When they are used for identification, there is a strict one-to-one correspondence between the numbers and the objects. In such case the numbers do not reflect the amount of the characteristic possessed by the objects. The only permissible operation on the numbers in a nominal scale is in count data. For example, the nationality of different employee's lies in certain company can be identified as: Nepali (1), Chinese (2), Indian (3), Americans (4) then the count of each nationality can be collected.

ii. **Ordinal scale**: Any numerical presentation that maintains the ranked ordering of the responses, when they can be ranked according to certain priorities or criteria such measurement system is called ordinal scale. A ranking scale in which numbers are assigned to objects to indicate the relative extent to which the objects possess some characteristic. The ordinal scale can determine whether an object has more or less of a characteristic than some other object, but not how much more or less. In this scale any series of numbers can be assigned that preserves the ordered relationships between the objects.

Example: How would rate the facilities?

Excellent (1)

Good (2)

Fair (3)

Poor (4)

The only valid statistics that can be calculated from an ordinal scale are descriptive measures that are based on the ranking process. Hence only appropriate measure of central tendency is the median.

iii. **Interval scale**: An interval scale assigns each measurement to one of an unlimited number of categories that are equally spaced i.e. it possess constant interval size are called interval scale. This scale has no true zero or absolute point, 0 degrees does not mean there is no temperature, zero degree has significant value. For example income, distance travelled, temperature etc) are numerical measurements and are said to have an interval scale. In interval data the intervals between values are the same. For example, in the Fahrenheit temperature scale, the difference between 70 degrees and 71 degrees is the same as the difference between 32 and 33 degrees. But the scale is not a ratio Scale. 40 degrees Fahrenheit is not twice as much as 20 degrees Fahrenheit.

iv. **Ratio scale**: Ratio data are observations measured on a scale so constructed that equal numbers imply equal distances between the values of the characteristic being measured. Ratio scale has an absolute zero such as in weight and height. In this scale measurements can be added, divided and multiplied. Examples of ratio data are: age of a person (0-at birth, a 50-year old person is much older than 20-year old. This scale has true zero point.

The data values in ratio data do have meaningful ratios, for example, age is a ratio data, and someone who is 40 is twice as old as someone who is 20. Both interval and ratio data involve measurement. Most data analysis techniques that apply to ratio data also apply to interval data. Therefore, in most practical aspects, these types of data (interval and ratio) are grouped under metric data.

Exercise 1.1

1. Describe the history and meaning of statistics.
2. Distinguish between descriptive and inferential statistics.
3. Describe the importance of statistics in the fields of engineering.
4. Explain the uses and limitations of statistics.
5. What are primary data? Explain various methods of collecting primary data.
6. What are the differences between primary and secondary data?
7. What is sampling? Distinguish between probability and non probability sampling.
8. Explain different types of measurement scale of data.

1.2 Classification and presentation of data

Classification means the process of arranging the data into different groups or classes according to their similarities. Classification of data helps to change the voluminous and heterogeneous data into homogenous groups so that its important characteristics can be easily grasped.

1.2.1 Objectives of classification

- To change the complex and voluminous data into simpler form.
- It helps to enable a statistical treatment of information collected.
- It helps to prepare the background of tabulation, analysis and interpretation of data.
- It helps to facilitate comparison.

1.2.2 Different types of classification

According to the nature of data collected, the following are the ways of classification.

a. **Geographical classification**: This type of classification is done to study for those variables that are extremely affected by geographic location. While classifying the data on the basis of geographic location, it is essential that what the geographical boundary of the data is. In most of the cases it is found that some particular data is included twice, to avoid this type of error on the basis of geography by which the data are to be collected should be well defined. If the data are classified according to the difference between the various items due to geographical location, positions such as countries, zones, districts etc then it is called geographical classification. For example:

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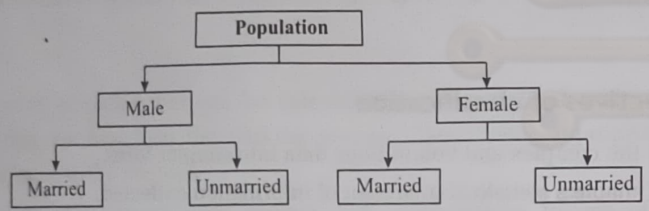
Bag	[illegible]
Narayani	14,38,803
Koshi	14,23,624
Gandaki	11,07,569
Mechi	9,32,625

b. **Chronological classification**: If the data are classified according to time periods like, yearly, quarterly, monthly, daily, hourly etc then it is called chronological classification. For example the number of students studied in a certain college in different years is.

Year	No. of students
2010	675
2011	940
2012	822
2013	1221

c. **Qualitative classification**: If the data are classified according to some quality such as honesty, beauty, literacy, intelligence, caste, religion etc then it is called qualitative classification. In this classification quality under study cannot measured quantitatively and the data are classified according to the presence or absence of quality in the given unit.



d. **Quantitative classification**: If the data are measured and classified according to some characteristics which can be measured quantitatively is known as quantitative classification. The marks, height, weight, body temperature, income etc can be measured in numerical value and use this types of classification.

Frequency: The number of times that the value of the variable occurs is called frequency.

Frequency distribution: It is simply a table in which the collected and classified data are presented into different classes and the number of cases i.e. frequencies which fall in each class is recorded. It shows the frequency of occurrence of different values of a single characteristic. There are two types of frequency distribution; they are univariate and bivariate frequency distribution.

Univariate frequency distribution: The univariate frequency distribution is of following three different series.

a. **Individual series**: If the value of the variable has the single observation in each case, then we can form individual series from such data, event wise observations are placed in this series. For example the maximum temperature of Pokhara valley in last week of December 2014 is given as.

Day	Sun.	Mon.	Tue.	Wed.	Thr.	Fri.	Sat.
Tem°C	20.5	22	21.4	19.8	20.3	23	22.5

b. **Discrete series**: The series formed from the discrete variable is known as discrete series. It is also known as discrete frequency distribution or ungrouped frequency distribution. In this series the data are presented in such a way that exact measurements of units are clearly indicated with their respective frequency. The example of discrete series is given below.

No. of households(X):	72	90	45	22	18	12
No. of cars (f):	0	1	2	3	4	5

c. **Continuous series**: The series formed from the continuous variable is termed as continuous series or continuous frequency distribution. In this series the variate value always lies between two numbers known as upper limit and lower limit and the difference between these two limits are class interval.

1.2.3 Construction of continuous frequency distribution/ series

To construct a continuous series from raw data of the continuous variable, first we need to find the possible number of classes, class interval or class width, for it the following rule is used.

Number of classes: The number of classes largely depends on the number of observation occurred in the raw data, the nature of the data and accuracy derived in the data. The number of classes is generally denoted by ***k***. According to the professor "Struges" the approximate number of classes is given by the relation:

$$K = 1 + 3.322 \log_{10} n,$$ where n is the total number of observations

The number of classes should lie between 5 and 15 i.e.($5 \le k \le 15$). If there are less than 5 classes then the series loses the some information of the data and further statistical analysis is not appropriate. If classes are more than 15 than the further analysis is tedious. Classes should be mutually exclusive.

Class interval: When approximate number of classes is known then the approximate range or the class interval denoted by *'i' or 'h'* can be find by the relation:

$$\text{'i' or 'h'} = \frac{\text{(Largest item - smallest item)}}{\text{No. of classes}} = \frac{(L - S)}{k}$$

For shake of simplicity if possible class interval should be 5 or multiple of 5.

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Class Limits: The class limits should be chosen in such a way that the mid value of the class interval and actual average of the observations in that class interval are as near to each other as possible. If possible class limits should be located at the points which are multiple of 0, 2, 5, 10, ..., etc, the figures capable of easy and simple analysis.

Example 1. The following table gives the electricity consumed (in KWH) by a production company in a sample of 50 different days of the year.

17.0 17.7 15.9 15.2 16.2 17.1 15.7 17.3 13.5 16.3 14.6 15.8 15.3 16.4 13.7 16.2 16.4 16.1
17.0 15.9 14.0 16.2 16.4 14.9 17.8 16.1 15.5 18.3 15.8 16.7 15.9 15.3 13.9 16.8 15.9 16.3
17.4 15.0 17.5 16.1 14.2 16.1 15.7 15.1 17.4 16.5 14.4 16.3 17.3 15.8

Construct a frequency distribution of the above data having a suitable number of classes and class interval.

Solution

According to Struge's rule the approximate number of classes is given by

$K = 1 + 3.322 \log_{10} n$,

Here n = 50,

Then

$k = 3.322 \log 50$,

or $k = 3.322 \times 1.6989$

$= 5.64$

$\therefore \quad k = 6$.

Then the class interval,

$i = \frac{(L - S)}{k}$

$= \frac{(18.3 - 13.5)}{5.64} = 0.85 = 1.0$

Therefore the continuous frequency distribution is given by:

Class Interval	Tally	Frequency
13.0 – 14.0	|||	3
14.0 – 15.0	𝍸	5
15.0 – 16.0	𝍸 𝍸 𝍸	15
16.0 – 17.0	𝍸 𝍸 𝍸 |	16
17.0 – 18.0	𝍸 𝍸	10
18.0 – 19.0	|	1

The grouped frequency distribution for the electricity consumption in the 50 days is given below. The given frequency distribution is called exclusive frequency distribution or exclusive continuous series, it is called exclusive series because only the lower limit and the value less than upper limit is included in each class. The upper limit is excluded in any specific class so it is called exclusive series.

Class Interval (power consumption KWH)	Frequency (no. of days)
13.0 – 14.0	3
14.0 – 15.0	5
15.0 – 16.0	15
16.0 – 17.0	16
17.0 – 18.0	10
18.0 – 19.0	1

Inclusive Continuous series: If both lower and upper limits are included in every class of the series, then such type of series is called inclusive continuous series. Some of the information may lose in these types of series when using further statistical analysis, so it needs to change inclusive series into exclusive continuous series.

Example 2. In our previous example of power consumption of 50 observations the inclusive series can be formed as given below.

Class Interval	Tally	Frequency
13.0 – 13.9	|||	3
14.0 – 14.9	𝍸	5
15.0 – 15.9	𝍸 𝍸 𝍸	15
16.0 – 16.9	𝍸 𝍸 𝍸 |	16
17.0 – 17.9	𝍸 𝍸	10
18.0 – 18.9	|	1

Solution

Class Interval (power consumption)	Frequency (no. of days)
13.0 – 13.9	3
14.0 – 14.9	5
15.0 – 15.9	15
16.0 – 16.9	16
17.0 – 17.9	10
18.0 – 18.9	1

To change the inclusive continuous series into exclusive series we have to use the correction factor (c.f.)

$$\text{c.f.} = \frac{(\text{Lower limit of the second class} - \text{Upper limit of the first class})}{2}$$

Thus in our case,

$$\text{c.f.} = \frac{(14.0 - 13.9)}{2} = 0.05$$

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Therefore, True lower limit = given lower limit - c.f.

True upper limit = given upper limit + c.f.

Class Interval	True Class Interval	Mid-point	Frequency
13.0 – 13.9	12.95 - 13.95	13.45	3
14.0 – 14.9	13.95 - 14.95	14.45	5
15.0 – 15.9	14.95 - 15.95	15.45	15
16.0 – 16.9	15.95 - 16.95	16.45	16
17.0 – 17.9	16.95 - 17.95	17.45	10
18.0 – 18.9	17.95 – 18.95	18.45	1

Cumulative frequency: If the frequencies are cumulated either from top or bottom then so formed frequencies are called cumulative frequencies (c.f.). There are two types of cumulative frequencies they are less than c.f. and more than c.f.

Relative frequency and Percentage frequency: Relative frequency $= \frac{\text{frequency}}{n}$

Percentage frequency = Relative frequency × 100%

Class Interval	Frequency	Cumulative Frequency	Relative Frequency	Cumulative Relative Frequency	Percentage Frequency	Cumulative Percentage Frequency
13.0 – 13.9	3	3	0.06	0.06	6%	6%
14.0 – 14.9	5	8	0.10	0.16	10%	16%
15.0 – 15.9	15	23	0.30	0.46	30%	46%
16.0 – 16.9	16	39	0.32	0.78	32%	78%
17.0 – 17.9	10	49	0.20	0.98	20%	98%
18.0 – 18.9	1	50	0.02	1.00	2%	100%

Open ended series: If the lower limit of the first class or the upper limit of the last class or both i.e. lower limit of the first class and upper limit of last class is not defined then such types of series are known as open ended series.

1.2.4 Displaying frequency distributions

Graphical presentation of data: For representing frequency (or relative frequency or percentage frequency) distributions, we may use one of the following graphs:

The Histogram: Histogram is a set of rectangles whose areas are proportional to its widths. When the absolute frequencies are plotted on the Y-axis and class interval is drawn in the X-axis, the rectangular shape formed by joining its frequency with its respective class interval is called histogram. There should be no gap in between the class intervals, if there are gaps, and then the class intervals are changed into the class boundaries. Histograms are used to find the value of mode.

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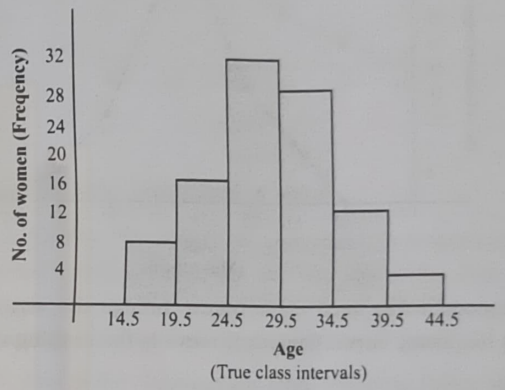
Example 3. Consider the following frequency distribution of the ages of 100 women.

True Class Interval (age)	Frequency (No. of women)	Less than c.f.	Mid points
14.5 - 19.5	8	8	17
19.5 - 24.5	16	24	22
24.5 - 29.5	32	56	27
29.5 - 34.5	28	84	32
34.5 - 39.5	12	96	37
39.5 - 44.5	4	100	44
Total		$n=100$	



The Frequency Polygon: When the distribution is discrete frequency polygon is obtained by plotting the points with the variable as the x-coordinate and corresponding frequencies as the y-coordinate and joining these lines by straight lines.

In case of continuous series frequency polygon can be drawn by two ways. i) Using histogram: Find the centre points of each of top of histogram, and join these respective points by straight lines, the curve obtained is called frequency polygon. ii) Without using histogram: Find the middle points of the every class interval and plot these points on X-axis, and respective frequencies in the Y-axis, the curve formed by joining the respective points are frequency polygon. Mode can be found by using frequency polygon.

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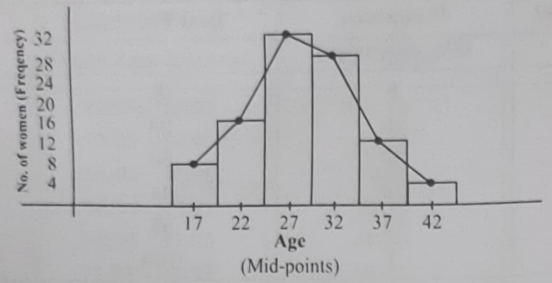
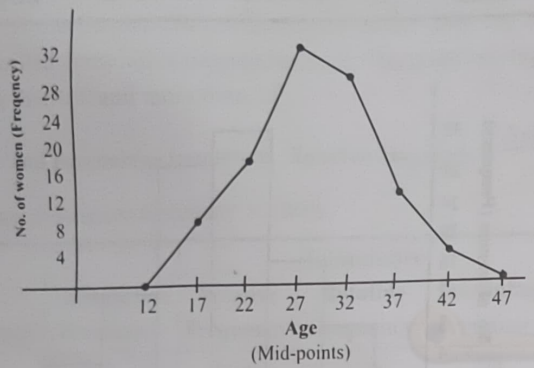
Example 4. Frequency polygon: Using the data of above histogram.



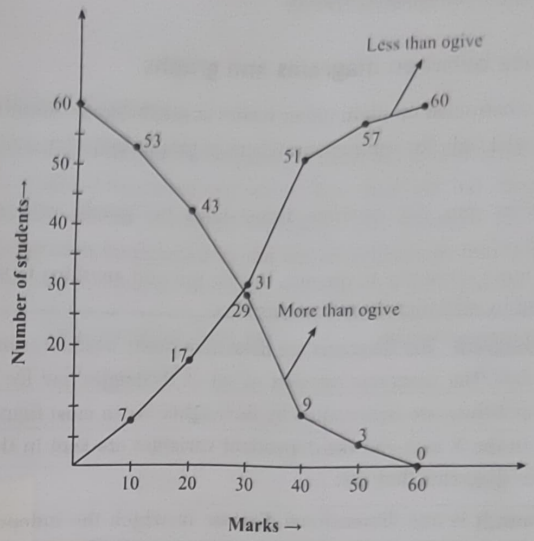


Frequency curve: A smooth free hand curve drawn through the vertices of the frequency polygon is known as frequency curve. Frequency curve is the limiting form of the frequency polygon.

Ogive: First plot the value of variable in X-axis and corresponding cumulative frequency in Y-axis, then find the points common to both these, the curve obtained by joining these points are called ogive. There are two types of ogive. i) Less than ogive: It is drawn from less than cumulative frequencies and ii) More than ogive: it is drawn from more than cumulative frequencies. The intersecting point of these two curves gives median value.

Example 5. Draw the ogive of the following series.

Marks (X):	No. of students (f)	Less than c.f.	More than c.f.
0 - 10	7	7	60
10 - 20	10	17	53
20 - 30	14	31	43
30 - 40	20	51	29
40 - 50	6	57	9
50 - 60	3	60	3



1.2.5 Diagrammatic representation of data

After the collection of the data, they should be presented in a proper form. The proper form refers to the appropriate method which can visualize about the data in the proper way. Depending upon the type of data, they can be shown in two forms i.e. graphical presentation of data as mentioned earlier and diagrammatic presentation of data.

While representing the data in the diagrammatic form, following things should be considered.

- **Title:** Title represents the name of which the table is associated with and it should be given at the top of the diagram.
- **Scale:** The scale should be properly done so that it does not make any confusion on reading the data. Generally it is done in multiples of ten. But it can be changed looking upon the nature of the data.
- The footnotes.
- **The index:** Any abbreviations or the colour note is referred as the index of the table. Index should be given properly.

The diagrams are used in statistics because of the following reasons

- The diagrams help to get the clear vision of the variables
- Diagrams help to predict the past or future trend of the change of variables
- Diagrams help in comparison of the data.

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1.2.6 Difference between diagrams and graphs

- Diagrams are constructed on plain paper where as graphs are on graph paper.
- Diagrams are used only for comparison where as graphs helps for studying mathematical relationship.
- Diagrams may be one, two or three dimensional but graphs will generally be of two dimensional.
- Diagrams are more attractive to the eye, but do not add anything to the data, graphs are frequently used by statisticians and researchers.

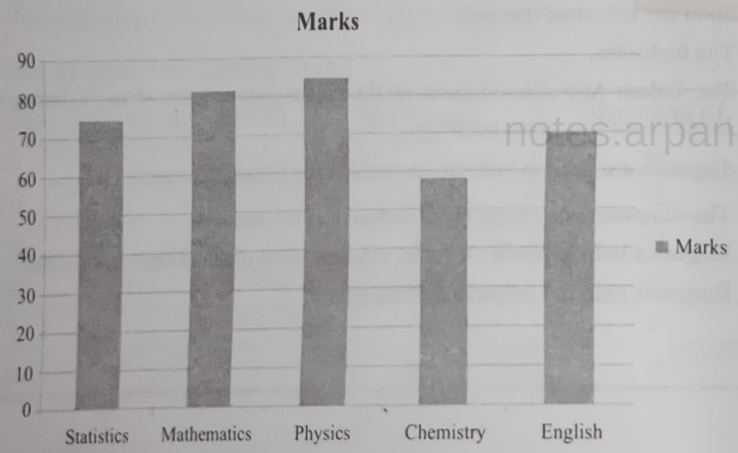
One dimensional diagram: Bar diagrams are most commonly used one dimensional diagrams of presenting the data. Bar diagrams consists of set of rectangles one for each set of data in which magnitude or values are represented by its heights. As in most figures the independent variables are kept in the X axis and the dependent variables are kept in the Y axis. There are various forms of bar diagrams they are:

Simple bar Diagram: It is one dimensional diagram in which the independent variables are kept in the X axis and the dependent variables are kept in the Y axis. For example the marks obtained by a student in the final examination in the following subjects are given below.

Subject:	Marks secured:
Statistics	75
Mathematics	82
Physics	85
Chemistry	58
English	70

It can be presented in a simple bar diagram as follows.

Simple bar diagram of marks obtained by a student:

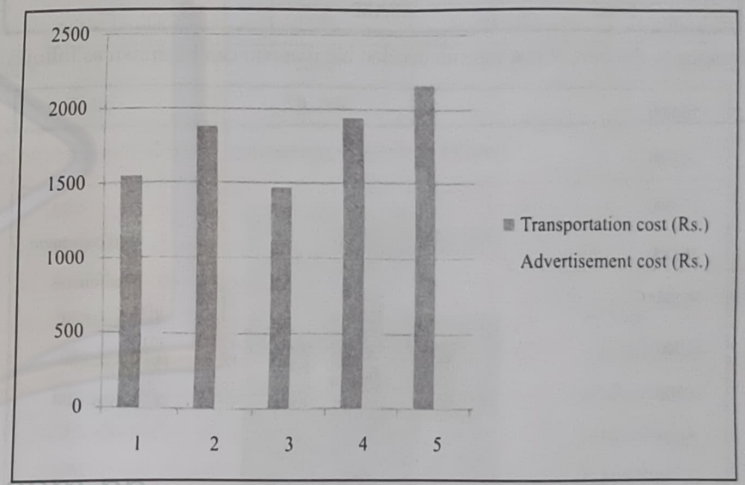


1.2.7 Multiple bar diagram

It is an extended form of a simple bar diagram. If there are two or more dependent variables associated with a single independent variables then multiple bar diagram is drawn. This diagram helps to compare the variables, separately. For example The Company wants to compare the costs between transportation and the advertisement that it had spent in the last five years. With the help the data given below, construct the multiple bar diagram.

Year	Transportation cost(Rs)	Advertisement cost(Rs.)
1985	1543	675
1986	1873	876
1987	1463	453
1988	1940	786
1989	2164	456

The multiple bar diagram for the above set of data is given below.



1.2.8 Sub divided bar diagram

It is another method of expressing the data through bar diagram. In subdivided bar diagram there are subdivisions within the bar so that each subdivision represent the components within this. The subdivided bar diagram facilitates in the comparison between the components and itself in a whole.

The expenditure of the family under different heads are given below. Express the following data in the form of subdivided bar diagram.

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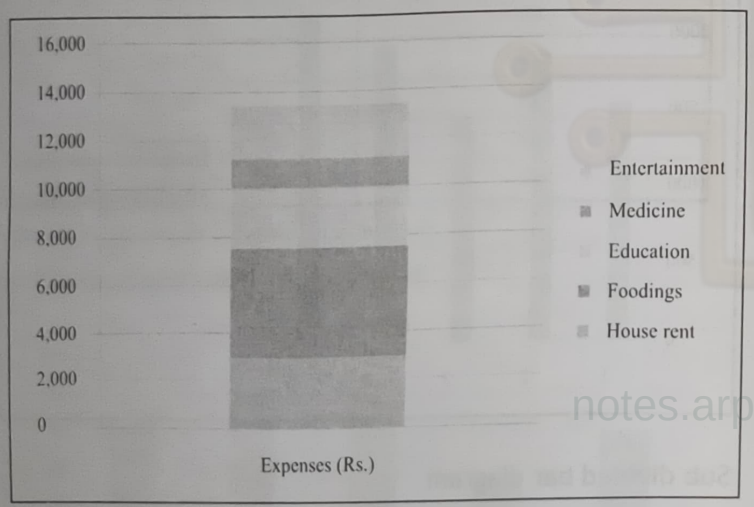
Particulars	Expenses (Rs.)
House rent	3,000
Foodings	4500
Education	2500
Medicine	1250
Entertainment	2175

The table for the cumulative expenses is given below

Particulars	Expenses (Rs.)	Cumulative Expenses
House rent	3,000	3,000
Foodings	4500	7,500
Education	2500	10,000
medicine	1250	11,250
Entertainment	2175	13,425
Total	13,425	

With reference to the data above the sub divided bar diagram can be drawn as follows:



Percentage bar Diagram: In the percentage diagram, all the variables are converted in terms of hundred. Then the subdivided bar diagram is constructed out of the converted data. In percentage bar diagram, the total value is always considered to be 100. This facilitates in comparison if there are two values to be compared. This can be shown by the example given below.

Example 6. The household expenditure of a family is given below. Construct the percentage bar diagram from the given figures

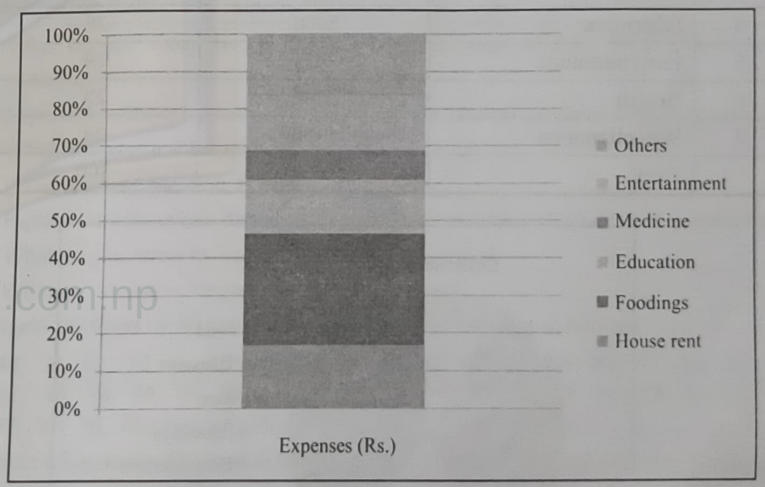
Particulars	Expenses (Rs.)
House rent	2,500
Foodings	4500
Education	2198
medicine	1250
Entertainment	2175
Others	2543

Solution

Converting the given information into percentages we get,

Particulars	Expenses (Rs.)	% expenditure	cumulative %
House rent	2,500	16.484	16.484
Foodings	4500	29.672	46.156
Education	2198	14.493	60.649
medicine	1250	8.242	68.891
Entertainment	2175	14.341	83.232
others	2543	16.768	100.000
Total	**15,166**		

This can be represented in the percentage diagram as follows



Two dimensional diagrams: In two dimensional diagrams the length as well as width of the bars is considered. Thus the area of the bar represents the given data. Rectangles, square and Pie-charts are two dimensional diagrams.

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1.2.9 Pie-diagram or pie chart

It is the circular representation of the data. In pie chart, the total sum of the variables is 360 degree. It is also the convenient tool for the comparison of the data. Each variable are converted into degrees by using the formula:

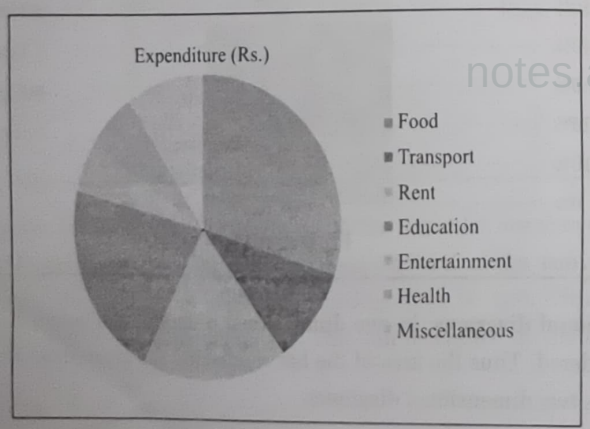
$$\text{Angle for the certain constrain} = \frac{\text{Value taken by the variable}}{\text{Sum of the variables}} \times 360$$

Example 7. The table below shows the monthly expenditure of a family. Construct the pie chart to represent the given information.

S.No.	Particulars	Expenditure (Rs.)
1	Food	3500
2	Transport	1200
3	Rent	2100
4	Education	2500
5	Entertainment	620
6	Health	750
7	Miscellaneous	1100

Solution

S.No.	Particulars	Expenditure(Rs.)	degrees
1	Food	3500	107
2	Transport	1200	37
3	rent	2100	64
4	Education	2500	76
5	Entertainment	620	19
6	Health	750	23
7	Miscellaneous	1100	34
	Total	11770	360



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Stem leaf display: It is an effective method of representing individual data into classes. In this method, the original data is well retained in the sense that the original data can be identified from diagram, unlike the frequency table. It is used to find the maximum concentration of data in the stem and minimum concentration as well.

Example 8. Consider the following marks obtained by 20 students draw the stem leaf display from the data.

22, 75, 54, 76, 53, 47, 45, 38, 17, 84, 72, 54, 19, 09, 39, 33, 51, 54, 55, 65, 66, 11, 13, 05, 26.

Solution

Stem	leaf	f
0	59	2
1	1379	4
2	26	2
3	389	3
4	57	2
5	134445	6
6	56	2
7	256	3
8	4	1

Thus above stem leaf display shows that the stem 5 has maximum concentration, i.e. 5 has leaf 6, and minimum concentration on the stem 8, i.e. 8 has leaf 1.

Exercise 1.2

1. a. What do you mean by classification and tabulation of data? How it is important in statistical analysis?
   b. Explain the difference between continuous and discrete variables with examples.
   c. What is histogram? Explain its uses.
   d. What is ogive? Mention its uses.
   e. Explain the role of one dimensional and two dimensional diagrams two present the data.
   f. What do you mean by inclusive and exclusive series?
   g. What is cumulative frequency? How it is formed?
2. The marks obtained by 50 students in computer sciences were as follows.

   40, 48, 50, 52, 52, 45, 48, 48, 52, 57, 48, 48, 47, 55, 57, 49, 47, 54, 55, 58, 54, 56, 32, 38, 59, 58, 34, 38, 58, 62, 36, 39, 71, 22, 62, 64, 76, 72, 26, 84, 64, 76, 28, 67, 27, 10, 85, 67, 67, 15.

   Construct a frequency distribution having suitable class and class interval. Draw a histogram of the so formed series and find mode graphically.

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3. Draw a suitable diagram to represent patient flow at Tilganga eye center since 1999 to 2003

Year:	1999	2000	2001	2002	2003
No. of patient:	84888	87313	1,27,017	1,20,955	1,21,098

4. Surface tension at 25 degree Celsius of various solvents is given below.

Solvent:	Ethyl Acetate	Cyclohexane	Benzene	Diethyl ether	chloroform
γ (Dyn/cm):	23.2	19.8	28.2	16.5	26.5

Represent the above diagram in the suitable diagram.

5. The following data are the number of residential houses was completed in first three months of the three different years of newly planned city.

Year/month	January	February	March
2021	5520	5678	5784
2022	8490	10,030	12290
2023	1294	1511	1720

Represent the above data by a subdivided bar diagram.

6. Types of waste generation on health care institutions in a valley are given below.

Types of waste	Waste in kg/day
Infections	289.28
Cotton, tissue, surgical procedures etc	240.90
Sharp needle only	48.38
Non Infections	1516.19
Saline Bottles	382.0

Represent the above information by Pie-diagram.

7. The following are the ages of 30 engineers who worked in a mega project. Construct stem and leaf display from these data.

35, 33, 46, 32, 45, 22, 21, 23, 38, 43, 64, 35, 39, 10, 56, 60, 34, 36, 27, 57, 12, 36, 34, 45, 55, 37, 44, 43, 27, 33.

8. The marks obtained by 25 students in the Java program are given below.

12, 23, 32, 45, 48, 46, 42, 36, 38, 33, 22, 11, 13, 8, 9, 24, 24, 25, 17, 16, 22, 24, 48, 45, 21.

Represent the above data in stem and leaf display.

9. Draw the pie-chart from the following data.

Expenditure Dollar

Items	Family A	Family B
Food	1200	1700
Clothing	300	500
Fuel and light	400	600
Education	450	500
Miscellaneous	250	300

10. The weight of 40 school children is recorded as below:

20, 37, 32, 29, 21, 40, 27, 35, 23, 23, 25, 41, 24, 42, 27, 39, 30, 44, 38, 28, 31, 23, 31, 40, 27, 35, 20, 41, 28, 42, 30, 22, 35, 37, 26, 28, 31, 25, 28,48.

Prepare the statistical table with following characteristics. a) A frequency distribution of with class interval 5. b) Less than and more than cumulative frequency c) Relative frequency distribution

11. The maximum temperature in degree Celsius of certain city recorded in 42 days of year is recorded as follows.

27.4, 23.5, 21.9, 28.6, 31.0, 30.9, 30.2, 27.3, 34.2, 27.4, 24.7, 22.7, 28.9, 25.6, 36.6, 22.2, 25.7, 22.9, 25.4, 27.3, 37.1, 21.3, 21.3, 23.1, 24.8, 37.8, 36.9, 22.3, 34.9, 28.8, 24.2, 32.6, 27.5, 28.8, 27.1, 29.5, 25.9, 23.4, 23.1, 38.8, 24.2, 20.8.

Construct a frequency distribution having suitable number of class and class interval from the above data, and find median and mode graphically.

12. Construct a grouped frequency distribution of the following data on the amount of time (in hours) that 80 college students devoted to leisure activities during a typical school week:

23, 24, 18, 14, 20, 24, 24, 26, 23, 21, 16, 15, 19, 20, 22, 14, 13, 20, 19, 27, 29, 22, 38, 28, 34, 32, 23, 19, 21, 31,16, 28, 19, 18, 12,27, 15, 21, 25, 16, 30, 17, 22, 29, 29, 18, 25, 20, 16, 11, 17, 12, 15, 24, 25, 21, 22, 17, 18, 15,21, 20, 23 18, 17, 15, 16, 26, 23, 22, 11, 16, 18, 20, 23,19, 17, 15, 20 and 10

After constructing a frequency distribution find cumulative frequencies, relative frequency and also draw the ogive and histogram of the so formed frequencies.

13. Draw a histogram and frequency polygon from the following data.

X:	5 - 9	10 - 14	15 - 19	20 - 24	25 - 29	30 - 34	39 - 39	40 - 44
f:	1	7	11	10	7	3	1	2

14. Draw the histogram of the following data.

X:	0 - 5	5 - 10	10 - 20	20 - 30	30 - 40	40 - 50	50 - 60	60 - 90	90 - 150
f:	1	3	11	25	41	36	16	15	9

(Note: Frequency density- frequency divided by its respective class interval is called frequency density. In the above problem first find f.d. as 0.5, 0.6, 1.1..... etc and put it into Y axis and variate values in X-axis then draw histogram).

15. Draw a stem leaf display from the following information.

6.4 7.9 9.5 5.2 7.9 7.6 10.2 9.8 9.2 10.7 5.9 8.0 7.1 5.7 7.9 6.5 8.6 6.8 7.5 6.9 8.0 8.9 8.5 6.2

16. The most popular colors for compact and sports cars in recent years are given in the table.

Colors:	Silver	Gray	Blue	Black	White	Brown	Green	Other
Percentage:	19	17	14	18	13	6	5	8

Use appropriate graphical display to describe these data.

17. A batch of cotton wool produced by the company was shown to have 79% with on defects, 15% slightly chipped, 10% discolored and 55% dirt. Construct pie chart using these data.

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18. The status of students in statistics class noted as below.

Status:	Freshman	Sophomore	Junior	Senior	Grad student	Total
Frequency:	5	23	32	35	10	105

Present the above data in appropriate diagram.

1.3 Measures of central tendency, location and variability

1.3.1 Measures of central tendency

Study of concentration of observation in the central part series is known as measure of central tendency. It is used to describe the middle or centre value of the data series and it also facilitate to compare two or more sets of data. To get single value that represents the entire data, to facilitate the comparison and to compute various statistical measures are the main objective of central tendency. The various measures of central tendency are:

a. Arithmetic mean (a.m.)
b. Median
c. Mode
d. Harmonic mean (H.M.)
e. Geometric mean (G.M.)

Arithmetic mean

Arithmetic mean is simple to calculate and it is commonly used of all averages. It is used to calculate the average value of the quantitative data and the mean for a group of observation is simply the sum of numerical values of each and every observation divided by the total number of observation.

Individual series

Let $x_1, x_2, \ldots, x_n$ be the set of n observation, then arithmetic mean (a.m.) denoted by $\bar{X}$ is defined by:

$$\bar{X} = \frac{\sum_{i=1}^{n} x_i}{n}, \text{ or simply } \bar{X} = \frac{\sum x}{n}.$$

Short cut method: Let A be the any assumed or arbitrary mean and $d = X - A$, then a.m. is given by

$$\bar{X} = A + \frac{\sum d}{n}$$

Discrete series

Let $x_1, x_2, \ldots, x_n$ be the set of observations, with frequencies $f_1, f_2, \ldots, f_n$ respectively then a.m. is given by $\bar{X} = \frac{\sum fx}{N}$

Short cut method: Let A be the any assumed or arbitrary mean and $d = X - A$, then a.m. is given by $\bar{X} = A + \frac{\sum fd}{N}$

Continuous series

In case of continuous series, let $x_1, x_2, \ldots, x_n$ be the set of mid value of the observations, with frequencies $f_1, f_2, \ldots, f_n$ respectively then a.m. is given by:

$\bar{X} = \frac{\sum fx}{N}$, here x is the mid value of the observations.

Short cut method: Change of origin and scale

Let $d_i = \frac{x_i - A}{h}$

or, $hd_i = x_i - A$

or, $hf_id_i = f_ix_i - A$

or, $h\frac{1}{N}\sum_{i=1}^{n} f_i d_i = \frac{1}{N}\sum_{i=1}^{n} f_i x_i - A\frac{1}{N}\sum_{i=1}^{n} f_i$

or, $h\frac{1}{N}\sum_{i=1}^{n} f_i d_i = \bar{x} - A$

$\therefore \quad \bar{x} = A + h\frac{1}{N}\sum_{i=1}^{n} f_i d_i$

Properties of Arithmetic mean

The sum of deviation taken from arithmetic mean is zero.

$$\sum_{i=1}^{n} (x_i - \bar{x}) = 0$$

Prrof: L.H.S. $= \sum_{i=1}^{n} (x_i - n\bar{x})$

$$= \sum_{i=1}^{n} x_i - n \cdot \frac{1}{n} \sum_{i=1}^{n} x_i$$

$$= 0$$

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Combined Mean

Let a series having sizes n_1 and its a.m. $\bar{X}_1$, another series having sizes n_2 and arithmetic mean $\bar{X}_2$ then combined a.m. denoted by $\bar{X}_{12}$ is defined as:

$$\bar{X}_{12} = \frac{n_1\bar{X}_1 + n_2\bar{X}_2}{n_1 + n_2}$$

Can be generalized for n sets of data

$$\bar{x}_{12..n} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2 + \ldots + n_n\bar{x}_n}{n_1 + n_2 + \ldots + n_n}$$

Weighted arithmetic mean

In case of grouped data, whether continuous or discrete if the frequencies of occurrence of items are highly different, in such a case we use proper weights to the items according to their relative importance and we compute weighted arithmetic mean. Let $x_1, x_2, \ldots, x_n$ be the set of observations and $w_1, w_2, \ldots, w_n$ be their respective weights, and then weighted arithmetic mean is given by:

$$\bar{X}_w = \frac{\sum wx}{\sum w}$$

Merits of Arithmetic mean

- It is clearly defined.
- It is easy to calculate and unique.
- It is based on all observations.
- It is useful for further statistical procedures.
- It is least affected by fluctuations of sampling values.

Demerits of mean

- It is affected by extreme values that are not representative of the rest of the data.
- It cannot be used for open ended series.
- It cannot be appropriate in qualitative data.
- It cannot locate graphically.

Example 9. The marks obtained by 8 students in statistics are given below.

72, 38, 84, 55, 60, 65, 24 and 75, calculate the mean marks of the students.

Solution

We have,

$$\bar{X} = \frac{\sum x}{n} = \frac{473}{8} = 59.12$$

Example 10. Calculate the arithmetic mean from the following frequency distribution.

X:	20	25	30	35	40	45	50
f:	3	8	4	5	9	6	7

Solution

The arithmetic mean in case of discrete series is given by:

$$\bar{X} = \frac{\sum fx}{N}$$

After calculation we have,

$\Sigma fx = 1535$, $N = \Sigma f = 42$

$$\therefore \quad \bar{x} = \frac{1535}{42} = 36.55$$

Example 11. The life in months of 50 randomly selected bulbs from a company is given below.

Life in month:	0 - 4	4 - 8	8 - 12	12 - 16	16 - 20	20 - 24
No. of bulb:	6	12	10	5	9	8

Calculate the mean life of the bulbs.

Solution

we have,

$\bar{X} = \frac{\sum fx}{N}$, where x is the mid value.

Life in month:	Mid value (x)	f	fx	d = (x -10)/4	fd
0 - 4	2	6	12	-2	-12
4 - 8	6	12	72	-1	-12
8 - 12	10	10	100	0	0
12 - 16	14	5	70	1	5
16 - 20	18	9	162	2	18
20 - 24	22	8	176	3	24
Total		50	592		23

$$\therefore \quad \bar{x} = \frac{592}{50} = 11.84$$, hence mean life of the bulbs produced by the company = 11.84 months.

Short cut method:

$$\therefore \quad \bar{x} = A + h\frac{1}{N}\sum_{i=1}^{n} f_i d_i$$

$$= 10 + 4 \times \frac{23}{50}$$

$$= 10 + 1.84$$

$$= 11.84 \text{ months.}$$

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Median

The variate value which divides the given series into two equal parts is called median and denoted by M_d. The numbers of observation between two sides of median are equal. Median is suitable in case of qualitative data and open ended series.

Individual series:

To find the median in case of individual series, first arrange the given series either ascending or descending order, then use

$$M_d = \text{Size of } \frac{(n+1)^{th}}{2} \text{ item.}$$

Example 12. Find median of the following series:

13, 17, 23, 29, 31, 37, 41

Solution

Here, n = 7

M_d = Size of $\frac{(7+1)^{th}}{2}$ item = 4^{th} item

Then median = 29,

If there are only 6 observations: 13, 17, 23, 29, 31 then

M_d = Size of 3.5^{th} item,

$M_d = \frac{(23+29)}{2} = 26$

Discrete series

In case of discrete series to find median,

First find the cumulative frequency (c.f.), then use,

$$M_d \text{ Position} = \text{Size of } \frac{(N+1)^{th}}{2} \text{ item, where } N = \Sigma f.$$

See the c.f. column and note the value corresponding to the c.f. equal or greater than $\frac{(N+1)}{2}$, which gives the median.

Example 13. Find the median from the following data.

x	f	cf
250	25	25
300	16	41
450	20	61
600	24	85
100	15	100
	100	

We have,

M_d = Size of $\frac{(N+1)^{th}}{2}$ item

= Size of $\frac{(100+1)^{th}}{2}$ = Size of 50.5^{th} item

$\therefore$ $M_d = 450 = 450.$

Continuous series

To find the median in case of continuous series, proceed as follows:

Prepare less than c.f.

To find the median class use M_d = Size of $\frac{N^{th}}{2}$ item.

See c.f. column equal to or just greater than $\frac{N^{th}}{2}$, which gives the median class.

Then use, $M_d = L + \frac{\frac{N}{2} - \text{c.f.}}{f} \times h$

Where L = lower limit of the M_d class, c.f. = cumulative frequency preceding the M_d class, f = simple frequency of the M_d class, h = class interval or class width

Example 14. Find the M_d of the following series.

x	f	c.f.
20 - 30	10	10
30 - 40	20	30
40 - 50	10	40
50 - 60	25	65
60 - 70	10	75
70 - 80	50	125
80 - 90	40	165
90 - 100	30	195
Total	**195**	

We have,

M_d class = Size of $\frac{N^{th}}{2}$ item

= Size of $\frac{195^{th}}{2}$

= Size of 97.5^{th} item

$\therefore$ M_d class = 70 – 80

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$L = 70,\ h = 10,\ f = 50,\ c.f. = 75$

$$M_d = L + \frac{\frac{N}{2} - c.f.}{f} \times h$$

$$= 70 + \frac{\frac{195}{2} - 75}{50} \times 10$$

$\therefore\ M_d = 74.5$

Merits of Median

- It is clearly defined, easy to calculate and understand.
- It is not affected by extreme observations.
- It can be computed for the data with open ended classes.
- It can be located graphically.
- It is suitable for qualitative data also.

Demerits of Median

- It is not based on all observations.
- It needs to make data arrangements.
- It has no implication in further mathematical treatment.
- It is affected by fluctuations of sampling than mean

Mode

The variate value which have maximum frequency (i.e. which is repeated a maximum number of times in a given series) is called mode. It can be used in case of qualitative as well as quantitative data set. It is more appropriate in nominal data series. In case of individual series there is no frequency so there is no mode i.e. mode is ill-defined. The series may be of unimodel, biomodel and of multimodel.

Discrete series

From definition we can get mode easily.

Example 15. Find mode of the given series.

X:	10	20	30	40	50
f:	2	6	9	4	3

Solution

Here maximum frequency is 9, whose variate value is 30.

Hence by definition, Mode = 30

Continuous series:

In case of continuous series model class can be finding by using definition. Then exact mode is found by using the following relation:

$$M_0 = L + \frac{\Delta_1}{\Delta_1 + \Delta_2} \times h$$

Where,

L = Lower limit of the model class

$\Delta_1 = f_1 - f_0, \qquad \Delta_2 = f_1 - f_2,$

f_0 = frequency preceding the model class

f_1 = frequency of the model class,

f_2 = frequency succeeding the model class

h = class interval

Example 16. Find mode from the following distribution.

x:	0 - 10	10 - 20	20 - 30	30 - 40	40 - 50	50 - 60	60 - 70
f:	4	12	15	20	32	14	13

Solution

Here the mode lies in the class (40 – 50), because it corresponds to the maximum frequency 32.

$f_1 = 32,\ f_2 = 14,\ f_0 = 20,\ h = 10$

$\Delta_1 = f_1 - f_0 = 32 - 20 = 12$

$\Delta_2 = f_1 - f_2 = 32 - 14 = 18$

We have,

$$M_0 = L + \frac{\Delta_1}{\Delta_1 + \Delta_2} \times h$$

$$= 40 + \frac{12}{12 + 18} \times 10 = 44$$

Merits of mode

- It is simple and easy to calculate.
- In some case it can be found by inspection.
- It can be computed for the data with open ended classes.
- It can be estimated by the help of histograms.
- It is not affected by extreme observations.

Demerits of mode

- It is ill defined in some individual series.
- It is not based on all observation.
- It is affected by fluctuations of sampling.
- It is not suitable for further mathematical analysis.

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Geometric Mean

Geometric mean of set of non- negative and none zero n observations is the n^{th} root of its product.

Individual series

Let $x_1, x_2, \ldots, x_n$ be the set of n none zero and non-negative observations. Then geometric mean denoted by G.M. is defined as:

$$G.M. = \sqrt[n]{(x_1, x_2, x_3, \ldots, x_n)}$$

$$\log G.M. = \frac{1}{n}(\log x_1 + \log x_2 + \ldots + \log x_n)$$

$$= \frac{\Sigma \log x_i}{n}$$

$$\therefore \quad G.M. = A.L.\left\{\frac{\Sigma \log x_i}{n}\right\}$$

Discrete series

Let $x_1, x_2, \ldots, x_n$ be the set of observations, with frequencies $f_1, f_2, \ldots, f_n$ respectively then G.M. is given by:

$$G.M. = A.L.\left\{\frac{f_i \log x_i}{N}\right\}$$

Continuous series

In case of continuous series G.M. is same as that of discrete series, but here x is the mid values.

Harmonic mean (H.M.)

Harmonic mean of a set of observations is the reciprocal of the arithmetic mean of the reciprocal of the given values.

Individual series

Let $x_1, x_2, \ldots, x_n$ be the set of n observations then harmonic mean (H.M.) is given as:

$$H.M. = \frac{n}{\sum_{i=1}^{n} \frac{1}{x_i}}$$

In case of **discrete series** H.M. is given as:

$$H.M. = \frac{N}{\sum_{i=1}^{n} \frac{f_i}{x_i}}$$

In case of discrete series also H.M. is same as that of discrete series.

Relation between different measures of central tendency

In case of symmetrical distribution mean, median and mode coincide.

i.e., mean = median = mode,

In case of moderately asymmetrical distribution, **Mode = 3 median – 2 mean**

The relation of a.m., G.M. and H.M. is a.m. ≥ G.M. ≥ H.M.

1.3.2 Partition values

The variate value which divides the given series into number of equal number of parts is called the partition values. The frequently partition values are quartiles, deciles and percentiles.

Quartiles

The variate values which divide the given series into four equal parts is called quartiles. There are three points to divide the series into four equal parts. The points are Q_1, Q_2 and Q_3.

Deciles

There are nine points to divide the given series into ten equal points. The points are denoted by $D_1, D_2, \ldots, D_9$..

Percentiles

Similarly there are ninety nine points to divide the given series into hundred equal points. The points are denoted by $P_1, P_2, \ldots, P_{99}$.

Partition values in Individual series

To find the different partition values in the individual series proceed as follows:

First arrange the series either ascending or descending.

Then Use,

$$\text{Position of } Q_i = \text{Size of } \frac{i(n+1)}{4}^{th} \text{ item. } (i = 1,2,3.)$$

$$\text{Position of } D_i = \text{Size of } \frac{i(n+1)}{10}^{th} \text{ item. } (i = 1,2,3.....,9)$$

$$\text{Position of } P_i = \text{Size of } \frac{i(n+1)}{100}^{th} \text{ item. } (i = 1,2,.....,99)$$

After using the above relation partition values can be calculated.

Discrete series

To find the partition values in the case of discrete series, first find c.f. of the given series.

Then use,

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Position of Q_i = Size of $\frac{i(N+1)}{4}^{th}$ item. (i = 1,2,3)

Position of D_i = Size of $\frac{i(N+1)}{10}^{th}$ item. (i = 1,2,.....,9)

Position of P_i = Size of $\frac{i(N+1)}{100}^{th}$ item. (i = 1,2,.....,99)

Check the c.f. column exactly equal or just greater than the value given by the above formula, that variate value of that class is the required partition values.

Continuous series

To find the partition values in the case of continuous series, first find c.f. of the given series. Then find the position of the partition class, using:

Position of Q_i = Size of $\frac{i(N)}{4}^{th}$ item. (i = 1,2,3)

Position of D_i = Size of $\frac{i(N)}{10}^{th}$ item. (i = 1,2,.....,9)

Position of P_i = Size of $\frac{i(N)}{100}^{th}$ item. (i = 1,2,.....,99)

After finding the partition class, use the following formula to find the exact partition values.

$$Q_i = L + \frac{\frac{iN}{4} - c.f.}{f} \times h$$

$$D_i = L + \frac{\frac{iN}{10} - c.f.}{f} \times h$$

$$P_i = L + \frac{\frac{iN}{100} - c.f.}{f} \times h$$

Where,

L = lower limit of the partition class

c.f. = Cumulative frequency preceding the partition class

f = simple frequency of the partition class

h = class width of the partition class

N = total number of the observation in the given series

Example 17. Calculate the upper and lower quartile of the following series:

20, 18, 15, 16, 19, 25, 12, 14, 22, 24

Solution

First we can arrange the series as, 12, 14, 15, 16, 18, 19, 20, 22, 24

Then we have,

Q_1 = Size of $\frac{i(n+1)}{4}^{th}$ item.

Q_1 = Size of $\frac{1(9+1)}{4}^{th}$ item

Q_1 = Size of 2.5th item

= 2^{nd} item + 0.5(3^{rd} item – 2^{nd} item)

= 14 + 0.5(15 -14)

= 14.5

Again,

Q_3 = Size of $\frac{3(9+1)}{4}^{th}$ item

= Size of 7.5^{th} item

= 7^{th} item + 0.5(8^{th} item – 7^{th} item)

= 20 + 0.5(22 – 20)

= 21

Example 18. From the following distribution of marks of 500 engineering students of university, find the minimum pass marks obtained, if the 20% of the students had failed and also find the minimum marks obtained by the top 25% of the students.

Marks in mathematics	0 - 20	20 - 40	40 - 60	60 - 80	80 -100
No. of students	50	100	240	60	50

Solution

x:	f	c.f.
0 - 20	50	50
20 - 40	100	150
40 - 60	240	390
60 - 80	60	450
80 - 100	50	500
Total	**500**	

If 20 % of the students had failed, means 80% students had passed. The minimum pass marks obtained by the students will be P_{20}.

We have,

20^{th} percentile lies in = Size of $20\left(\frac{N}{100}\right)^{th}$ 100^{th} item

= Size of $\frac{(20 \times 500)}{100^{th}}$ item

= 100^{th} item

$\therefore$ P_{20} lies in the class = 20 - 40

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$$P_i = L + \frac{\frac{iN}{100} - c.f.}{f} \times h$$

$$P_{20} = L + \frac{\frac{iN}{100} - c.f.}{f}$$

$$P_{20} = 20 + \frac{\frac{20 \times 500}{100} - 50}{100}$$

$$= 30$$

∴ Required minimum pass marks = 30.

Minimum marks obtained by top 25% students will be P_{75}.

P_{75} lies in the class $= \text{Size of } \left(\frac{75N}{100}\right)^{th} \text{ item}$

$= \text{Size of } \left(\frac{75 \times 500}{100}\right)^{th} \text{ item}$

$= 375^{th} \text{ item}$

∴ P_{75} lies in the class $= 40 - 60$

$$P_{75} = 40 + \frac{\frac{75 \times 500}{100} - 150}{240} \times 20$$

$$= 58.75$$

∴ Required minimum marks obtained by top 25% students = 58.75

Exercise 1.3

1. The marks obtained by 60 students in mathematics are given below.

Marks:	20	30	40	50	60	70
No. of students:	8	12	20	10	6	4

2. The life time in hours of randomly selected ten samples of certain types of battery is as follows.

123, 116, 122, 110, 175, 126, 125, 111, 118, 117

Find the mean and median of life time of battery.

3. The arithmetic mean of the given series is 30.5, find the missing value.

Mid-value:	10	20	?	40	50
Frequency:	8	10	20	15	7

4. From the following series find the missing frequency when mean is 15.38.

Mid-value	10	12	14	16	18	20
Frequency:	3	7	?	20	8	5

5. The following table presents the data for the marks obtained by 120 students of certain college, if the median mark of the students is 42.5, and then find the missing frequencies.

Marks:	20 - 25	25 - 30	30 - 40	40 - 45	45 - 50	50 - 55	55 - 60	60 - 70	70 - 80
f:	6	12	?	30	10	?	8	5	2

6. Find the missing frequency from the following series, if the model incomes of 47 workers of certain company are Rs. 27 thousands.

Income ('000 Rs.):	0 - 10	10 - 20	20 - 30	30 - 40	40 - 50
No. of workers:	5	?	15	?	7

7. From the following series calculate the mode and median.

X:	Below 10	10 - 12	12 - 14	14 - 16	16 - 18	18 - 20
f:	3	15	27	20	3	2

8. The temperature in degree Celsius in a chemical reactor was measured in every half hour under the same conditions and the results was noted as:

78.1, 79.2, 78.9, 80.2, 78.3, 78.8, 79.4

Calculate the mean, median, lower quartile and upper quartile.

9. The cold start engine ignition time of an automobile engine is being investigated by gasoline manufactured. The following time in second were obtained for a test vehicle.

1.79, 1.92, 2.62, 2.36, 3.09, 3.15, 2.53, 1.91

Calculate lower quartile, upper quartile and median of the series.

10. The marks obtained by some students are given below.

Mid-value:	10	20	30	40	50	60	70
No. of students:	7	8	13	29	35	9	3

If 78 students were passed in the examination, find the minimum pass marks.

11. One hundred Students appeared in the examination of statistics, the results of those who secured less than 80% marks are given below, if mean marks of all students were 52. Then find the mean marks of the students, who secured more than 80%.

Marks:	0 – 20	20 - 40	40 - 60	60 - 80
No. of students:	16	24	20	10

12. The mean marks obtained by 300 students in Numerical methods are 45. The mean of top 100 students was found to be 70 and mean of last 100 students was known to be 20. What are the mean marks of the remaining students?

13. One hundred twenty Students appeared for IQ test, the score obtained are given below.

Score:	0 – 20	20 – 40	40 - 60	60 – 80	80 – 100
Number of students:	10	30	36	30	14

Find the i) limits marks of centre 30% students ii) number of students who fail in IQ test if 35 score are required for passing the test iii) the percentage of students getting score more than 75.

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15. ... mode and median are 20 and 24 respectively, find the value of mean.

16. The mean of 200 observations was 50, later on it was found that the two observations are wrongly inserted 92 and 8 instead of 192 and 88, find the correct mean.

17. Calculate the mean, median and mode of the following series and describe the shape of the series.

X:	21 - 22	22 - 23	23 - 24	24 - 24	24 - 25
f:	7	13	22	10	8

18. In certain examination if 60% of the students pass the exam, find the minimum marks secured by the students who passed the exam and median marks secured by students.

Marks more than:	0	10	20	30	40	50
No. of students:	50	46	40	20	10	3

19. From the following frequency distribution find the limits of income of centre 70% workers and median income of the workers.

Income Rs.	500 - 600	600 - 700	700 - 800	800 - 900	900 - 1000
No. of workers:	150	300	500	200	50

20. 120 students appeared in centre entrance examination of M.sc engineering, the following are the marks distribution secured by them.

Marks:	0 - 20	20 - 40	40 - 60	60 - 80	80 - 100
No. of students:	10	30	36	30	14

Find: a) the limits of marks of middle 30% students b) the number of students failed, if 35 marks are required to pass the exam c) the percentage of students getting marks more than 75.

Answers

1. 41 **2.** 124.3, 120 **3.** 30 **4.** 12 **5.** 27, 20
6. 8, 12 **7.** Median = mode = 13.26 **8.** 78.98, 78.9, 78.3, 79.4
9. 1.91 sec, 2.85 sec, 2.44 sec. **10.** 33.46 **11.** 87.34 **12.** 45
13. 41.11 and 61.33, 33 students, 17.9% or 18.33 **14.** 28 **15.** 26
16. 50.9 **17.** 2547, 2541, 2536, the series is not symmetrical.
18. Minimum pass marks (fortieth percentile) =25, median = 27
19. Limits for centre 70% workers (P85 –P15) =Rs. 225, Median Rs 730
20. a. (41.11 – 61.33) **b.** 33 **c.** 17.9%

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1.3.3 Measure of dispersion

The study of concentration of observation in the central part of the series is known as measure of central tendency, which gives the approximate central single value that represents the entire data. The measure of central tendency does not explain how the values of observation are deviated or scattered from the central part of the series, such type of scatterings or deviation of observation from central part is studied in measure of dispersion.

The measure of scatterness or the deviation of the observation from the central part i.e. from the central value of the series is known as measure of dispersion. It provides the amount of variability present in the data series. Measure of dispersion is used to determine the reliability of averages, to compare two or more series with regard to their variability, quality control theory and hypothesis testing etc.

When dispersion is expressed in terms of original units of series, then it is called absolute measure. In case of single series or distribution absolute measure of dispersion is used. The relative measure of dispersion is the pure number expressed as a ratio or percentage and to compare two or more series relative measure of dispersion is used.

Characteristics of ideal measure of dispersion

- To be an ideal measure of dispersion following characteristics should be satisfied.
- It should be rigidly defined.
- It should be easy to calculate and understand.
- It should be based on all observations.
- It should be appropriate for further mathematical treatment.
- It should be least affected by fluctuations of sampling.

1.3.4 Various measures of dispersion

The following are the commonly used important measures of dispersion.

a. Range
b. Quartile deviation
c. Mean deviation
d. Standard deviation

Range

The difference between largest item and smallest item is called range. It is crudest or simplest measure of dispersion. The range is appropriate in case of meteorological data relating to temperature rainfall etc. The range has a significant role in describing the variability of a data set, if there are no outliers, if outlier exist the value of range can be misleading.

Range = Largest item – smallest item

$R = L - S$

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Coefficient of range

Range is absolute measure of dispersion, that's why to compare dispersion of two or more series range cannot be used. In such a case we need relative measure corresponding to the range, which is known as coefficient of range.

$$\text{Coefficient of range} = \frac{L - S}{L + S}$$

Quartile deviation

Quartile deviation is a measure of dispersion or variability based on quartiles. The absolute difference between the upper quartile and lower quartile is called interquartile range. The half of the interquartile range is called semi interquartile range or quartile deviation (Q.D.). The smaller the quartile deviation, the more closely packed the data is around the median.

$$\text{Q.D.} = \frac{Q_3 - Q_1}{2}$$

It is better measure of dispersion than range of the distribution. It makes use of 50% of the data and ignores remaining 50% so it is also not a reliable measure of dispersion. For relative measure of dispersion we use coefficient of Q.D.

$$\text{Coefficient of Q.D.} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

Merits

- It is easy to calculate and understand.
- It is not affected by extreme values.
- It can be calculated in the case of open-ended series, so it is appropriate measure of dispersion in case of open ended series.

Demerits

- It is not suitable for further mathematical calculation.
- It is affected by fluctuations of sampling.
- It is not based on all observations.

Example 19. To find the quartile deviation and coefficient of Q.D. of incubation period of small pox in 10 patients: 8 14 13 11 15 10 7 9 12 10

Solution

The given series can be arranged as

7 8 9 10 10 11 12 13 14 15

We have,

$$Q_1 = \text{Size of } \left(\frac{i(n+1)}{4}\right)^{th} \text{ item}$$

$$Q_1 = \text{Size of } \left(\frac{11}{4}\right)^{th} \text{ item}$$

$$= 2.75^{th} \text{ item,}$$

$$= 8.75$$

$$Q_3 = \text{Size of } \left(\frac{33}{4}\right)^{th} \text{ item}$$

$$= 8.25^{th} \text{ item}$$

$$Q_3 = 13.25$$

$$\text{Q.D.} = \frac{(Q_3 - Q_1)}{2}$$

$$= \frac{(13.25 - 8.75)}{2}$$

$$= 2.25$$

$$\text{Coefficient of Q.D.} = \frac{(Q_3 - Q_1)}{(Q_3 + Q_1)}$$

$$= \frac{4.5}{22} = 0.20$$

Mean Deviation

The sum of the deviation of the items either from the mean, median or mode when all observations are considered positive is called mean deviation (M.D.)

Individual series

Let $\bar{X}$ and M_d be the arithmetic mean and median of the arithmetic series of size n, here mode (M_0) is not defined, and then mean deviation is defined as:

$$\text{M.D.} = \frac{\Sigma |X - \bar{x}|}{n}$$

$$\text{M.D.} = \frac{\Sigma |x - M_d|}{n}$$

Discrete series:

$$\text{M.D.} = \frac{\Sigma f|x - \bar{x}|}{N}$$

$$\text{M.D.} = \frac{\Sigma f|x - M_d|}{N}$$

$$\text{M.D.} = \frac{\Sigma f|x - M_0|}{N}$$

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Continuous series

In case of continuous series M.D. is same as that of discrete series but here x is the mid values.

Coefficient of M.D. = (M.D. from mean/mean)
= (M.D. from median/median)
= (M.D. from mode/mode)

Merits

- It is based on all observations.
- M.D. is least when the deviations of the items are taken from median, but in practice M.D. from mean is mostly used.
- It is better than range and quartile deviation.

Demerits

- It is not appropriate in case of open ended series.
- It ignores the negative sign.
- M.D. form mode cannot calculate.

Standard Deviation

The positive square root of the sum of square of deviation taken from arithmetic mean is called standard deviation (s.d.) and it is generally denoted byσ . Standard deviation is more useful tool to measure the variation in statistics. The amount of pressure on a construction to calculate the points of breaking, the mobility of a group of visitors in a building, crowd management, water pressure at different levels, variation in machine or human production, precision of variation in manufacture process etc standard deviation is useful.

Individual series:

Direct Method: Let X be the variate takes $x_1, x_2, \ldots, x_n$ as the set of n observations then and $\bar{X}$ be their a.m. then s.d. σ is given by:

$$\sigma = \sqrt{\left\{\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2\right\}}$$

or, $$\sigma = \sqrt{\left\{\frac{\Sigma x^2}{n} - \bar{x}^2\right\}}$$

Variance:

$$\sigma^2 = \left\{\frac{\Sigma x^2}{n} - \bar{x}^2\right\}$$

Short cut method: Let A be the assumed mean and d = X – A (be the deviations) then s.d. can be calculated as:

$$\sigma = \sqrt{\left\{\frac{\Sigma d^2}{n} - \left(\frac{\Sigma d}{n}\right)^2\right\}}$$

Discrete series:

Direct method: In case of discrete series the s.d. can be computed as:

$$\sigma = \sqrt{\left\{\frac{1}{n}\sum_{i=1}^{n} f(x_i - \bar{x})^2\right\}}$$

or, $$\sigma = \sqrt{\left\{\frac{fx}{N} - \bar{x}^2\right\}}$$

Short cut:

$$\sigma = \sqrt{\left\{\frac{\Sigma fd^2}{N} - \left(\frac{\Sigma fd}{N}\right)^2\right\}}, \text{ where } d = X - A$$

Variance: $\sigma^2 = \left\{\frac{\Sigma fx^2}{N} - \bar{x}^2\right\}$ }

Continuous series:

$$\sigma = \sqrt{\left\{\frac{1}{n}\sum_{i=1}^{n} f(x_i - \bar{x})^2\right\}}$$

or, $$\sigma = \sqrt{\left\{\frac{\Sigma fx^2}{N} - \bar{x}^2\right\}}$$

Variance: $\sigma^2 = \left\{\frac{\Sigma fx^2}{N} - \bar{x}^2\right\}$

Short cut or step deviation method: Let A be the assumed mean and $d' = \frac{(X - A)}{h}$, h be class interval then s.d. can be calculated as:

$$\sigma = h \times \sqrt{\left\{\frac{\Sigma fd'^2}{N} - \left(\frac{\Sigma fd'}{N}\right)^2\right\}}$$

The Variance

The square of s.d. is called variance. The variance is one of the most important measures of dispersion. The variance is a measure that uses the mean as a point of reference.

- The variance of the data is small when the observations are close to the mean.
- The variance of the data is large when the observations are spread out from the mean.
- The variance of the data is zero (no variation) when all observations have the same value (concentrated at the mean).

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The variance is given by

$$\sigma^2 = \sum \frac{(X - \bar{X})^2}{n}$$

Coefficient of Variation (C.V.):

The ratio of standard deviation and arithmetic mean of the series is called coefficient of variation, and it is always expressed in percentage. It is a relative measure of variation.

$$C.V. = \frac{\sigma}{\bar{X}} \times 100\%$$

C.V. is unit-less. To determine the variability of two or more sets of data (i.e. to determine whether which set is more variation) and which set is more uniform or consistent C.V. is used. The series having more C.V. has more variation and the series having least C.V. is more uniform or consistent.

Example 20. Two persons participated in five shooting competitions and were able to hit the target correctly out of 15 shots as given below.

Competition A: 6 12 12 10 7

Competition B: 12 15 7 7 4

Find which of the competitor is more consistent in performance.

Solution

For A		For B	
X	X^2	Y	Y^2
6	36	12	144
12	144	15	225
12	144	7	49
10	100	7	49
7	49	4	16
47	473	45	483

$$\bar{X} = \frac{47}{5}$$

$$= 9.4$$

$$\bar{Y} = \frac{45}{5}$$

$$= 9$$

$$\sigma_x = \sqrt{\left\{\frac{X^2}{n} - \bar{X}^2\right\}}$$

$$= \sqrt{\left\{\frac{473}{5} - 88.36\right\}}$$

$$= 2.49$$

$$\sigma_Y = \sqrt{\left\{\frac{483}{5} - 81\right\}}$$

$$= 3.94$$

$$\text{C.V. for A} = \frac{\sigma x}{\bar{X}} \times 100\%$$

$$= \frac{2.49}{9.4} \times 100\%$$

$$= 26.48\%$$

$$\text{C.V. for B} = \frac{\sigma x}{\bar{Y}} \times 100\%$$

$$= \frac{3.94}{9} \times 100\%$$

$$= 43.77\%$$

Here C.V for A is less than C.V for B, hence A is more consistent than B.

Combined Standard deviation

Let $\bar{x}_1$ and $\bar{x}_2$ be the mean, σ_1 and σ_2 be the standard deviations of the two series having sizes n_1 and n_2, then combined s.d. denoted by σ_{12} is defined as:

$$\sigma_{12} = \sqrt{\frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{n_1 + n_2}}$$

Where, $d_1 = \bar{x}_1 - \bar{x}_{12}$, $d_2 = \bar{x}_2 - \bar{x}_{12}$

Example 21. A student obtained mean and standard deviation of 100 observations as 40 and 5 respectively. Later on it was discovered that he has wrongly copied as an observation 50 instead of 40. Calculate the correct mean and standard deviation.

Solution

$n = 100$, incorrect $\bar{X} = 40$, incorrect $\sigma = 5$

We have, $\bar{X} = \frac{\Sigma X}{n}$ or incur. $\Sigma x = n \times \bar{X} = 100 \times 40 = 4000$

or cor. $\Sigma x = 4000 - 50 + 40 = 3990$

Cor. $\bar{X}$ = cor. $\frac{\Sigma X}{n} = \frac{3990}{100} = 39.9$

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Again, $\sigma^2 = \frac{\Sigma x^2}{n} - \bar{X}^2$

or $\Sigma x^2 = n(\sigma^2 + \bar{X}^2)$ or incur. $\Sigma x^2 = 100(5^2 + 40^2) = 162500$

Or cur. $\Sigma x^2 = 162500 - 50^2 + 40^2 = 161600$

$\therefore$ cor. $\sigma^2 = \text{cor.} \frac{\Sigma x^2}{n} - \text{cor.}\ \bar{X}^2$

$= \frac{161600}{100} - 1592.01$

$= 23.99$

$\therefore$ correct $\sigma = 4.89$

Example 22. The average electricity consumption and standard deviation of two sets of households are as follows.

Statistic	set I	set II
Mean:	65 units	72 units
Standard deviation:	13 units	18 units
No. of households:	50	40

Which set is more variable in consumption?

Calculate the c.v. of the combined set.

Solution

i) For set I

C.V. $= \frac{\text{s.d.}}{\text{mean}} \times 100\%$

$= \frac{13}{65} \times 100$

$= 20\%$

For set II

C.V. $= \frac{18}{72} \times 100$

$= 25\%$

Set II has more C.V. hence set II is more variable than set I.

ii) $\bar{x}_1 = 65,\ \bar{x}_2 = 72, \sigma_1 = 13, \sigma_2 = 18, n_1 = 50, n_2 = 40$

Then combined s.d. denoted by σ_{12} σ_{12} is defined as:

$$\sigma_{12} = \sqrt{\frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{n_1 + n_2}}$$

Where,

$d_1 = \bar{x}_1 - \bar{x}_{12},\ d_2 = \bar{x}_2 - \bar{x}_{12}$

$$\bar{X}_{12} = \frac{n_1\bar{X}_1 + n_2\bar{X}_2}{n_1 + n_2}$$

Combined C.V. $= \frac{\sigma_{12}}{\bar{X}_{12}} \times 100\%$.

Then complete the solution by substituting the above values.

1.3.5 Box and Whisker plot

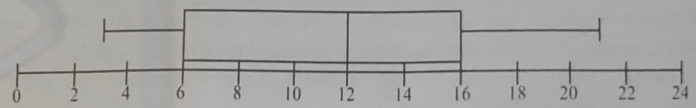
A box and whisker plot or box plot is a diagram constructed from the five number summary of a data set. To construct box plot we first draw equal interval scale on which to make our box plot. It is a visual representation of the distribution of data, larger distances in the diagram should correspond larger distances between values. Using equal interval scale we draw a rectangular box with one end at Q_1 and the other end at Q_3. Then we draw a vertical segment at the median value. Finally, we draw two horizontal segments on each side of the box, one down to the minimum value and one up to the maximum value; these segments are called the Whiskers.

Example 23. Draw a box-and-whisker plot for the data set: 3, 7, 8, 5, 12, 14, 21, 13, 18.

Solution

From given series, we had the five-number summary:

Minimum: 3, Q_1 : 6, Median: 12, Q_3 : 16, and Maximum: 21.



Notice that in any box-and-whisker plot, the left-side whisker represents where we find approximately the lowest 25% of the data and the right-side whisker represents where we find approximately the highest 25% of the data. The box part represents the interquartile range and represents approximately the middle 50% of all the data. The data is divided into four regions, which each represent approximately 25% of the data.

Exercise 1.4

1. From the following data calculate the range, lower quartile and upper quartile.
   31, 17, 29, 37, 23, 41, 47, 13
2. Calculate the mean deviation from mean and standard deviation from the following series.
   a. 10, 8, 4, 12, 16, 6, 20 b. 10, 11, 12, 13, 14, 15, 16
3. Calculate the range, mean deviation from mean and standard deviation of the following series.

X:	25	32	24	36	42
f:	8	5	7	10	4

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4. Calculate the quartile deviation and s...

Weight (in kg):	0 - 4	4 - 8	8 - 12	12 - 16	16 - 20	20 - 24
No. of children:	8	10	5	6	7	2

5. The following frequency distribution is the number tails obtained in tossing of four coins with experiment repeated 25 times, calculate the standard deviation and variance.

Tails occurred (x):	0	1	2	3	4
Frequency (f):	1	7	10	6	1

6. The expected life of two types of bulbs in months produced by two companies A and B and their sample sizes are given below.

Life of bulbs (in months):	5 - 10	10 - 15	15 - 20	20 - 25	25 - 30	30 - 35
Sample size from A:	2	9	29	44	11	5
Sample size from B:	9	11	18	32	17	13

It is assumed that the bulbs having uniform life will have long life time and the prices of the bulbs produced by two companies are equal, which companies bulbs would be preferred to buy the buyer, mention your reason.

7. Calculate the appropriate measure of dispersion from the following series.

Marks:	Below 35	35 - 37	38 - 40	41 - 43	Over 43
No. of students:	14	60	95	24	7

8. The mean weight of 150 students is 60 kg; the mean weight of boys is 70 kg with standard deviation of 10 kg and respective values for girls is 55 kg and 15 kg. Find the combined mean and number of boys and girls.

9. The mean weekly income and variance of 250 workers in community A and community B are Rs. 560 and 5497 respectively. The mean and variance of income of 100 workers of community A are 650 and 121 respectively. Find the variance of income in the community B.

10. The mean and variance of 200 observations are found to be 60 and 400. Later on it was found that the two observations are wrongly copied as 3 and 67 instead of 13 and 17. Find the correct mean and standard deviation.

11. A student obtained the mean and standard deviation of 100 observations as 40 and 5.1 respectively. It was later found that the two observations were wrongly copied 40 and 55 instead of 50 and 65 find the correct mean and standard deviation when the wrong observations are omitted and corrected observations are inserted.

12. A distribution consists of three components with frequencies 200, 250 and 100 having means 25, 10 and 15 with standard deviations 3, 4 and 5 respectively. Find the combined mean and standard deviation of the series.

13. Calculate the coefficient of variation for the following distribution giving 300 calls according to their duration in seconds.

Duration in sec:	0 - 30	30 - 60	60 - 90	90 - 120	120 - 150	150 - 180	180 -210
No. of calls:	9	17	43	82	81	44	24

Calculate the coefficient of variation in calls.

14. The score of two golfers for 10 rounds each are:

A:	58	59	60	54	65	66	52	75	69	52
B:	84	56	92	65	86	78	44	54	78	68

Which player is more consistent in performance and why?

15. The rainfall recorded in various places of five districts in a week are given below.

Rainfall (in mm):	45	50	55	60	65	70
No. of places	5	13	4	9	5	4

Find the variances in rain fall in different places.

16. The measurements of the diameters (in cm) of the plates prepared in factory are given below. Find standard deviation.

Diameter (cm)	21 - 24	25 - 28	29 - 32	33 - 36	37 - 40	41 -44
No. of plates	15	18	20	16	8	7

17. The average electricity consumption in two localities A and B are given in the table. Which localities have more variation in electricity consumption? Also calculate the coefficient of variation of the combined localities.

Measures:	Locality I	Locality II
Mean:	65 units	72 units
Standard deviation:	13 units	18 units
Number of households:	50	40

18. A factory produces two types of car batteries i.e. battery A and battery B. An experiment shows the life of the batteries in days which were recorded as follows.

Life in days:	500 - 700	700 - 900	900 - 1100	1100 - 1300	1300 - 1500
Battery A:	5	11	26	10	8
Battery B:	4	30	12	8	6

Compare the variability of two types of batteries using coefficient of variation.

19. The following are the number of minutes that a person had to wait for the train to work on 15 working days of the month.

10, 4, 13, 9, 5, 9, 2, 10, 3, 8, 6, 17, 2, 10, 15

Draw a box blot and interpret the findings.

20. Find the standard deviation of the data: 2, 3, 5, 7, 8. Then multiply each data by 4, and compare the standard deviation of new values.

21. Find the standard deviation of the observations: 7, 4, 8, 10, 11. And add3 to all the observations then find the standard deviation of new values and compare the results.

22. The daily wages of men and women workers in a certain village are given below.

Particulars	Women	Men
Number of workers:	60	40
Mean daily wages Rs.:	320	400
Variance of wages Rs.:	1600	3600

a. Men and women which have greater stability in daily wages? b) What will be the combined coefficient of variation in wages?

23. The number of road accidents occurred during the period of 10 months in two different cities and number of person injured in the accidents are recorded as follows.

Month:	City A	Number injured	City B	Number injured
Jan.	23	9	27	10
Feb.	29	2	31	8
Mar.	28	5	30	5

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Apr.	22	10	29	4
May	30	1	28	6
June	21	4	33	5
July	25	11	34	5
August	26	6	35	11
Sep.	27	3	36	2
Oct.	24	9	32	4

If the homogeneity in the number of road accidents occurred is the appropriate factor to minimize the accidents, which of the cities will easier for traffic?

24. The following data give the one way communicating times (in minutes) from home to the construction site for all 12 engineers working at a construction company.

35, 10, 22, 38, 31, 27, 50, 40, 16, 44, 25, 12

Calculate the coefficient of variation based on the given data.

25. Scientists are doing research on a special type of nutrition which helps to increase the weights of infants. They randomly select 65 infants from certain community and used them to fed special nutrition during the periods of two weeks. The weights gained in grams (X) and by the number of infants (Y) after the after 2 weeks were recorded as follows.

X:	0 - 5	5 - 10	10 - 15	15 - 20	20 - 25	25 - 30	30 - 35
Y:	3	7	10	15	20	6	4

The scientists are confident that if the variation in weight gain is less than 30%, they will be successful in developing the special nutrition. Do you think that the scientists are successful in their endeavor?

Answers:

1. 34, 18.5, 40 **2. a.** 4.12, 5.21 **b.** 1.72, 2 **3.** 8.05, 9.25
4. Quartiles = 4.6, 8.8, 15.67, s.d. =6.30 **5.** 0.92, 0.84
6. For A, C.V. = 24.4%, for B: C.V. = 33.7; Buyer preferred to buy manufacture of A, since it have least C.V.
7. Q.D. = 1.8 **8.** 15.28, 50, 100 **9.** 81 **10.** 59.8 and 20.09
11. 4.2, 5.56 **12.** 16, 8 **13.** 33.5%
14. C.V. for A = 11.97, C.V. for B = 21.13; A is more consistent than B
15. 60.21 **16.** 6
17. C.V. for I = 20%, C.V. for II =25%; locality II has more variation in consumption of electricity, Combined C.V. = 23.21%
18. C.V. for A = 21.67%, C.V. for B = 23.4%; battery B has more variation in life
20. Initial s.d. = 2.28, after multiplying by 4 s.d. = 9.12
21. s.d. = 2.45, after adding 3 to all s.d. = 2.45; no change in standard deviations
22. **a.** stability in wages of women **b.** 17.82%
23. for A: C.V. = 9.16%, for B: C.V. = 9.5% ; city A is easier then city B
24. 41.89% **25.** C.V. = 40.45%, scientists are not successful.

Unit 2

Probability

2.1 Probability theory

The concepts of chance and uncertainty are old as human civilizations. People have always uncertainty in each steps of daily life and either knowingly or unknowingly they have been using probability. The modern theory of probability has been developed since the seventeenth century and has been applied in diverse fields of study. Nowadays probability theory has become indispensable tool in the fields of engineering, information technology, medicine, science, agriculture, economics, management, etc.

The statistical technique of studying the chance of occurrence or not occurrence / happening or not happening of certain event or phenomenon is called probability. A probability is a numerical value that ranges from 0 to 1. Zero for an event that cannot occur and 1 for an event that is certain to occur.

2.1.1 Basic terminologies used in probability

i. **Trial and events:** Performing a random experiment is a trial and getting the results from the experiment is the events. Rolling a die is a trial and getting the faces 1, 2, 3, 4, 5 and 6 are the events.

ii. **Random experiment:** An experiment is said to be random if its outcome cannot be predicted before performing an experiment. Tossing a coin, rolling a die, drawing a card from pack of cards, etc. are all random experiments.

iii. **Sample space:** The set of all possible outcomes of the random experiments is called sample space and it is denoted by S. For example:

Tossing a coin : $S = \{H, T\}$

Rolling a die : $S = \{1, 2, 3, 4, 5, 6\}$ faces.

iv. **Simple events:** When the occurrences of only single event is considered, then it is called simple event. Occurrences of head (H) or tail (T) in a tossing a coin is simple event.

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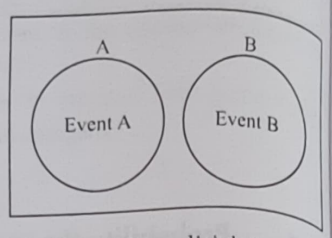
v. **Compound events:** An event is said to be compound when the occurrences of two or more events are considered. When a coin is tossed and die is rolled simultaneously, then occurrences of H1, H2, H3, ..., T4, T5 and T6 are compound events.

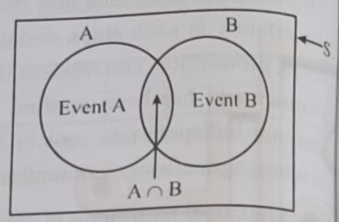
vi. **Equally likely events:** Two or more events are said to be equally likely, if the chance of occurrences of each of the events are equal. In rolling a unbiased die, any number faces out of 1, 2, 3, 4, 5, 6 is equally likely to come up.

vii. **Mutually exclusive events:** Two or more events are said to be mutually exclusive, if they cannot occur at the same time. Tossing a unbiased coin, head and tail cannot occur at the same time, either head (H) or tail (T) are mutually exclusive events.



Event A and event B are mutually exclusive if $A \cap B = \phi$. i.e., A and B are disjoint events, they have no common occurrences.

viii. **Not mutually exclusive (intersecting) event:** Two or more events are said to be not mutually exclusive or intersecting events, if they have common occurrence or can occur at the same time. A card is drawn from a pack of cards, the occurrences of king and red cards, queen and black cards are not mutually exclusive events, because they have common occurrences.



Let A and B be two events, it can be presented as diagram.

Here A and B are not mutually exclusive because they have common occurrences.

ix. **Independent events:** Two or more events are said to be independent, if the probability of occurrences of first does not affect the probability of second and so on. In tossing a fair coin several times head and tail are independent in each of the tosses. Probability of occurrences of head does not affect for the occurrences of head in any other tosses of such fair coin.

x. **Dependent events:** Two or more events are said to be dependent, if the probability of occurrences first affects the probability of occurrence of second and so on. A card drawn from pack of cards and without replacing second card is drawn, in this case probability of first draw of any one of 52 cards affects the probability of occurrence of any one of the cards in second draw. This is the example of dependent events.

xi. **Favourable cases or events:** The total number of outcomes of a random experiment which results in the happening or desired occurring event is called favourable events. For example, drawing a card from pack of card, the number of cases favourable to drawing a king is 4, the number of cases favourable to drawing a red card is 26. It is denoted by m.

xii. **Exhaustive cases of total number of cases:** Total number of possible outcomes of any random experiment is called exhaustive cases or total number of cases and it is denoted by n. For example, a die is rolled exhaustive cases $n = 6$, a coin is tossed exhaustive cases $m = 2$.

xiii. **Prime number:** The natural number which has only two divisors 1 and itself are called prime numbers. For example, 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, ..., etc.

xiv. **Impossible event:** The empty set ϕ is a subset of sample space S. So, it is impossible event.

2.1.2 Counting rules

When calculating probabilities, we always use favourable number of cases and total number of cases. When the random experiment results larger number of possible outcomes, the listing of outcomes becomes complicated. To solve such complications, the following counting rule is used.

i. **Multiplicative rule:** Let a random experiment is performed in a sequence of n times, in which first trial has k_1 possibilities. Second trial has k_2 possibilities, third trial has k_3 possibilities, and so on. Then

Total possible outcomes $= k_1 \cdot k_2 \cdot k_3, \ldots k_n$.

Let a coin is tossed and die is rolled simultaneously, then

Total possible outcomes $= k_1 \cdot k_2$

$= 2 \cdot 6$ [$\because$ For coin, $k_1 = 2$; for die $k_2 = 6$]

$= 12$

ii. **Combination:** Combinations of n things taken r at a time is denoted by ${}^nC_r$ and defined as:

$$ {}^nC_r = \frac{n!}{(n-r)!\, r!}, \; r \le n. $$

It does not consider the order of objects.

iii. **Permutation:** Permutation of n things taken r at a time is denoted by ${}^nP_r$ and defined as:

$$ {}^nP_r = \frac{n!}{(n-r)!}. $$

It considers the order of objects.

Example 1. Let we have 4 objects: A, B, C and D. Two objects are taken at a time.

Solution

Then using combination:

AB, AC, AD, BC, BD, CD; i.e., in 6 ways.

Using permutation:

AB, BA, AC, CA, AD, DA, BC, CB, BD, DB, CD, DC; i.e., in 12 ways.

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2.1.3 Approaches to probability

The different approaches to probability are:

i. **Classical / priori / mathematical approaches to probability:** Let n be mutually exclusive, equally likely exhaustive cases among them m are favourable to the happening or occurring event A, then probability of A is:

$$P(A) = \frac{\text{Favourable number of cases}}{\text{Total number of cases}}$$

or, $P(A) = \frac{m}{n}$

$\therefore P = \frac{m}{n}$

$p + q = 1, \quad 0 \le p \le 1.$

where, p = probability of success

q = probability of failure.

Classical / mathematical approach will fail to find the probability if the random experiment is performed in a sequence of n times and exhaustive number of cases in a trial is infinite.

ii. **Empirical / statistical approaches to probability:** If an experiment is repeated a large number of times under the same conditions, then the ratio of the number of repetition of an event to the total number of trials are calculated. Let an event A occurs m times in n repetition of random experiment, in limiting case when n becomes sufficiently large, the probability of happening event A is:

$$P(A) = \lim_{n \to \infty} \frac{m}{n}.$$

iii. **Modern / axiomatic approach to probability:** Let S be the simple space of random experiments and A be any event is S, then probability function which is defined in S and function P(A) satisfying the following axioms:

- P(A) is defined real and non-negative, i.e., $P(A) \ge 0$.
- P(S) = 1, axiom of certainty
- If $A_1, A_2, \ldots, A_n$ is any finite or infinite sequence of disjoint events of S, then

$$P(A_1 \cup A_2 \cup \ldots A_n) = P(A_1) + P(A_2) + \ldots + P(A_n)$$

$$\text{or, } \left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$

The above axioms are known as axiom of positiveness, certainty and additive respectively. The above mentioned three different approaches of probabilities are also known as objective approach of probability.

2.1.4 Subjective probability

Subjective approach of probability is based on individual's past experience, personal opinion and the analysis of particular situation or phenomenon. This type of personal guess is used in forecasting and making decisions in the situation when objective probability cannot be applied.

Example 2. A card is drawn from a pack of well-shuffled cards. What is the probability of getting (a) a King (b) a red Queen?

Solution

There are 52 cards in a pack of cards.

n = 52

a. There are 4 Kings, i.e., (n) = 4.

$$P(\text{King}) = \frac{m}{n} = \frac{4}{52} = \frac{1}{13}$$

b. There are 2 red queens, i.e., m = 2.

$$P(\text{red Queen}) = \frac{2}{52} = \frac{1}{26}$$

Example 3. There are 30 marbles numbered 1 to 30 in a bag. A marble is drawn at random, then what is the probability that it will be:

a. a prime number b. an even number c. a multiple of 3 or 6

Solution

Total number of cases (n) = 30

a. Prime numbers are

2, 3, 5, 7, 11, 13, 17, 19, 23, 29

m = 10

$$P(\text{it is a prime number}) = \frac{m}{n} = \frac{10}{30} = \frac{1}{3}.$$

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b. There are 15 even numbers, i.e., m = 15.

$$P(\text{it is an even number}) = \frac{m}{n} = \frac{15}{30} = \frac{1}{2}.$$

c. Events of getting multiple of 3 or 6 are:
3, 6, 9, 12, 15, 18, 21, 24, 17, 30, i.e., m = 10.

$$P(\text{it is a multiple of 3 or 6}) = \frac{m}{n} = \frac{10}{30} = \frac{1}{3}.$$

Example 4. 2 bulbs are chosen at random from a lot containing 12 bulbs, of which 4 are defective. What is the probability of choosing (a) both defective bulbs (b) both non-defective bulbs?

Solution

Total number of cases (n) = $^{12}C_2 = \frac{12!}{(12-2)!\,2!} = 66$.

a. Favourable cases for both defective (m) = $^4C_2 = 6$.

$\therefore$ P(Both defective bulbs) = $\frac{m}{n} = \frac{6}{66} = \frac{1}{11}$.

b. Favourable cases for both non-defective (m) = $^8C_2 = 28$.

$\therefore$ P(Both non-defective bulbs) = $\frac{m}{n} = \frac{28}{66} = \frac{14}{33}$.

Example 5. From a group of 3 scientists, 4 engineers and 5 economists, a sub-committee of 4 persons are to be formed randomly. What is the probability that the sub-committee will consist of (a) 2 scientists and 2 engineers (b) all economists?

Solution

4 persons can be selected from 12; (3 + 4 + 5) persons in $^{12}C_4$ ways.

$\therefore$ $n = {}^{12}C_4 = \frac{12!}{(12-4)!\,4!}$.

a. P(2 scientists and 2 engineers) = $\frac{{}^3C_2 \times {}^4C_2}{{}^{12}C_4} = 0.036$.

b. P(All 4 economists) = $\frac{{}^5C_4}{{}^{12}C_4} = 0.01$.

Example 6. A bag contains 4 white, 8 black, 6 red and 2 green balls. If 3 balls are drawn random, then find the probability that (a) all white balls (b) 1 white, 1 black and 1 green ball (c) 2 red and 1 black ball.

Solutions

Total number of balls in a bag = 4 + 8 + 6 + 2 = 20

3 balls can be selected from 20 balls (n) = $^{20}C_3$.

a. P(All white balls) = $\frac{{}^4C_3}{{}^{20}C_3} = \frac{4}{1140} = 0.0035$.

b. P(1W, 1B and 1G balls) = $\frac{{}^4C_1 \times {}^8C_1 \times {}^2C_1}{{}^{20}C_3} = \frac{64}{1140} = 0.056$.

c. P(2R and 1B balls) = $\frac{{}^6C_2 \times {}^8C_1}{{}^{20}C_3} = \frac{120}{1140} = 0.1052$.

2.1.5 Laws of probability

i. Probability of impossible event is zero. i.e., $P(\phi) = 0$.

ii. Probability of the complementary event is given as:

$P(\bar{A}) = 1 - P(A)$

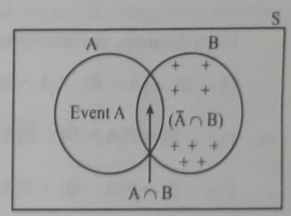
$P(A) = 1 - P(\bar{A})$

iii. For any two events A and B, defined in sample space S:

$P(\bar{A} \cap B) = P(B) - P(A \cap B)$.

Proof:

Two events A and B intersecting each other as shown in adjoining figure, we can write that:



$B = (A \cap B) \cup (\bar{A} \cap B)$.

or, $P(B) = P(A \cap B) \cup P(\bar{A} \cap B)$

In figure $(A \cap B)$ and $(\bar{A} \cup B)$ are mutually exclusive or disjoint events,

$P(B) = P(A \cap B) + P(\bar{A} \cap B)$

$\therefore$ $P(\bar{A} \cap B) = P(B) - P(A \cap B)$

Proved

Similarly, we can prove that $P(A \cap \bar{B}) = P(A) - P(A \cap B)$.

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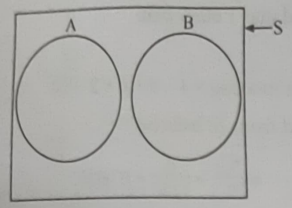
iv. Additive law of probability:

Case (a) when two events are mutually exclusive



Let A and B be two mutually exclusive and equally likely events in S, then

$P(A \text{ or } B) = P(A \cup B)$

$P(A \cup B) = P(A) + P(B)$

For three mutually inclusive events:

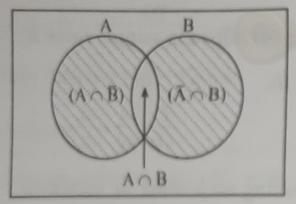
$P(A \cup B \cup C) = P(A) + P(B) + P(C)$

Case (b) when events are not mutually exclusive

Let A and B are two not mutually exclusive and equally likely events in S, then probability of at least one event– (Probability of A or B) is:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Proof:



From diagram, we can write

$(A \cup B) = (A \cap \bar{B}) \cup (A \cap B) \cup (\bar{A} \cap B)$

or, $P(A \cup B) = P(A \cap \bar{B}) \cup P(A \cap B) \cup P(\bar{A} \cap B)$

$\therefore$ $P(A \cup B) = P(A \cap \bar{B}) + P(A \cap B) + P(\bar{A} \cap B)$

Because all three events of R.H.S. are mutually exclusive,

or, $P(A \cup B) = P(A) - P(A \cap B) + P(A \cap B) + P(B) - P(A \cap B)$

$\therefore$ $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ *proved*

Similarly, for events A, B and C which are not mutually exclusive

$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$

Alternate: Probability of at least one event

Let A, B, C, D, ... be finite or infinite mutually exclusive and equally likely events in S, then

$P(\text{at least one event}) = 1 - P(\bar{A}) \cdot P(\bar{B}) \cdot P(\bar{C}) \cdot P(\bar{D}) \ldots$

Example 7. A box contains 5 white, 2 red, 8 yellow and 10 green bulbs. A bulb is drawn at random, what is the probability that it will be either white or red or yellow bulb?

Solution

Probability of drawing a white bulb $P(W) = \frac{5}{25}$

Probability of drawing a red bulb $P(R) = \frac{2}{25}$

Probability of drawing a yellow bulb $P(Y) = \frac{8}{25}$

Since the events are mutually exclusive

$$P(W \cup R \cup Y) = P(W) + P(R) + P(Y)$$

$$= \frac{5}{25} + \frac{2}{25} + \frac{8}{25}$$

$$= \frac{15}{25}$$

$$= \frac{3}{5}$$

Example 8. The chances of winning two race horses are $\frac{1}{3}$ and $\frac{1}{6}$ respectively. What is the probability that at least one will win, when the horses are running (a) in different races (b) in the same race?

Solution

Let A and B denote the two horses, then $P(A) = \frac{1}{3}$ and $P(B) = \frac{1}{6}$.

a. $P(\text{in different races}) = P(A \cup B)$

$= P(A) + P(B) - P(A \cap B)$

$= P(A) + P(B) - P(A) \cdot P(B)$ [$\because$ A and B are independent]

$= \frac{1}{3} + \frac{1}{6} - \frac{1}{3} \cdot \frac{1}{6}$

$= \frac{6 + 3 - 1}{18}$

$= \frac{8}{18}$

$= 0.44$

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$= \frac{1}{3} + \frac{1}{6}$

$= \frac{1}{2}$

Example 9. Three newspaper A, B and C are published in a certain city. It is estimated from a survey that 20% read A, 16% read B, 14% read C, 8% read A and B, 5% read A and C, 4% read B and C, 2% read all three papers. A person is chosen at random, then what is the probability that he (a) does not read any paper (b) does not read paper C (c) read A but not B?

Solution

$P(A) = 0.2$, $P(B) = 0.16$, $P(C) = 0.14$

$P(A \cap B) = 0.08$, $P(A \cap C) = 0.05$, $P(B \cap C) = 0.04$

$P(A \cap B \cap C) = 0.02$

a. $P(\text{Does not read any paper}) = P(\overline{A \cup B \cup C})$

$= 1 - P(A \cup B \cup C)$

or, $P(\overline{A \cup B \cup C}) = 1 - [P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)]$

$= 1 - [0.2 + 0.16 + 0.14 - 0.08 - 0.04 - 0.05 + 0.2]$

$= 0.65$

b. $P(\text{Does not read C}) = 1 - P(C)$

$= 1 - 0.14$

$= 0.86$

c. $P(\text{Read A but not B}) = P(A) - P(A \cap B)$

$= 0.2 - 0.08$

$= 0.12$

Example 10. Three groups of children contain 3 girls and 2 boys, 2 girls and 2 boys and 1 girl and 2 boys respectively. One child is selected from each group, find the probability that the selected children will be 1 girl and 2 boys.

Solution

1 girl and 2 boys can be selected from 3 groups as:

Cases	Group I	Group II	Group III
A	Girl	Boy	Boy
B	Boy	Girl	Boy
C	Girl	Boy	Girl

$P(A) = \frac{3}{5} \times \frac{2}{4} \times \frac{3}{4} = \frac{18}{80}$

$P(B) = \frac{2}{5} \times \frac{2}{4} \times \frac{3}{4} = \frac{12}{80}$

$P(C) = \frac{2}{5} \times \frac{2}{4} \times \frac{1}{4} = \frac{4}{80}$

$\therefore$ $P(\text{Selecting 1 girls and 2 boys}) = P(A \cup B \cup C)$

$= P(A) + P(B) + P(C)$

$= \frac{18}{80} + \frac{12}{80} + \frac{4}{80}$

$= \frac{17}{40}$

Example 11. In a company there are 2 civil engineers, 4 IT officers and 2 accountants. A committee of 4 members has to be formed. What is the probability that a committee contains (i) 1 civil engineer and 3 IT officers (ii) no civil engineers (iii) at least one IT officer?

Solution

i. $P(\text{1 civil engineer and 3 IT officers}) = \frac{{}^2C_1 \times {}^4C_3}{{}^8C_4}$

$= \frac{2 \times 4}{70}$

$= \frac{4}{35}$

ii. $P(\text{A committee contains no engineers}) = \frac{{}^2C_0 \times {}^4C_4 \times {}^2C_0}{{}^8C_4} + \frac{{}^2C_0 \times {}^4C_3 \times {}^2C_1}{{}^8C_4} + \frac{{}^2C_0 \times {}^4C_2 \times {}^2C_2}{{}^8C_4}$

$= \frac{1}{70} + \frac{4 \times 2}{70} + \frac{6 \times 1}{70}$

$= \frac{15}{70}$

$= \frac{3}{14}$

iii. $P(\text{At least one IT officers}) = 1 - P(\text{No IT officers})$

$= 1 - P(\text{2 civil engineers and 2 accountants})$

$= 1 - \frac{{}^2C_2 \times {}^2C_2}{{}^8C_4}$

$= 1 - \frac{1}{70}$

$= \frac{69}{70}$

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Example 12. Probability that the price of cement will rise, $P(A) = \frac{1}{2}$ and the probability that the price of cement will decrease $P(B) = \frac{1}{3}$. Find the probability that the price of cement will remain constant $P(C) = ?$

Solution

$P(A) = \frac{1}{2}$, $P(B) = \frac{1}{3}$, $P(C) = ?$

Since A, B and C are mutually exclusive

$P(A) + P(B) + P(C) = 1$

or, $P(C) = 1 - \frac{1}{2} - \frac{1}{3}$

$= \frac{1}{6}$

Hence the probability of the price of the cement remaining constant is $\frac{1}{6}$.

2.1.6 Multiplicative law of probability

a. When events are independent

Let A and B are any two independent events, then the probability of occurrence of A and B in two successive trials is equal to the product of the individual probabilities of the occurrence of A and B in two trials.

i.e., $P(A \text{ and } B) = P(A \cap B)$

$= P(A) \cdot P(B)$

If A, B and C are three independent events, then

$P(A \cap B \cap C) = P(A) \cdot P(B) \cdot P(C)$

b. When events are dependent

Let A and B be two dependent events, the probability of occurrence of A, affects the probability of occurrence of B and vice-versa, this is the case of conditional probability.

The condition probability of the event A that B has already occurred is denoted by P(A/B); read as probability of A given B is defined as:

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

or, $P(A \cap B) = P(B) \cdot P(A/B)$

Similarly, probability of B given A is

$$P(B/A) = \frac{P(A \cap B)}{P(A)}$$

or, $P(A \cap B) = P(A) \cdot P(B/A)$

For three dependent events A, B and C

$P\{(A \cap B) \cap C\} = P(A \cap B) \cdot P(C/A \cap B)$

$= P(A) \cdot P(B/A) \cdot P(C/A \cap B)$

or, $P(A \cap B \cap C) = P(B) \cdot P(A/B) \cdot P(C/A \cap B)$

$= P(A) \cdot P(C/A) \cdot P(B/A \cap C)$

$= P(C) \cdot P(B/C) \cdot P(A/B \cap C)$, etc.

Independence

An event A is said to be independent of another event B if

$P(A \cap B) = P(A) \cdot P(B)$

If conditional probability of A given B is equal to the unconditional probability of A, then A is said to be independent of B.

i.e., $P(A/B) = A$

$P(B/A) = B$

Example 13. A bag contains 5 red and 3 white balls. Two balls are drawn randomly one by one without replacement. Find the probability of (i) both balls are red (ii) first red and second ball is white.

Solution

Let R and W denotes the white balls and red balls respectively.

Total number of balls in a bag = 5 + 3 = 8.

Since the balls are drawn without replacement it is the case of dependent events.

i. $P(\text{Both balls are red}) = P(R_1 \cap R_2)$

$= P(R_1) \cdot P(R_2/R_1)$

$= \frac{5}{8} \times \frac{4}{7}$

$= \frac{5}{14}$

ii. $P(\text{First red and second white}) = P(R \cap W)$

$= P(R) \cdot P(W/R)$

$= \frac{5}{8} \cdot \frac{3}{7}$

$= \frac{15}{56}$

Example 14. Two cards are drawn one after another with replacement from a well shuffled pack of cards. Find the probability that both cards are King.

Solution

Since the cards are drawn with replacement it is the case of independent events.

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$$P(K_1) = \frac{4}{52}, P(K_2) = \frac{4}{52}$$

$$P(K_1 \cap K_2) = P(K_1) \cdot P(K_2)$$

$$= \frac{4}{52} \cdot \frac{4}{52}$$

$$= \frac{1}{169}$$

Example 15. *If six bullets of which three are blank are randomly inserted into a gun, what is the probability that the first three bullets fired will be blank?*

Solution

Total number of bullets = 6

Number of blank bullets = 3

P(The first three bullets fired will be blank)

$$= P(B_1 \cap B_2 \cap B_3)$$

$$= P(B_1) \cdot P(B_2/B_1) \cdot P(B_3) / (B_1 \cap B_2)$$

$$= \frac{3}{6} \cdot \frac{2}{5} \cdot \frac{1}{4}$$

$$= \frac{1}{20}$$

$$= 0.05$$

Example 16. Two balls are drawn in succession (one after another) from a bag containing 6 white and 2 black balls. What is the probability that the both balls are of same colour?

Solution

Let W_1, W_2 and B_1, B_2 be the white and black balls drawn in the first and second draw.

$$P(W_1) = \frac{6}{8}, \quad P(W_2/W_1) = \frac{5}{7}$$

$$P(B_1) = \frac{2}{8}, \quad P(B_2/B_1) = \frac{1}{7}$$

P(Both balls are of same colour) $= P(W_1 \cap W_2)$ or $P(B_1 \cap B_2)$

$$= P(W_1) \cdot P(W_2/W_1) + P(B_1) \cdot P(B_2/B_1)$$

$$= \frac{6}{8} \times \frac{5}{7} + \frac{2}{8} \times \frac{1}{7}$$

$$= \frac{30}{56} + \frac{2}{56}$$

$$= \frac{32}{56}$$

$$= 0.571$$

Example 17. If the probability that a communication system will have high fidelity is 0.81 and the probability that it will have high fidelity and high selectivity is 0.18, what is the probability that a system with high fidelity will also have high selectivity?

Solution

$P(F) = 0.81$

$P(F \cap S) = 0.18$

$P(S / F) = ?$

We have,

$$P(S/F) = \frac{P(F \cap S)}{P(F)}$$

$$= \frac{0.18}{0.81}$$

$$= 0.223$$

Example 18. A problem of statistics is given to three students; Hari, Rita and Gita; whose chances of solving it are $\frac{1}{2}, \frac{1}{3}$ and $\frac{1}{4}$ respectively. Find the probability that (i) all of them can solve the problem (ii) none of them can solve (iii) problem will be solved.

Solution

$$P(H) = \frac{1}{2}, \quad P(R) = \frac{1}{3}, \quad P(G) = \frac{1}{4}$$

i. P(All of them can solve) $= P(H \cap R \cap G)$

$$= P(H) \cdot P(R) \cdot P(G)$$

$$= \frac{1}{2} \cdot \frac{1}{3} \cdot \frac{1}{4}$$

$$= \frac{1}{24}$$

ii. P(None of them can solve) $= P(\bar{H} \cap \bar{R} \cap \bar{G})$

$$= P(\bar{H}) \cdot P(\bar{R}) \cdot P(\bar{G})$$

$$= \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4}$$

$$= \frac{6}{24}$$

$$= \frac{1}{4}$$

iii. P(Problem will be solved) $= 1 - P(\bar{H} \cap \bar{R} \cap \bar{G})$

$$= 1 - \frac{1}{4}$$

$$= \frac{3}{4}$$

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Example 19. From a group of 8 members, 5 engineers and 3 nurses; 3 members are selected at random. Calculate the probability that the selected members contains (i) only one nurse (ii) only one particular nurse (iii) at least one nurse.

Solution

Here, engineers = 5, nurses = 3, total = 8

i. $P(\text{Only one nurse}) = \frac{{}^5C_2 \times {}^3C_1}{{}^8C_3}$

$= \frac{15}{28}$

ii. $P(\text{Only one particular nurse}) = \frac{1}{3} \times \frac{15}{28}$

$= \frac{5}{28}$

iii. $P(\text{At least one nurse}) = \frac{{}^5C_2 \times {}^3C_1}{{}^8C_3} + \frac{{}^5C_1 \times {}^3C_2}{{}^8C_3} + \frac{{}^5C_0 \times {}^3C_3}{{}^8C_3}$

$= \frac{30}{56} + \frac{15}{56} + \frac{1}{56}$

$= \frac{23}{28}$

Example 20. A bag contains 12 balls of which 4 are white, five balls are drawn at random and laid aside, their colour unnoted. Another ball is drawn what is the probability that it is white?

Solution

White ball = 4, Unknown colour = 8

Let E_i; i = 0, 1, 2, 3, 4, denote the white ball drawn in first draw.

$P(E_0) = \frac{{}^4C_0 \times {}^8C_5}{{}^{15}C_5}$

$= \frac{7}{99}$

$P(E_1) = \frac{{}^4C_1 \times {}^8C_4}{{}^{12}C_5}$

$= \frac{35}{99}$

$P(E_2) = \frac{{}^4C_2 \times {}^8C_3}{{}^{12}C_5}$

$= \frac{42}{99}$

$P(E_3) = \frac{{}^4C_3 \times {}^8C_1}{{}^{12}C_5}$

$= \frac{14}{99}$

$P(E_4) = \frac{{}^4C_4 \times {}^8C_1}{{}^{12}C_5}$

$= \frac{1}{99}$

Let A be the event of drawing single white ball in second draw.

$P(A/E_0) = \frac{4}{7}, \quad P(A/E_1) = \frac{3}{7}, \quad P(A/E_2) = \frac{2}{7}$

$P(A/E_3) = \frac{1}{7}, \quad P(A/E_4) = 0$

$\therefore$ P(White ball)

$P(A) = P(E_0) \cdot P(A/E_0) + P(E_1) \cdot P(A/E_1) + P(E_2) \cdot P(A/E_2) + P(E_3) \cdot P(A/E_3) + P(E_4) \cdot P(A/E_4)$

$= \frac{7}{99} \times \frac{4}{7} + \frac{35}{99} \times \frac{3}{7} + \frac{42}{99} \times \frac{2}{7} + \frac{14}{99} \times \frac{1}{7} + \frac{1}{99} \times 0$

$= \frac{237}{693}$

$= \frac{1}{3}$

Example 21. If the probability that a married man will vote in given election is 0.5 and woman is 0.6. The probability that a woman will vote in the given election given her husband will vote is 0.90. Find the probability that (i) both husband and wife will vote in the election (ii) at least one of them will vote in the election.

Solution

Let E_1 and E_2 denote the married man and woman who will vote in election.

$P(E_1) = 0.5, \quad P(E_2) = 0.6, \quad P(E_2/E_1) = 0.90$

i. $P(\text{Both will vote}) = P(E_1 \cap E_2)$

$= P(E_1), P(E_2/E_1)$

$= 0.5 \times 0.9$

$= 0.45$

ii. $P(\text{At least one of them will vote}) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$

$= 0.5 + 0.6 - 0.45$

$= 0.65$

Example 22. Probability that a new Pokhara Airport will get an award for its design is 0.16, the probability that it will get an award for the efficient use of the materials is 0.24 and the probability that it will take both awards is 0.11. What is the probability that (i) it will get at least one of the award? (ii) it will get only one of the two awards?

Solution

Let E_1 and E_2 denote the events that airport will get an award for its design and efficient use of materials.

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$P(E_1) = 0.16, \quad P(E_2) = 0.24$

$P(E_1 \cap E_2) = 0.11$

i. P(Getting at least one of the awards) $= P(E_1 \cup E_2)$

$= P(E_1) + P(E_2) - P(E_1 \cap E_2)$

$= 0.16 + 0.24 - 0.11$

$= 0.29$

ii. The probability of getting one of the award

$P(E_1 \cap \bar{E}_2)$ or $P(\bar{E}_1 \cap E_2) = P(E_1 \cap \bar{E}_2) + P(\bar{E}_1 \cap E_2)$

$= P(E_1) \cdot P(\bar{E}_2) + P(\bar{E}_1) \cdot P(E_2)$

$= 0.16 \times 0.76 + 0.84 \times 0.24$

$= 0.32$

Example 23. A construction company is bidding for two contracts A and B. The probability that the company will get contract A is $\frac{3}{5}$. The probability that the company will get contract B is $\frac{1}{3}$ and the probability that the company will get both the contracts is $\frac{1}{8}$. What is the probability that the company will get contract A or B?

Solution

We have,

$P(A) = \frac{3}{5}, \quad P(B) = \frac{1}{3}, \quad P(A \cap B) = \frac{1}{8}$

$P(A \text{ or } B) = P(A \cup B)$

$= P(A) + P(B) - P(A \cap B)$

$= \frac{3}{5} + \frac{1}{3} - \frac{1}{8}$

$= 0.81$

2.1.7 Baye's theorem

Let $E_1, E_2, \ldots, E_n$ be mutually exclusive events of sample space S with $P(E_i) \neq 0$, 1, 2, ..., n then form any arbitrary event A, which is subset of S such that $P(A) > 0$, we have

$$P(E_r/A) = \frac{P(E_r) \cdot P(A/E_r)}{\sum_{i=1}^{n} P(E_i)\, P(A/E_i)}, \quad 1 \le r \le n$$

Proof:

Let S be the sample space and A be any arbitrary events in S. From Diagram we can write A is subset of S and

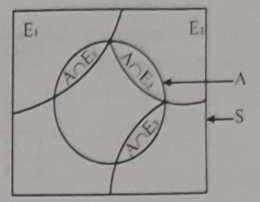
$S = (E_1 \cup E_2 \cup E_3 \ldots \cup E_n)$

Also,

$A = A \cap S$

$= A \cap [E_1 \cup E_2 \cup E_3 \ldots \cup E_n)$

$= (A \cap E_1) \cup (A \cap E_2) \cup (A \cap E_3) \ldots \cup (A \cap A_n)$



Here $(A \cap E_1), (A \cap E_2) \ldots (A \cap A_n)$ are mutually exclusive so,

$P(A) = P[(A \cap E_1) \cup (A \cap E_2) \cup (A \cap A_3) \ldots \cup (A \cap A_n)]$

$= P(A \cap E_1) + P(A \cap E_2) + P(A \cap E_3) \ldots + P(A \cap A_n)$

$\therefore \quad P(A) = \sum_{i=1}^{n} P(A \cap E_i)$

$= \sum_{i=1}^{n} P(E_i) \cdot P(A/E_i) \qquad \ldots.(1)$

For any particular event E_r, $1 \le r \le n$ is

$P(A \cap E_r) = P(A) \cdot P(E_r/A)$

$\therefore \quad P(E_r/A) = \frac{P(A \cap E_r)}{P(A)}$

Using (1) we can write

$$P(E_r/A) = \frac{P(E_r) \cdot P(A/E_r)}{\sum_{i=1}^{n} P(E_i)\, P(A/E_i)} \quad \textit{proved.}$$

Baye's theorem defines the probability of an event based on the prior knowledge of the conditions related to the events. In case if we know the conditional probability (prior) we can easily find the reverse (posterior) probabilities using Baye's theorem. If we know the conditional probability of an event than Baye's theorem is used to find reverse probabilities.

Example 24. A man is known to speak the truth 2 out of 3 times. He throws a die and reports that the number obtained is 4. Find the probability that the number obtained in actually a 4.

Solution

Let E_1 be the event that four is obtained and E_2 be the event four is not obtained. Let A be the arbitrary event that the man reports that number four is obtained.

$P(E_1) = \frac{1}{6}$ [∵ When a die is rolled there are 6 faces; Probability that four occurs is $\frac{1}{6}$]

$P(E_2) = 1 - \frac{1}{6} = \frac{5}{6}$ [Probability that four does not occur]

$P(A/E_1) = \frac{2}{3}$ [∵ P(Man reports 4 and it is actually 4) = $\frac{2}{3}$]

$P(A/E_2) = \frac{1}{3}$ [∵ P(Man reports 4 and it is not a 4) = $\frac{1}{3}$]

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Probability that number obtained is actually four

$$P(E_1/A) = \frac{P(E_1) \cdot P(A/E_1)}{P(E_1) \cdot P(A/E_1) + P(E_2) \cdot P(A/E_2)}$$

$$\therefore \quad P(E_1/A) = \frac{\frac{1}{6} \times \frac{2}{3}}{\frac{1}{6} \times \frac{2}{3} + \frac{5}{6} \times \frac{1}{3}}$$

$$= \frac{\frac{1}{9}}{\frac{1}{9} \times \frac{5}{18}}$$

$$= \frac{1}{9} \times \frac{18}{7}$$

$$= \frac{2}{7}$$

$\therefore$ Required probability $= \frac{2}{7}$.

Example 25. It is observed that 50% of mails are spam. There is a software that filters spam mail before reaching the inbox. The accuracy for detecting a spam mail is 99% and chance of tagging a non-spam mail as spam mail is 5%. If certain mail is tagged as spam, what is the probability that it is not a spam mail?

Solution

Let E_1 and E_2 denote the event of spam and non-spam mail. A denote the software detecting a spam mail.

Here,

$P(E_1) = 50\% = 0.5$ (probability of spam mail)

$P(E_2) = 0.5$ (probability of non-spam mail)

$P(A/E_1) = 0.99$

$P(A/E_2) = 0.05$

$\therefore$ Probability that it is not spam mail given it is spam mail

$$P(E_2/A) = \frac{P(E_2) \cdot P(A/E_2)}{P(E_1) \cdot P(A/E_1) + P(E_2) \cdot P(A/E_2)}$$

$$= \frac{0.05 \times 0.5}{0.5 \times 0.99 + 0.5 \times 0.05}$$

$$= \frac{5}{104}$$

$\therefore \quad P(E_2/A) = 4.8\%$.

Example 26. An insurance company has insured 4000 doctors, 8000 skilled workers and 12000 businessmen. The chances of doctors, skilled workers and businessman dying before the age of 68 is 0.01, 0.03 and 0.05 respectively. If one of the insured people dies before 68, find the probability that he is doctor.

Solution

$$P(D) = \frac{4000}{24000} = \frac{1}{6}$$

$$P(S) = \frac{8000}{24000} = \frac{1}{3}$$

$$P(B) = \frac{12000}{24000} = \frac{1}{2}$$

Let A be the event of death of insured person

$P(A/D) = 0.01$

$P(A/S) = 0.03$, $P(A/B) = 0.05$

Probability of doctor dying given he is insured

$$P(D/A) = \frac{P(D) \cdot P(A/D)}{P(D) \cdot P(A/D) + P(S) \cdot P(A/S) + P(B) \cdot P(A/B)}$$

$$= \frac{0.01 \times \frac{1}{6}}{0.01 \times \frac{1}{6} + 0.03 \times \frac{1}{3} + 0.05 \times \frac{1}{2}}$$

$$= \frac{1}{22}$$

$$= 0.045$$

Example 27. If three persons A, B and C have chances of being selected as a chief engineer of a cement company is in the ratio 4 : 1 : 2. The chances for them to introduce new strategies are 0.3, 0.8 and 0.5 respectively. If a change has taken place, find the probability that, it is due to selection of B.

Solution

We have,

$$P(A) = \frac{4}{7}, P(B) = \frac{1}{7}, P(C) = \frac{2}{7}$$

Let S denote the event of new strategies introduced.

Then $P(S/A) = 0.3$, $P(S/B) = 0.8$, $P(S/C) = 0.5$

The change has taken place, probability it is due to B

$$P(B/S) = \frac{P(B) \cdot P(S/B)}{P(A) \cdot P(S/A) + P(B) \cdot P(S/B) + P(C) \cdot P(S/C)}$$

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$$= \frac{0.3 \times \frac{1}{7}}{0.8 \times \frac{4}{7} + 0.3 \times \frac{1}{7} + 0.5 \times \frac{2}{7}}$$

$\therefore$ $P(B/S) = \frac{0.0429}{0.6431} = 0.066.$

Example 28. The production plants A and B make wire cables that are sent to a common distributors. 40% of the cables are sent to distributors from plant A and remaining 60% from plant B. Among the cables at plant A 99% meet the strength specification and at plant B 98% meet the strength specification. The distributors selects one cable at random from among all cables in the stock and found to meet the strength specifications. What is the probability that the cable was produced at plant A? (ii) Let A and B be the event that cables manufactured from plant A and Plant B and Y be the event that meets the strength specification, N be the event that does not meet the strength specifications; are events A and Y independent? Give your reason.

Solution

Let A and B be the events produced by two plants and S denote they meet the strength specification.

$P(A) = 0.4, \quad P(B) = 0.6$

$P(S/A) = 0.99, \quad P(S/B) = 0.98$

i. P(A randomly selected cable was produced by plant A is

$$P(A/S) = \frac{P(A) \cdot P(S/A)}{P(A) \cdot P(S/A) + P(B) \cdot P(S/B)}$$

$$= \frac{0.4 \times 0.99}{0.4 \times 0.99 + 0.6 \times 0.98}$$

$\therefore$ $P(A/S) = 0.40$

ii. Since A and B denote the events that a cable manufactured from plant A and B respectively.

$P(A) = 0.4, \quad P(B) = 0.6$

Also given that Y denote the event that they meet the strength specification and N do not meet strength specification.

$P(Y/A) = 0.99, \quad P(N/A) = 0.01$

$P(Y/B) = 0.98, \quad P(N/B) = 0.02$

We have,

$$P(A/Y) = \frac{P(A) \cdot P(Y/A)}{P(A) \cdot P(Y/A) + P(B) \cdot P(Y/B)}$$

$$= 0.4$$

Since $P(A/Y) = P(A)$, so the event A and Y are independent.

Example 29. A company has two plants to manufacture the screws. Out of 100 screws, plant I produces 80% and plant II manufacturers 20% of the screws. Where as production from plant I and II screws rated 85% and 65% as standard quality. (i) One screw is selected at random from the whole consignment and was found to be standard quality, what is that probability that it was manufactured by plant I? (ii) What is the probability that the selected screws being defective?

Solution

Let E_1 and E_2 be the events that screws manufactured by plant I and II respectively.

$P(E_1) = 0.8, \quad P(E_2) = 0.2$

Let A denote event that screws are rated quality.

i. Probability that selected screws found to be standard quality was produced by plant I

$$P(E_1/A) = \frac{P(E_1) \cdot P(A/E_1)}{P(E_1) \cdot P(A/E_1) + P(E_2) \cdot P(A/E_2)}$$

$$= \frac{0.8 \times 0.85}{0.8 \times 0.85 + 0.2 \times 0.65}$$

$$= 0.84$$

ii. Probability that selected screws found to be standard quality i.e., non-defective

$$= P(E_1) \cdot P(A/E_1) + P(E_2) \cdot P(A/E_2)$$

$$= 0.8 \times 0.85 + 0.2 \times 0.65$$

$$= 0.81$$

$\therefore$ The probability that the selected screws being defective = 1 – 0.81 = 0.19.

Example 30. The HAL Cooperation wishes to improve the resistance of its personal computers to disk-drive and keyboard failures. At present the design of the computer is such that disk-drive failure occur only one-third as often as keyboard failures. The probability of simultaneous disk-drive and keyboard failure is 0.05. (i) If the computer is 80% resistant to disk-rive and / or keyboard failure, how low must the keyboard failure be? (ii) If the keyboard is improved so that it fails only twice as often as the disk-drive (and the simultaneous failure probability is still 0.05), will the disk-drive failure probability part (i) yield a resistance to disk-drive and / or keyboard failure higher or lower than 90%?

Solution

Let A and B denote the events of computer has a disk-drive and keyboard failure respectively.

$P(A) = \frac{1}{3} \cdot P(B)$

$\therefore$ $P(B) = 3P(A)$

Also given, $P(A \cap B) = 0.05$

i. Since the computer is 80% resistant to disk-drive / or keyboard failure

$P(A \cup B)^c = P(\bar{A} \cap \bar{B}) = 0.8$

or, $P(A \cup B) = 1 - P(A \cup B)^c = 1 - 0.8 = 0.2$

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We have

$P(A \cap B) = P(A) + P(B) - P(A \cap B)$

or, $0.2 = P(A) + 3P(A) - 0.05$

$\therefore \quad P(A) = \frac{0.25}{4} = 0.0625$

$\therefore$ Probability of disk-drive failure = 0.0625.

ii. $P(B) = 2P(A) = 2 \times 0.0625 = 0.125$

Also $P(A \cap B) = 0.05$

Then $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$= 0.0625 + 0.125 - 0.05$

$\therefore P(A \text{ or } B) = 0.1375$

Hence the probability of resistance to disk-drive or / and keyboard failure is

$P(A \text{ or } B)^C = P(\bar{A} \text{ and } \bar{B}$ [$\because (A \cup B)^C = (A^C \cap B^C)$ Demorjan's Law]

$= 1 - P(A \cup B)$

$= 1 - 0.1375$

$= 0.8625$

= 86.25, which is lower than 90%

Example 31. A company has 2 machines that produce PVC pipes. An older machine produces 23% defective pipes, which the new machine produces only 8% defective pipes. In addition, the new machine produces 3 times as many pipes as the older machine does. (i) What is the probability that randomly chosen pipe produced by the company is defective? (ii) Given that a randomly chosen pipe was tested and found to be defective, what is the probability that the new machine produced it?

Solution

Let A and B be two machines, i.e., older and new machines that produces PVC pipes.

Given their new machine produces 3 times pipes as older does.

i.e., $B = 3A$

or, $\frac{B}{A} = \frac{3}{1}$

or, $P(B) = \frac{3}{3+1} = 0.75$

$P(A) = \frac{1}{4} = 0.25$

Probability of older machine P(A) = 0.25

Probability of new machine P(B) = 0.75

Let D denote the defective pipe.

$P(D/A) = 0.23, \qquad P(D/B) = 0.08$

i. P(Randomly chosen pipe produced is defective

$P(D) = P(A) \cdot P(D/A) + P(B) \cdot P(D/B)$

$= 0.25 \times 0.23 + 0.75 \times 0.08$

$= 0.1175$

ii. P(Randomly chosen defective pipe from new machine)

$$P(B/D) = \frac{P(B) \cdot P(D/B)}{P(A) \cdot P(D/A) + P(B) \cdot P(D/B)}$$

$$= \frac{0.75 \times 0.08}{0.25 \times 0.23 + 0.75 \times 0.08}$$

$$= \frac{0.06}{0.1175}$$

$= 0.5106$

$\therefore$ Randomly chosen pipe found defective from new machine = 51.06%.

Example 32. A salesman has a 60% chance of making sale to each customer. The behaviour of successive customer is independent. If two customers A and B enter, what is the probability that the salesman will make a sale to A or B?

Solution

Since a salesman has 60% chance of making sale to each customer

$P(A) = 0.6, \qquad P(B) = 0.6$

$P(A \cap B) = 0.6 \times 0.6 = 0.36$ [$\because$ A and B are independent]

We have,

$P(A \text{ or } B) = P(A \cup B)$

$= P(A) + P(B) - P(A \cap B)$

$= 0.6 + 0.6 - 0.36$

$\therefore \quad P(A \text{ or } B) = 0.84$

Example 33. An oil explorer orders seismic tests to determine whether oil is likely to be found in a certain drilling area. The seismic tests have a known reliability. When oil does exist in the testing area, the test will indicate so 85% of time; when oil doe not exist in the test area 10% of the time the test will indicate that it does exist. The explorer believes that the probability of existence of oil deposit in the area is 0.4. If the test is conducted and indicates the presence of oil, what is the probability that an oil deposit really exists?

Solution

Let T denote the seismic test positive and O for the area exists oil.

$P(T/O) = 0.85, \quad P(T/\bar{O}) = 0.1$

$P(O) = 0.4, \qquad P(\bar{O}) = 0.6$

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$$\therefore \quad P(O/T) = \frac{P(O) \times P(T/O)}{P(O) \times P(T/O) + P(\bar{O}) \times P(T/\bar{O})}$$

$$= \frac{0.4 \times 0.85}{0.4 \times 0.85 + 0.6 \times 0.1}$$

$$= 0.85$$

Note:

If odds in favour of A be m : n, then $P(A) = \frac{m}{n+n}$

If odds in against of A be m : n, then $P(A) = \frac{n}{m+n}$

Example 34. The odds against A solving a problem as 8 to 6 and odds in favour of B solving the same problem is 14 to 10. What is the probability that (i) both A and B will solve it? (ii) A solves it but B fails to solve it?

Solution

$$P(A) = \frac{6}{8+6} = \frac{6}{14}$$

$$P(B) = \frac{14}{14+10} = \frac{14}{24}$$

i. $P(A \cap B) = P(A) \cdot P(B)$

$$= \frac{6}{14} \cdot \frac{14}{24} = \frac{1}{4}$$

ii. $P(A \cap \bar{B}) = P(A) \cdot P(\bar{B})$

$$= \frac{6}{14} \times \frac{10}{24} = \frac{5}{28}$$

Example 35. Ram speaks truth in 70% of the cases and Shyam 80% of the cases. In what percentages of cases they likely contradict each other in stating the same fact?

Solution

$P(R) = 0.7, \quad P(\bar{R}) = 0..3$

$P(S) = 0.8, \quad P(\bar{S}) = 0.2$

P(Ram and Shyam contradict to each other stating the same fact)

$P(R\ \bar{S})$ or $P(\bar{R} \cap S) = P(R) \cdot P(\bar{S}) + P(\bar{R}) \cdot P(S)$

$= 0.7 \times 0.2 + 0.3 \times 0.8$

$= 0.14 + 0.24$

$= 0.38$

$\therefore$ Required % of contradiction to each other = 38%.

Example 36. It is known that 25% of the males and 10% females are unmarried in a certain village consisting of equal number of males and females. A person is selected at random, what is the probability that (i) he is male? (ii) she is unmarried and female? (iii) he is married and male?

Solution

	Married	Unmarried	Total
Male	75	25	100
Female	90	10	100
Total	165	35	200

Let A denote married, B unmarried, M for male and F for female. Then

i. $P(M) = \frac{100}{200} = \frac{1}{2}$

ii. $P(F \cap B) = \frac{10}{200} = \frac{1}{20}$

iii. $P(M \cap A) = \frac{75}{200} = \frac{3}{8}$

iv. P(She is female given that she is married)

$$P(F/A) = \frac{P(F \cap A)}{P(A)}$$

$$= \frac{\frac{90}{200}}{\frac{165}{200}}$$

$$= \frac{90}{165}$$

$$= 0.667$$

Example 37. A physical therapist at certain college knows that volleyball team will play 40% of its games on artificial turf this season. He also knows that volleyball player's chance of incurring a knee injury are 60% higher if he is playing on artificial turf instead of grass. If player's probability of knee injury grass is 0.40, what is the probability that (i) A randomly selected volleyball player incure a knee injury? (ii) A randomly selected volleyball player with knee injury incurred the injury playing on grass?

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Solution

Let T be the event that volleyball team play on artificial turf. G be the event that team plays on grass.

Here,

$P(T) = 0.30, \quad P(G) = 0.75$

and E be the event that the player has knee injury.

$P(E/G) = 0.40$

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It is given that, $P(E/T) = P(E/G) + 60\%$ of $P(E/G)$

$= 1.6\, P(E/G)$

$= 1.6 \times 0.40$

$= 0.64$

$\therefore \quad P(E/T) = 0.64$

i. P(Randomly selected player incurs knee injury)

$P(E) = P(T) \cdot P(E/T) + P(G) \cdot P(E/G)$

$= 0.3 \times 0.64 + 0.7 \times 0.40 = 0.472$

ii. P(Randomly selected player with knee injury, incurred injury playing on grass)

$$P(G/E) = \frac{P(G \cap E)}{P(E)}$$

$$= \frac{P(G) \cdot P(E/G)}{P(E)}$$

$$= \frac{0.70 \times 0.40}{0.472} = 0.59$$

Example 38. The probability that a contractor will be plumbing contract is $\frac{2}{3}$ and the probability that he will not get an electric contract is $\frac{5}{9}$. If the probability of getting at least one contract is $\frac{4}{5}$, what is the probability that he will get both the contracts?

Solution

Let A and B denote the contract will get plumbing and electric contract. Then

$$P(A) = \frac{2}{3}$$

$$P(B) = 1 - P(\bar{B}) = 1 - \frac{5}{9} = \frac{4}{9}$$

$P(A \cup B) = P(\text{Gets at least one contract}) = \frac{4}{5}$ (given)

We have

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$$= \frac{2}{3} + \frac{4}{9} - \frac{4}{5}$$

$$= \frac{30 + 20 - 36}{45}$$

$$= \frac{14}{45}$$

Hence, the contractor will get both the contract $= \frac{14}{45}$.

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Exercise 2.1

1. A bag contains 9 red, 7 white and 4 black balls, a ball is drawn at random, what is the probability of drawing a red balls?
2. The probability that an integrated circuit chip will have defective etching is 0.12, the probability that it will have a crack defect is 0.29 and the probability that it has both defects is 0.07. Are the etching defect and crack defect of integrated circuit statistically independent?
3. A small city has fire engine and one ambulance available, for emergencies. The probability that the fire engine is available when needed is 0.98, and the probability that the ambulance is available when called is 0.92. In the event of an injury resulting from a burning building, find the probability that both the ambulance and fire engine will be available, assuming they operate independently.
4. Suppose A and B are mutually exclusive events for $P(A) = 0.3$ and $P(B) = 0.5$. What is the probability that (i) Either A or B occurs? (ii) A occurs but B does not ? (iii) Both A and B occurs?
5. A statistics class for engineers consists of 25 industrial, 10 mechanical, 10 electrical and 8 civil engineering students. If a student is selected at random by the teacher to answer a question, find the probability that the student chosen is (i) an industrial engineering major (ii) a civil engineering or an electrical engineering major?
6. Ram and Shyam are interested to attend in the international seminar. The chance of attending a seminar by Ram is 0.6 and that of Shyam is 0.3. The probability both of them attend the seminar is 0.18, what is the probability that at least one of them will attend the seminar?
7. A man and his wife appear in an interview for two vacancies in the same post. The probability of husband's selection is $\frac{1}{7}$ and probability of wife's selection is $\frac{1}{5}$. What is the probability that (i) only one of them is selected? (ii) both of them are selected?
8. A researcher has to consult a recently published book. The probability of its being available is 0.5 for library A and 0.7 for library B. Assuming the two events are independent what is the probability that the book being available in library A and not in library B?
9. A piece of electronic equipment has two essential parts A and B. In the past part A failed 30% of the times and B failed 20% of times, and both parts failed simultaneously 5% of the time. Assuming both parts must operate to enable the equipment to function, what is the probability that the equipment will function?
10. The odds that A speaks the truth are 3 : 2 and the odds that B speaks the truth are 5 : 3. In what percentage of cases are they likely to contract each other on an identical point?
11. A can hit a target 4 times with 5 shots. B can hit it 3 times with 4 shots and C can hit it 3 times with 5 shots. Each fire a volley, what is the probability that (i) only two shots hit the target? (ii) only one shot hit the target?
12. The probability that a person stopping at a petrol pump will get his tyre checked is 0.12, the probability that he will get oil checked is 0.29 and the probability that he will get both checked is 0.07. Find the probability that a person stopping at this pump (i) will have neither his tyres nor oil checked (ii) his oil checked will also have his tyres checked.

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13. During the study of bike accidents, the Highway Safety Council found that 60% of all accidents occur at night, 52% are alcohol related and 37% occur at night and alcohol related. What is the probability that an accident occurred at night given that it was alcohol related?

14. In Kathmandu city, 25% of all cars emits excessive amount of pollutants. If the probability is 0.99, that a car emitting excessive amounts of pollutants will fail the city vehicular test and the probability is 0.17, that a car not emitting excessive amounts of pollutants will nevertheless fail the test. What is the probability that car which fail the test actually emits exclusive amounts of pollutants?

15. A company produces electric scooters of which 10% are defective. Each scooter is inspected before being exported, but the inspector will incorrectly classify an item 10% of the time (only items classified goods are shipped). What proportion of items shipped is defective?

16. A company producing electric relays has three manufacturing plants producing 50, 30 and 20 percentage respectively of its product. Suppose that the probabilities that a relay manufactured by these plants is defective are 0.02, 0.05 and 0.01 respectively.
    i. If a relay is selected at random from the output of the company, what is the probability that it is defective?
    ii. If a relay is selected at random is found to be defective, what is the probability that it was manufactured by plant 2?

17. Toyota Car Company produces cars at two factories labelled A and B. 10% of the cars produced by factory A are found to be defective, while 5% of the cars produced at factory B are defective. If A produces 100,000 cars per year and factory B produces 50,000 cars per year. Find:
    i. The probability of purchasing a defective car from the manufacture.
    ii. If a car purchased from the manufacture is defective, what is the probability that it comes from factory B?

18. There are three machines A, B and C producing 1000, 2000 and 3000 articles per hour respectively. These machines are known to be producing 10, 40 and 90 defective items per hour respectively. One article is selected at random from an hour production of three machine and found to be defective. From which machine the defective article is produced?

19. A company buys tyres from two suppliers I and II. Supplier I has a record of delivering tyres containing 10% of the defective, whereas supplier II has a defective rate of only 5%. Suppose 40% of the current supply come from supplier I. If a tyre is taken from his supply and found to be defective, find the probability that it is from supplier I?

20. The chances of A, B and C becoming a Chief Engineer of a certain hydropower company are 4 : 2 : 3. The probabilities that the new transmission line will be introduced in A, B and C becomes Chief Engineer are 0.3, 0.5 and 0.8 respectively. If the new transmission line has been introduced, what is the probability that Mr. A is appointed as Chief Engineer?

21. In a class if 75 students, 15 students were considered to be very intelligent, 45 as medium and rest below average. The probability that a very intelligent, medium and below average students will fail in examination is 0.005, 0.05 and 0.15 respectively. If a randomly selected students is known to have passed the examination, what is the probability that he is below the average?

22. Tree roads construction firms X, Y and Z bid for certain contracts from past experience. It is estimated that the probability that X will be awarded the contract is 0.40, while for Y and Z the probabilities are 0.35 and 0.25. If X does receive the contract, the probability that the work will be satisfactorily complete on time is 0.75 for Y and Z there probabilities are 0.80 and 0.70. (i) What is the probability that the work will be completed satisfactorily? (ii) The work was completed satisfactorily, what is the probability that Y was awarded the contract?

23. A problem of Mathematics is given to three students A, B and C whose chances of solving it are $\frac{1}{3}, \frac{1}{4}$ and $\frac{1}{5}$ respectively. What is the probability that (i) all of them can solve? (ii) None of them can solve? (iii) Exactly one can solve ? (iv) Problem will be solved?

24. A student knew only 60% of the questions in a test each with 5 answers. He simply guessed while answering the questions in the test, what is the probability that he knew the answer to a question given that he answers correctly.

    Hint: Let E_1 and E_2 be the event that he knew the correct answer and guessing the correct answer and A denote the correct answer.

    $P(E_1) = 0.6, P(E_2) = 0.4$

    $P(A/E_1) = 1, P(A/E_2) = \frac{1}{5} = 0.2$

    $P(E_1/A) = ?$, use Baye's theorem

25. Three groups of children contain respectively 3 girls and 1 boy, 2 girls 2 boys, 1 girl 2 boys. One child is selected from each group, what is the probability that the selected children will be of 1 girl and 2 boys?

26. There are two machines in a factory, the machine M_1 working 24 hours without failure is 0.8 and probability of machine M_2 working 24 hours without failure is 0.5. Find the probability that (i) both machine will work 24 hours without failure. (ii) only one machine will work for 24 hours without failure. (iii) at least one of them will work 24 hours without failure.

27. A software engineer applied for a job in two company X and Y. He estimates that the probability of his being selected in company X is 0.7 and being rejected in company Y is 0.5 and the probability of at least one of his application being rejected is 0.6. What is the probability that he will be selected in one of the company?

28. Mr. Sunil feels that the probability that he will fail in Mathematics course is 0.2, and probability that her girl friend will pass in the same course is 0.9 and the probability that both will pass the course is 0.72. What is the probability that (i) at least one of them will pass the course? (ii) both will fail in the course?

29. The probabilities that three drivers will be able to drive home safely after drinking are $\frac{1}{2}, \frac{1}{4}$ and $\frac{1}{5}$ respectively. If they set out to drive home after a party, what is the probability that at least one driver will drive home safely?

30. The probability that Nepal wins a cricket match against Pakistan is given to $\frac{2}{3}$. If Nepal and Pakistan play three matches, what is the probability that (i) Nepal will lose all three matches? (ii) Nepal will win at least one of the matches? (iii) Nepal will win all three matches?

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31. There are two bags, first bag contains 4 white and 6 black balls and second bag contains 5 white and 7 black balls. A ball is transferred from first bag to second bag, then a ball is drawn from second bag. What is the probability that it is a white ball?

32. There are two identical boxes containing 3 white and 4 black balls. A box is selected at random and a ball is drawn, what is the probability that (i) it is a white ball? (ii) the white ball is drawn from first box?

33. The human resource department of a software company has records which show the following analysis of 200 IT workers.

Age years	Bachelor degree	Master's degree
Under 30	90	10
30 - 40	20	30
Over 40	40	10

If one of the IT worker is selected at random from the company, find the probability that (i) he has a bachelor degree. (ii) he is a master degree given that he is over 40. (iii) he is under 30 and have bachelor degree.

34. An advertising executive is studying the television viewing habit of married men and women, during prime time hours. On basis of part viewing records, the executive has determined that during prime time, husbands are watching TV 60% of the time. When the husband is watching TV, 40% of the time, the wife is also watching TV, when the husband is not watching TV, 35% of the time the wife is also watching TV. Find the probability that (i) if the wife is watching TV the husband is also watching TV? (ii) the wife is watching TV in prime time.

35. Two sets of candidates are competing for the position on the board of directors of a company. The probabilities that the first and second set will win 0.6 and 0.4 respectively. If the first set wins the probability of introducing a new product is 0.9 and the corresponding probability if the second set wins is 0.4. What is the probability that a new product will be introduced?

Answers

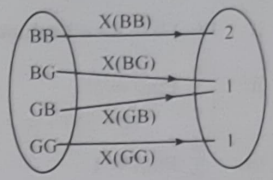
1. 0.45 2. No 3. 0.9016 4. 0.8, 0.3, 0 5. $\left(\frac{25}{53}, \frac{18}{53}\right)$
6. 0.72 7. $\left(\frac{2}{7}, \frac{1}{35}\right)$ 8. 0.15 9. 0.55 10. $\frac{19}{40}$
11. 0.45, 0.17 12. 0.66, 0.24 13. 0.74 14. 0.2837 15. 0.121
16. 0.27, 0.556 17. 0.0833, 0.8 18. Machine C = 0.641 19. 0.5714
20. 0.26 21. 0.18 22. 0.755, 0.37 23. $\left[\frac{1}{60}, \frac{2}{5}, \frac{13}{30}, \frac{3}{5}\right]$
24. 0.88 25. $\frac{13}{22}$ 26. 0.4, 0.5, 0.9 27. 0.8 28. 0.98, 0.02
29. $\frac{7}{10}$ 30. $\frac{1}{27}, \frac{26}{27}, \frac{8}{27}$ 31. 0.4154 32. 0.564, 0.379
33. $\frac{3}{4}, \frac{1}{5}, \frac{9}{20}$ 34. 0.632, 0.38 35. 0.70

2.2 Random variables and probability distribution

2.2.1 Random variable

In any trial or experiment, the outcome will occur randomly. We can assign a real number to each outcome of such experiment. Thus "a variable whose values are determined by the outcomes of the random experiment is called random variable."

Let a couple is planning to have two children, where B denotes boy and G denotes a girl. The possible outcomes will be BB, BG, GB, GG. Let X denote the number of occurrences of boy in sample space, then the values of X will be 0, 1 and 2.



Symbolically,

$$X = X(W)\begin{cases} 2 & \text{if } W = \text{BB} \\ 1 & \text{if } W = \text{BG or GB} \\ 0 & \text{if } W = \text{GG} \end{cases}$$

Definition: A random variable (r.v.) is a real valued function X(W), defined on sample space S (domain) and range $(-\infty, \infty)$ in random experiment. There are two types of random variables; they are discrete and continuous random variables. They are generally denoted by capital letters X, Y, Z, ..., etc.

2.2.2 Probability distribution

A table listing of all the possible outcomes of a random experiment and the probability associated with each of the outcome is called probability distribution.

Let a coin is tossed 3 times, the possible outcomes will be: HHH, HHT, HTT, TTT, TTH, THH, THT, HTH; where H denote the face head and T be the face tail. Let X denote the number of occurrences of head, then the values of X will be: 0, 1, 2, 3. The mentioned experiment and their outcome can be shown as probability distribution of X.

Event	X: (No. of head)	P(X)
(No head) TTT	0	$\frac{1}{8}$
(One head) HTT	1	$\frac{3}{8}$
(Two head) HTT	2	$\frac{3}{8}$
Three head (HHH)	3	$\frac{1}{8}$

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2.2.3 Discrete random variable

A random variable which takes only whole number or integers values then it is called discrete random variable. Discrete random variables are defined on discrete sample space.

2.2.4 Probability mass function

Let X be a discrete r.v. having values $x_1, x_2, \ldots, x_n$ with probabilities $P(x_1), P(x_2), \ldots, P(x_n)$. Then $P(x = x_i)$ or $P(x_i)$, $1 = 1, 2, \ldots, n$ is called probability function (pmf) if it satisfies the following conditions:

i. $P(x_i) \geq 0$, for all values of i.

ii. $\sum_{i=1}^{n} P(x_i) = 1$

or, $\sum_{i=1}^{\infty} P(x_i) = 1$

2.2.5 Probability distribution function

Let X be a discrete r.v. associated with values $x_1, x_2, \ldots$, with respective probabilities $P(x_1)$, $P(x_2), \ldots$, respectively. Then cumulative properties of X is called probability distribution function and denoted by F_x.

$$F(x_i) = P(x \leq x_i)$$

$$= P(x_o) + P(x_1) + \ldots + P(x_i) = \sum_{i=0}^{j} P(x_i).$$

2.2.6 Properties of probability distribution function

i. If a and b are two constant values $a < b$ and F is the distribution function, then $P(a < x < b) = F(b) - F(a)$.

ii. $0 \leq F(x_i) \leq 1$, $i = 1, 2, \ldots, \infty$, $F(x_i) < F(x_j)$ if $x_i \leq x_j$.

iii. $F(-\infty) = \lim_{x_i \to -\infty} F(x_i) = 0$.

iv. $F(+\infty) = \lim_{x_i \to -\infty} F(x_i) = 1$.

Example 39. A random variable X has the following probability mass functions:

Values pf x: x:	0	1	2	3	4	5	6	7
P(x)	0	K	2K	2K	3K	K^2	$2K^2$	$7K^2 + 2$

a. Find K.

b. Evaluate $P(x < 6)$, $P(x \geq 5)$, $P(0 < x < 5)$

c. Find the probability distribution function of X.

Solution

a. We have

$$\sum_{i=0}^{n} P(x_i) = 1$$

So, $0 + K + 2K + 2K + 3K + K^2 + 2K^2 + 7K^2 + K = 1$

or, $10K^2 + 9K - 1 = 0$

or, $10K^2 + 10K - K - 1 = 0$

or, $(K + 1)(10K - 1) = 0$

or, $K = -1$, which is impossible.

$\therefore K = \frac{1}{10} = 0.1$

b. $P(x < 6) = P(x = 0) + P(x = 1) + P(x = 2) + P(x = 3) + P(x + 4) + P(x = 5)$

$= 0 + 0.1 + 0.2 + 0.2 + 0.3 + 0.01$

$= 0.81$

$P(x \geq 5) = P(x = 5) + P(x = 6) + P(x = 7)$

$= K^2 + 2K^2 + 7K^2 + K$

$= 0.01 + 2(0.01) + 7(0.01) + 0.1$

$= 0.01 + 0.02 + 0.07 + 0.1$

$= 0.2$

$P(0 < x < 5) = P(x = 1) + P(x = 2) + P(x = 3) + P(x = 4)$

$= 0.1 + 0.2 + 0.2 + 0.3$

$= 0.8$

c. The probability distribution function of X is:

X:	0	1	2	3	4	5	6	7
P(x):	0	0.1	0.2	0.2	0.3	0.01	0.02	0.17
Fx:	0	0.1	0.3	0.5	0.8	0.81	0.83	1

Example 40. A random variable X has the following probability function

X:	-2	-1	0	1	2	3
P(x):	0.1	K	0.2	2K	0.3	K

a. Find the value of K.

b. Find $P(x < 2)$, $P(-2 < x < 2)$, $P(0 \leq x < 2)$.

c. Find $F(-2)$, $F(0)$, $F(3)$

d. Find Fx.

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Solution

a. We have, $\sum P(x) = 1$

or, $0.1 + K + 0.2 + 2K + 0.3 + K = 1$

or, $4K = 0.4$

$\therefore K = 0.1$

b. $P(x < 2) = P(x = -2) + P(x = -1) + P(x = 0) + P(x = 1)$

$= 0.1 + 0.1 + 0.2 + 0.2$

$= 0.6$

$P(-2 < x < 2) = P(x = 1) + P(x = 0) + P(x = 1)$

$= 0.1 + 0.2 + 0.2$

$= 0.5$

$P(0 \le x < 2) = P(x = 0) + P(x = 1)$

$= 0.2 + 0.2$

$= 0.4$

c. $F(-2) = P(x = -2) = 0.1$

$F(0) = P(x = -2) + P(x = -1) + P(x = 0)$

$= 0.1 + 0.1 + 0.2$

$= 0.4$

$F(3) = P(x = -2) + P(x = -1) + \ldots + P(x = 3)$

$= 0.1 + 0.1 + 0.2 + 0.2 + 0.3 + 0.1$

$= 1.0$

d. Here, F(3) = fx, i.e., cumulation of all probabilities.

Example 41. A newly married couple is planning to have 3 children. If X denote the r.v. that the children being boys. Then find probability distribution function (Fx).

Solution

The family have 3 children, the probability of occurrences of boys and girls are equal. The possible outcomes will be: BBB, BBG, BGG, GGG, GGB, GBB, GBG, BGB–8 cases. Where, B denote the boy and G for girl.

Here X denote the number that these children being boys. Then

X	P(x)	Fx
0	$\frac{1}{8}$	$\frac{1}{8}$
1	$\frac{3}{8}$	$\frac{4}{8}$
2	$\frac{3}{8}$	$\frac{7}{8}$
3	$\frac{1}{8}$	1

Example 42. If a r.v. x takes the values 1, 2, 3, 4 such that $2P(x = 1) = 3P(x = 2) = P(x = 3) = 5P(x = 4)$ then find probability distribution of x.

Solution

Let $P(x = 3) = K$, then from question

$P(x = 1) = \frac{K}{2}$, $P(x = 2) = \frac{K}{3}$ and $P(x = 4) = \frac{K}{5}$

Since total probability is equal to 1, we have

$P(X = 1) + P(X = 2) + P(X = 3) + P(x = 4) = 1$

or, $\frac{K}{2} + \frac{K}{3} + K + \frac{K}{5} = 1$

or, $K = \frac{30}{61}$

Hence the probability distribution of X is

X:	1	2	3	4
P(x):	15/61	10/61	60/61	6/61
Fx:	15/61	25/61	55/61	1

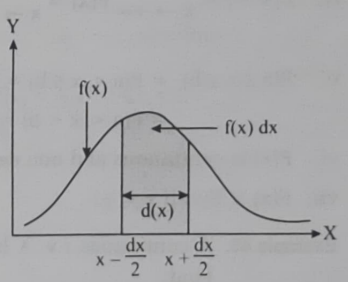
2.2.7 Continuous random variable

A continuous r.v. is a variable that is used to model the continuous data and its value lies between certain range. Height, weight, temperature, area, volume, acceleration, etc. are continuous variable.

2.2.8 Probability density function (p.d.f.)

Let a continuous r.v. which takes value x and f(x) be any function of x. Consider a small interval $\left[x - \frac{dx}{2}, x + \frac{dx}{2}\right]$ of length dx round the point x, dx is small value of x. The probability that the small interval $\left[x - \frac{dx}{2}, x + \frac{dx}{2}\right]$ is represented by f(x) dx given by

$$P\left[x - \frac{dx}{2}, x + \frac{dx}{2}\right] = f(x)\,dx.$$



In figure f(x) dx represent that area bounded by the curve y = f(x) at the ordinate points $x - \frac{dx}{2} + x + \frac{dx}{2}$. The function f(x) is defined as probability function of r.v. X. f(x) dx usually written as d F(x) is known as probability differential and the curve y = f(x) is known as probability density curve.

The probability density function f(x) satisfies the following properties:

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i. $f(x) \geq 0$, for all $x \in R$.

ii. $\int_{-\infty}^{\infty} f(x)\,dx = 1$.

iii. For any particular event E, if X falls in interval (a, b)

$P(E) = P(a \leq x \leq b)$

$= \int_{a}^{b} f(x)\,dx$

2.2.9 Continuous probability distribution function

Let X be a continuous r.v. with probability density function f(x) then continuous probability distribution function denoted by F(x) is simply cumulative distribution function of X.

$$F(x) = P(x \leq x) = \int_{-\infty}^{x} f(x)\,dx, -\infty < x < \infty$$

2.2.10 Properties of F(x)

i. $0 \leq F(x) = 1, -\infty < x < \infty$.

ii. $d\,F(x) = f(x)\,dx$, i.e., $F'(x) = f(x) \leq 0$.

iii. $F(-\infty) = \lim_{x \to -\infty} F(x) = \lim_{x \to -\infty} \int_{-\infty}^{x} f(x)\,dx = \int_{-\infty}^{\infty} f(x)\,dx = 0$.

iv. $F(+\infty) = \lim_{x \to +\infty} F(x) = \lim_{x \to -\infty} \int_{-\infty}^{x} f(x)\,dx = \int_{-\infty}^{\infty} f(x)\,dx = 1$.

v. $P(a \leq x \leq b) = P(a < x \leq b) = P(a \leq x < b)$

$= P(a < x < b) = F(b) - F(a)$

vi. F(x) is continuous and non decreasing function.

vii. $F(x) < F(y)$ if $x < y$.

Example 43. A continuous r.v. X has a probability density function $f(x) = k(1 + x)$, $2 \leq x \leq 5$. Find:

a. k

b. $P(x \leq 4)$.

Solution

a. We have,

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

or, $k\int_{2}^{5} (1 + x)\,dx = 1$

Integrating, we have

$$k\left[x + \frac{x^2}{2}\right]_2^5 = 1$$

or, $k\left\{(5 - 2) + \frac{25 - 4}{2}\right\} = 1$

or, $k\left\{3 + \frac{21}{2}\right\} = 1$

or, $\frac{27k}{2} = 1$

$\therefore \quad k = \frac{2}{27}$

Hence probability density function of X is

$f(x) = \frac{2}{27}(1 + x)$, $2 \leq x \leq 5$

b. $P(x < 4) = \int_{2}^{4} \frac{2}{27}(1 + x)\,dx$

$= \frac{2}{27}\left\{\int_{2}^{4} dx + \int_{2}^{4} x\,dx\right.$

$= \frac{2}{27}\left\{[x]_2^4 + \left[\frac{x^2}{2}\right]_2^4\right.$

On solving, we get $P(x < 4) = \frac{16}{27}$.

Example 44. A continuous r.v. x follows the probability law $f(x) = A\,x^2$, $0 \leq x < 1$.

a. Determine A b. $P(0.2 < x < 0.5)$

Solution

a. We have

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

or, $\int_{0}^{1} A\,x^2\,dx = 1$

or, $A = \left[\frac{x^3}{3}\right]_0^1 = 1$

$\therefore \quad A = 3$

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b. $P(0.2 < x < 0.5) = \int_{0.2}^{0.5} 3x^2\,dx$

or, $P(0.2 < x < 0.5) = [x^3]_{0.2}^{0.5} = 0.017$

Example 45. If a random variable X has the density function $f(x) = \begin{cases} \frac{1}{4} & -2 < x < 2 \\ 0 & \text{otherwise} \end{cases}$. Find:

a. $P(X < 1)$ b. $P(|X| > 1)$ c. $P(2X + 3 > 5)$

Solution

a. We have

$$\int_{-\infty}^{\infty} f(x)\,dx$$

$$P(X < 1) = \int_{-\infty}^{1} f(x)\,dx$$

$$= \int_{-\infty}^{-2} f(x)\,dx + \int_{-2}^{1} f(x)\,dx$$

$$= 0 + \int_{-2}^{1} \frac{1}{4}\,dx$$

$$= 0 + \frac{1}{4}[x]_{-2}^{1}$$

$$= \frac{3}{4}$$

b. $P(|X| > 1) = 1 - P(1 \times 1 \le 1)$

$$= 1 - P(-1 \le X \le 1)$$

$$= 1 - \int_{-1}^{1} \frac{1}{4}\,dx$$

$$= 1 - \frac{1}{4}[x]_{-1}^{1}$$

$$= 1 - \frac{1}{2}$$

$$= \frac{3}{4}$$

c. $P(2x + 3 > 5) = P(X > 1)$

$$= \int_{1}^{\infty} f(x)\,dx$$

$$= \int_{1}^{2} \frac{1}{4}\,dx + \int_{2}^{\infty} f(x)\,dx$$

$$= \frac{1}{4}[x]_{1}^{2} = \frac{1}{4}$$

Example 46. Let X be a r.v. with p.d.f.

$f(x) = ax, \quad 0 \le x \le 1$

$= a, \quad 1 \le x \le 2$

$= -ax + 3a, \quad 2 \le x \le 3$

$= 0, \quad$ otherwise

Find (a) a (b) $P(X \le 1.5)$

Solution

a. We have

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

or, $\int_{-\infty}^{\infty} f(x)\,dx + \int_{0}^{1} f(x)\,dx + \int_{1}^{2} f(x)\,dx + \int_{2}^{3} f(x)\,dx + \int_{3}^{\infty} f(x)\,dx = 1$

or, $0 + \int_{0}^{1} ax\,dx + \int_{1}^{2} a\,dx + \int_{2}^{3} (-ax + 3a)\,dx + 0 = 1$

or, $\left[\frac{ax^2}{2}\right]_{0}^{1} + [ax]_{1}^{2} + \left[-\frac{ax^2}{2} + 3ax\right]_{2}^{3} = 1$

On solving, we get $a = \frac{1}{2}$

b. $P(X \le 1.5) = \int_{0}^{1} ax\,dx + \int_{1}^{1.5} a\,dx$

$$= \frac{1}{2}\left[\frac{x^2}{2}\right]_{0}^{1} + \frac{1}{2}[x]_{1}^{1.5}$$

$= \frac{1}{2}$, on solving

Example 47. A discrete r.v. X has the p.m.f

$P(x) = \frac{x}{15}, \quad x = 1, 2, 3, 4, 5$

$= 0, \quad$ Otherwise

Find

a. $P(X = 1 \text{ or } 2)$ b. $P\left\{\frac{1}{2} < x < \frac{5}{2} / x > 1\right\}$

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Solution

a. $P(X = 1 \text{ or } 2) = P(X = 1) + P(X = 2)$

$$= \frac{1}{15} + \frac{2}{15} = \frac{1}{5}$$

b. $P\left\{\frac{1}{2} < x < \frac{5}{2} / x > 1\right\} = \frac{P\left(\frac{1}{2} < x < \frac{5}{2}\right) \cap (x > 1)}{P(x > 1)} = \frac{P\left(\frac{1}{2} < x < \frac{5}{2}\right)}{1 - P(x \le 1)}$

$$= \frac{P(x = 2)}{1 - \{P(x = 0) + P(x = 1)\}}$$

$$= \frac{2/15}{1 - \left\{0 + \frac{1}{15}\right\}}$$

$$= \frac{1}{7}$$

Exercise 2.2

1. A random variable X has the following probability distribution:

Value of X; x:	0	1	2	3	4	5	6	7	8
P(x)	a	3a	5a	7a	9a	11a	13a	15a	17a

   a. Determine the value of a.

   b. Find $P(X \le 3)$, $P(0 < X < 6)$, $F(4)$ and $F(7)$

   c. Find the distribution function of r.v. X.

2. A random variable X has the following probability distribution:

x:	−3	−1	0	1	2	3	5	8
P(x)	0.1	0.2	0.15	0.2	0.1	0.15	0.05	0.05

   a. Complete the probability when X is negative.

   b. Find $P(1 < x \le 8)$.

   c. Find $P(X = 3 / x \le 0)$ and $P(x \ge 3 / x > 0)$.

   Hint:

$$P(X = 3/X \le 0) = \frac{P\{(X = 3) \cap (x \le 0)\}}{P(X \le 0)} = 0$$

and $P(X \ge 3 / P(X > 0) = \frac{P\{(X \ge 3) \cap (x > 0)\}}{P(X > 0)}$

$$= \frac{P(3 \le x \le 8)}{1 - P(X \le 0)}, \text{ then use value and solve.}$$

3. From a lot of 10 items containing 3 defective, a sample of 4 items is drawn at random. Let X denote the number of defective items in the sample, when sampling is drawn without replacement. Then find

   a. $P(X \le 1)$, $F(2)$, $F(3)$ b. Find probability distribution of X.

4. A random variable X has the density function $f(x) = a + bx^2$, $0 < x < 1$. Determine a and b so that $a + b = 2$.

5. The p.d.f. of random variable X is $f(x) = \begin{cases} Kx(1-x), & \text{for } 0 < x < 1 \\ 0, & \text{Otherwise} \end{cases}$. Find:

   a. K b. $P(0.3 < X < 0.5)$

   c. Show that $P(X < 0.4) = 0.352$

6. A continuous r.v. X having pd.f. as $f(x) = \begin{cases} \frac{1}{16}(x+3)^2, & \text{for } -3 \le x < -1 \\ \frac{1}{16}(6 - 2x^2), & \text{for } -1 \le x \le 1 \\ \frac{1}{16}(3-x)^2, & \text{for } 1 \le x \le 3 \end{cases}$. Find:

   a. Verify the area under the curve is unity.

   b. Determine $P(-0.5 < x < 1)$.

7. Find K so that the following can serve as probability density function of a random variables.

$$f(x) = \begin{cases} 0, & \text{for } x \le 0 \\ Kxe^{-4x^2}, & \text{for } x > 0 \end{cases}.$$

   a. Determine K b. Determine F(120)

8. A probability density function of X is

$$f(x) = \frac{1}{6}(x+2), \quad 0 \le x < 2$$
$$= 0 \quad \text{Otherwise}$$

. Find:

   a. $P(X > 0)$ b. $F(1.5)$ c. $P(0 < x \le 1)$

9. A continuous r.v. X has p.d.f. $f(x) = \frac{2(1+x)}{21}$, $1 \le x \le 4$. Find:

   a. The distribution function f(3) b. $P(2 < x < 3)$

Answers

1. a. $\frac{1}{81}$ b. $\frac{16}{81}, \frac{35}{81}, \frac{25}{81}, \frac{66}{81}$

   c.

X:	0	1	2	3	4	5	6	7	8
P(x):	1/81	3/81	5/81	7/81	9/81	11/81	13/81	15/81	17/81
Fx:	1/81	4/81	9/81	16/81	25/81	36/81	49/81	64/81	1

2. a. 0.3 b. 0.9 c. 0.284

3. a. $\frac{2}{3}, \frac{29}{30}, 1$

   b.

X:	0	1	2	3
P(x):	1/6	1/2	3/10	1/30

4. a. $\frac{1}{2}$ b. $\frac{1}{3}$ 5. a. $k = 6$ b. 0.2844 6. b. 0.5156

7. a. 8 b. $(1 - e^{-400})$ 8. a. 1 b. 0.6875 c. $\frac{5}{12}$

9. a. 0.8571 b. 0.334

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2.3 Mathematical expectation

2.3.1 Expectation of discrete random variables

Let X be a discrete r.v. which takes values $x_1, x_2, x_3, \ldots, x_n$ with probabilities $P(x_1), P(x_2), \ldots, P(x_n)$ respectively. Then mathematical expectation of X, denoted by E(X), read as expected X is defined as:

$$E(X) = \sum_{i=1}^{n} x_i P(x_i)$$

$$= \sum X P(x)$$

Properties of expectation

i. Expectation of a constant is constant itself.

E(c) = c, E(20) = 20

ii. E(aX + b) = a E(X) + b.

iii. $E(X_1 + X_2 + \ldots + X_n) = E(X_1) + E(X_2) + \ldots + E(X_n)$.

iv. $E(X_1 \cdot X_2 \cdot X_3, \ldots, X_n) = E(X_1) \cdot E(X_2), \ldots, E(X_n)$

v. Variance of a random variable

a. $V(X) = E(X^2) - [E(X)]^2$

Where, $E(X^2) = \Sigma X^2 \cdot P(X)$, $E(X) = \Sigma X \cdot P(X)$

b. Variance of constant is equal to zero, i.e., V(c) = 0

c. $V(aX + b) = a^2V(X)$

d. $V(aX \pm by) = a^2 V(X) \pm b^2V(Y)$

vi. Covariance

The measure of simultaneous variation between the random variables X and Y which is denoted by Cov (X, Y) or σ_{XY} is give as:

$$Cov(X, Y) = E\{(X - E(X))(Y - E(Y))\}$$

$$= E(XY) - E(X) \cdot E(Y) - E(X) \cdot E(Y) + E(X) \cdot E(Y)$$

or, $$Cov(X, Y) = E(XY) - E(X) \cdot E(Y)$$

or, $$Cov(X, Y) = \sum_x \Sigma XY\, P(X, Y) - \left[\sum_x \cdot P(X)\right]\left[\sum_y \cdot P(Y)\right]$$

If X and Y are independent then

$$Cov(X, y) = E(X) \cdot E(Y) - E(X) \cdot E(Y) = 0$$

2.3.2 Expectation of continuous random variable

Let X be a continuous r.v. with probability density function f(x), then mathematical expectation of X is:

$$E(X) = \int_{-\infty}^{\infty} X \cdot f(x)\, dx.$$

Let f(x) be the probability density function of X and X is defined in [a, b], then

$$\text{Arithmetic mean } E(X) = \int_a^b X \cdot f(X)\, dx.$$

$$V(X) = E(X^2) - [E(X)]^2$$

$$= \int_a^b X^2 f(x)\, dx - \left\{\int_a^b X \cdot f(x)\, dx\right\}^2$$

2.3.3 Covariance

$$Cov(X< Y) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xy\, f(x, y)\, dx\, dy - \int_{-\infty}^{\infty} x \cdot f(x)\, dx \int_{-\infty}^{\infty} y \cdot f(y)\, dy$$

Example 48. A random variable X has the following probability function:

X:	−2	−1	0	1	2	3
P(x)	0.1	K^2	0.2	$2K^2$	0.3	K^2

Find:

a. Value of K^2 b. E(X) c. V(X)

Solution

a. We have, $\Sigma P(x) = 1$

or, $4K^2 + 0.6 = 1$

$\therefore$ $K^2 = 0.1$

b. $E(X) = \Sigma x \cdot P(x)$

$= (-2) \times 0.1 + (-1) \times 0.1 + 0 \times 0.2 + 1 \times 0.2 + 2 \times 0.3 + 3 \times 0.1$

$= -0.2 - 0.1 + 0 + 0.2 + 0.6 + 0.3$

$= 0.8$

$E(X^2) = \Sigma X^2 \cdot P(x)$

$= 4 \times 0.1 + 1 \times 0.1 + 0 + 1 \times 0.2 + 4 \times 0.3 + 9 \times 0.1$

$= 0.4 + 0.1 + 0.2 + 1.2 + 0.9$

$= 2.8$

$V(X) = E(X^2) - [E(x)]^2$

$= 2.8 - (0.8)^2$

$= 2.8 - 0.64$

$= 2.16$

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Example 49. A die is thrown at random, what is the expectation of number on it?

Solution

After throwing a die, let X be a random variable which can take values 1, 2, 3, 4, 5, 6 with equal probability $\frac{1}{6}$.

X:	1	2	3	4	5	6
P(x):	1/6	1/6	1/6	1/6	1/6	1/6

$E(X) = \Sigma x \cdot P(x)$

$$= 1 \times \frac{1}{6} + 2 \times \frac{1}{6} + 3 \times \frac{1}{6} + 4 \times \frac{1}{6} + 5 \times \frac{1}{6} + 6 \times \frac{1}{6}$$

$$= \frac{7}{2}$$

$$= 3.5$$

Example 50. An appliance dealer sells different models of freezer having capacity 13.5, 15.9 and 19.1 cubic feet of storage space respectively. Let X be the amount of storage space purchased by the next customer to buy a freezer and P(x) be the probability mass function given as:

X: x	13.5	15.9	19.1
P(x)	0.2	0.5	0.3

a. Complete mean and variance of X.

b. If the price of the freezer having capacity x cubic feet is 25x – 8.5, what is the price paid by next customer to buy freezer?

c. What is the variance of the price 25x – 8.5 paid by the next customer?

Solution

a. $E(X) = \Sigma x P(x)$

$= 13.5 \times 0.2 + 15.9 \times 0.5 + 19.1 \times 0.3$

$= 16.38$

$E(X^2) = \Sigma x^2 \cdot P(x)$

$= 13.5^2 \times 0.2 + 15.9^2 \times 0.5 + 19.1^2 \times 0.3$

$= 272.298$

$V(X) = 272.298 - (16.38)^2$

$= 3.99$

b. We have, $E(X) = 16.38$

Expected price paid by next customer $= 25\,E(X) - 8.5$

$= 25 \times 16.38 - 8.5$

$= 401$

c. To find variance of the price 25x – 8.5 purchased next customer, let us introduced new variable y = 25x – 8.5.

X : x	Y : y	P(y)
13.5	329	0.2
15.9	389	0.5
19.1	469	0.3

$E(y) = \Sigma y \cdot P(y)$

$= 329 \times 0.2 + 389 \times 0.5 + 469 \times 0.3$

$= 401$

$E(Y^2) = \Sigma y^2 \cdot P(y)$

$= 329^2 \times 0.2 + 389^2 \times 0.5 + 469^2 \times 0.3$

$= 163297$

$\therefore \quad V(Y) = E(Y^2) - [E(Y)]^2$

$= 163297 - 401^2$

$= 2496$

$\therefore$ Variance of price paid by next customer = 2496.

Example 51. A software engineer spends Rs. 3000 to prepare for a bid on constructing software package of certain organization. After deducting expenses on constructing software package and the cost of bidding will yield a profit of Rs. 25,000 if the bid is won. The chances of winning the bid is 10%, compute his expected profit and state the decision whether to bid or not to bid.

Solution

Profit if bid won = Rs. 25000

P(Winning bid) = 10% = 0.1

P(Losing the bid) = 0.9

Expenses on preparing bid = Rs. 3000

Expected profit $= \Sigma x \cdot P(x)$

$= 25000 \times 0.1 + (-3000) \times 0.9$

$= -$ Rs. 200

Since the engineer's expected profit is negative, better he should not bid.

Example 52. A company makes electronic gadgets. One out of every 50 gadgets is faulty, but the company does not know which one are faulty until a buyer complains. Suppose the company makes a $3 profit on the sale of any working gadgets but suffers a loss of $80 for every faulty gadgets because they have to repair the unit. Check whether the company can expect a profit in the long term.

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Solution

$$\text{Expected profit} = \frac{49}{50} \times 3 + \frac{1}{50} \times (-80)$$

$$= \frac{147}{50} - \frac{80}{50} = 1.34$$

Since expected profit is positive, the company can expect a profit. On average they make a profit of \$1.34 per gadget produced.

Example 53. A player tosses two fair coins, he wins Rs. 5 if 2 head occurs, Rs. 2 if 1 head occurs and Rs. 1 if no head occurs.

a. Find his expected gain.

b. How much should he pay the game if it is to be fair?

Solution

a. Let X be the number of occurrences of head, when two coin is tossed.

Events	X	P(X)	Gain (Y)	P(Y) = P(X)
TT	0	1/4	1	0.25
HT, TH	1	2/4	2	0.50
HH	2	1/4	5	0.25

Expected gain of player $E(Y) = \Sigma Y \cdot P(Y)$

$= 1 \times 0.25 + 2 \times 0.50 + 5 \times 0.25$

= Rs. 2.50

b. The game is said to be fair, if mathematical expectation of the player is zero. Hence he should play Rs. 2.50 to play the game if this is to be fair.

Example 54. Three bag contains 3 green 2 white balls, 5 green 6 white balls and 2 green 4 white balls. One ball is drawn from each bag. Find the expected number of white balls drawn out.

Solution

3G, 2W	5G, 6W	2G, 4W
I	II	III

Let X_1, X_2, X_3 denote the drawing a white ball from bag I, II and III in single draw.

For Bag I

$P(X_1 = 1) = \frac{2}{5}$

$P(X_1 = 0) = \frac{3}{5}$

$E(X_1) = 1 \times \frac{2}{5} + 0 \times \frac{3}{5}$

$= \frac{2}{5}$

For Bag II

$P(X_2 = 1) = \frac{6}{11}$

$P(X_2 = 0) = \frac{5}{11}$

$E(X_2) = \frac{6}{11}$

For Bag III

$P(X_3 = 1) = \frac{4}{6}$

$P(X_3 = 0) = \frac{2}{6}$

$E(X_3) = \frac{4}{6}$

$\therefore$ Expected number of white balls drawn out $E(X) = E(X_1) + E(X_2) + E(X_3)$

$\therefore E(X) = \frac{2}{5} + \frac{6}{11} + \frac{4}{6}$

$= \frac{266}{165}$

$= 1.612$

Example 55. A computer train arrives punctually at a station every 25 minutes. Each morning a computer leaves his house and casually walks to the train station. Let X denote the amount of time in minutes that computer waits for the train the time he reaches the train station. It is know that p.d.f. of X is $f(x) = \begin{cases} \frac{1}{25}, & \text{for } 0 < x < 25 \\ 0, & \text{Otherwise} \end{cases}$. Obtain the mean value of the r.v.

Solution

We have,

$$E(X) = \int_{-\infty}^{\infty} x \cdot f(x)\, dx$$

$$= \int_0^{25} x \cdot \frac{1}{25}\, dx$$

$$= \frac{1}{25}\left[\frac{x^2}{2}\right]_0^{25}$$

$$= 12.5$$

$\therefore$ Expected waiting time of computer = 12.5 minutes.

Example 56. If $f(x)$ is defined as $f(x) = Ke^{-2x}$, $0 \le x < \infty$ is a p.d.f., determine K and also find mean of the distribution.

Solution

We have,

$$\int_{-\infty}^{\infty} f(x)\, dx = 1$$

or, $K\int_{-\infty}^{\infty} e^{-2x}\, dx = 1$

or, $K\left[\frac{e^{-2x}}{-2}\right]_0^{\infty} = 1$

$\therefore K = 2$

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Again,

$$E(X) = \int_{-\infty}^{\infty} x\, f(x)\, dx$$

$$= \int_{0}^{\infty} x \cdot 2e^{-2x}\, dx$$

$$= 2\left\{\left[x\frac{e^{-2x}}{2}\right]_0^{\infty} - \int_0^{\infty}\frac{e^{-2x}}{-2}\right\}$$

$$= \int_0^{\infty} e^{-2x}$$

$$= \frac{1}{2}$$

Example 57. The time to failure in thousand of hours of an important piece of electronic equipment used in the manufactured DVD player has the density function.

$$f(x) = \begin{cases} 3e^{-3x}, & x > 0 \\ 0 & \text{Otherwise} \end{cases}$$

Find the expected life of piece of equipment.

Solution

$$E(X) = \int_0^{\infty} x \cdot 3e^{-3x}\, dx$$

$$= 3\left\{\left[x\frac{e^{-3x}}{3}\right]_0^{\infty} - \int_0^{\infty}\frac{e^{-3x}}{-3}\right\} dx$$

$$= \int_0^{\infty} e^{-3x}$$

$$= \frac{1}{3}$$

Example 58. Find the mean and variance of the random variable with probability density function: $f(x) = xe^{-x}, x \geq 0$.

Solution

We have

$$E(X) = \int_{-\infty}^{\infty} x \cdot f(x)\, dx$$

$$= \int_0^{\infty} x \cdot xe^{-x}\, dx$$

$$= \int_{-\infty}^{\infty} e^{-x} x^{3-1}\, dx \qquad \left[\because \int_0^{\infty} e^{-x} x^{\alpha-1}\, dx = \lceil\alpha \right.$$
$$\left. \lceil\alpha = (\alpha-1)! \right]$$

$$\therefore \quad E(X) = \lceil 3 = 2$$

$$(EX^2) = \int_0^{\infty} x^3 e^{-x}\, dx$$

$$= \int_0^{\infty} x^3 e^{-x}\, dx$$

$$= \int_{-\infty}^{\infty} x^{4-1} e^{-x}\, dx$$

$$= \lceil 4$$

$$= 3 \times 2 \times 1$$

$$= 6$$

$$\therefore \quad V(X) = E(X^2) - [E(X)]^2$$

$$= 6 - 2^2$$

$$= 2$$

Example 59. The p.d.f. of a continuous r.v. is given as:

$f(x) = A\, x(1-x)^2, 0 \leq x \leq 1$, elsewhere

Find E(X) and V(X).

Solution

We have,

$$\int_{-\infty}^{\infty} f(x)\, dx = 1$$

$$\text{or,} \quad A\int_0^1 x(1-x)^2\, dx = 1$$

$$\text{or,} \quad A\left\{\int_0^1 (x^2 - 2x^2 + x)\, dx = 1\right.$$

After integration, we get $A = 12$

$$E(X) = \int_{-\infty}^{\infty} x \cdot f(x)\, dx$$

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$$= 12\int_0^1 x^2(1 - 2x + x^2)\,dx$$

$$= 12 \times \frac{2}{60}$$

$$= \frac{2}{5}$$

$$E(X^2) = \int_{-\infty}^{\infty} x^2 \cdot f(x)\,dx$$

$$= 12\int_0^1 x^2(1 - 2x + x^2)\,dx$$

$$= 12 \times \frac{1}{60}$$

$$= \frac{1}{5}$$

$$\therefore \quad V(X) = E(X^2) - [E(X)]^2$$

$$= \frac{1}{5} - \frac{2}{25}$$

$$= \frac{1}{25}$$

Exercise 2.3

1. A dean of the certain college has determined the probability distribution of X, that denote the number of student absent on any given day is as follows:

X:	0	1	2	3	4	5	6	7
P(x)	0.005	0.025	0.310	0.340	0.080	0.220	0.019	0.001

Compute E(X) and V(X).

2. The following table presents the number of different brands of cell phone sold, with their probabilities from the certain store in new year 2023.

Sales number	10	20	30	40	50
P(x)	0.08	0.24	0.28	0.30	0.10

Compute the mean and variance of the number of sold phones.

3. The number of calls arriving at an exchange during any given minute between noon and 1:00 pm on a week day is a random variable X, with the following probability distribution.

Number of calls (X):	0	1	2	3	4	5
P(x):	0.25	0.15	2K	0.20	K	0.16

Find the K, mean and standard deviation of the distribution.

4. The probability distribution of a random variable is given as:

X:	−2	−1	0	1	2
P(x)	0.2	0.1	0.3	0.3	0.1

Find (a) E(X) (b) E(2x − 3) (c) V(X) (d) V(2x − 3)

5. A family has 3 children, find the mathematical expectation of these children being girls.
6. A coin is tossed until a tail appears, what is the expectation of the number of tosses required?
7. Five bags contain respectively 2 black and 2 white balls, 7 black and 6 white balls, 2 black and 1 white ball, 5 black and 2 white balls, 2 black and 4 white balls. One ball is drawn from each bag, find the expected number of white balls drawn out.
8. A business venture a man can make a profit of Rs. 50,000 or incur a loss of Rs. 20,000. The probability of making profit or incurring loss from the past experience is known to be 0.75 and 0.25 respectively. What is his expected profit?
9. A continuous r.v. X has following p.d.f.

   $f(x) = AX(2 - x),\ 0 \le x \le 2.$

   Find (a) A (b) E(X) (c) V(X)
10. A random variable X has the density function: $f(x) = a + bx^2,\ 0 < x < 1$. Determine a and b so that mean is $\frac{2}{3}$. Also find the variance of the distribution.
11. The number of miles an automobile tire last before it reaches a critical point in tread wear can be represented by a probability density function:

$$f(x) = \begin{cases} \frac{1}{30} e^{-x/30}, & \text{for } x > 0 \\ 0 & \text{Otherwise} \end{cases}$$

Find the expected number of miles a tire would last until it reaches the critical tread wear point.

12. A continuous r.v. X, has the following p.d.f.

    $f(x) = K x^3(4 - x)^2,\ 0 < x < 1$

    Find (a) K (b) E(X) (c) V(X).
13. Consider a r.v. X, with p.d.f.

$$f(x) = \begin{cases} 4x^3, & \text{if } 0 < x < 1 \\ 0 & \text{Otherwise} \end{cases}$$

Then find mean and variance of thee distribution.

14. Suppose a random variable X, whose p.d.f. is given as

$$f(x) = \begin{cases} c(4x - 2x^2), & \text{if } 0 < x < 2 \\ 0 & \text{Otherwise} \end{cases}$$

a. Determine the value of c.
b. Determine cumulative distribution of X.
c. Find $P(X > 1)$.
d. Find E(X) and V(X).

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15. A continuous r.v. X, range (–3, 3) is defined as:

$$f(x) = \frac{1}{16}(3 + x)^2, \quad -3 \le x \le -1$$

$$= \frac{1}{16}(6 - 2x^2), \quad -1 \le x \le 1$$

$$= \frac{1}{16}(3 - x)^2, \quad 1 \le x \le 3$$

a. Verify the area under the curve is unity.

b. Find mean and variance.

Answers

1. 3.066, 1.18
2. 31, 11.18
3. 0.08, 2.19, 6.87
4. 0, –3, 1.6, 6.4
5. $\frac{3}{2}$
6. 2
7. 2.0329
8. Rs. 32.5000
9. $\frac{1}{2}$, 1, $\frac{1}{5}$
10. $\frac{1}{3}$, 2, 0.0666
11. 30
12. $\frac{30}{77}$, $\frac{422}{539}$, 0.0295
13. $-\frac{4}{5}$, $\frac{2}{75}$
14. $C = \frac{3}{8}$, $F(x) = P(X \le x) = \begin{cases} \frac{3x^2}{4} - \frac{x^3}{4}, & 0 < x < 2 \\ 0 & \text{Otherwise} \end{cases}$, $P(X > 1) = \frac{1}{2}$, $E(X) = 1$, $V(x) = 1.2$
15. 0, 1

Unit 3

Theoretical Discrete Probability Distribution

3.1 Introduction

A discrete distribution is a probability distribution that describes the occurrences of discrete (countable) outcomes. In discrete random variables are distributed accordingly to some definite probability rule, which can be expressed in some mathematical model. In this chapter we shall study some important univariate probability distribution.

3.1.1 Bernoulli trial

Many of the random experiment that we carry out have only two outcomes that are failure or successes. A student appeared in the exam he will pass or fail the exam, the product produced by certain production process can be defective or non-defective. These types of independent trial which have only two possible outcomes are known as Bernoulli's trial. The Bernoulli's trial must satisfy the following assumptions:

- The number of trial should be finite.
- Each trial should have exactly two outcomes, i.e., success or failure.
- The probability of success or failure does not change for each trial.
- The trials must be independent.

Consider flipping a coin with 25% probability of heads. A single toss of coin has Bernoulli distribution with P = 0.25. If you toss a coin 3 times and record the number of heads this is known as binomial distribution with P = 0.25 and n = 3. We can use probability rules to determine the probability of 0, 1, 2 or 3 heads. So we can explain binomial distribution in detail as below.

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3.1.2 Binomial distribution

Binomial distribution was developed by James Bernoulli. Binomial distribution is a graphical representation of the results of Bernoulli trial. It gives the idea of probability events throughout the experiment successions. This can also be said as frequency distribution of the probability of a given number of success in a random number of repeated Bernoulli trial. If Bernoulli's trial are the experiment, the binomial distribution can be said as result or finding of the experiment.

Binomial distribution was developed by using all the assumptions that are used in Bernoulli trial.

Definition: A discrete random variable X is said to follow binomial distribution if its probability mass function (p.m.f.) is given as:

$$P(X = x) = P(x) = {}^nC_x\, p^x\, q^{n-x}, \text{ if } x = 0, 1, 2, \ldots, n$$
$$= 0, \text{ Otherwise}$$

Note:

i. $X \sim B(n, P)$, it means discrete r.v. X, follows binomial distribution with parameter n and P. Where n is the number of trial, P is probability of success, q is the probability of failure and x is the number of observation.

ii. Total probability of binomial distribution

$$\sum_{x=0}^{n} P(x) = \sum_{x=0}^{n} {}^nC_x\, p^x q^{n-x}$$
$$= {}^nC_0\, p^0 q^n + {}^nC_1 p^1 q^{n-1} + {}^nC_2 p^2 q^{n-2} + \ldots + {}^nC_n p^n q^0$$
$$= (q + p)^n$$
$$= 1$$

Mean and variance of binomial distribution

Mean

$$E(X) = \sum_{x=0}^{n} x \cdot P(x)$$
$$= \sum_{x=0}^{n} x \cdot \frac{n!}{(n-x)!\, x!}\, p^x\, q^{n-x}$$
$$= \sum_{x=0}^{n} x \cdot \frac{n\,(n-1)!}{(n-x)!\, x(x-1)!}\, p \cdot p^{x-1} q^{n-x}$$
$$= np \sum_{x=0}^{n} {}^{n-1}C_{x-1}\, p^{x-1} q^{n-x}$$
$$= np\, \{{}^{n-1}C_0\, p^0 q^{n-1} + {}^{n-1}C_1\, p^1 q^{n-2} + \ldots + {}^{n-1}C_{n-1}\, p^{n-1} q^0\}$$
$$= np\, (q + p)^{n-1}$$

$$\therefore E(X) = np$$

$$E(X^2) = \sum_{x=0}^{n} x^2\, p(x)$$
$$= \sum_{x=0}^{n} \{(x-1) + x\} \frac{n!}{(n-x)!\, x!}\, p^x q^{n-x}$$
$$= \sum_{x=0}^{n} x(x-1) \cdot \frac{n(n-1)\,(n-2)!}{(n-x)!\, x(x-1)\,(x-2)!}\, p^1 p^{x-2} q^{n-x} + \sum_{x=0}^{n} x \cdot \frac{n!}{(n-x)!\, x!}\, p^x q^{n-x}$$
$$= n(n-1) p^2 \sum_{x=2}^{n} {}^{n-2}C_{x-2}\, p^{x-2} q^{n-x} + np$$
$$= n(n-1)\, p^2 + np \qquad [\because (q + p)^{n-2} = 1]$$

Variance: $V(X) = E(X^2) - [E(X)]^2$
$$= n^2p^2 - np^2 + np - n^2p^2$$
$$= np(1 - p)$$
$$= npq$$

Moment generation function of binomial distribution

$$M_X(t) = E(e^{tx}) \cdot P(x)$$
$$= \sum_{x=0}^{n} e^{tx}\, {}^nC_x\, p^x q^{n-x}$$
$$= \sum_{x=0}^{n} (e^t p)^x\, {}^nC_x q^{n-x}$$
$$= {}^nC_0 (e^t p)^0\, q^n + {}^nC_1 (e_t p)^1\, q^{n-1} + {}^nC_2 (e^t p)^2\, q^{n-2} + \ldots + (e^t p)^n$$

$$\therefore M_X(t) = (q + pe^t)^n$$

Example 1. The phone lines to an airline reservation system are occupied 40% of the time. Assume that the events the lines are occupied on successive calls are independent. Assume that 10 calls are placed to the airline, then what is the probability that (a) for exactly 3 calls the line are occupied (ii) for at least one call the call lines are not occupied. Also find the expected number of calls in which the lines are occupied?

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Solution

Let X denote the r.v. the phone lines occupied

$P = 40\% = 0.4$, $q = 1 - p = 0.6$, $n = 10$

a. $P(X = 3) = {}^nC_x p^x q^{n-x}$
$= {}^{10}C_3 (0.4)^3\, (0.6)^{10-3}$
$= 0.215$

b. For not occupied situation

$p = 0.6$, $q = 0.4$, $n = 10$

$P(x \geq 1) = 1 - P(X < 1)$

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$= 1 - P(X = 0)$

$= 1 - {}^{10}C_0 (0.6)^0 (0.4)^{10}$

$= 0.998$

Expected number of calls = $np = 10 \times 0.4 = 4$.

Example 2. There are four fused bulbs in a lot of 10 good bulbs. If three bulbs are drawn at random, find the probability distribution of fused bulbs.

Solution

$p(\text{Drawing fused bulbs}) = \frac{4}{10+4} = \frac{2}{7}$

$q(\text{Drawing good bulbs}) = 1 - \frac{2}{7} = \frac{5}{7}$, $n = 3$

Let X be the event of drawing 0, 1, 2, 3 fused bulbs.

$P(x) = {}^nC_x p^x q^{n-x}$

$P(X = 0) = {}^3C_0 \left(\frac{2}{7}\right)^0 \left(\frac{5}{7}\right)^{3-0} = \frac{125}{343}$

$P(X = 1) = {}^3C_1 \left(\frac{2}{7}\right)^1 \left(\frac{5}{7}\right)^{3-1} = \frac{150}{343}$

$P(X = 2) = {}^3C_2 \left(\frac{2}{7}\right)^2 \left(\frac{5}{7}\right)^{3-2} = \frac{60}{343}$

$P(X = 3) = {}^3C_3 \left(\frac{2}{7}\right)^3 \left(\frac{5}{7}\right)^{3-3} = \frac{8}{343}$

The required probability of fused bulbs is

X	0	1	2	3
P(x)	125/343	150/343	60/343	8/343

Example 3. An airline sells 65 tickets for plane with capacity of 60 passengers. This is done because it is possible for some people to not show up. The probability of a person not showing up for the flight is 0.1, all the passengers believe independently. Find the probability of the event that the airlines does not have to arrange separate tickets for excess.

Solution

$p(\text{passengers show up}) = 1 - 0.1 = 0.9$

$q(\text{passengers not show up}) = 0.1$, $n = 65$

$P(X \le 60) = 1 - p(X > 60)$

$= 1 - \{P(X = 61) + P(X = 62) + P(X = 63) + P(X = 64) + P(X = 65)\}$

$= 1 - \{{}^{65}C_{61} (0.9)^{61} (0.1)^{65-61} + \ldots + {}^{65}C_{65} (0.9)^{65} (0.1)^0\}$

$= 1 - 0.2091$

$= 0.7909$

$\therefore$ Probability of the airlines doe not have to arrange separate tickets excess = 0.7909.

Example 4. Out of 8,000 families in a certain city with 4 children each, what percentage would be expected to (a) 1 boy and 3 girl. (b) at least one boy. (c) find the expected number of families having 2 boys. Assume boys and girls having equal probabilities.

Solution

$P(\text{boys}) = \frac{1}{2}$, $P(\text{Girls}) = \frac{1}{2}$, $n = 4$

Let X denote the number of occurrences of boys.

a. $P(1 \text{ boy } 3 \text{ girls}) = P(1 \text{ boy})$

$P(X) = {}^nC_x p^x q^{n-x}$

$P(X = 1) = {}^4C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^{4-1} = 4 \times \frac{1}{16} = \frac{1}{4}$.

Percentage of families with 1 boy and 3 girls = $\frac{1}{4} \times 100 = 25\%$.

b. $P(\text{At least one boy}) = 1 - P(X = 0)$

$= 1 - {}^4C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^{4-0}$

$= 1 - \frac{1}{16}$

$= \frac{15}{16} \times 100$

$= 93.75\%$

c. $N = 8{,}000$

$P(\text{At most 2 boys}) = \{(X \le 2)$

$= P(X = 0) + P(X = 1) + P(X = 2)$

$= {}^4C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^4 + {}^4C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^2 + {}^4C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^2$

$= \frac{1}{16} + \frac{4}{16} + \frac{6}{16}$

$= \frac{11}{16}$

Expected number of families having at most 2 boys

$E(X) = N \cdot P(X \le 2) = 8{,}000 \times \frac{11}{16} = 5500$.

Example 5. Probability of a bomb hitting the target is $\frac{1}{5}$. Two bombs are enough to destroy a bridge. If six bombs are aimed at the bridge, find the probability that the bridge is destroyed.

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Solution

$p(\text{Bomb hitting the target}) = \frac{1}{5}$

$q(\text{Not hitting the target}) = \frac{4}{5}$, $n = 6$

Let X be a r.v. that denotes the hitting the target.

Out of 6 bombs at least 2 bombs destroyed the bridge

$$P(X \geq 2) = 1 - P(X < 2)$$
$$= 1 - \{P(X = 0) + P(X = 1)\}$$
$$= 1 - \left\{ {}^6C_0 \left(\frac{1}{5}\right)^0 \left(\frac{4}{5}\right)^{6-0} + {}^6C_1 \left(\frac{1}{5}\right)^1 \left(\frac{1}{5}\right)^{6-1} \right\}$$
$$= 1 - \left(\frac{4}{5}\right)^6 - 6 \cdot \left(\frac{1}{5}\right)\left(\frac{4}{5}\right)^5$$
$$= 0.345$$

Example 6. If the mean of binomial distribution is 6 and standard deviation is 2, find the probability of at most 2 successes.

Solution

$E(X) = np = 6$

$\sigma = \sqrt{npq} = 2$

$V(X) = 4$, i.e., $npq = 4$

We can write,

$\frac{npq}{np} = \frac{4}{6} \Rightarrow q = \frac{2}{3}$

$p = 1 - q = \frac{1}{3}$

$n = \frac{6}{1/3} = 18$

$$P(X \leq 2) = 1 - P(X > 2)$$
$$= 1 - \{P(X = 0) + P(X = 1)\}$$
$$= 1 - \left\{ {}^{18}C_0 \left(\frac{1}{3}\right)^0 \left(\frac{2}{3}\right)^{18-0} + {}^{18}C_1 \left(\frac{1}{3}\right)^1 \left(\frac{2}{3}\right)^{18-1} \right\}$$
$$= 1 - \left(\frac{2}{3}\right)^{18} - 6\left(\frac{1}{3}\right)^{17}$$
$$= 0.99$$

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Example 7. Six dice are thrown 729 times, how many times do you expect at least 3 die show a five or six?

Solution

Let X denote the r.v. falling 1, 2, 3, 4, 5,6

$P(5 \text{ or } 6) = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}$

$q = \frac{2}{3}$

$n = 6$

$P(x) = {}^nC_x p^x q^{n-x}$

$\therefore$ $P(X \geq 3) = 1 - 0.680 = 0.32.$

$P(\text{At least 3 die show 5 or 6})$

$\therefore$ Required expected number $= N \cdot P(X \geq 3)$
$= 729 \times 0.32$
$= 233$

3.2 Poisson distribution

Poisson distribution is a important discrete probability distribution, which was developed by French Mathematician Simeon Denis Poisson in 1837. Poisson distribution is the limiting case of binomial distribution, under the following conditions:

i. The number of trials is indefinitely large, i.e., $n \to \infty$.

ii. Probability of successes for each trial is sufficiently small, i.e., $p \to 0$.

iii. The product $np = \lambda$(constant, say)

Here $p = \frac{\lambda}{n}$, $q = 1 - \frac{\lambda}{n}$.

Prove that Poisson distribution is the limiting case of binomial distribution.

Proof:

Poisson distribution is the limiting case of binomial distribution under the following conditions:

i. $n \to \infty$

ii. $p \to 0$

iii. np = constant (λ, say)

$p = \frac{\lambda}{n}$, $q = 1 - \frac{\lambda}{n}$

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We have,

Probability of x successes out of n independent trial in a binomial distribution

$$b(x; n, p) = {}^nC_x p^x q^{n-x},\ x = 0, 1, 2, \ldots, n$$

Then using above conditions

$$\lim_{n\to\infty} b(x; n, p) = \lim_{n\to\infty} \frac{n!}{(n-x)!\,x!} p^x q^{n-x}$$

$$= \lim_{n\to\infty} \frac{n(n-1)(n-2)\ldots(n-x+2)(n-x+1)(n-x)!}{(n-x)!\,x!} \left(\frac{\lambda}{n}\right)^x \left(1-\frac{\lambda}{n}\right)^{n-x}$$

$$= \lim_{n\to\infty} \frac{n^x\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\ldots\left(1-\frac{x-2}{2}\right)\left(1-\frac{x-1}{n}\right)}{x!} \frac{\lambda^x}{n^x}\left(1-\frac{\lambda}{n}\right)^{n-x}$$

$$= \frac{\lambda^x}{x!} \lim_{n\to\infty}\left\{\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\ldots\left(1-\frac{x-2}{n}\right)\left(1-\frac{x-1}{n}\right)\right. \lim_{n\to\infty}\left(1-\frac{\lambda}{n}\right)^n \left.\lim_{n\to\infty}\left(1-\frac{\lambda}{n}\right)^{-x}\right.$$

$$= \frac{\lambda^x}{x!}(1.1 \ldots 1.1)e^{-\lambda} \cdot 1 \left[\because \begin{matrix}\lim\limits_{n\to\infty}\left(1-\frac{f}{n}\right)^n = e^{-f} \\ \lim\limits_{n\to\infty}\left(1-\frac{f}{n}\right)^{-x} = 1\end{matrix}\right]$$

$$\therefore \lim_{n\to\infty} b(x; n, p) = \frac{e^{-\lambda}\lambda^x}{x!} \textit{ proved}$$

Definition: A discrete random variable X, is said to follow Poisson distribution with parametric λ if its p.m.f is given as

$$P(x) = \frac{e^{-\lambda}\lambda^x}{x!},\ \text{if } x = 0, 1, 2, \ldots, \infty$$

$$= 0, \text{ Otherwise}$$

In brief, we can write it as $X \sim P(\lambda)$ to denote X is a Poisson variate with parameter λ.

Where, x = number of observation

e = 2.71828 natural logarithm

λ = constant or average or mean

Note: Total probability is 1.

$$\sum_{x=0}^{\infty} P(x) = \sum_{x=0}^{\infty} \frac{e^{-\lambda}\lambda^x}{x!}$$

$$= e^{-\lambda}\left\{1 + \frac{\lambda}{1!} + \frac{\lambda^2}{2!} + \frac{\lambda^3}{3!} + \ldots\right\}$$

$$= e^{-\lambda} \cdot e^{\lambda}\left[\because 1 + \frac{\lambda}{1!} + \frac{\lambda^2}{2!} + \frac{\lambda^3}{3!} + \ldots = e^{\lambda}\right]$$

$$= 1$$

Mean and variance of Poisson distribution

$$E(X) = \sum_{x=0}^{\infty} x \cdot P(x)$$

$$= \sum_{x=0}^{\infty} x \cdot \frac{e^{-\lambda}\lambda^x}{x!}$$

$$= e^{-\lambda} \sum_{x=0}^{\infty} x \cdot \frac{\lambda \cdot \lambda^{x-1}}{x(x-1)!}$$

$$= \lambda \cdot e^{-\lambda} \sum_{x=0}^{\infty} \frac{\lambda^{x-1}}{x(x-1)!}$$

$$= \lambda\, e^{-\lambda}\left(1 + \frac{\lambda}{1!}\frac{\lambda^2}{2!} + \frac{\lambda^3}{3!} + \ldots\right)$$

$$= \lambda e^{-\lambda} e^{\lambda}$$

$\therefore$ Mean $E(X) = \lambda$

$$E(X^2) = \sum_{x=0}^{\infty} x^2 P(x)$$

$$= \sum_{x=0}^{\infty} \{x(x-1) + x\} \cdot \frac{e^{-\lambda}\lambda^x}{x!}$$

$$= \sum_{x=0}^{\infty} x(x-1)\frac{e^{-\lambda}\lambda^x}{x!} + \sum_{x=0}^{\infty} x \cdot \frac{e^{-\lambda}\lambda^x}{x!}$$

$$= \sum_{x=0}^{\infty} x(x-1)\frac{e^{-\lambda}\lambda^2\lambda^{x-2}}{x(x-1)(x-2)!} + \lambda$$

$$= \lambda^2 e^{-\lambda} \sum_{x=0}^{\infty} \frac{\lambda^{x-2}}{(x-2)!} + \lambda$$

$$= \lambda^2 \cdot e^{-\lambda} \cdot e^{\lambda} + \lambda$$

$$= \lambda^2 + \lambda$$

$\therefore$ $V(X) = E(X^2) - [E(X)]^2$

$$= \lambda^2 + \lambda - \lambda^2$$

$$= \lambda$$

Moment generating function

$$M_x(t) = E(e^{xt})$$

$$= \sum_{x=0}^{\infty} e^{tx} \cdot \frac{e^{-\lambda}\lambda^x}{x!}$$

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$$= e^{-\lambda} \sum_{x=0}^{\infty} (e^{t-1}) \frac{X}{x!}$$

$$= e^{-\lambda} \left\{ 1 + \frac{\lambda e^t}{1!} + \frac{(\lambda e^t)^2}{2!} + \frac{(\lambda e^t)^3}{3!} + \ldots \right\}$$

$$= e^{-\lambda} \cdot e^{\lambda e^t}$$

$$= e^{\lambda(e^t - 1)}$$

Example 8. Find the probability that at most 5 defective fuse will be found in a box of 200 fuses. Previous experience shows that 2% of such fuses are defectives.

Solution

Here $p = 0.02$, $n = 200$, $\lambda = np = 200 \times 0.02 = 4$

Let X denote the number of defective fuses in a box.

$P(X \leq 5) = ?$

$$P(X) = \frac{e^{-\lambda}\lambda^x}{x!}$$

$$P(X \leq 5) = P(X = 0) + P(X = 1) = \ldots + P(X = 5)$$

$$= \frac{e^{-4}4^0}{0!} + \frac{e^{-4}4^1}{1!} + \frac{e^{-4}4^2}{2!} + \frac{e^{-4}4^3}{3!} + \frac{e^{-4}4^4}{4!} + \frac{e^{-5}4^5}{5!}$$

$$= e^{-4}\left[1 + \frac{4}{1!} + \frac{4+2}{2!} + \frac{4^3}{3!} + \frac{4^4}{4!} + \frac{4^5}{5!}\right]$$

$$= 0.183\left(\frac{643}{15}\right)$$

$$= 0.785$$

Example 9. The number of visitors of web server per minute follows a Poisson distribution. If the average number of visitors per minute is four, what is the probability that (a) there are two or fewer visitors in one minute? (b) there are exactly two visitors in 30 seconds?

Solution

a. Let X denote the number of visitors to a web page per minute and $\lambda = 4$

$$P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2)$$

$$= \frac{e^{-4} \cdot 4^0}{0!} + \frac{e^{-4}4^1}{1!} + \frac{e^{-4}4^2}{2!}$$

$$= e^{-4}(1 + 4 + 8)$$

$$= 0.238$$

b. Let X denote number of visitors to a web server per 30 seconds. Then

$$\lambda = \frac{4}{2} = 2$$

$$P(X = 2) = \frac{e^{-2}e^2}{2!} = 0.2707.$$

Example 10. The number of road accidents in a year attributed to a taxi driver follows a Poisson distribution with mean 3 out of 1000 taxi drivers, find approximately the number of tax drivers with (a) no accidents in a year. (b) more than three accidents in a year.

Solution

Let X denote the r.v. that the number of road accidents in a year.

$\lambda = 3$, $N = 1000$

a. $$P(X = 0) = \frac{e^{-\lambda}\lambda^x}{x!}$$

$$= \frac{e^{-3}3^0}{0!}$$

$$= \frac{1}{e^3}$$

Approximate number of taxi drivers with no accidents in year

$$= N \cdot P(X = 0)$$

$$= 1000 \times \frac{1}{e^3}$$

$$= 50$$

b. $$P(X \geq 3) = 1 - P(X < 3)$$

$$= 1 - \{P(X = 0) + P(X = 1) + P(X = 2)\}$$

$$= 1 - \left\{\frac{e^{-3}3^0}{0!} + \frac{e^{-3}3^1}{1!} + \frac{e^{-3}3^2}{2!}\right\}$$

$$= 1 - 0.05 - 0.15 - 0.224$$

$$= 1 - 0.424$$

$$= 0.576$$

$\therefore$ Approximate number of taxi drivers with more than 3 accidents in a year

$= 1000 \times 0.576 = 576.$

Example 11. An office switch board receives telephone calls at the rate of 3 calls per minute. What is the probability of receiving (a) no call in one minute interval? (b) at most 3 calls in a 5 minute interval?

Solution

Let X be the r.v. denotes the number of calls received

$$P(x) = \frac{e^{-\lambda}\lambda^x}{x!}$$

a. On an average 3 calls per minute

$$P(X = 0) = \frac{e^{-3}3^0}{0!} = e^{-3} = 0.0498.$$

b. In one minute interval number of calls = 3

Then in five minute interval number of calls $= 3 \times 5 = 15$

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$$P(X \le 3) = P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3)$$

$$= \frac{e^{-15}15^0}{0!} + \frac{e^{-15}15^1}{1!} + \frac{e^{-15}15^2}{2!} + \frac{e^{-15}15^3}{3!}$$

$$= e^{-15}\left(1 + \frac{15}{1!} + \frac{15^2}{2!} + \frac{15^3}{3!}\right)$$

$\therefore$ $P(X \le 3) = 0.0002114$

Example 12. Vehicles arrive an intersection at the rate of 10 per minute. A traffic light cycle lasts 45 seconds. If the number of vehicles that arrive per cycle follows a Poisson distribution, what is the probability that exactly 6 vehicles will arrive in cycle?

Solution

Traffic light cycle lasts = 45 sec.

The rate of vehicles arrives in intersection $= 10$ / minute

$$= \frac{10}{60} \times 45 \text{ per seconds}$$

$\therefore$ $\lambda = 7.5$ per second

$$P(x) = \frac{e^{-\lambda}\lambda^x}{x!}$$

$$P(X = 6) = \frac{e^{-7.5}(7.5)^6}{6!} = 0.1376.$$

Example 13. In a certain factory turning out optical lenses, there is a small chance $\frac{1}{500}$ for any lens to be defective. The lens are supplied in a packets of 10 each. What is the probability that the packet will contain (a) no defective? (b) at most 2 defective lenses?

Solution

Let X denote the number of defective lenses in a packet of 10, then $n = 10$, $p = \frac{1}{500}$.

$$\lambda = 10 \times \frac{1}{500} = 0.02$$

$$P(x) = \frac{e^{-\lambda}\lambda^x}{x!}$$

a. $$P(X = 0) = \frac{e^{-0.02}(0.02)^0}{0!}$$

$$= \frac{1}{e^{0.02}} = 0.002$$

b. $$P(X \le 2) = P(X = 0) + P(X = 1) + P(X = 2)$$

$$= \frac{e^{-0.02}(0.02)^0}{0!} + \frac{e^{-0.02}(0.02)^1}{1!} + \frac{e^{-0.02}(0.02)^2}{2!}$$

$$= 0.99$$

3.3 Hypergeometric distribution

Hypergeometric distribution is used to determine the probability of obtaining certain number of success without replacement from a specific sample size. Let a box contains N items, it can be broken into two types; i.e., $N = N_1 + N_2$ of which N_1 are of first type and N_2 are of second type. If sample size n is drawn from the box without replacement then hypergeometric distribution is used, we have to find the probability of x number of items of type first and $n - x$ number of items of type second.

Definition: A discrete r.v. X, is said to follow hyper geometric distribution with parameter n, N_1 and N_2 if its probability mass function is given by

$$P(x) = \frac{\binom{N_1}{x}\binom{N_2}{n-x}}{\binom{N}{n}}, \text{ if } x = 0, 1, 2, \ldots, (N_1 \le n).$$

$$= 0, \text{ otherwise}$$

Mean and variance

Mean

$$E(X) = \frac{n N_1}{N}$$

Variance

$$V(X) = \frac{n N_1(N - N_1)(N - n)}{N^2(N - 1)}$$

Example 14. A box contains 20 bulbs, among them 6 are defective. If 5 bulbs are drawn at random without replacement, what is the probability that exactly 4 bulbs are defective?

Solution

Let X be a r.v. that number of defective bulbs drawn.

Here $N = 20$, $N_1 = 6$, $N_2 = 14$, $n = 5$

Probability mass function of hypergeometric distribution is

$$P(x) = \frac{\binom{N_1}{x}\binom{N_2}{n-x}}{\binom{N}{n}}$$

$$P(X = 4) = \frac{C(6, 4) \times C(14, 1)}{C(20, 5)}$$

$$= \frac{15 \times 14}{15504}$$

$\therefore$ $P(X = 4) = 0.0135.$

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Example 15. Three defective items are mixed with seven good ones. If three items are drawn at random, find the probability distribution of defective items. Also find the mean and variance of the distribution.

Solution

Let X denote the number of defective items.

Here, $N = 3 + 7 = 10$, $N_1 = 3$, $N_2 = 7$, $n = 3$

$$P(x) = \frac{C(N_1, x) \cdot C(N_2, n - x)}{C(N, n)}$$

$$P(X = 0) = \frac{C(3, 0) \times C(7, 3)}{C(10, 3)}$$

$$= \frac{3 \times 21}{120}$$

$$= \frac{21}{40}$$

$$C(X = 1) = \frac{C(3, 1) \times C(7, 2)}{C(10, 3)}$$

$$= \frac{3 \times 21}{120}$$

$$= \frac{21}{40}$$

$$P(X = 2) = \frac{C(3, 2) \times C(7, 1)}{C(10, 3)}$$

$$= \frac{7 \times 3}{120}$$

$$= \frac{7}{40}$$

$$P(X = 3) = \frac{C(3, 3) \times C(7, 0)}{C(10, 3)}$$

$$= \frac{1}{120}$$

Hence probability distribution of X is:

X:	0	1	2	3
P(x):	7/24	21/40	7/40	1/120

Mean

$$E(X) = \Sigma x \cdot P(X)$$

$$= 0 \times \frac{7}{24} + 1 \times \frac{21}{40} + 2 \times \frac{7}{40} + 3 \times \frac{1}{120}$$

$$= 0.9$$

$$E(X^2) = \Sigma x^2 \cdot P(x)$$

$$= 0^2 \times \frac{7}{24} + 1^2 \times \frac{21}{40} + 2^2 \times \frac{7}{40} + 3^2 \times \frac{1}{120} = 1.3$$

$\therefore$ Variance $V(X) = E(X^2) - [E(X)]^2 = 1.3 - (0.9)^2 = 0.49$

Example 16. A committee of 5 members is to be selected from a group of 6 engineers and 9 chemists. If the selection is made at random, what is the probability that the committee consists of at least one engineer?

Solution

Let X be a r.v. which denote the selection of number of engineers.

Here $N = 6 + 9 = 15$, $N_1 = 6$, $N_2 = 9$, $n = 5$

$$P(x) = \frac{C(N_1, x) \cdot C(N_2, n - x)}{C(N, n)}$$

$$P(X \geq 1) = 1 - P(X = 0)$$

$$= 1 - \frac{C(6, 0) \cdot C(9, 5)}{C(15, 5)}$$

$$= 1 - \frac{126}{3003}$$

$$= 0.958$$

$\therefore$ P(At least one engineer) = 0.958

Example 17. A particular industrial product is shipped in lots of 20. Testing to determine whether an item is defective is costly, hence the manufacturer samples production rather than using 100% inspection plan. A sampling plan constructed to minimize the number of defective shipped to customers calls for sampling five items from each lot and rejecting the lot if more than one defective is observed. (If the lot is rejected each item in the lot is then tested). If a lot contains four defective, what is the probability that it will be accepted?

Solution

Let x be the number of defectives in the sample.

Here, $N = 20$, $N_1 = 4$, $N_2 = 20 - 4 = 16$, $n = 5$.

The lot will be rejected if $x = 2, 3$ or 4, then

$$P(\text{Accept the lot}) = P(x \leq 1)$$

$$= P(x = 0) + P(x = 1)$$

$$P(X) = \frac{C(N_1, x) \cdot C(N_2, n - x)}{c(N, n)}$$

$$P(X \leq 1) = \frac{C(4, 0) \times C(16, 5)}{C(20, 5)} + \frac{C(4, 1) \times C(16, 4)}{C(20, 5)}$$

$$= \frac{91}{323} + \frac{455}{469}$$

$$= 0.7513$$

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3.4 Negative binomial distribution

Negative binomial distribution is a probability distribution of a number of occurrences of success and failures in a sequence of independent trials before specific number of success occurs. The negative binomial experiment consists of independent trials which are repeated until desirable number of success occur. Negative binomial distribution is used if the experiment consists of x repeated and sequence of independent trials, each trial has only two possible outcomes and their probabilities for each trial is fixed and same.

Let x failures preceding the r^{th} success in r + x independent trials. In this situation the last trial must be success whose probability is p. There must be (r – 1) successes out of x + r – 1 trials.

Definition: A discrete random variable X is said to follow negative binomial distribution with parameter r and p if its probability mass function is given by

$$P(x) = \binom{x+r-1}{r-1} p^r q^x, \text{ if } x = 0, 1, 2, \ldots$$

$$= 0, \text{ otherwise}$$

where, x + r is the total number of trail, r is the number of success.

x is the number of failure.

Mean and variance

Mean: $E(X) = \frac{rq}{p}$

Variance: $V(X) = \frac{rq}{p^2}$

Example 18. You are surveying people existing from a polling booth and asked them if they voted independently in the election. The probability that a person voted independent is 20%, what is the probability that 15 people you must be asked before you can find 5 people who voted independently?

Solution

Probability that a person voted independent (p) = 0.2

Probability that a person did not vote independent (q) = 0.8

Let X denote the number of failure to vote, i.e., x = 15 – 5 = 10

The number of success to vote (r) = 5

$$P(x) = \binom{x+r-1}{r-1} p^r q^x$$

$$P(x + r = 15) = \binom{10+5-1}{5-1} (0.2)^5 (0.8)^{10} = {}^{14}C_4 (0.2)^5 (0.8)^{10}$$

$$P(X = 5) = 0.034$$

∴ The probability among 15 people 5 people vote independent = 0.034 or 3.4%.

Example 19. An item is produced in large number, the machine is known to produced 5% defective. A quality control inspector is examining the items by taking them at random. What is the probability that at least 4 items are to be examined in order to get 2 defective? Also find the mean and variance.

Solution

Probability of producing defective items (P) = 0.05

Total number of trials (x + r) = 4

Number of defective (i.e., successes(r = 2, x = 4 – 2 = 2

If 2 defective items are to be obtained then it can happen in 2 or more trials.

$$P(x + r \geq 4) = P(X \geq 2)$$

$$= 1 - P(X < 2)$$

$$P(X \geq 2) = 1 - \{P(X = 0) + P(X = 1)\}$$

$$= 1 - \left\{\binom{0+2-1}{2-1} (0.05)^2 (0.95)^0 + \binom{1+2-1}{2-1} (0.05)^1\right\}$$

$$\therefore \quad P(X \geq 2) = 0.992$$

Mean

$$E(X) = \frac{rq}{p} = \frac{2 \times 0.95}{0.05} = 38$$

$$V(X) = \frac{rq}{p^2} = \frac{2 \times 0.95}{(0.05)^2} = 760$$

Example 20. A student has taken a 5 multiple choice orally. He continues to answer questions until he gets 3 correct answers. What is the probability that he gets them at most 7 questions, if he guesses at each answer?

Solution

Probability of getting right answer (p) = $\frac{1}{5} = 0.2$

Probability of getting wrong answer (q) = 0.8

Maximum number of correct answer (r) = 3

Let X be a r.v. denote the number of wrong answer of questions.

$$P(X \leq x) = P(X \leq 4)$$

$$P(x) = \binom{r+r-1}{r-1} p^r q^x$$

$$= \binom{0+3-1}{3-1} (0.2)^3 (0.8)^0 + \binom{1+3-1}{3-1} (0.2)^3 (0.8)^1 + \binom{2+3-1}{3-1} (0.2)^3 (0.8)^2 +$$

$$\binom{3+3-1}{3-1} (0.2)^3 (0.8)^3 + \binom{4+3-1}{3-1} (0.2)^3 (0.8)^4$$

$$= {}^2C_2 (0.2)^3 + {}^3C_2 (0.2)^3 (0.8) + {}^4C_2 (0.2)^3 (0.8)^2 + {}^5C_2 (0.2)^3 (0.8)^3 + {}^6C_2 (0.2)^3 (0.8)^4$$

$$= 0.1480$$

∴ Required probability = 0.1480.

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Example 21. Mariya is writing an exam with multiple choice questions and his probability of attempting question with right answer is 60%, what is the probability that Mariya gives the third correct answer for the fifth attempted question?

Solution

Probability of being right answer (p) = 0.6

Probability of being wrong answer (p) = 0.4

Let X denote the events of number of wrong answer.

$x + r = 5, r = 3$

$x = 2$

$$P(x, r = 3) = \binom{x + r - 1}{r - 1} p^r q^x$$

$$= \binom{2 + 3 - 1}{3 - 1} (0.6)^3 (0.4)^2$$

$$= 0.207$$

Example 22. A oil company conducts a geological study indicates that an exploratory oil use should have a 20% chance of striking oil. (a) What is the probability that third strike comes on the seventh well drilled? (b) What is the mean and variance of the number of wells? What must be drilled if oil company wants to set up 3 providing wells?

Solution

Probability of striking oil (p) = 0.2

Probability of not striking oil (q) = 0.8

Let X be the r.v. that denotes the number of not well drill.

a. $x + r = 3$

$r = 1, x = 2$

$$P(X = 2) = \binom{x + r - 1}{r - 1} p^r q^x$$

$$= \binom{2 + 1 - 1}{1 - 1} (0.2)^1 (0.8)^2$$

$$= {}^2C_0 (0.2) (0.8)^2 = 0.128$$

$\therefore$ In any case there is about 13% chance that the first strike.

b. $x + r = 7$

$r = 3, x = 4$

$$P(x = 4) = \binom{4 + 3 - 1}{3 - 1} (0.2)^3 (0.8)^4$$

$$= {}^6C_2 (0.2)^3 (0.8)^4 = 0.49$$

$\therefore$ There is about 5% chance that the third strike comes on the 7th well drill.

c. $r = 3$

$$E(X) = \frac{r}{p} = \frac{3}{0.20} = 15$$

$$V(X) = \frac{rq}{p^2} = \frac{3 \times 0.8}{0.20^2} = 60$$

Exercise 3.1

1. What are the conditions under which binomial distribution can be used?
2. Explain the characteristics of binomial distribution.
3. The mean variance of a binomial distribution are 40 and 30 respectively, then find n and p.
4. The mean of a binomial distribution is 4 and its standard deviation is $\sqrt{3}$. What are the values of n, p and q?
5. In a binomial distribution consisting of 5 independent trials, probabilities 1 and 2 successes are 0.4096 and 0.2048 respectively. Find parameter p, mean and variance of the distribution.
6. The average number of defective pieces in the manufacturing of an article is 1 in 10. Find the probability that of getting exactly 3 defective articles in a packet of 10 articles selected at random.
7. If the chance that pack of cargo goods arrives at any airport is 0.9, find the probability that out of 5 packs of cargo goods expected at least 4 will arrive safely?
8. It is known that 5% of the screws manufactured by automatic machine are defective. If sample of 20 screws is selected at random, what is the probability that the sample contains
   a. exactly 2 defective? b. at least 2 defective screws?
9. Food packing apparatus under fills 10% of the containers. Find the probability that for any five containers the number of under filled will be
   a. exactly 3. b. zero c. at least one
10. Warranty records show that the probability that a new car needs warranty repairs in the first 90 days is 0.20. If a sample of six new car is selected, what is the probability of that in the first year (a) none needs a warranty repair? (b) at least one needs a warranty repair?
11. Among 12 solar collectors on display at a trade show 9 are flat plate collectors and others are concentrating collectors. If a person visiting the show randomly selects 4 of the solar collectors to check out, what is the probability that 3 of them will plate collectors?
12. In particular college in Nepal it has been found that 20% of the students withdraw without completing engineering course. Assume that 15 students have register in a new semester, what is the probability that (a) none will withdraw the course? (b) at least two will withdraw the course? (c) Find the mean and variance of the distribution.
13. On average 88% of the items produced by the machine are non-defective. What is the probability that out of random sample of 20 items produced by the machine 5 are defective?
14. The average percentage of failures in CMAT examination of TU is 30, what is the probability that out of 5 students at least 2 passed in the examination?

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15. Truck tyres are tested over rough terrain. Trucks have a 25% chance of failing to complete the test run without a blowout. Fifteen trucks go through the test, find the probability that (a) fewer than four truck's have blowout. (b) more than two trucks have blowout.
16. A multiple choice has 5 questions having four options in each question. If a student is participated in test, then what is the probability of answering correct for all questions and for at least four questions.
17. Explain the conditions that are used in Poisson distribution.
18. Explain the difference between binomial and Poisson distribution.
19. Suppose that the number of claims for missing baggage average 6 per day. Find the probability that on a given day there will be (a) no claim. (b) at least 2 claims.
20. A manufacturer of keyboard of the computer knows that on an average 2% of the production is defective. He sells 100 boxes of keyboard and guarantee that not more than two keyboard will be defective. What is the probability that a box randomly selected is (a) will meet the guaranteed quality? (b) will not meet the guaranteed quality?
21. Past experience shows that 4 bike accidents per day on an average in Kathmandu valley. Out of 100,000 bike registered, find how many of them will have (a) no accidents? (b) at least one accident?
22. If on average three customers arrived per minute at the bank during the lunch hour, what is the probability that in a given minute exactly two customers will arrive? Also what is the chance that more than two customers will arrive in a given minute?
23. If a mean number serious accidents per year in a large factory is four. Find the probability that in a current year there will be at least 10 accidents?
24. An office switch board receives an average of 4 emergency calls in 10 minutes interval, what is the probability that (a) there is no calls in a 10 minute interval? (b) there is at least 2 calls in 5 minute interval?
25. At a checkout counter customer arrive at an average of 1.5 per minute. Find the probability that (a) at most four will arrive in any given minute. (b) at least 3 will arrive during an interval of 2 minutes. (c) at most 5 will arrive during an interval of 6 minute.
26. Six coins are toss ed 64 times. Using Poisson distribution find the approximate probability of getting 6 heads 1 time.

Hint: $\lambda = np = 64 \times \left(\frac{1}{2}\right)^6$, $P(X = 1) = 0.3678$

27. An aircraft has 116 seats, from long experience that on an average 2.5% of the people who have bought the tickets for the flight do not arrive for the flight. The airline sells 120 tickets for a particular flight. What is the probability that (a) more than 116 people arrive for flight? (b) there is no empty seats for a flight?

Hint: $\lambda = np = 120 \times 0.025 = 3$

i. P(More than 116 arrive for flight) = P(Less than 4 do not arrive)
$P(X < 4) = 0.647$

ii. P(Empty seats) = P(Less than 116 people arrive)
= P(5 or more people do not arrive)
$P(x \geq 5)$ = $1 - P(X < 5) = 0.185$

28. A manufacturer company who produces mineral water bottles, finds that 0.1% of the bottles are defective. Bottles are packed in boxes containing 500 bottles. A businessman buys 100 boxes from the producer company, find how many boxes will contain (a) no defective? (b) at least 2 defectives?
29. The number of misprints distributed according to Poisson law such that 100 misprints are distributed randomly throughout 100 pages of books. What is the probability that a page observed at random will contain (a) no misprint? (b) at most three misprint? (c) not less than one misprint?
30. In a certain city, there is a small chance of road accidents during a month is $\frac{1}{120}$. Find the approximate vehicles with at least two accidents, if 20 vehicles are selected at random. It is recorded that 1000 vehicles run in the road of the city during the month.
31. In a certain locality there are 121 houses in which 65 of them are new, find the probability that 7 houses are old out of 25 houses selected randomly.
32. A box contains 25 fuses in which 15 of them are defective. Find the probability of getting at least 5 non-defective fuses, when 12 fuses are drawn randomly from the box.
33. 6 of 18 new car in a highway violate the traffic rule. What is the probability that a traffic who randomly selects 4 of the new car for inspection will catch (a) none of the car violate the rule? (b) at least 3 of the new car violate the rule?
34. A car rental company has 12 Toyota and 8 Nisan cars. If five of these cars are in workshop for repairs, Toyota is as likely to be in for repairs as Nisan, what is the probability that (a) 3 of them are Toyota and 2 are Nisan? (b) at least 3 of them are Toyota? (c) all 5 of them are of the same make?
35. A person of army fired bullet from his rifle at a target, what is the probability that his 10^{th} throw is his 5^{th} hit, if the probability of hitting the target and not hitting the target are equal.
36. An item is produced in large number, the machine is known to produce 5% defective. A quality control inspector is examining the items by taking them at random. What is the probability that at least 4 items are to be examined in order to get 2 defective? Also find the mean and variance.

Hint: $x + r = 4, r = 2$
$P(X + r \geq 4) = P(X + 2 \geq 4)$
or, $P(X \geq 2) = 0.993$
$E(X) = 38, V(X) = 760$

37. Assume that each of your calls to a popular radio station has a probability of 0.02 of connecting that is not obtaining a busy signal. Assume that your calls are independent. (a) What is the probability that your first call that connects your 10^{th} call? (b) What is the probability that it requires more than 5 calls for you to connect?

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Answers

3. 160, 0.25 4. $16, \frac{1}{4}, \frac{3}{4}$ 5. 0.2, 1, 0.8 6. 0.0574 7. 0.9185
8. 0.315, 0.97 9. 0.0081, 0.59, 0.4095 10. 0.2621, 0.7378 11. 0.4218
12. 0.035, 0.881, 3, 2.4 13. 0.0567 14. 0.9692 15. 0.461, 0.764
16. 0.001, 0.0156 19. 0.0025, 0.983 20. $P(X \leq 2) = 0.6765$, $P(X) > 2 = 0.3335$
21. 1932, 98168 22. 0.224, 0.576 23. 0.0081 24. 0.018, 0.68 25. 0.98, 0.576, 0.116
26. 0.3678 27. 0.647, 0.185 28. 61, 10 29. 0.367, 0.981, 0.632
30. Approx 13 vehicles 31. 0.0219 32. 0.596 33. 0.167, 0.084
34. 0.397, 0.704, 0.055 35. 0.123 36. 0.993 37. 0.0167, 0.981

❐

Unit 4

Theoretical Continuous Probability Distribution

4.1 Normal distribution

The normal distribution was first studied when the patterns in errors of measurement were observed to follow a symmetrical, bell shaped distribution. In 1733, De-Moivre obtained the mathematical equation for this distribution while dealing with problems arising in the game of chance. Karl Friedrich Gauss (1777-1855) used this distribution to describe the accidental errors of measurement, then after normal distribution also called Gaussian distribution. Nowadays most of the data relating to economics, business, social science, health science, physical sciences confirm to this distribution and has so widely useful.

Definition: A continuous random variable X is said to follow normal distribution, with mean μ and variance σ^2 if its probability density function (p.d.f.) is given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, \text{ if } -\infty < x < \infty,$$

where, X = Continuous r.v.

σ = Population standard deviation

μ = Population mean

e = 2.71828 (Natural logarithm)

$\pi = \frac{22}{7}$ (Constant)

The mean (μ) and standard deviation (σ) are also called the parameters of the distribution. It is denoted by $X \sim N(\mu, \sigma^2)$, i.e., continuous r.v. X, follows normal distribution with mean μ and variance σ^2.

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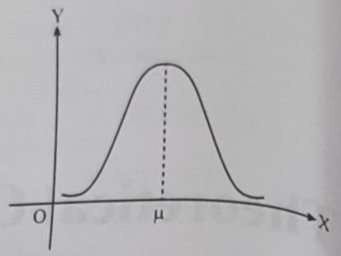
4.2 Curve of the normal distribution

The graph of the normal curve is symmetrical, bell shaped about line $X = \mu$, that extends infinitely in both directions approaching closer to the horizontal axis without touching it. Since the total area under the curve is equal to 1, symmetry implies that area to the right $\mu = 0.5$ and left μ is also 0.5. The shape of curve is determined by σ, larger values of σ reduce the height of the curve and increases flatness.



4.2.1 Properties of normal distribution

- The normal curve is bell shaped and symmetrical about the line $x = \mu$.
- In normal distribution mean = median = mode.
- Total area under the curve is equal to 1, perpendicular bisector divides the curve in to equal halves.
- When μ increases the curve becomes flatten out and for small value of σ. it has sharp peak.
- $f(\mu + x) = f(u - x)$, the density is symmetric about μ.
- The maximum value of f occurs at $x = \mu$.
- The points of inflation of f are at $x = \mu \pm \sigma$.
- It is a unimodel, i.e., the curve has single peak.
- The quartiles are equidistant from median, i.e., $Q_1 + Q_3 = 2\,Md = 2\mu$.
- No portion of the curve lies below the x-axis, since f(x) is non-negative.
- Linear combination of independent-normal variate is also normal variate.
- The maximum probability is occurred at $x = \mu$.

4.2.2 Importance of normal distribution

- The normal distribution is used to analyze the data when there is equally likely chance of being above or below the mean for continuous data.
- Many distributions, which are not normal can be reduced to normal by simple transformations.
- The normal distribution can be used to approximate the binomial and Poisson distribution.
- For large values of n central limit theorem follow the normal distribution.
- It is used in estimation theory, to estimate the population parameter with the help of sample statistics.
- It is used in hypothesis testing, to test the validity of statistical hypothesis.

- In exact sample test like t, F, χ^2-test is based on the assumption that sample is drawn from the parent population follows normal distribution.
- It is used in statistical quality control, for setting control limits for the control chart.
- The normal distribution has numerous mathematical population which make it popular and comparatively easy to manipulate.
- Mean $E(X) = \mu$, and variance $V(X) = \sigma^2$.

4.3 Standard normal distribution

The normal distribution is completely specified by its mean and standard deviations, there may be infinite values of μ and σ, so there may also infinite bell-shaped normal curve. By making a simple transformations the infinite distributions of normal curves can convert into one standard form, which is called standard normal distribution.

If $X \sim N(\mu, \sigma^2)$, then introduce a new variate Z whose mean is 0 and variance 1, i.e., $E(Z) = 0$ and $V(Z) = 1$, then Z is given as

$$Z = \frac{x - \mu}{\sigma}$$

Here $Z \sim N(0, 1)$ and its probability density function is given as

$$F(Z) = \frac{1}{\sigma\sqrt{2}} e^{-\frac{z^2}{2}}, -\infty < Z < \infty$$

Here $Z = \frac{x - \mu}{\sigma}$ is called standard normal variate.

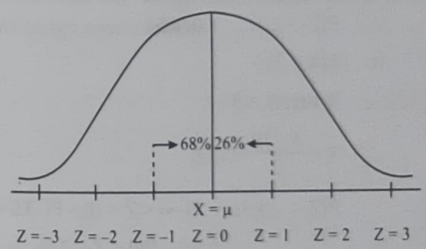
4.3.1 Area property of normal / standard normal distribution

Standard normal variate

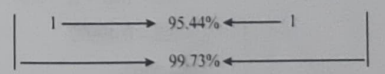
$$Z = \frac{X - \mu}{\sigma}$$

When $X = \mu + \sigma$
$Z = 1$

When $X = \mu - \sigma$
$Z = -1$



In the graph, we indicated the areas between the regions as follows.



$\mu - \sigma \le Z \le \mu + \sigma \quad \Rightarrow -1 \le Z \le 1 = 68.27\%$

$u - 2\sigma \le Z \le \mu + 2\sigma \quad \Rightarrow -2 \le Z \le 2 = 95.45\%$

$u - 3\sigma \le Z \le \mu + 3\sigma \quad \Rightarrow -3 \le Z \le 3 = 99.73\%$

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It comes from as

$$\frac{1}{\sqrt{2\pi}}\int_{-1}^{1} e^{-\frac{z^2}{2}}\,dZ = 0.6827, \frac{1}{\sqrt{2\pi}}\int_{-2}^{2} e^{-\frac{z^2}{2}}\,dZ = 0.9544, \frac{1}{\sqrt{2\pi}}\int_{-3}^{3} e^{-\frac{z^2}{2}}\,dZ = 0.09973$$

The total are from $-\infty \le Z \le \infty$ is equal to 1.

4.4 The normal / Z table

The areas under the curve Z = 0 and any positive value of Z can be checked in the Z or normal table. From this table the area under the standard normal curve between any two ordinates can be found by using the symmetry of the curve about Z = 0.

Example 1. Let X be a normally distributed randomly distributed with mean of 10 and standard deviation 2, find the probability that:

a. X is greater than 15 b. X is less than 3

c. X lies between 11 and 13.6

Solution

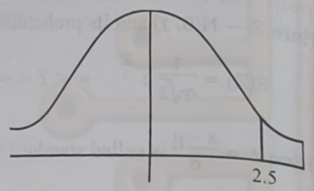
Here, $\mu = 10, \sigma = 2$

We have $Z = \frac{X-\mu}{\sigma}$

a. $P(X > 15) = ?$

When X = 15

$$Z = \frac{15-10}{2} = 2.5$$



$P(Z > 2.5) = P(0 < Z < \infty) - P(0 < Z < 2.5)$

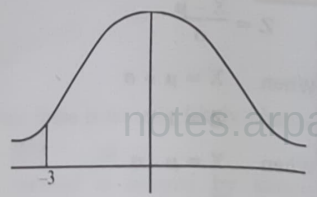
$= 0.5000 - 0.4946$

$\therefore P(Z > 2.5) = 0.0054$

b. $P(X < 3)$

When X = 3

$$Z = \frac{3-10}{2} = -3.5$$



$P(Z < -3.5) = P(-\infty < Z < 0) - P(-3.5 < Z < 0)$

$= 0.5 - P(0 < Z < 3.5)$ [$\because$ Z curve is symmetric]

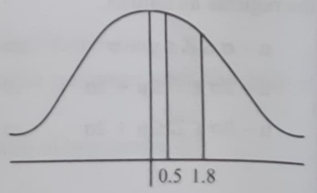
$= 0.5 - 0.49977$

$= 0.0003$

$P(11 < X < 13.6)$

When X = 11

$$Z_1 = \frac{11-10}{2} = 0.5$$



When X = 13.6

$$Z_2 = \frac{13.6-10}{2} = 1.8$$

$P(0.5 < Z < 1.8) = P(0 < Z < 1.8) - P(0 < Z < 0.5)$

$= 0.4641 - 0.191$

$= 0.2726$

$\therefore P(0.5 < Z < 1.8) = 0.2726$

Example 2. Studies show that gasoline use for compact cars sold in the United States is normally distributed with a mean of 35.5 miles per gallon (mpg) and standard deviation of 4.5 mpg. What percentage of compact cars get 40 mpg or more?

Solution

$\mu = 35.5$ mpg, $\sigma = 4.5$ mpg

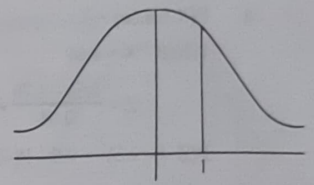
$P(X > 40) = ?$

We have, $Z = \frac{X-\mu}{\sigma}$

When X = 40

$$Z = \frac{40-35.5}{4.5}$$

$= 1$



$P(Z > 1) = P(0 < Z < \infty) - P(0 < Z < 1)$

$= 0.5 - 0.3413$

$\therefore P(Z > 1) = 0.1587$

The percentage exceeding 40 mpg is 15.87%.

Example 3. The pitch diameter of the thread on fitting normally distributed with a mean of 0.4008 cm and standard deviation of 0.0004 cm. The design specifications are 0.4000 ± 0.0010, find the probability that lie in design specifications.

Solution

Here, $\mu = 0.4008, \sigma = 0.004$

$P(0.3990 < X < 0.4010) = ?$

We have, $Z = \frac{X-\mu}{\sigma}$

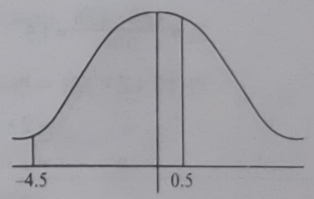
When X = 0.3990

$$Z_1 = \frac{0.3990-0.4008}{0.0004} = -4.5$$

When X = 0.4010

$$Z_1 = \frac{0.4010-0.4008}{0.0004}$$

$= 0.5$



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$P(-4.5 < Z < 0.5) = P(-4.5 < Z < 0) + P(0 < Z < 0.5)$

$= P(0 < Z < 4.5) + P(0 < Z < 0.5)$

$= 0.0 + 0.1915$

$= 0.1915$

$\therefore$ Required probability = 0.1915

Example 4. The burning time of an exponential rocket is a random variable having the normal distribution with mean 4.76 seconds and standard deviation 0.04 second. What is the probability that this kind of rocket will burn:

a. less than 4.66 seconds? b. more than 4.8 seconds?

c. anywhere from 4.7 to 4.82 seconds?

Solution

Let X denote the time of burning rocket.

$\mu = 4.76$ sec, $\sigma = 0.04$ sec

We have, $Z = \dfrac{X - \mu}{\sigma}$

a. $P(X < 4.66) = ?$

When $X = 4.66$

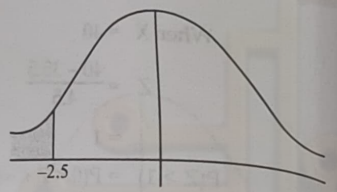
$$Z = \frac{4.66 - 4.76}{0.04} = -2.5$$

$P(Z < -2.5) = 0.5 - P(-2.5 < Z < 0)$

$= 0.5 - P(0 < Z < 2.5)$

$= 0.5 - 0.4938$

$= 0.0062$



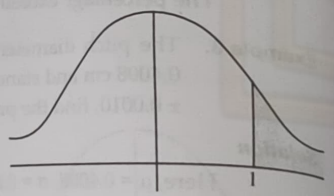
b. $P(X > 4.80) = ?$

$$Z = \frac{4.80 - 4.76}{0.04} = 1$$

$P(Z > 1) = 0.5 - P(0 < Z < 1)$

$= 0.5 - 0.3413$

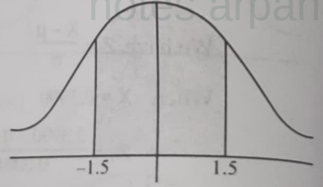
$= 0.1587$



c. $P(4.70 < x < 4.82) = ?$

$$Z_1 = \frac{4.70 - 4.76}{0.04} = -1.5$$

$$Z_2 = \frac{4.82 - 4.76}{0.04} = 1.5$$



$P(-1.5 < Z < 1.5) = P(-1.5 < Z < 0) + P(0 < Z < 1.5)$

$= 2 \times P(0 < Z < 1.5)$ [$\because$ Normal curve is symmetric]

$= 2 \times 0.4332$

$= 0.8664$

Example 5. In BE entrance examination, participated 900 students secured mean marks of 50 with standard deviation 20.

a. Find the number of students securing between 30 and 70.

b. Find the value of score exceeded by the top 90 students.

Solution

Let X denote the marks secured by the students.

$\mu = 50$, $\sigma = 20$, $N = 900$

We have, $Z = \dfrac{X - \mu}{\mu}$

a. $P(30 < X < 70) = P\left(\dfrac{30 - 50}{20} < Z < \dfrac{70 - 50}{20}\right)$

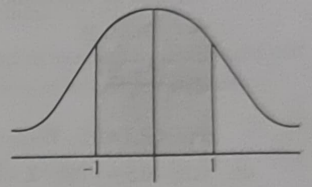
$= P(-1 < Z < 1)$

$P(-1 < Z < 1) = P(-1 < Z < 0) + P(0 < Z < 1)$

$= 2\,P(0 < Z < 1)$

$= 2 \times 0.3413$

$= 0.6826$



$\therefore$ Required number of students = $900 \times 0.6826 = 614.3400 = 614$.

b. Let x_1 be the minimum score of top 90 students

$P(X > x_1) = \dfrac{90}{900} = 0.1$

We have, $Z_1 = \dfrac{x_1 - 50}{20}$ ...(1)

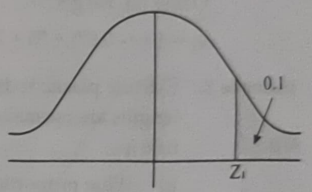
$P(Z > Z_1) = P(0 < Z < \infty) - P(0 < Z < Z_1)$

or, $P(0 < Z < Z_1) = 0.5 - 0.1 = 0.4000$

$\therefore$ $Z_1 = 1.28$

Using (1), we get

$x_1 = 1.28 \times 20 + 50 = 76.$



Example 6. A set of examination marks is approximately normally distributed with mean of 75 and standard deviation of 5. If top 5% student get grade A and bottom 25% get grade F, what marks is the lowest of A and what marks is the highest of F?

Solution

Let X denote the marks secured by the students.

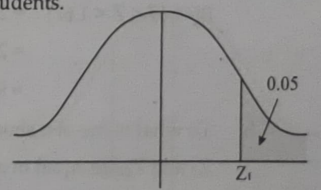
$\mu = 7.5$, $\sigma = 5$.

Let x_1 be the minimum marks obtained by top 5% students.

$P(X \geq x_1) = 0.05$

We have, $Z = \dfrac{X - \mu}{\sigma}$

$Z_1 = \dfrac{x_1 - 75}{5}$ ...(1)



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$P(Z \geq Z_1) = 0.5 - P(0 < Z < Z_1)$

or, $0.05 = 0.5 - P(0 < Z < Z_1)$

or, $P(0 < Z < Z_1) = 0.4500$

$\therefore$ $Z_1 = 1.64$ from normal table.

Using (1), we get

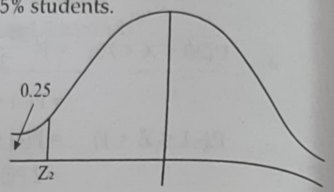
$x_1 = 5 \times 1.64 + 75 = 83.2$.

Again,

Let x_2 be the maximum marks obtained by bottom 25% students.

$P(X \leq 2) = 0.25$

$$Z_2 = \frac{x_2 - 75}{5} \quad \ldots(2)$$



$P(Z < Z_2) = 0.5 - P(-Z_2 < Z < 0)$

or, $0.25 - 0.5 = - P(0 < Z < Z_2)$

or, $P(0 < Z < Z_2) = 0.2500$

$\therefore$ $Z_2 = - 0.67$

Using (2), we get

$x_2 = 5 \times (- 0.67) + 75 = 71.65$.

Example 7. Extrude plastic rods are automatically cut into normally length of 6 inches. Actual lengths are normally distributed about mean of 6 inch and their standard deviation 0.06 inc.

a. What proportion of the rod exceeds tolerance limits of 5.9 to 6.1 inches?

b. To what value does the s.d. need to reduce if 99% of the rods must be with in tolerance?

Solution

a. Let X be the actual length of plastic rods.

Here, $\mu = 6$, $\sigma = 0.06$

We have, $Z = \frac{X - \mu}{\sigma}$, $Z_1 = \frac{5.9 - 6}{0.06}$, $Z_2 = \frac{6.1 - 6}{0.06}$

$P(5.9 < X < 6.1) = ?$

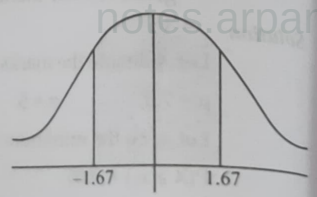
$$P(5.9 < X < X < 6.1) = P\left(\frac{5.9 - 6}{0.06} < Z < \frac{6.1 - 6}{0.06}\right)$$

$= P(-1.67 < Z < 1.67)$

$P(-1.67 < Z < 1.67) = 2\,P(0 < Z < 1.67)$

$= 2 \times 0.4525$

$= 95.5\%$



b. To what value of σ should reduced for 99% tolerance limits.

Z_1 and Z_2 are equal in magnitude but opposite in direction.

$2\,P(0 < Z < Z_2) = 0.9900$

$P(0 < Z < Z_2) = 0.4950$

$\therefore$ $Z_2 = 2.57$ from normal table.

We also have,

$$Z_2 = \frac{6.1 - 6}{\sigma}$$

or, $\sigma = \frac{0.1}{2.57} = 0.0389$.

$\therefore$ For 99% tolerance limits around 3.89% value of s.d. should be reduced.

Example 8. The incomes of group of 10,000 persons we found to be normally distributed with mean Rs. 750 and s.d. Rs. 50. Find the lowest income of 10% richest persons.

Solution

μ = Rs. 750, σ = Rs. 50, N = 10,000

Let lowest income of richest 10% person = x_1

$P(X > x_1) = 0.1$

$$Z_1 = \frac{x_1 - 750}{50} \quad \ldots(1)$$

$P(Z > Z_1) = 0.5 - P(0 < Z < Z_1)$

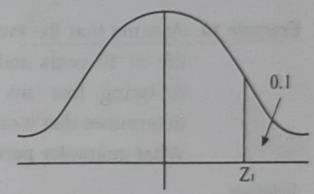
or, $0.1 - 0.5 = - P(0 < Z < Z_1)$

or, $P(0 < Z < Z_1) = 0.4000$

$Z_1 = 1.28$ from normal table.

Using (1), we get

$x_1 = 1.28 \times 50 + 750 = 814$.



$\therefore$ Lowest income of richest 10% persons is Rs. 814.

Example 9. The breaking strength of plastic bags used for packaging produce is normally distributed with mean of 5 lbs per square inch and a standard deviation of 1.5 pounds per square inch. What portion of the bags has of breaking strength of:

a. at least 3.6 pounds per square inch?

b. between what two values symmetrically distributed around the mean will 95% of the breaking strength fall?

Solution

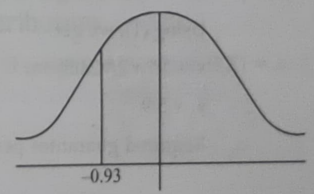
Let r.v. X, denote the breaking strength of plastic bags.

$\mu = 5$ lbs, $\sigma = 1.5$ lbs

We have, $Z = \frac{X - \mu}{\sigma}$

a. $P(X \geq 3.6) = ?$

$$Z = \frac{3.6 - 5}{1.5} = -0.93$$



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$P(Z \geq -0.93) = P(-0.93 < Z < 0) + 0.5$

$= P(0 < Z < 0.93) + 0.5$

$= 0.3238 + 0.5$

$= 0.8238$

$= 82.38\%$

b. Let 95% breaking strength will fall with in x_1 and x_2.

$Z_1 = \frac{x_1 - 5}{1.5}$ ...(1)

$Z_2 = \frac{x_1 - 5}{1.5}$ ...(2)

Z_1 and Z_2 are equal in magnitude, but opposite in direction. So,

$2P(0 < Z < Z_1) = 0.9500$

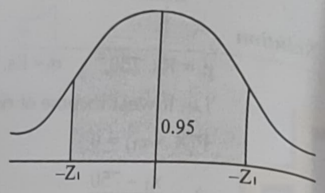
$P(0 < Z < Z_1) = 0.4750$

$\therefore$ $Z_1 = -1.96, Z_2 = 1.96$

Using (1) and (2), we get

$x_1 = 1.5(-1.96) + 5 = 2.06$

$x_2 = 1.5(1.96) + 5 = 7.94$



Hence, 95% of breaking strength will fall between 2.06 and 7.94 pounds.

Example 10. Assume that the average life of a small motor is normally distributed with mean life of 10 years and standard deviation 2 years. The company has a policy of replacing free any motor that fails whilst under guarantee. The company determines that it can only afford to freely replace up to 2% of motors it produces. What guarantee period should be set?

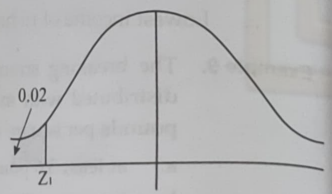
Solution

Let x_1 be the required guarantee period.

$\mu = 10$ years, $\sigma = 2$ years

$P(X < x_1) = 0.02$

$Z_1 = \frac{x_1 - 10}{2}$ ...(1)



$P(Z < Z_1) = 0.5 - P(-Z_1 < Z < 0)$

or, $0.02 - 0.5 = -P(0 < Z < Z_1)$ [$\because$ By symmetric property]

or, $P(0 < Z < Z_1) = 0.4800$

$\therefore$ $Z_1 = -2.05$ using Z-table.

Using (1), we get

$x_1 - 10 = 2(-2.05)$

$x_1 = 5.9$

$\therefore$ Required guarantee period to be set is 5.9 years.

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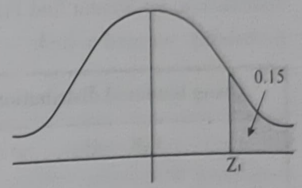
Example 11. In an examination 15% of the student got first class 60 marks or above, while 40% securing below 40 marks. Assuming the marks are normally distributed, estimate the mean and standard deviation.

Solution

Let X be a r.v. denote the marks obtained by student in an examination.

$P(X \geq 60) = 15\%$

$Z_1 = \frac{60 - \mu}{\sigma}$ ...(1)



$P(Z < Z_1) = 0.5 - P(0 < Z < Z_1)$

or, $0.15 - 0.5 = -P(0 < Z < Z_1)$

or, $P(0 < Z < Z_1) = 0.3500$

$\therefore$ $Z_1 = 1.02$

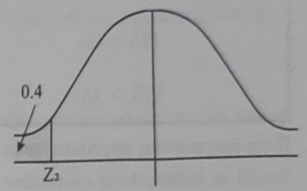
From (1), we get

$60 - \mu = 1.02\,\sigma$

or, $1.02\,\sigma + \mu = 60$ ...(2)

Again, $P(X < 40) = 0.4$

$Z_2 = \frac{40 - \mu}{\sigma}$ ...(3)



$P(Z < Z_2) = 0.5 - P(-Z_2 < Z < 0)$

or, $P(0 < Z < Z_2) = 0.1000$

$\therefore$ $Z_2 = -0.25$ from Z-table.

From (3), we get

$60 - \mu = 0.25\,\sigma$

or, $0.25\,\sigma - \mu = -60$ ...(4)

Solving equation (2) and (4), we get

$\mu = 43.94$

$\sigma = 15.74$

4.5 Relation between binomial and normal distribution

If the number of trails be infinitely very large, i.e., $n \to \infty$ and neither p or q is very small, then

$$Z = \frac{X - \mu}{\sigma} = \frac{X - np}{\sqrt{npq}}.$$

4.5.1 Relation between Poisson and normal distribution

Normal distribution is a limiting case of Poisson distribution if mean $\lambda = \infty$, where $E(X) = \lambda$.

$$Z = \frac{X - \lambda}{\sqrt{\lambda}}.$$

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4.5.2 Continuity correction

Continuity correction is applied when we want to use a normal distribution to approximate a binomial distribution. To make a discrete random variable X, as continuous variable we simply add or subtract 0.5 to a discrete value X.

Suppose a coin is tossed 100 times, we want to find the probability that head occur less than or equal to 45 time, i.e., $P(X \leq 45)$. To use normal distribution to approximate binomial distribution, we should find $P(X \leq 45.5)$. When we should add or subtract 0.5, based on the probability, we need to find:

Using binomial distribution	Using normal distribution with continuity correction
$P(X = 45)$	$P(44.5 < X < 45.5)$
$P(X \leq 45)$	$P(X < 45.5)$
$P(X < 45)$	$P(X < 44.5)$
$P(X \geq 45)$	$P(X > 44.5)$
$P(X > 45)$	$P(X > 45.5)$

If we use normal approximation to binomial or Poisson distribution, we would necessary have to add or subtract the correction factor.

Example 12. If 10% of the electric bulbs manufactured by the company are defective, find the probability that in a sample of 200 bulbs

a. 40 or more bulbs are defective.

b. Exactly 30 bulbs are defective.

Solution

$p = 0.1$, $n = 200$ since n is large, we use Poisson distribution

$\lambda = np$ (= mean = variance) $= 200 \times 0.1 = 20$.

Let X denote the number of defective bulb. We have,

$$Z = \frac{X - \mu}{\sigma} = \frac{X - \lambda}{\sqrt{\lambda}} \sim N(0, 1).$$

a. $P(X \geq 40) = P(X > 40 - 0.5)$ using continuity correction

$P(X > 39.5) = ?$

$$Z = \frac{39.5 - 20}{\sqrt{20}} = 4.36$$

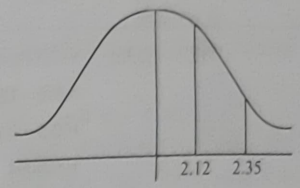
$P(Z > 4.36) = 0.$

b. $P(X = 30) = P(X = 30 \pm 0.5)$ Using continuity correction

$P(29.5 < X < 30.5) = ?$

$$Z_1 = \frac{29.5 - 20}{\sqrt{20}} = 2.12,$$

$$Z_2 = \frac{30.5 - 20}{\sqrt{20}} = 2.35$$



$P(2.12 < Z < 2.35) = P(0 < Z < 2.35) - P(0 < Z < 2.12)$

$= 0.4906 - 0.4830$

$= 0.0076$

Example 13. In international flight, an airline has the three different choices on its desert menu, ice-cream, apple juice and chocolate cake. Based on past passed experience, the airline feels that each desert is equally chosen. If a random sample of 90 passengers is selected, what is the probability that:

a. At least 20 will choose ice-cream for desert?

b. Less than 20 will chosen ice-cream for desert? Use normal approximation to binomial and Poisson distribution separately.

Solution

Let X be a random variable that number of passenger choose ice-cream as desert menu.

For normal distribution to binomial distribution

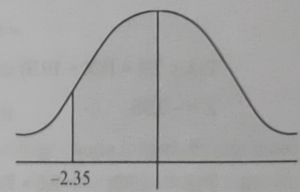
$$P = \frac{1}{3}, \quad q = \frac{2}{3}, \quad n = 90$$

$\mu = np = 30$

$V(X) = npq = 20$

a. $P(X \geq 20) = P(X > 19.5)$ Using continuity correction

$$Z = \frac{X - np}{\sqrt{npq}} = \frac{19.5 - 30}{\sqrt{20}} = -2.35.$$



$P(Z > -2.35) = P(-2.35 < Z < 0) + 0.5$

$= P(0 < Z < 2.35) + 0.5$

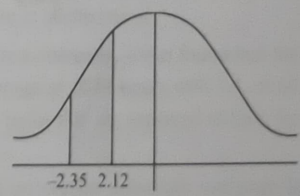
$= 0.4906 + 0.5$

$= 0.9906$

b. $P(X = 20) = P(19.5 < X < 20.5)$

$$Z_1 = \frac{19.5 - 30}{\sqrt{20}} = -2.35$$

$$Z_2 = \frac{20.5 - 30}{\sqrt{20}} = -2.12$$



$P(-2.35 < Z < -2.12) = 0.4906 - 0.4830 = 0.0072$

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c. $P(X < 20) = P(X < 19.5)$

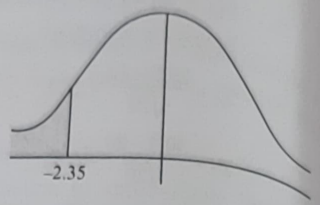
$Z = -2.35$

$P(Z < -2.35) = 0.5\ P(-2.35 < Z < 0)$

$= 0.5 - P(0 < Z < 2.35)$

$= 0.5 - 0.4906$

$= 0.0094$



Again, using normal distribution to Poisson distribution

Mean $(\lambda) = np = 30$

Variance $(\lambda) = 30$

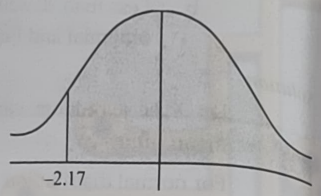
i. $P(X \geq 20) = P(X > 19.5)$

$$Z = \frac{X - \lambda}{\sqrt{\lambda}} = \frac{19.5 - 30}{\sqrt{30}} = -2.17.$$

$P(Z > -2.17) = P(-2.17 < Z < 0) + 0.5$

$= 0.4850 + 0.5$

$= 0.9850$



ii. $P(X = 20) = P(19.5 < X < 20.5)$

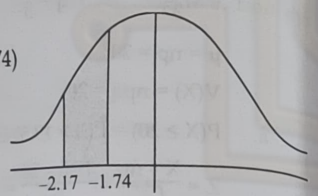
$$Z_1 = \frac{19.5 - 30}{\sqrt{30}} = -2.17$$

$$Z_2 = \frac{20.5 - 30}{\sqrt{30}} = -1.74$$

$P(-2.17 < Z < -1.74) = P(0 < Z < 2.17) - P(0 < Z < 1.74)$

$= 0.4850 - 0.4591$

$= 0.0299$



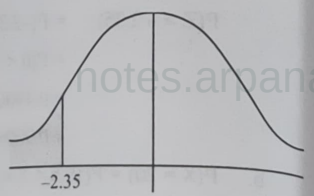
$P(X < 20) = P(X < 19.5)$ using c.c.

$Z = -2.35$

iii. $P(Z < -2.35) = 0.5 - P(0 < Z < 2.35)$

$= 0.5 - 0.4850$

$= 0.015$



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Exercise 4.1

1. A normal random variable x has mean 10 and standard deviation 2. Find the probability associated with each of the following intervals.
   a. $x > 13.5$ b. $x < 8.2$ c. $9.4 < x < 10.6$
2. The breaking strength (in Newton) of a synthetic fabric is normally distributed with mean 800 N and s.d. 144 N. The purchaser of the fabric to have a strength of at least 772 N, what is its probability?
3. The marks secured by students in entrance exam is normally distributed with mean 150 and variance 10. Find the percentage of students who secured (a) less than 172? (b) more than 174? (c) between 150 and 180?
4. The mean weight of 500 male students at certain college is 65.6 kg with standard deviation of 10 kg. Assuming that the weights are normally distributed, how many students weight (a) more than 75.5 kg? (b) between 55.5 and 75.5 kg?
5. The mean of the inner diameter (inches) of the sample of 400 tubes produced by a machine is 0.502 and the standard deviation is 0.05. The purpose for which these tubes are produced allows a maximum tolerance of 0.496 to 0.508, otherwise the tubes are considered defective. What percentage of tubes produced by the machine is expected to be defective if the diameters are found to be normally distributed?

   Hint: Required probability $= [1 - P90.496 \leq x \leq 0.508)] = 99.44\%$
6. The amount of fuel consumed by an engine of jet flight between two cities is normally distributed with mean 5.7 tonnes with standard deviation of 0.5 tonnes. Carrying to much fuel is inefficient as it shows the plane. If however, too little fuel is loaded on the plane, an emergency loaded may be necessary. The airline would like to determine the amount of fuel to load so that there will be a 0.99 probability that the plane will arrive at its destination.

   Hint: Level of confident = 0.99
   Level of risk $(\alpha) = 0.01$
   For two tail risk $Z_{\alpha/2}$ (1%) = 2.57
   Required limits are: $P(\mu - 2.57\ \sigma < X < \mu + 2.57\ \sigma) = 0.99$
7. The time needed to complete a final examination in a certain college course is normally distributed with a mean of 80 minutes and standard deviation of 10. What is the probability that (a) the exam is completing one hour or more? (b) a student will complete the exam in no more than 75 minutes? (c) assume that the class has 60 students and the examination period is 90 minutes in length, how many students do you expect will be able to complete the exam in allotted time?
8. As a result of tests on 20,000 electric bulbs manufactured by a company, it was found that the life time of the bulb was normally distributed with an average of 2040 hours with s.d. of 60 hours. On the basis of information, estimate the number of bulbs that are expected to burn for less than 1960 hours?
9. From past experience it is known that time to download on certain university webpage is normally distributed with mean 7 seconds and standard deviation 2 seconds. What is the probability that the download will be completed under 3.5 seconds?

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10. A Trucking Company determined that on an basis the distance travelled per truck is normally distributed with mean of 50 thousand miles and s.d. of 12 thousand miles. (a) What proportion of trucks can be expected to travel between 34 thousand and 50 thousand miles in a year? (b) What proportion of trucks can be expected to travel either below 30 or above 60 thousand miles in a year?

11. A statistical analysis of 1,000 long distance telephone calls made from the headquarters of the NTC computer cooperation indicates that the length of these calls is normally distributed worth mean 240 seconds and s.d. 40 seconds. (a) What percentage of calls lasted less than 180 seconds? (b) What is the probability that a particular call lasted between 180 and 300 seconds? (c) How many calls lasted less than 180 seconds or more than 130 seconds? (d) What is the length of particular call if only 1% of all calls are shorter?

12. Specification for contain job all for washers with an inside diameter of 0.300 ± 0.005 inch. If the inside diameters of the washers supplied by a given company may be looked upon a random variable having normal distribution with mean 0.302 inch and s.d. 0.003 inch. What percentage of these washers will meet specification?

13. In certain examination, the percentage of passes and distinctions were 45 and 9 respectively. Estimate the average marks and standard deviation obtained by the candidates. Assume the distribution of marks to be normal. The minimum pass marks and distinction marks being 40 and 75.

14. In a distribution exactly normal, 7% of the items are under 35 and 89% are under 63. What are the mean and standard deviation of the distribution?

15. In a normal distribution 31% of the items are under 45 and 8% are over 64. Find the mean and s.d. of the distribution.

16. Incomes of a group of 2000 employees in certain company were found to be normally distributed with mean Rs. 55,000 and standard deviation Rs. 10,000. Find the expected number of employees with income (a) below 40 thousand. (b) above 80 thousand. (c) between 60 and 70 thousand. (d) the highest income of low ranked 10% of the employees. (e) the lowest income of high ranked 15% of the employees.

17. A set of final examination grads in statistic course was found to be normally distributed with mean of 73 and standard deviation of 8. (a) What percentage of students scored between 65 and 89? (b) What is the probability of getting grade no higher than 91 in this exam? (c) Only 5% of the students taking the test scored higher than what grade?

18. Assume that the mean height of the students of certain college be 5.5 feet with variance of 2.5 feet. How many students in that college of 1000 would you expect to be over 6 feet tall? Assume the height of students is normally distributed?

19. A particular product has a declared weight of 60 gm. Assume the process mean is 61.8 gm and standard deviation is know to be 1.5 gm. Assume the weight to be normally distributed. (a) What proportion of the product will be above the declared weight? (b) Determine the proportion of product below 95% of the declared weight. (c) Determine the proportion of product between 60 gm and 63 gm.

20. In a normal distribution with standard deviation of 5.0, the probability that an observation selected at random exceeds 21 is 0.4. Find the mean of the distribution.

21. One of the components produced by the certain firm 40% are faulty. What is thee probability that 50 components contains 10 defective? By using binomial distribution and normal approximation to binomial distribution.

22. There are 600 bulbs in a box, out of them 60 are defective. What is the probability that out of sample of 20 bulbs (a) exactly 5 are defective? (b) at most 4 are defective, by using normal approximation to Poisson distribution?

Answers

2. 0.9713 4. 81, 341 5. 99.44% 7. 0.977, 0.308, 10
8. 1836 9. 0.401 10. 0.408, 0.252 11. 0.67, 0.866, 930, 147.2
12. 0.8314 13. 36, 29 14. 60.97, 14.87 15. 50, 10
16. 139, 12, 483, 42, Rs. 220 and Rs. 65,400 17. 0.818, 0.978, 86.16
18. 375 approx 19. 0.884, 0.007, 0.673 20. 19.7 21. 0.0015, 0.019

4.6 Rectangular or uniform distribution

Definition: A continuous random variable X is said to follow uniform distribution over an interval $[a, b]$ if its probability density function is given by

$$f(x) = \frac{1}{b-a}, \text{ if } a < x < b = 0, \text{ otherwise}$$

Mean and variance

$$\text{Mean } E(X) = \int_a^b x\, f(x)\, dx$$

$$= \frac{1}{b-a}\left[\frac{x^2}{2}\right]_a^b$$

$$= \frac{a+b}{2}$$

$$\therefore \quad E(X) = \frac{a+b}{2}.$$

$$E(X^2) = \frac{b^2 + ab + a^2}{3}$$

$$\text{Variance } V(X) = \frac{(b-a)^2}{12}$$

Example 14. Suppose the time it take a data collection operator to fill out an electronic form for a database is uniformly distributed between 1.5 and 2.2 minutes. (a) What is the probability that it will take less than 2 minutes to fill out the form? (b) What is the mean and variance of the time? (c) Determine the cumulative distribution function of the time it takes to fill out the form?

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Solution

Let X be a r.v. denote the time taken to fill out form.

a. $P(X < 2) = ?$

We have,

$$f(x) = \frac{1}{b-a}$$

$$P(X < 2) = \int_{1.5}^{2} \frac{1}{b-a}\,dx$$

$$= \frac{1}{0.7}[x]_{1.5}^{2}$$

$$= 0.714$$

b. $E(X) = \frac{a+b}{2}$

$$= \frac{1.5+2.2}{2}$$

$V(X) = \frac{(b-a)^2}{12}$

$$= \frac{(2.2-1.5)^2}{12}$$

$$= 0.041$$

c. Here, $f(x) = \frac{1}{b-a}$

$$= 1.43$$

Cumulative distribution function

$$F(X) = \frac{x-a}{b-a}, a < x < b$$

$$= \frac{x-1.5}{0.7}, 1.5 < x < 2.2$$

Example 15. The train on a certain route run every half hour in a day time. What is the probability that a man entering the station at a random time during this period will have to wait at least 15 minutes?

Solution

Let X be a r.v. denote waiting time. The man waits in the station (0, 30) minutes.

$$f(x) = \frac{1}{b-a}, a < x < b$$

$$= \frac{1}{30}, 0 < x < b$$

$$P(X \geq 15) = \int_{15}^{30} \frac{1}{30}\,dx$$

$$= \frac{1}{30}[x]_{15}^{30}$$

$$= \frac{1}{30}(30-15)$$

$$= \frac{15}{30}$$

$$= \frac{1}{2} = 0.5$$

4.7 Gamma distribution

Gamma distribution is a continuous probability distribution which is defined as two parameters, i.e., shape parameter and inverse scale parameter.

Definition: A continuous variable X, is said to follow gamma distribution with parameter α and β if its probability density function is given as

$$f(x) = \frac{1}{b^{\alpha}\lceil\alpha} \cdot x^{\alpha-1} e^{-x/\beta}, \text{ if } x > 0, \alpha > 0, \beta > 0$$

$$= 0, \text{ otherwise}$$

Mean and variance

Mean: $E(X) = \alpha\beta$

Variance $V(X) = \alpha\beta^2$

Remarks:

1. The gamma function $\lceil\alpha$ (read as gamma alpha), given as:

$$\lceil\alpha = \int_0^{\infty} x^{\alpha-1} e^{-x}\,dx, \qquad \frac{\lceil\alpha}{a^{\alpha}} = \int_0^{\infty} x^{\alpha-1} e^{-ax}\,dx$$

$$\lceil\alpha = (\alpha-1)!, \qquad \left\lceil\frac{1}{2} = \sqrt{\pi}\right.$$

2. If $\alpha = 1, \beta = \frac{1}{\lambda}$ gamma distribution reduces to exponential distribution.

3. Gamma distribution tends to normal distribution as $\lambda = \frac{1}{\beta} \to \infty$.

4. Gamma is useful in the study of electrical supply in certain area, distribution of petrol. waiting time in life testing, waiting time, etc.

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Example 16. In a certain industrial area the daily consumption of electric power (in millions unit) can be described as a random variable having gamma distribution with $\alpha = 3$ and $\beta = 2$. If the power plant if this area has a daily capacity of 12 million units, what is the probability that this power supply is not sufficient on any given day?

Solution

Let r.v. X denotes daily consumption of electricity (kwh, i.e., units).

Here, $\alpha = 3, \beta = 2$

The p.d.f. of gamma distribution is

$$f(x) = \frac{1}{\beta^{\alpha}\lceil\alpha} x^{\alpha-1} e^{-x/\beta}, \; x > 0$$

or, $$f(x) = \frac{1}{2^3\lceil 3} x^{3-1} e^{-x/2}$$

$$= \frac{1}{16} x^{2} e^{-x/2}$$

Probability that the power supply is not sufficient,

$$P(X > 12) = 1 - P(X \le 12)$$

$$= 1 - \frac{1}{16}\int_0^{12} x^2 e^{-x/2}\, dx$$

$$= 1 - \frac{1}{16}\left[\frac{x^2 \cdot e^{-x/2}}{(-1/2)} - \frac{2x \cdot e^{-x/2}}{(-1/2)(-1/2)} + \frac{2e^{-x/2}}{(-1/2)(-1/2)(-1/2)}\right]_0^{12}$$

$$= 1 - \frac{1}{16}\left[-2x^2e^{-x/2} - 8xe^{x/2} - 16e^{-x/2}\right]_0^{12}$$

$$P(X > 12) = 0.0619$$

Example 17. In Bhaktapur the daily consumption of water (in millions of galloons) follows approximately gamma distribution with $\alpha = 2$ and $\beta = 3$. If daily capacity of this district is 9 million galloons of water, show that the probability that on any given day the water supply is inadequate i.e., $4e^{-3}$.

Solution

Let X be r.v. which denotes daily consumption of water (in millions of gallons).

$\alpha = 2, \beta = 3$

We have,

$$f(x) = \frac{1}{\beta^{\alpha}\lceil\alpha} x^{\alpha-1} e^{-x/\beta}$$

$$= \frac{1}{3^2\lceil 2} x^{2-1} e^{-x/3}$$

$$= \frac{1}{9} x\, e^{-x/3}$$

The probability that the water is inadequate

$$P(X > 9) = 1 - P(X \le 9)$$

$$= 1 - \int_0^9 \frac{1}{9} x e^{-x/3}\, dx$$

$$= 1 - \frac{1}{9}\left\{\left[\frac{x\, e^{-x/3}}{-1/3}\right]_0^9 - \int_0^9 \frac{e^{-x/3}}{(-1/3)}\, dx\right\}$$

$$= 1 - \frac{1}{9}\left\{-27e^{-3} - 0 + 3\left[\frac{e^{-x/3}}{-1/3}\right]_0^9\right\}$$

$$= 1 + 3e^{-3} + e^{-3} - 1$$

$$= 4e^{-3}$$

4.8 Beta distribution

The beta distribution is a continuous probability distribution often used to model the uncertainty about the probability of success of an experiment.

Definition: A continuous random variable X is said to follow beta distribution with parameters α and β if its probability density function is given as;

$$f(x) = \frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1} \text{ if } \alpha > 0, \beta > 0, 0 < x < 1$$

$$= 0, \text{ Otherwise}$$

where, $B(\alpha, \beta)$ is the beta function, the random variable having beta distribution is also called beta random variable.

$$B(\alpha, \beta) = \int_0^1 x^{\alpha-1} (1-x)^{\beta-1}\, dx$$

also $$B(\alpha, \beta) = \frac{\lceil\alpha \; \lceil b}{\lceil(\alpha + \beta)}$$

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Note: Total probability of beta distribution is

$$\int_0^1 f(x)\, dx = \int_0^1 \frac{x^{\alpha-1}(1-x)^{\beta-1}\, dx}{B(\alpha, \beta)}$$

$$= \frac{1}{B(\alpha, \beta)} \cdot B(\alpha, \beta)$$

$$= 1$$

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Mean and variance

$$\text{Mean: } E(X) = \frac{\alpha}{\alpha + \beta}$$

$$\text{Variance: } V(X) = \frac{\alpha\beta}{(\alpha + \beta)^2 (\alpha + \beta + 1)}$$

Example 18. If the annual proportion of erroneous income tax refunds filed with the IRS can be looked upon as a random variable having a beta distribution with $\alpha = 2$ and $\beta = 3$, what is the probability that in any given years, there will be fewer than 10% erroneous returns?

Solution

Let X denote the proportion of erroneous income tax return. $\alpha = 2, \beta = 3$.

We have,

$$f(x) = \frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1 - x)^{\beta-1}$$

$$= \frac{\lceil(\alpha + \beta)}{\lceil\alpha \lceil\beta} x^{2-1} (1 - x)^{3-1}$$

$$= \frac{\lceil(5)}{\lceil 2 \lceil 3} x(1 - x)^2$$

$$= \frac{(5 - 1)!}{(2 - 1)! (3 - 1)!} x(1 - x)^2 \qquad \text{[4 factorial]}$$

$$= 12x(1 - x)^2$$

Probability that in any given year there will be fewer than 10% return

$$P(X < 0.1) = \int_0^{0.1} 12x(1 - x)^2 \, dx$$

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$$= 12 \int_0^{0.1} (x - 2x^2 + x^3) \, dx$$

$$= 12 \left\{ \left[\frac{x^2}{2}\right]_0^{0.1} - 2\left[\frac{x^3}{3}\right]_0^{0.1} + \left[\frac{x^4}{4}\right]_0^{0.1} \right\}$$

$$= 0.0523$$

Example 19. In a certain country, the proportion of highway sections requiring repairs in any given year is a random variable having the beta distribution with $\alpha = 5$ and $\beta = 2$. Find (a) On the average what percentage of highway sections required repairs in any given years? (b) The probability that at most half of the highway section will required repairs in any given year?

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$$P(X \le 0.25) = \int_0^{0.025} f(x)\,dx$$

$$= 12\int_0^{0.025} (x^2 - x^3)\,dx$$

$$= 12\left[\frac{x^3}{3} - \frac{x^4}{4}\right]_0^{0.25}$$

$$= 0.051$$

4.9 Exponential distribution

The exponential distribution is one of the widely used continuous distributions. It is often used to model the time elapsed between events.

Definition: A continuous random variable X is said to follow exponential distribution with parameter λ, if its probability density function is given by:

$f(x) = \lambda e^{-\lambda x}$, for $x \ge 0$

$= 0$, Otherwise

Mean: $E(X) = \frac{1}{\lambda}$

Variance: $V(X) = \frac{1}{\lambda^2}$

4.9.1 Memory loss property of exponential distribution

Suppose $x > s$ and $x > s + t$ are two events, where X is a random variable which follows exponential distribution with parameter λ.

Then

$$P(X > s) = \int_s^{\infty} f(x)\,dx$$

$$= \int_s^{\infty} \lambda e^{-\lambda x}\,dx$$

$$= \lambda\left[\frac{e^{-\lambda x}}{-\lambda}\right]_s^{\infty}$$

$$= -e^{-\lambda\infty} + e^{-\lambda s}$$

$\therefore \quad P(X > s) = e^{-\lambda s}$

Again,

$$P(X > s + t) = \int_{s+t}^{\infty} \lambda e^{-\lambda x}\,dx$$

$$= \lambda\left[\frac{e^{-\lambda x}}{-\lambda}\right]_{s+t}^{\infty}$$

$$= e^{-\lambda(s+t)}$$

Now,

$$P\left(X > \frac{s+t}{X > s}\right) = \frac{P(X > s + t)}{P(X > s)}$$

$$= \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}}$$

$$= e^{-\lambda t}$$

$$= P(X > t)$$

Here, $P\left(X > \frac{s+t}{X > s}\right) = P(X > t)$

Which is memory loss property of exponential distribution.

Example 21. Suppose that the average life of a bulb is a r.v. having exponential distribution with $\lambda = 0.5$. What is the probability that such bulb will be still in operating condition after 4000 hours?

Solution

Let X be a r.v. denotes the service life of a bulb. $\lambda = 0.5$

$$P(X > 4000) = 1 - P(X \le 4000)$$

$$= 1 - \int_0^{4000} f(x)\,dx$$

$$= 1 - \int_0^{4000} \lambda e^{-\lambda x}\,dx$$

$$= 1 - \frac{\lambda}{(-\lambda)}\left[e^{-\lambda x}\right]_0^{4000}$$

$$= 1 + [e^{-2000} - 1]$$

$$= \frac{1}{e^{2000}}$$

$$= 0$$

Example 22. The mileage in thousands of miles which car owners get with a certain kind of type is a r.v. having exponential distribution with $\lambda = 0.005$. Find the probability that the mileage of car will be (a) at most 10,000 miles. (b) at least 30,000 miles. (c) anywhere from 16,000 to 24,000 miles.

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Solution

Let r.v. X denote the mileage of car in thousand of miles.

$\lambda = 0.005$

We have, $f(x) = \lambda e^{-\lambda x}$

a. $P(X \le 10{,}000) = \int_0^{1000} \lambda e^{-\lambda x}\, dx$

$= -[e^{-\lambda x}]_0^{10000}$

$= 1 - e^{-5}$

$= 0.993$

b. $P(X \ge 30{,}000) = 1 - P(X > 30{,}000)$

$= 1 + [e^{-\lambda x}]_0^{30000}$

$= 1 + [e^{-15} - e^{0}]$

$= e^{-15}$

c. $P(16{,}000 < X < 24{,}000) = \int_{16000}^{24000} \lambda e^{-\lambda x}\, dx$

$= -[e^{-\lambda x}]_{16000}^{24000}$

$= -[e^{-12} - e^{-8}]$

$= e^{-8} - e^{-12}$

Example 23. The amount of time that a surveillance camera will run without having to be reset is a r.v. having exponential distribution with $\lambda = \frac{1}{50}$ day. Find the probability that such a camera will have to reset in less than 20 days.

Solution

$P(X < 20) = \int_0^{20} f(x)\, dx, \lambda = 0.02$

$= \int_0^{20} \lambda e^{-\lambda x}\, dx$

$= -[e^{-\lambda x}]_0^{20}$

$= -[e^{-0.4} - e^{0}]$

$= 1 - e^{-0.4}$

$= 0.3296$

Example 24. The probability density function of exponential distribution is

$f(x) = 0.25\, e^{-0.25x}$ if $0 < x$

$= 0$, Otherwise

Find the probability that (a) the time to observe a particle is more than 200 microseconds. (b) the time to observe a particle is less than 10 microsecond.

Solution

a. $P(X \ge 200) = 1 - P(X < 200)$

$= 1 - \int_0^{200} 0.25\, e^{-0.25x}\, dx$

$= 1 - 0.25\left[\frac{e^{-0.25x}}{-0.25}\right]_0^{200}$

$= 1 + [e^{-50} - e^{0}]$

$= e^{-50}$

b. $P(X < 10) = -[e^{-0.25x}]_0^{10}$

$= e^{-2.5} + 1 = 0.92$

4.10 Log-normal distribution

A continuous random variable X is said to follow log-normal distribution, if $\ell n(x)$ follows normal distribution with mean μ and variance σ^2, then its probability density function is given as:

$$f(x, \mu, \sigma) = \frac{1}{x\,\sigma\sqrt{2\pi}}\, e^{-\frac{1}{2}\left(\frac{\ell n\, x - \mu}{\sigma}\right)^2};\ 0 < x < \infty$$

$= 0$, otherwise

Where, $\ell n\, x = \log e^x$, natural logarithm of x.

Mean: $E(X) = e^{\mu + \frac{\sigma^2}{2}} = e^{\mu + \frac{\sigma^2}{2}}$

Variance $V(X) = e^{(2\mu + \sigma^2)} \cdot (e^{\sigma^2} - 1)$

Log-normal distribution are used to model the growth rates that are independent of size. It also models time to failure in reliability studies, rainfall amounts, species abundance, etc.

Example 25. The length of time (in seconds) that a user views a page on website before moving to another page is log-normal random variable with parameters $\mu = 0.5$ second $\sigma^2 = 1$. (a) What is the probability that a page is viewed more than 10 seconds? (b) What is the length of time that 50% of users view the page? (c) What is the mean and standard deviation of time until a user moves from the page?

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Solution

Let X denote the time in second, follows log-normal distribution with $\mu = 0.5$ second and $\sigma = 1$ second.

a. $P(X > 10) = P(Z > \ell n\,(10))$

$= P\left(Z > \dfrac{\ell n\,(10) - 0.5}{1}\right)$

$= P(Z > (2.30 - 0.5))$

$P(Z > 1.80) = 0.5 - P(0 < Z < 1.80)$

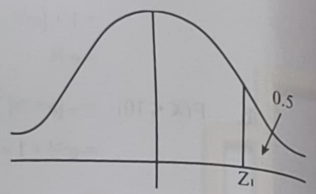
$= 0.5 - 0.4641$

$= .0359$

b. Let x_1 be the minimum length of time that 50% of the user view the page.

$P(X > x_1) = 0.5$

$P\left(Z_1 > \dfrac{\ell n x_1 - 0.5}{1}\right)$ ....(1)



$P(Z > Z_1) = 0.5 - P(0 < Z < Z_1)$

or, $0.5 = 0.5 - P(0 < Z < Z_1)$

or, $P(0 < Z < Z_1) = 0$

or, $\ell n\, x_1 - 0.5 = 0$

$x_1 = e^{0.5}$

$\therefore$ $x_1 = 1.65$ seconds

$\therefore$ Required time = 1.65 seconds.

$E(X) = e^{\mu + \frac{\sigma^2}{2}} = e^{0.5 + \frac{1}{2}} = e^1 = 2.7182$

$V(X) = e^{2\mu + \sigma^2} \cdot (e^{\sigma^2} - 1) = e^{(2(0.5)+1}\,(e^{1} - 1)$

or, $V(X) = e^2(e - 1) = 12.72$

$\therefore$ $S.d\,(\sigma) = \sqrt{Var\,(X)} = 3.56$

Example 26. Suppose that X follows log normal distribution with mean 50 and variance 4000. What is the probability that X is less than 150?

Solution

$E(X) = e^{\mu + \frac{\sigma^2}{2}}$

or, $\ell n\, 50 = \mu + \dfrac{\sigma^2}{2}$

or, $2\mu + \sigma^2 = 7.82$ ...(1)

or, $2\mu = 7.82 - 0.96$

$\therefore$ $\mu = 3.43$, $\sigma^2 = 0.96$

$V(X) = e^{2\mu + \sigma^2} \cdot (e^{\sigma^2} - 1)$

or, $4000 = e^{7.82}\,(e^{\sigma^2} - 1)$

or, $e^{\sigma^2} - 1 = 1.61$

or, $\sigma^2 = \ell n(2.61)$

$\therefore$ $\sigma^2 = 0.96$

$P(X < 150) = P\left(Z < \dfrac{\ell n\,(150) - 3.43}{0.96}\right)$

$= P(Z < 1.61)$

$= 0.9463$

Exercise 4.2

1. The time between arrivals of taxi at a busy intersection is exponentially distributed with mean of 10 minutes. (a) What is the probability that you wait longer than one hour for taxi? (b) Determine x such that the probability that you wait less than x minute is 0.10.
2. The distance between major cracks in highway follows an exponential distribution with mean of 5 miles. What is the probability that (a) there are no major cracks in a 10 mile stretch of the highway? (b) What is the standard deviation of the distance between two major cracks?
3. The lifetime of a semiconductor laser has a log-normal distribution with $\mu = 10$ hrs and S.d. 1.5 hrs. (a) What is the probability that the life time exceeds 10,000 hrs? (b) What life time is exceeded by 99% of the laser?
4. The time (in hours) required to repair a machine is exponentially distributed with parameter $\lambda = \dfrac{1}{3}$. What is the probability that the repair time exceeds 3 hours? Also find the probability that machine will be repaired in 3 hours?
5. Suppose that the proportion of defective ship by vendor, which varies somewhat from shipment to shipment may be looked upon a random variable having a beta distribution with $\alpha = 1$ and $\beta = 4$. Find the mean of this distribution, also find the probability that a shipment from this vendor will contain 25% or more defectives.
6. Suppose that the life time of a certain kind of computer is a random variable X, having the gamma distribution with $\alpha = 2$ and $\beta = 6$. (a) Find the mean lifetime of computers. (b) The probability that such computer will last more than 10 years?
7. The bus company A schedules a north bound bus every 30 minutes at a certain bus stop. A man comes to the stop at a random time. Let r.v. X count the number of minutes he has to wait for next bus. Assume X follows uniform distribution over the interval (0, 30). This is how we interpret the statement that he enters the station at random time. For each K = 5, 10, 15, 20, 30 compute the probability that he has to wait at least K minutes for the next bus.

Hint: $P(X \le K) = \int_0^K f(x)\,dx$

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8. In a certain experiments, the error made in determining the solubility of a substance is a random variable having uniform density with a = – 0.025 and b = 0.025. What are the probabilities that such an error will be (a) between 0.010 and 0.015? (b) between –0.012 and 0.012?

Hint: $f(x) = \frac{1}{b-a} = \frac{1}{0.05} = 20$

a. $P(0.010 < X < 0.015) = 0.1$

b. $P(-0.012 < X < 0.012) = 0.48$

Answers

1. 0.0025, 1.05 minute
2. $1 - P(X < 10) = 0.135$, S = 5
3. 0.695, 668.48 hrs
4. 0.368, 0.632
5. 0.2, 0.316
6. 12, 0.529
7. 0.167, 0.33, 0.5, 0.67, 1
8. 0.1, 0.48

❐

Unit 5

Joint Probability Distribution

5.1 Introduction

A statistical measure that calculates the combine probability of two events occurring together and at the same point in time is called joint probability. If two random variables that are defined on the same probability distribution the corresponding probability distribution on all pairs of outputs. The joint probability distribution can be considered for any given number of random variables. In this section we will discuss only bivariate probability distribution.

Let X and Y be two discrete random variables in which X takes values $X_1, X_2, \ldots, X_m$ with probabilities $P(x_1), P(x_2), \ldots, P(x_m)$ and Y takes values $y_1, y_2, \ldots, y_n$ with probabilities. Then joint probability of two discrete random variables denoted by $P(x_i, y_j) = P_{ij}$, i = 1, 2, ..., m; $j = 1,2, \ldots,$ n is given as

$$P(x_i, y_j) = P(X = x_i, Y = y_j)$$
$$= P(X = x_1 \text{ and } Y = y_j)$$
$$= P(X = x_1 \cap Y = y_j)$$

X → / Y ↓	X_1	X_2	...	X_m	
y_1	P_{11}	P_{21}	...	Pm_1	
y_2	P_{12}	P_{22}	...	Pm_2	
.					
.					
.					
y_n	P_{1n}	P_{2n}	...	P_{mn}	
					$\sum_i \sum_j P(x_i, y_j) = 1$

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5.1.1 Joint probability mass function

Let X be a discrete r.v. which takes values $X_1, X_2, \ldots, X_m$ with probability $P(x_1), P(x_2), P(x_m)$ and y be another discrete r.v. which takes values $y_1, y_2, \ldots, y_n$ with probabilities $P(y_1), P(y_2), \ldots, P(y_n)$, then $P(x_i, y_j)$ i = 1, 2, ..., m; j = 1, 2, ..., n is called joint probability mass function if it satisfies the following conditions.

i. $P(x_i, y_j) \geq 0$, for all i = 1, 2, ..., m; j = 1, 2, ..., n

ii. $\sum_{i}^{m} \sum_{j}^{n} P(x_i, y_j) = 1$.

5.1.2 Marginal probability mass function

Let $P(x_i, y_j) = P(X = x_i, Y = y_j)$, i = 1, 2, ..., m; j = 1, 2, ..., n be the joint probability mass function of two random variables X and Y; then marginal probability of X, denoted by $P(x_i)$, i = 1, 2, ..., m or simply P_i is defined as

$$P(X = x_i) = P(x_i) = \sum_{j=1}^{n} P(x_i, y_j)$$

or, $P(x_i) = P(x_i, y_1) + P(x_i, y_2) \ldots + P(x_i, y_n)$

Similarly, marginal probability of Y

$$P(y_j) = (Y = y_j) = \sum_{i=1}^{m} P(x_i, y_j)$$

or, $P(y_j) = P(x_1, y_j) + P(x_2, y_j) \ldots + P(x_m, y_j)$

5.1.3 Conditional probability

The conditional probability mass function of $P\left(\frac{X = x_i}{Y = y_j}\right)$ is defined as

$$P\left(\frac{X = x_i}{Y = y_j}\right) = \frac{P(X = x_i \cap Y = y_j)}{P(Y = y_j)}$$

$$P\left(\frac{X = x_i}{Y = y_1}\right) = \frac{P(X = x_i \cap Y = y_1)}{P(Y = y_1)}$$

$$P\left(\frac{X = x_m}{Y = y_n}\right) = \frac{P(X = x_m \cap Y = y_n)}{P(Y = y_n)}$$

Similarly,

$$P\left(\frac{Y = y_n}{X = x_m}\right) = \frac{P(X = x_m \cap Y = y_n)}{P(X = x_n)}$$

5.1.4 Independence

Two random variables X and Y are independent if joint probability mass function is equal to the product of marginal probabilities of X and Y.

i.e., $P(x_i, y_j) = P(x_i) \cdot P(y_j)$, i = 1, 2, ..., m; j = 1, 2, ..., n, otherwise X and Y are dependent.

5.2 Joint probability distribution function

Let $P(x_i, y_j) = P(X = x_i, Y = y_j)$ i = 1, 2, ..., m; j = 1, 2, ..., n be the joint probability mass function of two discrete r.v.'s X and Y, then joint probability distribution function denoted by F_{XY} is the cumulation of probabilities of X and Y at the given points. It is given as

$$F_{XY} = P(X \leq x_s, Y \leq y_t) = \sum_{i=1}^{s} \sum_{j=1}^{t} P(x_i, y_j)$$

5.3 Continuous random variables

5.3.1 Joint probability density function

Let X and Y be two continuous random variables, then f(x, y) is called the joint probability density function of X and Y if it satisfies the following condition:

a. $f(x, y) \geq 0$, for all x and $y \in R$.

n. $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y)\, dy\, dx = 1$

5.3.2 Marginal probability density function

Let f(x, y) be the joint probability density function of two continuous r.v.'s X and Y, then marginal probability density function of X, simply marginal of X is denoted by f(x) and defined as:

$$f(x) = \int_{-\infty}^{\infty} f(x, y)\, dy$$

Similarly marginal of Y

$$f(y) = \int_{-\infty}^{\infty} f(x, y)\, dx$$

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5.3.3 Conditional probability density function

Conditional probability density function of X given Y is:

$$f\left(\frac{x}{y}\right) = \frac{f(x, y)}{f(y)}$$

Similarly, $f\left(\frac{y}{x}\right) = \frac{f(x, y)}{f(x)}$

5.3.4 Joint probability distribution function

The joint probability distribution function denoted by F_{xy} of two continuous r.v.'s whose density function f(x, y) is defined as:

$$F_{XY} = P(X \le x_s, Y = y_t)$$

$$= \int_{-\infty}^{s}\int_{-\infty}^{t} f(x, y)\, dy\, dx$$

Example 1. The joint probability mass function of two variables are given as:

Y ↓ \ X →	1	2	3
1	0	0.2	0.1
2	0.15	0.1	0.05
3	0.3	0	0.1

Calculate the marginal probability of X and Y and E(X), E(Y).

Solution

Marginal of X:

$$P(X = x_i) = P(x_i) = \sum_{j=1}^{3} P(x_i, y_j)$$

$$P(X = 1) = P(x_1, y_1) + P(x_1, y_2) + P(x_1, y_3)$$
$$= 0 + 0.15 + 0.3 = 0.45$$

$$P(X = 2) = P(x_2, y_1) + P(x_2, y_2) + P(x_2, y_3)$$
$$= 0.2 + 0.1 + 0 = 0.3$$

$$P(X = 3) = P(x_3, y_1) + P(x_3, y_2) + P(x_3, y_3)$$
$$= 0.1 + 0.05 + 0.1 = 0.25$$

Similarly marginal of Y is:

$$P(Y = 1) = 0 + 0.2 + 0.1 = 0.3$$

$$P(Y = 2) = 0.15 + 0.1 + 0.05 = 0.3$$

$$P(Y = 3) = 0.3 + 0 + 0.1 = 0.4$$

$$E(X) = \sum_{i=1}^{3} x_i \cdot P(x_i)$$
$$= 1 \times 0.45 + 2 \times 0.3 + 3 \times 0.25 = 1.8$$

$$E(Y) = \sum_{j=1}^{3} y_j \cdot P(y_j)$$
$$= 1 \times 0.3 + 2 \times 0.3 + 3 \times 0.4$$
$$= 2.1$$

Example 2. The joint probability mass function of two discrete r.v.'s X and Y are given below:

Y ↓ \ X →	1	2	3
1	4/32	3/32	2/32
2	7/32	5/32	1/32
3	2/32	2/32	6/32

Find: a. Marginal probability of X
b. The conditional probability X given Y = 1.
c. E(X), E(Y), V(X), Cov(X, Y)
d. F(2, 3), F(2, 2), F(3, 1)

Solution

Marginal probability of X

a. $$P(x_i) = \sum_{j=1}^{3} P(x_i, y_j)$$
$$= P(x_i, y_1) + P(x_i, y_2) + P(x_i, y_3)$$

$$P(X = 1) = \frac{4}{32} + \frac{7}{32} + \frac{2}{32} = \frac{13}{32}$$

$$P(X = 2) = \frac{3}{32} + \frac{5}{32} + \frac{2}{32} = \frac{10}{32}$$

$$P(X = 3) = \frac{2}{32} + \frac{1}{32} + \frac{6}{32} = \frac{9}{32}$$

b. Conditional probability of X given Y = 1

$$P\left(\frac{X = x_i}{Y = 1}\right) = \frac{P(X = x_i \cap Y = 1)}{P(Y = 1)}$$

$$P\left(\frac{X = 1}{Y = 1}\right) = \frac{P(X = 1 \cap Y = 1)}{P(Y = 1)} = \frac{4/32}{9/32} = \frac{4}{9}$$

$$P\left(\frac{X = 2}{Y = 1}\right) = \frac{3/32}{9/32} = \frac{3}{9}$$

$$P\left(\frac{X = 3}{Y = 1}\right) = \frac{2/32}{9/32} = \frac{2}{9}$$

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c. $E(X) = \sum_{i=1}^{3} x_i \cdot P(x_i)$

$= 1 \times \frac{13}{32} + 2 \times \frac{10}{32} + 3 \times \frac{9}{32}$

$= \frac{60}{32} = 1.87$

$E(Y) = \sum_{j=1}^{3} y_j \cdot P(y_j)$

$= 1 \times \frac{9}{32} + 2 \times \frac{13}{32} + 3 \times \frac{10}{32}$

$= \frac{65}{32} = 2.03$

$Cov(X, Y) = E(XY) - E(X) \cdot E(Y)$

$E(XY) = \sum_{i=1}^{3}\sum_{j=1}^{3} x_i y_j P(x_i, y_j)$

$= \sum_{i} x_1y_1 P(x_i, y_1) + \sum_{i} x_iy_2 P(x_1, y_2) + \sum_{i} x_iy_3 P(x_1, y_3)$

$= x_1y_1 P(x_1, y_1) + x_2y_1 P(x_2, y_1) + x_3y_1 P(x_3, y_1) + x_1y_2 P(x_1, y_2) + x_2y_2 P(x_2, y_2) + x_3y_2 P(x_3, y_2) + x_1y_3 P(x_1, y_3) + x_2y_3 P(x_2, y_3) + x_3y_3 P(x_3y_3)$

$= 1 \cdot 1 \cdot \frac{4}{32} + 2 \cdot 1 \cdot \frac{3}{32} + 3 \cdot 1 \cdot \frac{2}{32} + 1 \cdot 2 \cdot \frac{7}{32} + 2 \cdot 2 \cdot \frac{5}{32} + 3 \cdot 2 \cdot \frac{1}{32} + 1 \cdot 3 \cdot \frac{2}{32} + 2 \cdot 3 \cdot \frac{2}{32} + 3 \cdot 3 \cdot \frac{6}{32}$

$= \frac{4}{32} + \frac{6}{32} + \frac{6}{32} + \frac{14}{32} + \frac{20}{32} + \frac{6}{32} + \frac{6}{32} + \frac{12}{32} + \frac{54}{32}$

$= \frac{128}{32}$

$= 4$

$\therefore Cov(X, y) = 4 - 1.87 \times 2.03$

$= 3.79$

$V(X) = E(X^2) - [E(X)]^2$

$E(X^2) = \sum_{i=1}^{3} \times i^2 P(x_i)$

$= 1^2 \times \frac{13}{32} + 2^2 \times \frac{10}{32} + 3^2 \times \frac{9}{32}$

$= \frac{13}{32} + \frac{40}{32} + \frac{81}{32}$

$= \frac{134}{32} = 4.19$

$V(X) = 4.19 - (1.87)^2$

$= 0.69$

d. $F(2, 3) = P(X \le x_2, Y \le y_3)$

$= \sum_{i=1}^{2}\sum_{j=1}^{3} P(x_i, y_j)$

$= \sum_{i} P(x_i, y_1) + \sum_{i} P(x_i, y_2) + \sum_{i} P(x_i, y_3)$

$= P(x_1, y_1) + P(x_2, y_1) + P(x_1, y_2) + P(x_2, y_2) + P(x_1, y_3) + P(x_2, y_3)$

$= \frac{4}{32} + \frac{3}{32} + \frac{7}{32} + \frac{5}{32} + \frac{2}{32} + \frac{2}{32}$

$= \frac{23}{32}$

$= 0.71$

$F(3, 3) = \sum_{i=1}^{3}\sum_{j=1}^{3} P(x_i, y_j)$

$= \frac{4}{32} + \frac{3}{32} + \frac{2}{32} + \frac{7}{32} + \frac{5}{32} + \frac{1}{32} + \frac{2}{32} + \frac{2}{32} + \frac{6}{32}$

$= \frac{32}{32}$

$= 1$

Example 3. The joint probability mass function of two random variable X and Y are given as

Y↓ \ X→	6	10
7	0.3	0.4
9	0.2	0.1

a. Find marginal probability of X and Y.

b. Find $P\left(X = \frac{x_i}{Y}\right)$

c. Find E(Y) and V(Y)

d. Find (2, 1), F(1, 2)

Solution

a. Marginal of X is

$P(X = x_i) = p(x_i) = \sum_{j=1}^{3} P(x_i, y_j)$

$P(X = 6) = P(x_1, y_1) + P(x_2, y_2) = 0.3 + 0.2 = 0.5$

$P(X = 10) = P(x_2, y_1) + P(X_2, y_2) = 0.4 + 0.1 = 0.5$

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Marginal of Y is

$P(Y = 7) = 0.3 + 0.4 = 0.7$

$P(Y = 9) = 0.2 + 0.1 = 0.1$

b. Conditional probability

$$P\left(\frac{X = 6}{Y = 9}\right) = \frac{P(X = 6 \cap Y = 9)}{P(Y = 9)} = \frac{0.2}{0.3} = \frac{2}{3}$$

$$P\left(\frac{X = 10}{Y = 9}\right) = \frac{P(X = 10 \cap Y = 9)}{P(Y = 9)} = \frac{0.1}{0.3} = \frac{1}{3}$$

c. $E(Y) = \sum_{j=1}^{2} P(y_j)$

$= 7 \times 0.7 + 9 \times 0.3 = 4.9 + 2.7 = 7.6$

$E(Y^2) = 49 \times 0.7 + 81 \times 0.3 = 34.3 + 24.3 = 58.6$

$\therefore V(Y) = 58.6 - (7.6)^2$

$= 0.84$

d. $F(2, 1) = \sum_{i=1}^{2} \sum_{j=1}^{1} P(x_i, y_j)$

$= \sum_{i} P(x_i, y_1)$

$= P(x_1, y_1) + P(x_2, y_1)$

$= 0.3 + 0.4 = 0.7$

$F(1, 2) = \sum_{i=1}^{1} \sum_{j=1}^{2} P(x_i, y_j)$

$= P(x_1, y_1) + P(x_1, y_2)$

$= 0.3 + 0.2 = 0.5$

Example 4. The probability density function of two continuous random variable are given as:

$$f(x, y) = \begin{cases} 6xy, & 0 \le x \le 1, \quad 0 \le y \le \sqrt{x} \\ 0, & \text{Otherwise} \end{cases}$$

Are X and Y independent?

Solution

$f(x, y) = 6xy$

$0 \le x < 1$

$0 \le y^2 \le x \Rightarrow y^2 \le x < 1$

We have,

$f(x) = \int_{-\infty}^{\infty} f(x, y)\, dy$

$= \int_{0}^{\sqrt{x}} 6xy\, dy$

$= 6x\left[\frac{y^2}{2}\right]_0^{\sqrt{x}}$

$= 6x \cdot \frac{x}{2} = 3x^2$

$f(y) = 6y \int_{y^2}^{1} x\, dx$

$= 6y\left[\frac{x^2}{2}\right]_{y^2}^{1}$

$= 3y\,(1 - y^4)$

Here $f(x, y) \neq f(x) \cdot f(y)$.

Hence X and Y are not independent.

Example 5. The joint p.d.f. of X and Y are

$$f(x, y) = \begin{cases} cx^2y, & 0 \le y \le x \le 1, \\ 0, & \text{Otherwise} \end{cases}$$

a. c

b. Find marginal of X and Y and check independence.

c. $P\left(Y \le \frac{x}{2}\right)$

d. $P\left(\frac{Y \le \frac{x}{4}}{Y \le \frac{x}{2}}\right)$

Solution

a. $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y)\, dy\, dx = 1$

or, $\int_{0}^{1} \int_{0}^{x} c\, x^2 y\, dy\, dx = 1$

or, $\int_{0}^{1} \frac{cx^4}{2}\, dx = 1$

or, $c\frac{[x^5]_0^1}{10} = 1$

$\therefore c = 10$

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b. $f(x) = \int_0^X 10x^2y\,dy$

$= 10x^2\left[\frac{y^2}{2}\right]_0^x$

$= 5x^4, 0 \le x \le 1$

$f(y) = \int_y^1 10x^2y\,dy$

$= 10y\,[x^3]_y^1$

$= \frac{10}{3}y(1 - y^3), 0 \le y \le 1$

Here $f(x, y) \neq f(x) \cdot f(y)$

Hence X and Y are not independent.

c. $P\left(Y \le \frac{x}{2}\right) = \int_0^1 \int_0^{x/2} 10x^2y\,dy\,dx$

$= \int_0^1 10x^2\left[\frac{y^2}{2}\right]_0^{x/2} dx$

$= \int_0^1 10x^2\frac{x^2}{8}dx$

$= \frac{5}{4}\int_0^1 x^4\,dx$

$\therefore P\left(Y \le \frac{x}{2}\right) = \frac{5}{4}\left[\frac{x^5}{5}\right]_0^1 = \frac{1}{4}$

Again,

d. $P\left(\frac{Y \le \frac{x}{4}}{Y = \le \frac{x}{2}}\right) = \frac{P\left(Y \le \frac{x}{4} \cap Y \le \frac{x}{2}\right)}{P\left(Y \le \frac{x}{2}\right)}$

$= 4P\left(Y \le \frac{x}{4}\right)$

$= 4\int_0^1 \int_0^{x/4} 10x^2y\,dy\,dx$

$= 4\int_0^1 \frac{5}{16}x^4\,dx$

$= \frac{5}{4}\left[\frac{x^5}{5}\right]_0^1$

$= \frac{5}{4} \times \frac{1}{5} = \frac{1}{4}$

$\therefore \left(\frac{Y \le \frac{x}{4}}{Y = \le \frac{x}{2}}\right) = \frac{1}{4}$

Example 6. The joint of X and y is given as

$$f(x) = \begin{cases} k(6 - x - y), & 0 < x < 2, \quad 2 < y < 4 \\ 0, & \text{Otherwise} \end{cases}$$

Then find:

a. k

b. $P(X < 1 \cap Y < 3), P(X + y < 3)$

c. Marginal p.d.f. of X and Y, check independence

d. Find E(X), V(X), E(Y).

Solution

a. $\int_0^2 \int_2^4 f(x, y)\,dy\,dx = 1$

or, $k\int_0^2 \left[6y - xy - \frac{y^2}{2}\right]_2^4 = 1$

or, $k\int_0^2 (12 - 2x - 6)\,dx = 1$

or, $k[12x - x^2 - 6x]_0^2 = 1$

$\therefore k = \frac{1}{8}$

b. $P(X < 1 \cap Y \le 3) = \int_0^1 \int_2^3 f(x, y)\,dy\,dx$

$= \frac{1}{8}\int_0^1 \int_0^3 (6 - x - y)dy\,dx$

$= \frac{1}{8}\int_0^1 \left[64 - xy - \frac{y^2}{2}\right]_2^3 dx$

$= \frac{1}{8}\int_0^1 (6 - x - 2.5)\,dx$

$= \frac{1}{8}\left[3.5x\frac{x^2}{2}\right]_0^1$

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$\therefore \quad P(X < 1 \cap Y < 3) = \frac{3}{8}$

$$P(X + Y < 3) = \int_0^2 \int_2^{3-x} \frac{1}{8}(6 - x - y)\, dy\, dx$$

$$= \frac{1}{8}\int_0^2 \left[6y - xy - \frac{y^2}{2}\right]_2^{3-x} dx$$

$$= \frac{1}{8}\int_0^2 \frac{(x^2 - 8x + 7)}{2} dx$$

or, $$P(X + Y < 3) = \frac{1}{16}\left[\frac{x^3}{3} - \frac{8x^2}{2} + 7x\right]_0^2$$

$$= \frac{1}{16} \times \frac{2}{3} = \frac{1}{24}$$

c. Marginal of X

$$f(x) = \frac{1}{8}\int_2^4 (6 - x - y)\, dy$$

$$= \frac{1}{8}\left[6y - xy - \frac{y^2}{2}\right]_2^4$$

$$= \frac{1}{8}[12 - 2x - 6]$$

$$= \frac{3 - x}{4}, \quad 0 < x < 2$$

$$f(y) = \frac{1}{8}\int_0^2 (6 - x - y)\, dy$$

$$= \frac{1}{8}\left[6x - \frac{x^2}{2} - xy\right]_0^2$$

$$= \frac{1}{8}[12 - 2 - 2y]$$

$$= \frac{5 - y}{4}, \quad 4 < y < 4$$

Here, $f(x, y) \neq f(x) \cdot f(y)$ are not independent.

d. $$E(X) = \int_0^2 x\, f(x)\, dx$$

$$= \frac{1}{4}\int_0^2 (3x - x^2)\, dx$$

$$= \frac{1}{4}\left[3 \cdot \frac{x^2}{2} - \frac{x^3}{3}\right]_0^2$$

$$= \frac{1}{4} \times \frac{10}{3} = \frac{5}{6}$$

$\therefore \quad E(X) = \frac{5}{6}$.

$$E(X^2) = \int_0^2 x^2 f(x)\, dx$$

$$= \int_0^2 x^2 \left(\frac{(3 - x)}{4}\right) dx$$

$$= \frac{1}{4}\int_0^2 (3x^2 - x^3)\, dx$$

$$= \frac{1}{4}\left[3\frac{x^3}{3} - \frac{x^4}{4}\right]_0^2$$

$$= \frac{1}{4}\left[8 - \frac{2^4}{4}\right]$$

$E(X^2) = 1$

$V(X) = E(X^2) - [E(X)]^2$

$$= 1 - \left(\frac{5}{6}\right)^2 = \frac{11}{36}$$

$$E(Y) = \int_2^4 y \frac{(5 - y)}{4}$$

$$= \frac{1}{4}\left[\frac{y^2}{2} \times 5 - \frac{y^3}{3}\right]_2^4$$

$$= \frac{1}{4} \times \frac{34}{3} = \frac{17}{6}$$

Example 7. The p.d.f. of two r.v.'s X and Y are given as:

$$f(x) = \begin{cases} 4xy, & 0 < x < 1, \quad 0 < y < 1 \\ 0, & \text{Otherwise} \end{cases}$$

a. Verify f(x, y) is a joint probability density function.

b. Find f(x) and f(y) and check the independence of X and Y.

c. Find conditional probability of X given Y = y.

d. Find $P(0 \leq x \leq 0.5, 0.5 \leq Y \leq 1)$

e. Find covariance between X and Y.

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Solution

a. We have,

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x, y)\, dy\, dx = 1$$

Then $\int_0^1\int_0^1 4xy\, dy\, dx = \int_0^1 4x\left[\frac{y^2}{2}\right]_0^1 dx$

$$= \int_0^1 2x\, dx$$

$$= 2\left[\frac{x^2}{2}\right]_0^1 = 1$$

$\therefore$ $f(x, y) = 4xy$ is a joint p.d.f.

b. Marginal p.d.f.

$$f(x) = \int_0^1 4xy\, dy = 4x\left[\frac{y^2}{2}\right]_0^1 = 2x,\ 0 < x < 1$$

$$f(y) = \int_0^1 4xy\, dy = 4y\left[\frac{x^2}{2}\right]_0^1 = 2y,\ 0 < y < 1$$

Here $f(x, y) = f(x) \cdot f(y)$

$\therefore$ X and Y are independent.

c. $f(X/Y = y) = \frac{f(x, y)}{f(y)} = \frac{4xy}{2y} = 2x$

d. $P(0 \le X < 0.5, 0.5 \le y < 1) = \int_0^{0.5}\int_{0.5}^1 dxy\, dy\, dx$

$$= \int_0^{0.5} 4x\left[\frac{y^2}{2}\right]_{0.5}^1 dx$$

$$= \int_0^{0.5} 2x\,(0.75)\, dx$$

$$= 1.5\left[\frac{x^2}{2}\right]_0^{0.5}$$

$$= 1.5 \times \frac{0.25}{2}$$

$$= 0.1875$$

e. $Cov(X, Y) = 0$ [$\because$ X and Y are independent]

Example 8. The joint p.d.f. of two random variables are:

$f(x, y) = \frac{9(1 + x + y)}{2(1 + x)^4 (1 + y)^4}$, $0 < x < \infty$, $0 < y < \infty$.

Find marginal p.d.f. of X and Y and conditional probability of Y given X = x.

Solution

$$f(x) = \int_0^{\infty} \frac{9(1 + x + y)}{2(1 + x)^4 (1 + y)^4}\, dy$$

$$= \frac{9}{2(1 + x)^4}\int_0^{\infty} \frac{(1 + x + y)}{(1 + y)^4}\, dy$$

$$= \frac{9}{2(1 + x)^4}\int_0^{\infty} \frac{x + (1 + y)}{(1 + y)^4}\, dy$$

$$= \frac{9}{2(1 + x)^4}\left[x\frac{(1 + y)^{-4+1}}{-4 + 1} + \frac{(1 + y)^{-3+1}}{-3 + 1}\right]_0^{\infty}$$

$$= \frac{9}{2(1 + x)^4}\left[\frac{x}{-3(1 + y)^3} - \frac{1}{2(1 + y)^2}\right]_0^{\infty}$$

$$= \frac{9}{2(1 + x)^4}\left[0 - \left(-\frac{x}{3} - \frac{1}{2}\right)\right]$$

$$= \frac{3(2x + 3)}{4(1 + x)^4}$$

Similarly marginal of Y

$$f(y) = \int_0^{\infty} f(xy)\, dx$$

$$= \int_0^{\infty} \frac{9(1 + x + y)}{2(1 + x)^4 (1 + y)^4}$$

$$= \frac{2(2y + 3)}{4(1 + y)^4}$$

Conditional probability

$$f(Y/X) = \frac{f(x, y)}{f(x)}$$

$$= \frac{9(1 + x + y)}{2(1 + x)^4 (1 + y)^4} \times \frac{4(1 + x)^4}{3(2x + 3)}$$

$$= \frac{6(1 + x + y)}{(2x + 3)(1 + y)^4}$$

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Exercise 5.1

1. The joint probability mass function of two random variables X and Y are given below:

Y↓ \ X→	1	2
1	0.32	0.08
2	0.15	0.45

a. Find marginal probability of X and Y.
b. Find conditional probability of X given Y = 2.

2. The joint probability mass function of X and Y are:

Y↓ \ X→	0	1	2
0	0.1	0.4	0.1
1	0.2	0.2	1

Find:
a. Marginal probability of X and Y.
b. Conditional probability of X given Y = 1.
c. E(X), E(Y), V(X), V(Y).
d. F(2, 1), F(1, 1)

3. The joint probability distribution of X and Y are:

Y↓ \ X→	−1	0	1
0	1/15	k	1/15
1	3/15	k	1/15
2	k	1/15	k

Find:
a. k
b. Marginal probability of X and Y
c. Are X and Y independent?
d. Find E(X), E(Y) and V(X)

4. The joint probability distribution of two random variables X and Y are:

Y↓ \ X→	1	2	3
1	5/27	4/27	2/27
2	1/27	3/27	3/27
3	3/27	4/27	2/27

Find:
a. E(X), E(Y), E(XY)
b. Cov(X, Y), E(X + Y), E(X² + Y²)

5. The joint probability distribution of two random variables are:

Y↓ \ X→	1	2	3
1	1/16	2/16	2/16
2	2/16	3/16	1/16
3	3/16	1/16	1/16

Find:
a. Marginal probability density function of X and Y.
b. Conditional probability of X given Y = 1.
c. E(X), E(Y), V(X) and Cov(XY).
d. F(1, 3), F(3, 3)

6. The joint probability distribution of X and Y are:

Y↓ \ X→	0	5	10	15
0	0.02	0.06	0.02	0.01
5	0.04	0.15	0.2	0.1
10	0.01	0.15	0.14	0.1

Find: E(X), E(Y), E(XY), E(X² + Y²)

7. The joint p.d.f. of two random variables X and Y are:

$$f(x, y) = \begin{cases} 2 - x - y, & 0 \le x < 1, \quad 0 \le y \le 1 \\ 0, & \text{Otherwise} \end{cases}$$

Find:
a. Marginal probability of X and Y.
b. V(X), V(Y)
c. Cov (X, Y)

8. The joint probability density function of X and Y are

$$f(x, y) = \begin{cases} k(4 - x^2 - y^2), & 0 < x < 2, \quad 0 < y < 2 \\ 0, & \text{Otherwise} \end{cases}$$

Find:
a. k
b. Marginal of X and Y and check independence.
c. Conditional probability of X given Y = y.
d. E(X), V(X).

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9. The joint p.d.f of two random variables are

$$f(x, y) = \begin{cases} \frac{2}{3}(x + 2y), & 0 < x < 1, \quad 0 < y < 1 \\ 0, & \text{Otherwise} \end{cases}$$

Find:

a. Marginal p.d.f. of X and Y.

b. Conditional probability of X given Y = y. Are X a Y independent?

10. The joint p.d.f of two random variables X and Y are:

$$f(x, y) = \begin{cases} k(2x + y), & 0 \le x < 2, \quad 0 \le y \le 3 \\ 0, & \text{Otherwise} \end{cases}$$

Find:

a. Marginal of X and Y

b. Are X and Y independent?

11. The joint p.d.f of two continuous r.v. X and Y are:

$$f(x, y) = \begin{cases} 4xye^{-(x^2+y^2)}, & x \ge 0, \quad y > 0 \\ 0, & \text{Otherwise} \end{cases}$$

a. Find marginal p.d.f. of X and Y.

b. Find the conditional p.d.f. of X given Y = y.

12. The joint p.d.f. of X and Y are:

$f(x, y) = e^{-(x+y)}, x \ge 0, y \ge 0$

$= 0$, Otherwise

Check whether X and Y are independent or not.

13. The joint p.d.f. of X and Y are:

$f(x, y) = (1 - e^{-x})(1 - e^{-y}), x > 0, y > 0$

$= 0$, Otherwise

Are X and Y independent?

14. The joint p.d.f. of X and Y are:

$f(x, y) = 8xy, 0 < x < y < 1$

a. Find $P\left(X < \frac{1}{2} \cap Y - \frac{1}{4}\right)$

b. Find marginal p.d.f. of X and Y.

15. The joint p.d.f. of two random variables are:

$$f(x, y) = \begin{cases} x + cy^2, & 0 \le x \le 1, \quad 0 \le y \le 1 \\ 0, & \text{Otherwise} \end{cases}$$

Find:

a. c

b. $P\left(0 \le X \le \frac{1}{2}, 0 \le y \le \frac{1}{2}\right)$

16. If two random variables X and Y have joint p.d.f. as:

$$f(x, y) = \begin{cases} \frac{6}{5}(x + y)^2, & 0 \le x \le 1, \quad 2 \le y \le 4 \\ 0, & \text{Otherwise} \end{cases}$$

Find:

a. $P(0.2 < X < 0.4 \cap 0.5 < Y < 1)$

b. $P(0 \le x \le 0.5, 0 \le y \le 0.5)$

b. Find marginal p.d.f. of X and Y.

Answers

1. Marginal of X and Y: 0.47, 0.53 and 0.40, 060; Conditional: 0.78, 0.80
2. a. 0.3, 0.6, 0.1: 0.6, 0.4 b. 0.5, 0.5, 0 c. 0.8, 0.4, 0.36, 0.24 d. 1, 0.9
3. a. $\frac{2}{15}$ b. for X and Y $= \frac{2}{5}, \frac{1}{3}, \frac{4}{15}; \frac{4}{15}, \frac{2}{5}, \frac{1}{3}$ c. not independent d. $-\frac{2}{15}, \frac{14}{15}, \frac{26}{225}$
4. a. 1.93, 1.93, 3.78 b. 0.05, 3.85, 8.74
5. b. $\frac{1}{5}, \frac{2}{5}, \frac{2}{5}$ c. 1.87, 0.69, 3.5 d. 0.37, 1 6. 5.5, 8.5, 44.2, 135.5
7. a. $\frac{3x - 2x^2}{2}, \frac{3y - 2y^2}{2}$ b. $\frac{9}{80}, \frac{19}{80}, \frac{5}{48}$
8. a. $\frac{16}{3}$ b. $\frac{32}{9}(8 - 3x^2), \frac{32}{9}(8 - 3y^2)$ c. $\frac{3(4 - x^2 - y^2)}{2(8 - 3y^2)}$ d. 10.61, – 106.24
9. a. $\frac{2(x + 1)}{3}, \frac{(4y + 1)}{3}$ b. $\frac{2x + 47}{4y + 1}$, dependent
10. a. $\frac{1}{30}$ b. $\frac{4x + 3}{20}, \frac{2 + y}{15}$, dependent 11. a. $2xe^{-x^2}, 2ye^{-x^2}$ b. $2xe^{-x^2}$
12. Independent 13. Not dependent
14. 0.0625, $4x(1 - x^2), 4y^3$ 15. a. $\frac{3}{2}$ b. $\frac{3}{32}$
16. a. 0.143 b. 0.139 c. $\frac{2}{5}(6x^2 + 24x + 56), \frac{2}{5}(1 + 3y + 3y^2)$

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Unit 6

Sampling and Estimation

6.1 Introduction

Researchers are always willing to find the true value of the population parameter, which is almost unknown. Sampling is the only one process to find the value of sample statistics, either the size of population may be finite or infinite. Before we learn sampling techniques in detail, we should know some terminologies which are given below:

- **Population:** The total number of things, goods or objects that may be living or non-living beings which is subject of investigation lies in defined area is called population. Population is the largest collection of elements or individuals, in which we are interested at a particular time, area about which we want to draw some conclusions. The total number of elements or objects of the population is called the population size and is denoted by N. Population may be of finite size and infinite. The number of planets lies in our solar system has finite size and number of star lies in the galaxy have infinite population size.
- **Sample:** A part or portion of population on which we collect data is called sample. Sample data is used to estimate the characteristics of population. The number of elements in the sample is called sample size and is denoted by n. Each unit of the population is called sampling unit. In sample we collect different sampling units.
- **Census:** Complete enumeration or counting of all population under study area is called census. The entire census operation has three strategies: (i) pre-enumeration–planning and preparatory work; (ii) enumeration– field work; (iii) post-enumeration–edition, coding, compilation, tabulation, analysis and publication of results, census requires more cost, time and manpower than sampling.
- **Sampling:** The process of selecting sample from the population is called sampling. Sampling involves the selection of number of study units from defined population. The process of partial enumeration is known as sample survey, the results and findings are then generalized and then made applicable to the whole field of inquiry. There are various method of sampling techniques used for the selection of sample from the population.

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- **Sampling frame:** For conducting any sample survey, it is essential to have a list or map identifying each sampling unit by a number, such type of complete list or map of sampling units is called sampling frame. Sampling frame should be up to date and there will be not duplication of items. A list of voters in election, list of all examiner in exam, list of households in certain city are examples of sampling frame.
- **Sample size (n):** After planning for any sample survey, it is important to know that how many samples, i.e., size of sample (n), should be collected in survey. If the study does not have an optimum sample size, the significance of the results in reality may not be detected, often, the study may fail to answer research question.

  Very large sample size (more than optimum size) involve extra patience, needs more cost, time and manpower, difficult to maintain quality of data. That's why sample size (n) should not be very small and not very large.
- **Parameters and statistics:** The numerical coefficients or constant which describes the characteristics of population is called parameters and it is function of the population. The population constants like mean (μ), standard deviation (σ), variance (σ^2), correlation coefficient (ρ), skewness (β_1), kurtosis (β_2), median (Md), etc. are parameters.

  The numerical coefficient or constant which describes the characteristics of sample is called statistics and it is function of the sample. The sample constants like mean ($\bar{x}$), standard deviation (S), variance (S^2), correlation coefficient (r), skewness (b_1), kurtosis (b_2), median (md), etc. are statistics.
- **Objectives of sampling:** The main objective of sampling is to obtain maximum information about the population with minimum time, cost and manpower. To estimate and to test the significance of the population parameter based on sample statistics.

6.2 Techniques in sampling

The sampling techniques are generally categorized in two groups; they are probability sampling and non-probability sampling.

6.2.1 Probability sampling

If the sampling process is carried out on the basis of some probabilistic rule, then it is called probability sampling. In this sampling all the units of population has equal chance being included in the sample. The commonly used probability sampling are as follows:

i. Simple random sampling (SRS)

Simple random sampling is one of the most commonly used probability / random sampling, in this sampling each unit in the sampling frame has an equal chance being included in the sample.

In simple random sampling, sampling can be done by two methods, i.e., lottery method and use of random number tables. If the size of the population is small, lottery method is better to use. In this method each unit in the population is given a number in a slip of a paper, these are put in a box and mixed and sample of required size can be drawn randomly from the box.

Another practical method of selecting random sample is by the use of random number tables. Random numbers can be obtained by the computer or random number tables. The various type of random number table can be used. Some of them are; Tippet's random number table, Fishers and Yates table, Kendall's table, etc.

If sample unit is selected and is not returned or mixed to the population before next drawing and so on, the process is called simple random sampling without replacement (SRSWOR). If the unit selected, noted and returned or mixed to the population before next drawing, the process is called simple random sampling with replacement (SRSWR).

Simple random sampling is more representative of the population, gives more efficient estimates as sample size increases and it save time, cost and manpower than other sampling.

ii. Stratified random sampling

When characteristics of population are heterogeneous and distributed through out the large area than best design to use is stratified random sampling. In this sampling population is divided into different groups called strata, in such a way that characteristics of units are homogeneous with in strata and heterogeneous between strata. Then simple random sampling is used to draw samples from each stratum. In stratified sampling allocation of sample size can be done in two ways, i.e., proportional allocation and disproportional allocation. In proportional stratified sampling the number of items drawn from each stratum is proportional to the size of strata, while disproportionate allocation, the number of item selected from each stratum is independent of its size.

iii. Cluster sampling

In cluster sampling population is divided into different groups called clusters in such a way that characteristics of units within the cluster are heterogeneous and between clusters are homogeneous, so that number of sampling units in each clusters should be approximately equal. After dividing the population into clusters, the required number of clusters can be selected, then all the units in selected clusters are enumerated to get the desired sample size. In cluster sampling, sampling frame is not available. The efficiency decreases with increases in cluster size. When the population size is large the enumeration of sampling units within the selected cluster is difficult. When sampling frame is not available cluster sampling is more appropriate to use.

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iv. Multistage sampling

This method of sampling is appropriate when the reference population is large and widely scattered and sampling procedure is carried out in several stages. First the whole universe or population is divided into groups called clusters, then specified number of clusters are selected, it is first stage sampling. When selected clusters are further sub-sampled, i.e., only some units of selected cluster, this is known as two stage sampling. The procedure is generalized to three or more stages, which is called multistage sampling. Sample size is reducing in each stage, so this technique saves time, cost and manpower. Sample in each stage should be taken carefully, otherwise the result may have error.

iv. Systematic sampling

In systematic sampling the first unit is selected by using simple random sampling and the rest units of samples are selected according to the pre-designed pattern of certain rule so called systematic sampling.

Let N = nK, where N be population size, n is sample size and the integer k be sampling interval, i.e., $K = \frac{N}{n}$.

The first sample is selected at random and other samples are chosen at regular interval of K from sampling frame until the desired sample size n. This method is simple and convenient to use, and the results obtained from this method is satisfactory.

6.2.2 Non-probability sampling

If the sampling procedure is carried out without following any probabilistic rule then it is called non-probability sampling. They are:

i. Purposive or judgemental sampling

If the researcher or investigator uses own judgement on selecting samples from universe then it is called purposive or judgemental sampling. This method depends upon entirely personal judgement of the person and better to use when sample size is small.

ii. Convenience sampling

In this sampling the samples are selected on the basis of the convenience of the researcher or investigator. This is also called chunk sampling, in which part of population being investigated by convenience. This method of sampling used for quick decision and better when sample size is small.

iii. Quota sampling

In quota sampling, certain number of sample units from different categories with specific characteristics known as quotas are set up and the sample items are selected on personal judgement to fulfill required number of quotas. Though it save time, cost and manpower, it may be biased because of personal beliefs and prejudices of researcher.

iv. Snow ball sampling

Snow ball sampling technique works like chain referral. After observing the initial sample units the researcher asks for assistance from the subject to help identify people with similar trait of interest. The researcher then observes the nominated subjects and continues in the same way until obtaining the sufficient number of subjects.

6.3 Errors in sampling

The errors occurred in the process of collecting, processing and analyzing data can be classified into two headings, they are:

6.3.1 Sampling errors

After sample survey, when sample statistics are used to estimate population parameter, there will be difference between true value and estimated value of the parameter, which is due to sampling errors. This error is inherent and unavoidable in any sampling technique. Sampling error can be minimized by increasing the sample size. Sampling errors occur due to; faulty selection of sample, substitution, faulty demarcation of sampling units and improper choice of the statistic.

6.3.2 Non-sampling errors

This type of errors arises at any stage of inquiry like planning and execution of survey and collection, processing, analysis of data. Non-sampling errors can occur at every stage of planning or execution of sample survey or census. Non-sampling errors occurs due to; faulty planning, response errors, non-response errors, compiling errors, publication errors.

6.4 Sampling distribution of mean

Sampling distribution is the probability distribution of different sample statistics computed from the sample drawn from the universe of the population which plays important role in inferential statistics. The various values of the sample statistics such as mean, median, standard deviation, proportion, etc., obtained can be arranged as a frequency distribution is known as sampling distribution.

The sampling distribution of mean is the probability distribution of all possible sample means which is generated from the population. Sample mean is used to estimate the population mean. Let $\bar{X}_1, \bar{X}_2, \ldots, \bar{X}_n$ be the sample mean of size n drawn from the population, then sampling distribution of mean denoted by $\bar{\bar{X}}$ is given as;

$$\bar{\bar{X}} = \frac{\bar{X}_1 + \bar{X}_2 + \bar{X}_3 + \ldots + \bar{X}_n}{\text{Total number of sample mean}}$$

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Example 1. Let four young students having ages 18, 20, 22 and 24 years were admitted in ABC Engineering Consultancy for part-time job. Let us draw a possible sample of size n = 2, with replacement.

Solution

The samples along their mean are given as:

1st draw / 2nd draw	18	20	22	24
18	(18, 18)	(18, 20)	(18, 22)	(18, 24)
Sample mean	18	19	20	21
20	(20, 18)	(20, 20)	(20, 22)	(20, 24)
Sample mean	19	20	21	22
22	(22, 18)	(22, 20)	(22, 22)	(22, 24)
Sample mean	20	21	22	23
24	(24, 18)	(24, 20)	(24, 22)	(24, 24)
Sample mean	21	22	23	24

The frequency distribution of mean will be:

$\bar{X}_i$	f	$f\bar{X}_i$	$\bar{X} - \bar{\bar{X}}$	$f(\bar{X} - \bar{\bar{X}})^2$
18	1	18	–3	9
19	2	38	–2	8
20	3	60	–1	3
21	4	84	0	0
22	3	66	1	3
23	2	46	2	8
24	1	24	3	9
	16	336		40

Now, sampling distribution of mean

$$\bar{\bar{X}} = \frac{\Sigma f\bar{X}_i}{\Sigma f} = \frac{336}{16} = 21$$

$$\text{Population mean } (\mu) = \frac{18 + 20 + 22 + 24}{4} = \frac{84}{4} = 21$$

From the above calculation, we found that sampling distribution of mean is unbiased estimate of population, i.e.,

$$\bar{\bar{X}} = \mu = 21$$

$\therefore$ $\bar{X}$ is unbiased estimate of population mean (μ).

For standard deviation,

Population standard deviation

$$\sigma = \sqrt{\frac{\Sigma(X - \bar{X})^2}{n}} = \sqrt{\frac{\Sigma(X - \mu)^2}{N}}$$

$$= \sqrt{\frac{(18 - 21)^2 + (20 - 21)^2 + (22 - 21)^2 + (24 - 21)^2}{4}}$$

$$= \sqrt{5}$$

$\therefore$ $\sigma = 2.23$

Standard deviation due to sampling distribution

$$\sigma_{\bar{X}} = \sqrt{\frac{\Sigma f(X - \bar{\bar{X}})^2}{\Sigma f}} = \sqrt{\frac{40}{16}}$$

$\therefore$ $\sigma_{\bar{X}} = 1.58$

Here, population s.d (σ) does not equal to standard deviation of sampling distribution.

i.e., $\sigma \neq \sigma_{\bar{X}}$.

Hence, sample standard deviation is biased estimate of population standard deviation.

The ratio of population standard deviation to the square root of sample size *n* give

$$\frac{\sigma}{\sqrt{n}} = \frac{2.23}{\sqrt{2}} = 1.58.$$

Which is exactly equal to the standard deviation of sampling distribution $\sigma_{\bar{X}}$.

$\therefore$ $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$.

6.4.1 Standard error

From previous example, it is clear that standard deviation can be computed both from all observations of the population and sampling distribution, there is average variability of observation and sampling distribution computed, this type of variability is due to error and known as standard error.

Standard error of mean

The variance in sampling distribution of mean is called standard error of mean and denoted by $\sigma_{\bar{X}}$.

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$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$ for infinite population

$= \frac{\sigma}{\sqrt{n}} \cdot \sqrt{\frac{N-n}{N-1}}$ for finite population of size N..

Standard error of difference of two means

$$\sigma_{\bar{X}_1 - \bar{X}_2} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

Standard error of proportion

$\sigma_p = \sqrt{\frac{PQ}{n}}$ for infinite population

$= \sqrt{\frac{PQ}{n}} \cdot \sqrt{\frac{N-n}{N-1}}$ for finite population of size N

Standard error of difference of two proportions

$$\sigma_{P_1 - P_2} = \sqrt{\frac{P_1Q_1}{n_1} + \frac{P_2Q_2}{n_2}}$$

6.4.2 Application of standard error

1. It is used to determine the precision of the sample estimate of population parameter.

   Precision of estimate $= \frac{1}{\text{S.E. of estimate}}$

   i.e., the larger the S.E. the less efficient (precise) of estimate.
2. S.E. is used to estimate the population parameter.
3. S.E. is used in hypothesis testing.
4. S.E. is used to test the significance of regression coefficients.
5. S.E. is also used in point and interval estimation.

6.5 Central limit theorem

The central limit theorem based on the concept of sampling distribution, which is the probability distribution of a statistic for a large number of samples taken from population.

Suppose we draw a random sample from a population and calculate sample mean. Again, we draw another random sample and calculate sample mean. We repeat many times and end up with large number of means is an example of sampling distribution of means. The central limit theorem states that, for large sample size, i.e., $n \geq 30$, the sampling distribution of mean will always normally distributed.

Let $\bar{X}$ be the mean of sample size n, drawn from the population having mean μ and standard deviation σ, then for large sample size (i.e., $n \geq 30$), the sampling distribution of mean $\bar{X}$ follows normal distribution with mean μ and standard deviation $\frac{\sigma}{\sqrt{n}}$. Mathematically,

$$Z = \lim_{n \to \infty} \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}.$$

Example 2. The records of liquor company claims that liquor bottling machine is set as normally distributed to produce a mean fill per bottle of 12.1 ounces with standard deviation of 0.2 ounce. What is the probability that the sample mean of 36 text bottle is less than 12 ounce?

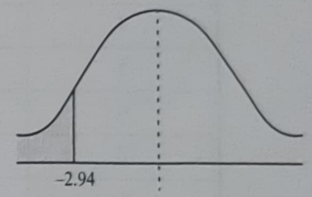
Solution

$\mu = 12.1$ ounce, $\sigma = 0.2$ ounce, $n = 36$

$P(\bar{X} < 12) = ?$

We have, from central limit theorem

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = \frac{12 - 12.1}{0.2/\sqrt{36}}$$



or, $Z = -2.94$

$P(Z < -2.94) = 0.5 - P(-2.94 < Z < 0)$

$= 0.5 - P(0 < Z < 2.94)$

$= 0.5 - 0.4984$

$= 0.0016$

Example 3. A company produces fluorescent lamps, whose mean life approximately follows normal distribution with mean life of 3000 hours with standard deviation of 60 hours. What is the probability that randomly selected 25 lamps will have average life than 3020 hours?

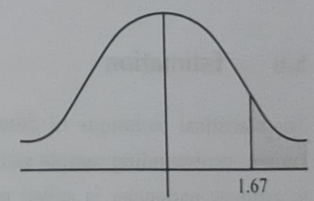
Solution

$\mu = 3000$ hours, $\sigma = 60$ hours, $n = 25$

$P(\bar{X} > 3020) = ?$

We have,

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = \frac{3020 - 3000}{60/\sqrt{25}}$$



or, $Z = 1.67$

$P(Z > 1.67) = P(0 < Z < \infty) - P(0 < Z < 1.67)$

$= 0.5 - 0.4525$

$= 0.047$

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Example 4. The population consists of value 1, 2, 3, 6, 8. Prove that sample mean of size 3 is unbiased estimate of population mean (without replacement).

Solution

$$\text{Population mean } (\mu) = \frac{1+2+3+6+8}{5} = 4.$$

Here, population size N = 5, sample size (n) = 3

The sample size 3, can be selected as follows.

Events	Observation	Sample mean ($\bar{X}$)	$P(\bar{X})$
1	(1, 2, 3)	2	1/10
2	(1, 2, 6)	3	1/10
3	(1, 2, 8)	3.666	1/10
4	(1, 3, 6)	3.334	1/10
5	(1, 3, 8)	4.	1/10
6	(2, 3, 6)	3.667	1/10
7	(2, 3, 8)	4.334	1/10
8	(2, 6, 8)	5.334	1/10
9	(3, 6, 8)	5.667	1/10
10	(2, 3, 8)	4.334	1/10

$E(\bar{X}) = \Sigma \bar{X}_i \cdot P(\bar{x}_i)$

$$= \frac{1}{10}(2 + 3 + 3.66 + 3.334 + \ldots + 5.667 + 4.334)$$

$$= \frac{1}{10} \times 40 = 4$$

Here, population mean = Sample mean = 4

Hence, sample mean is unbiased estimate of population mean.

6.6 Estimation

The statistical technique of determining unknown population parameter with the help of known corresponding sample statistic is called estimation. The sample statistic which is used to estimate parameter is called estimator. A specific numerical value of estimator is called estimate. Sample mean ($\bar{X}$), sample proportion (P), sample standard deviation (S) are the estimators of population mean (μ), population proportion (P), population standard deviation (σ), etc. There are two types of statistical estimation, they are; point estimation and interval estimation.

6.6.1 Point estimation

If a single value of population parameter is estimated or calculated from their respective sample statistic than the procedure is called point estimation.

Let $X_1, X_2, \ldots, X_n$ be a random sample of size n, drawn from the unknown population of parameter θ, here θ is the function X_i. Let $\hat{\theta}$ be the estimate of parameter θ, then $\hat{\theta} = f(X_1, X_2, \ldots, X_n)$. Therefore in point estimation, single value which is true value is calculated from the sample statistics to estimate the unknown population parameter.

Properties of good estimator

Some properties or criteria to select best estimator among the estimators, i.e., point estimation is said to be best estimator of parameter if it satisfies the following properties:

a. Unbiasedness
b. Consistency
c. Sufficiency
d. Efficiency

Unbiasedness

An estimator $\hat{\theta}$ is said to be unbiased estimate of θ, if expected value of $\hat{\theta}$ is equal to true value of θ. i.e., $E(\hat{\theta}) = \theta$.

Example 5. Sample mean is unbiased estimate of population mean. i.e., $E(\bar{X}) = \mu$.

Solution

Proof: Let $x_1, x_2, \ldots, x_n$ be a random sample of size n drawn from the population of mean μ and variance σ^2, then

$$\text{Sample mean}(\bar{X}) = \sum_{i=1}^{n} \frac{x_i}{n} \qquad \ldots.(1)$$

Taking expectation on both sides of (x), we get

$$E(\bar{X}) = \frac{1}{n} E\left(\sum_{i=1}^{n} x_i\right)$$

$$= \frac{1}{n}\left\{\sum_{i=1}^{n} E(x_i)\right\}$$

$$\therefore \quad E(\bar{X}) = \frac{1}{n} \cdot n\mu \qquad [\because E(x_i) = \mu, \forall, i = 1, 2, \ldots, n]$$

$$\because \quad E(\bar{X}) = \mu$$

i.e., sample mean is unbiased estimate of population mean.

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Consistency

An estimator t_n is said to be consistent estimator of parameter θ, if for large sample size, i.e., for $n \to \infty$, $t_n \to \theta$.

Efficiency

If t_1 and t_2 be two estimators of parameter, such that

$\text{Var}(t_1) < \text{Var}(t_2)$ for all values of n.

Then, t_1 is more efficient than t_2.

Sufficiency

An estimator $\hat{\theta}$ is said to be sufficient estimator of θ, if it is unbiased, consistent and efficient, i.e., it carries all the information in the sample regarding the parent population.

6.6.2 Interval estimation

An interval (C_1, C_2) which estimates the unknown population parameter θ with certain probability is called interval estimation. Here C_1 and C_2 are lower and upper points of the interval of the parameter θ. The values of C_1 and C_2 can be obtained when $(1 - \alpha)$ given probability so that:

$$P\{C_1 < \theta < C_2\} = 1 - \alpha$$

$(1 - \alpha)$ is called confidence level.

6.6.3 Confidence interval for mean (μ)

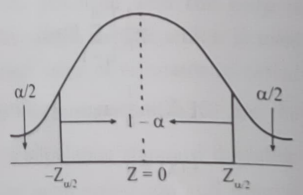
a. **Large sample case ($n \geq 30$):** Let random sample X which takes values $x_1, x_2, \ldots, x_n$ of large sample of size $n \geq 30$ be drawn from a normal population of unknown mean (μ) and variance σ^2. Then for large sample size sample mean $\bar{X}$ is also normally distributed with mean $E(X) = \mu$ and variance $\frac{\sigma^2}{n}$. Then from central limit theorem:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1).$$

Setting the confidence level $(1 - \alpha)$, the confidence interval for μ is given as;

From the normal curve,

$$P\{-Z_{\alpha/2} \leq Z \leq Z_{\alpha/2}\} = 1 - \alpha$$



or, $$P\left\{-Z_{\alpha/2} \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq Z_{\alpha/2}\right\} = 1 - \alpha$$

or, $$P\{\bar{X} - Z_{\alpha/2}\,\sigma/\sqrt{n} \leq \mu \leq \bar{X} + Z_{\alpha/2}\,\sigma/\sqrt{n}\} = 1 - \alpha$$

Hence, for $(1 - \alpha)$ confidence level, the confidence interval for μ is given as:

$$C.I.\text{ for } \mu = \bar{X} \pm Z_{\alpha/2}\,\sigma/\sqrt{n}$$

Taking negative sign we can get lower limit C_1 and taking positive sign, we get upper limit C_2 of mean μ.

b. **Large sample size and variance σ^2 is unknown:** When sample size is large $(n \geq 30)$ and population variance is not known, then sample variance (S^2) is unbiased estimator of population variance σ^2.

$$S^2 = \frac{\Sigma(x - \bar{x})^2}{n - 1} \approx \frac{\Sigma(x - \bar{x})^2}{n} = \sigma^2, \text{ i.e., } S = \sigma.$$

Then, $Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$, which is asymptotic standard normal variate.

Hence $(1 - \alpha)$ confidence level, confidence interval for μ is:

$$P\{\bar{X} - Z_{\alpha/2}\,S/\sqrt{n} \leq \mu \leq \bar{X} + Z_{\alpha/2}\,S/\sqrt{n}\} = 1 - \alpha$$

$$\therefore \quad C.I.\text{ for } \mu = \bar{X} + Z_{\alpha/2}\,S/\sqrt{n}$$

Confidence interval for mean (μ), small sample size ($n < 30$)

When the samples drawn from the normal population is of small size $(n < 30)$, then Z is not a standard normal variate. In this case, confidence interval for mean (μ) is determined by using student's t-statistic, t-statistic is defined as:

$$t = \frac{\bar{X} - \mu}{S/\sqrt{n}}, \; S = \text{Sample standard deviation.}$$

Which follows t–distribution with $(n - 1)$ degrees of freedom.

Then at $(1 - \alpha)$ confidence level, the confidence interval of μ is:

$$P\{-t_{\alpha/2} < t < t_{\alpha/2}\} = 1 - \alpha$$

or, $$P\{\bar{X} - t_{\alpha/2}\,S/\sqrt{n} \leq \mu \leq \bar{X} + t_{\alpha/2}\,S/\sqrt{n}\} = 1 - \alpha$$

$$\therefore \quad C.I.\text{ for } \mu = \left(\bar{X} \pm t_{\alpha/2}\frac{S}{\sqrt{n}}\right)$$

Note: If sample size n are drawn from finite normal population of size N, then for $(1 - \alpha)$ confidence level, confidence limits for μ is:

$$C.I.\text{ for } \mu = \bar{X} \pm Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\sqrt{\frac{N - n}{N - 1}}.$$

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Example 6. A random sample of 100 normal adult males of 25 years of certain community had a mean systolic blood pressure of 125 mm Hg, with standard deviation of 15 mm Hg. Find 95% confidence interval for the mean population blood pressure of the age group of the community.

Solution

Here, $n = 100$, $\bar{X} = 125$, $S \approx \sigma = 15$, $\alpha = 0.05$

$Z_{\alpha/2} = 1.96$ from normal table.

Then, C.I. for $\mu = \bar{X} + Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$

$$= 125 \pm 1.96 \times \frac{15}{\sqrt{100}}$$

Solving negative and positive sign separately, we get the limits for confidence interval are $C_1 = 115.7$ mm Hg and $C_2 = 134.3$ mm Hg.

Example 7. From a population of 540, a sample of 60 individual is taken. From this sample the mean is found to be 6.2 and standard deviation is 1.368. Construct 96% confidence interval for mean.

Solution

Population size (N) = 540, Sample size (n) = 60

Sample mean ($\bar{X}$) = 6.2, Sample s.d (S) = 1.368

$Z_{\alpha/2} = 2.05$

$\therefore$ C.I. for $\mu = \bar{X} + Z_{\alpha/2}\frac{S}{\sqrt{n}}\sqrt{\frac{N-n}{N-1}}$

$$= 6.2 \pm 2.05 \times \frac{1.368}{\sqrt{60}}\sqrt{\frac{540-60}{540-1}}$$

$$= 6.2 \pm 0.35$$

On solving, lower limit (c_1) = 5.85 and upper limit (c_2) = 6.55

Example 8. A sample of 16 girls of certain school had a mean weight of 71.5 lbs and standard deviation of 12 lbs respectively. Find 90% confidence interval for population mean weight.

Solution

Here, $n = 16$, $\bar{X} = 71.5$ lbs, $S = 12$ lbs

$\alpha = 0.1$, from t-table, $t_{\alpha/2}$ 10% = 1.75 (at 15 d.f)

$\therefore$ C.I. for $\mu = \bar{X} \pm t_{n-1,\alpha/2}\frac{S}{\sqrt{n}}$

$$= 71.5 \pm 1.75 \times \frac{12}{\sqrt{16}}$$

On solving, confidence interval for mean are $(c_1, c_2) = (66.24 \text{ lbs}, 76.75 \text{ lbs})$.

Example 9. The following are the eight observations from infinite population, with normal distribution.

76, 76, 83, 91, 80, 77, 84, 81

Construct 98% confidence interval for population mean.

Solution

We have, $\bar{X} = \frac{\Sigma X}{n} = \frac{647}{8} = 80.87$

$$S = \sqrt{\frac{\Sigma(X - \bar{X}^2)}{n-1}} = 5.22$$

From t-table

$t_{8-1,\alpha/2\,(2\%)} = 2.998$

$\therefore$ C.I. for $\mu = \bar{X} \pm t_{n-1,\alpha/2}\frac{S}{\sqrt{n}}$

$$= 8.87 \pm 2.998 \times \frac{5.22}{\sqrt{8}}$$

On solving confidence interval for mean are (73.34, 86.40).

Example 10. A random sample of 40 students is drawn from college and their height showed a mean of 68.6 inches and standard deviation of 2.5 inches. Construct 95% confidence interval for mean height of all the students of the college.

Solution

Given, $\bar{X} = 68.6$, $\sigma = 2.5$, $n = 40$, $\alpha = 0.005$, $Z_{\alpha/2}\,(5\%) = 1.96$

$\therefore$ Confidence interval for μ

C.I. for $\mu = \bar{X} + Z_{\alpha/2}\frac{S}{\sqrt{n}}$

$$= 68.6 \pm 1.96 \times \frac{2.5}{\sqrt{40}}$$

Solving by taking positive and negative sign, we get

$(c_1, c_2) = (67.72, 69.27)$ inches.

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Example 11. A company produces automobile tyre, a test sample of 64 tyres was taken and a test run showed average life of 50,000 miles. Find the 95% confidence limits for population mean, given that population standard deviation is 3000 miles.

Solution

$\bar{X} = 50{,}000$ miles, $\sigma = 3{,}000$ miles, $n = 64$, $\alpha = 0.05$

C.I. for $\mu = \bar{X} + Z_{\alpha/2}\frac{S}{\sqrt{n}}$

$$= 50{,}000 \pm 1.96 \times \frac{3{,}000}{\sqrt{64}}$$

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On solving, we get

Lower limit (c_1) = 50,000 – 735 = 49,265 miles

Upper limit (c_2) = 50,000 + 735 = 50,735 miles

Example 12. Specific gravity of aluminium from randomly selected 16 measurement samples from normal population had a mean of 2.705 with standard deviation of 0.059. Construct 95% confidence interval for actual mean specific gravity of aluminium.

Solution

$\bar{X} = 2.705$, Sample standard deviation (S) = 0.029

$n = 16$, $\alpha = 0.05$

$t_{16-1,\alpha/2} = 2.13$ from t-table

$$\therefore \text{ C.I. for } \mu = \bar{X} \pm t_{n-1,\alpha/2}\frac{S}{\sqrt{n}}$$

$$= 2.705 \pm 2.13 \times \frac{0.029}{\sqrt{16}}$$

Solving negative and positive sign, we get

lower limit (c_1) = 2.69

upper limit (c_2) = 2.72

6.7 Confidence limits for population proportion

Let p be the sample proportion of success and P be the population proportion of success, then (1 – α)% confidence interval for population proportion from large samples (i.e., SRSWR) are given as:

$$\text{C.I. for } p = p = \pm Z_{\alpha/2}\,\text{S.E}(p)$$

$$= p \pm Z_{\alpha/2}\sqrt{\frac{PQ}{n}}$$

where, p = population proportion of success.

If population proportion (p) is unknown, we can use unbiased estimate provided by sample proportion (p). Then,

$$\text{C.I. for } p = p \pm Z_{\alpha/2} \times \sqrt{\frac{pq}{n}} \quad [\because \hat{p} = p \text{ for large samples}]$$

In case of SRSWOR from finite population of size N, the (1 – α)% confidence limits of p are given as

$$\text{C.I. for } p = p \pm Z_{\alpha/2}\sqrt{\frac{PQ}{n}}\sqrt{\frac{N-n}{N-1}}$$

If P and Q are unknown, then

$$\text{C.I. for } P = p \pm Z_{\alpha/2}\sqrt{\frac{pq}{n}}\sqrt{\frac{N-n}{N-1}} \quad \left[\because \hat{P} \text{ and } \hat{Q} = q \text{ for large samples}\right]$$

Example 13. A sample survey of 500 households of certain city selected at random showed that 350 of them used fluorescent lamps and 150 of the household used filament lamps, construct a 95% confidence interval estimate of the proportion of all the household of the city who used fluorescent lamps.

Solution

Given,

$n = 500$, $p = \frac{350}{500} = 0.7$ [∵ p is unknown]

$q = 1 - p = 0.3$, $Z_{\alpha/2}(5\%) = 1.96$

Now, confidence limits for the household who use fluorescent lamp

$$\text{C.I. for } P = p \pm Z_{\alpha/2}\sqrt{\frac{pq}{n}}$$

$$= 0.7 \pm 1.96 \times \sqrt{\frac{0.7 \times 0.3}{500}}$$

$$= 0.7 \pm 0.0401$$

On solving,

lower limit (c_1) = 0.659

upper limit (c_2) = 0.7401

Example 14. A biased coin was thrown 400 times and heads were resulted 240 times, find the 90% confidence interval for population proportion of head.

Solution

Sample proportion of head (p) = $\frac{240}{400} = 0.6$, q = 0.4

$\alpha = 0.1$, $Z_{\alpha/2}(10\%) = 1.64$

$$\therefore \text{ C.I. for } P = p \pm Z_{\alpha/2}\sqrt{\frac{pq}{n}}$$

$$= 0.6 \pm 1.64 \times \sqrt{\frac{0.6 \times 0.4}{400}}$$

On solving, we get

Lower limit (c_1) = 0.56

Upper limit (c_2) = 0.64

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6.8 Determination of sample size by using mean

Let $\bar{X}$ be the mean of random samples of size n drawn from the population having mean μ and variance σ^2, then from central limit theorem, we have

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

or, $$Z = \frac{\sqrt{n}\,|\bar{X} - \mu|}{\sigma}$$

or, $$2\sigma = \sqrt{n}\,E \qquad [\because |\bar{X} - \mu| = E]$$

or, $$n = \frac{Z^2\sigma^2}{E^2}$$

For α level of significance in two tail,

$$n = \frac{Z^2_{\alpha/2}\,\sigma^2}{E^2}$$, gives required sample size

where, E = error, i.e., $E = |\bar{X} - \mu|$

6.9 Determination of sample size using proportion

Let p be the same proportion and P be the population proportion and n be the sample size, then we can write

$$Z = \frac{p - P}{\sqrt{\frac{PQ}{n}}}$$

Let the difference between sample proportion and population proportion be denoted by E, i.e., $E = |p - P|$. Then,

$$n = \frac{Z^2_{\alpha/2}\,PQ}{E^2}$$

If population proportion P are not given, then assume probability of success and failure are equal. (P = Q = 0.5)

Example 15. The mean and standard deviation of random sample were found to be 100 and 10 respectively. If an investigator wants to be 95% confident that the error is estimate of population mean should not exceed ± 2, how many observations are required?

Solution

$\bar{X} = 100, S = \hat{\sigma} = 10$

$\alpha = 0.05, Z_{\alpha/2} = 1.96, E = \pm 2$

We have,

$$n = \frac{Z^2_{\alpha/2}\,\sigma^2}{E^2}$$

$$= \frac{(1.96)^2 \times 10^2}{2^2}$$

$$= 96.04$$

$\therefore$ n = 96.

Example 16. A researcher wishes to estimate mean electricity consumed by the households in the defined community. Preliminary information is that mean consumption is about 150 units with standard deviation of 30 units. If sampling error up to 5 units in the estimate to be tolerated, how many households should be included at 95% confidence interval? If the community has 1000 households, what should be the sample size at 95% confidence interval?

Solution

Given,

$\bar{X}$ = 150 units, σ = 3- units, α = 5%, E = 5

Optimum sample size (n) = ?

We have,

$$n = \frac{Z^2_{\alpha/2}\,\sigma^2}{E^2} = \frac{(1.96)^2 \times 30}{5^2}$$

$\therefore$ n = 138

If the population (N) = 1000

$$n = \frac{Z^2_{\alpha/2}\sigma^2}{E^2 + (Z^2_{\alpha/2}\sigma^2)/N}$$

$$= \frac{(1.96)^2 \times (30)^2}{5^2 + (1.96)^2\,(30)^2/1000} = 12.45$$

$\therefore$ n = 121

Example 17. A team of software developer feels 95% confident that a new software they developed will cover 80% user of banks. How large should the sample size be for team with in ± 2% of the proportion of all cases that the software will cover?

Solution

P = 0.8, Q = 0.2, α = 5%, E = 0.02

We have,

$$n = \frac{Z^2_{\alpha/2}\,PQ}{E^2}$$

$$= \frac{(1.96)^2 \times 0.8 \times 0.2}{(0.02)^2}$$

$\therefore$ n = 1537

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Example 18. A researcher wants to estimate the proportion of smokers among the people of certain city. Previous records showed that proportion of smokers was 0.3. What sample size should be selected so as to have the proportion of smokers not to exceed by 10% with 98% confidence level?

Solution

Given that,

$P = 0.3, Q = 0.7$

$E = |0.1 - 0.3| = 0.2, \alpha = 0.02$

$Z_{\alpha/2} = 2.32$

We have,

$$n = \frac{Z^2_{\alpha/2}\, PQ}{E^2} = \frac{(2.32)^2 \times 0.3 \times 0.7}{(0.2)^2}$$

$\therefore \quad n = 28.$

6.10 Confidence interval for difference between means of two normal populations

6.10.1 Large sample case

a. When variances are known: Let $\bar{X}$ and $\bar{Y}$ be the sample means of two series having sizes n_1 and n_2 drawn from two unknown means μ_1 and μ_2 with known variance σ_1^2 and σ_2^2. Then confidence interval for difference of two population means at $(1 - \alpha)$ confidence level is given by:

$$\text{C.I. for } \mu_1 - \mu_2 = (\bar{X} - \bar{Y}) \pm Z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n^2}}$$

b. When variance are unknown and unequal: When population variances are not known, they must be estimated from sample variance S_1^2 and S_2^2. Then confidence interval estimate of $\mu_1 - \mu_2$ at $(1 - \alpha)$ confidence level is given as

$$\text{C.I. for } \mu_1 - \mu_2 = (\bar{X} - \bar{Y}) \pm Z_{\alpha/2} \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n^2}}$$

c. When variances are unknown but equal: When unknown population variance σ_1^2 and σ_2^2 are equal, i.e., $\sigma_1^2 = \sigma_2^2 = \sigma^2$, then unknown variance σ^2 can be computed by using combined sample variances. The combined sample variances denoted by Sp^2 is given as

$$Sp^2 = \frac{(n-1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$$

Hence, confidence interval for differences of two means $(\mu_1 - \mu_2)$ at $(1 - \alpha)$ level of confidence is:

$$\text{C.I. for } (\mu_1 - \mu_2) = (\bar{X} - \bar{Y}) \pm Z_{\alpha/2}\, Sp \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

6.10.2 Small sample case ($n_1 < 30$ and $n_2 < 30$)

Let $\bar{X}$ and $\bar{Y}$ be the sample means of two series having sizes n_1 and n_2 drawn from normal population with unknown mean μ_1 and μ_2 and variances σ_1^2 and σ_2^2, the variances must be estimated from sample variances S_1^2 and S_2^2. Then confidence interval for difference of two means at $(1 - \alpha)$ confidence level is given as:

$$\text{C.I. for } (\mu_1 - \mu_2) = (\bar{X} - \bar{Y}) \pm t_{n_1 + n_2 - 2, \alpha/2} \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}$$

where, $S_1^2 = \sum_{i=1}^{n} \frac{(x_i - \bar{x})^2}{n_1 - 1}$, $S_2^2 = \sum_{i=1}^{n_2} \frac{(y_i - \bar{y})^2}{n_2 - 1}$

Example 19. Two independent random samples of size 40 and 50 were drawn from two normal populations having variances 30 and 35. The sample means of two samples were found to be 350 and 340. Construct 90% confidence interval for $\mu_1 - \mu_2$.

Solution

$n_1 = 40, \quad n_2 = 50, \quad \alpha = 0.1$

$\sigma_1^2 = 30, \quad \sigma_2^2 = 35, \quad Z_{\alpha/2} = 1.64$

$\bar{X} = 350, \quad \bar{Y} = 340$

We have,

$$\text{C.I. for } \mu_1 - \mu_2 = (\bar{X} - \bar{Y}) \pm Z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{n_2^2}{n_2}}$$

$$= (350 - 340) \pm 1.64 \sqrt{\frac{30}{40} + \frac{35}{50}}$$

$$= 10 \pm 1.98$$

On solving, we get

C.I. for $(\mu_1 - \mu_2) = (8.02, 11.98)$.

Example 20. Two independent random samples of 70 boys and 55 girls from normal population were drawn and found their mean weighs were 62 kg and 46 kg with variances 16 and 12 respectively. Construct a 95% confidence interval for difference between mean weight of boys and girls.

Solution

Given,

$n_1 = 70, \quad n_2 = 55, \quad \alpha = 0.05$

$\bar{X} = 62$ kg, $\quad \bar{Y} = 46$ kg

$S_1^2 = 16, \quad S_2^2 = 12$

As the variances are unequal and sample sizes are larges, then

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$$\text{C.I. for } \mu_1 - \mu_2 = (\bar{X} - \bar{Y}) \pm Z_{\alpha/2}\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}$$

$$= (62 - 46) \pm 4.96\sqrt{\frac{16}{70} + \frac{12}{55}}$$

$$= 16 \pm 1.32$$

On solving, we get

C.I. for $(\mu_1 - \mu_2)$ = (14.68 kg, 17.32 kg).

Example 21. Two different catalysts are being compared to determine when they require different items for removal of identical quantities of photoresist material. 12 batches were run with first catalyst resulting sample mean emersion time of 24.6 minutes with standard deviation of 0.85 minutes. 15 batches were run with second catalyst resulting mean emersion time of 22.1 minutes with standard deviation of 0.98 minutes. Construct 95% confidence interval for difference of means.

Solution

Given,

$n_1 = 12$, $n_2 = 15$

$\bar{X}_1 = 24.6$ minutes $\bar{X}_2 = 22.1$ minutes

$S_1 = 0.85$ minutes $S_2 = 0.98$ minutes

$\alpha = 0.5$, $t_{\alpha/2}$ at 25 d.f. = 2.06

$$\therefore \text{C.I. for } \mu_1 - \mu_2 = (\bar{X} - \bar{Y}) \pm t_{n_1 + n_2 - 2, \alpha/2}\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}$$

$$= (24.6 - 22.1) \pm 2.06 \times \sqrt{\frac{0.85^2}{12} + \frac{0.98^2}{15}}$$

$$= 2.5 \pm 0.728$$

$\therefore$ C.I. for $(\mu_1 - \mu_2)$ = (1.74, 3.23 minutes)

Exercise 6.1

1. A population consists of value 7, 6, 8, 4, 10. Show that sample mean of size 3 is unbiased estimate of population mean.
2. A population consists of the values 3, 6, 8, 9, 12 and 15. Draw the sample of size 3 and show that unbiased estimate of population mean.
3. What do you mean by unbiased estimator? Consider a population of 5 units with 4, 2, 0, 6, 8. Show that sample mean of size 2 is unbiased estimator of population mean.
4. In a class of 60 students, their average marks in mathematics was 70 with standard deviation of 8. A sample of 25 students is taken at random, calculate the standard error of mean in case of finite and infinite population.

5. Pokhara University, School of Engineering is interested in knowing the average height of seniors at this school but they do not have enough time to examine the records of 430 seniors. They randomly selected 48 seniors and find the average height of 64.5 inches with standard deviation of 2.5 inches. Find the standard error of mean and construct 90% confidence interval for population mean.
6. In maintaining the quality control of the production of optical lenses, the production manager wants to estimate the mean thickness of the lens produced. A random sample of 100 lenses revealed a mean of 0.50 mm thickness and population standard deviation known to be 0.15 mm. Construct 99% confidence interval for the mean thickness of lens produced.
7. Twelve civil engineer were randomly selected from the construction site and it was found that, they made an average of 3.6 errors per day with standard deviation of 0.42. Construct 90% confidence interval for population mean of errors per day.
8. From a population having sizes 540, sample of 60 individual is taken and sample mean is found to be 6.2 with standard deviation of 1.36. Then find standard error of mean and construct 96% confidence interval for population mean.
9. From the central post office, 8 envelops were randomly selected and their weight (in gm) found to be: 12.1, 11.9, 12.3, 11.9, 12.1, 12.4, 12.1 and 11.5. Find the 99% fiducial limits for the mean weight of envelops received at that post office.
10. A random sample of 100 articles selected from large batch showing mean inner diameter of the articles is 0.354 and standard deviation is 0.048. Construct 98% confidence interval for the true mean of articles.
11. Find 90% confidence intervals for the defective in a large lot of material on the basis of random sample of 50 items, if the number of defective items found in the sample is 18.
12. A random sample of 500 bulbs was taken from the manufacturing company, among them 65 were found to be defective. Assign the limits in which the non-defective bulbs produced by the company at 95% confidence levels.
13. A sample poll of 100 voters were chosen at random from all voters in Kaski district indicated that 55% of them were in favor of particular party. Find the 95% confidence limits for the proportion of all the voters in favor of the party.
14. A school mathematics teacher surveyed top 150 students of the school and found that 42% of them were unable to add fractions correctly. Construct a 99% confidence interval for the true proportion of the top students who can not correctly add fractions.
15. In Pokhara city, 400 persons were randomly selected and asked whether they have intenet facility in their residence or not. Among them 35% answered they do not have internet facility. Construct 95% and 99% confidence interval for the population proportion who have internet facility in their residence. What happens when level of confidence increases?
16. An experiment was conducted in two types of engine A and B fuel mileage in miles per gallon was measured. Fifty experiments in type A gives 36 miles per gallon, and 75 experiments in type B gives 42 miles per gallon. Assume the population standard deviation are 6 and 8 for engine A and B respectively. Construct 96% confidence interval for the difference of true mean mileage between engine B and A.

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17. In a study of growth of children, one group of 100 children had of mean height of 60 cm and standard deviation 2.5 cm. While another group of 150 children had a mean height of 62 cm and standard deviation 3 cm. Construct 95% confidence interval for the difference of mean height in two groups.

18. Heights of teenage boys and girls were calculate, the mean height of 12 boys was 174 cm and variance 62 cm. For the sample of 15 girls the mean was 165 cm with variance of 64 cm. Find the confidence limits for the difference of mean height of boys and girls at 96% level of confidence.

19. The breaking strength of 14 glass tubes are tested. The mean and standard deviation obtained were 17.8 and 1.9 hundred pounds respectively. Construct 95% confidence interval for the average breaking strength of glass rods.

20. The following are the number of students attended in two different schools in any specific week.

School A:	107	148	123	165	102	119
School B:	134	115	112	151	33	129

Find 95% confidence limits of the difference in mean number of students attended in the school.

21. A study was to be conduced and on the percentage of homeowners who own at least two television sets is 20%. How large a sample is required if we wish to be 99% confident that the error in estimating this quantity is less than 0.03?

22. How large a sample is required if we want to be 95% confidence that the error is using $\hat{P}$ to estimate P is less than 0.05? Using $\hat{P} = 0.16$ as an initial estimate of P and level of confidence is 95%?

23. A researcher wants to estimate the population mean and finds that at 95% confidence level, the sample mean will not differ from true population mean by more than 25% of the standard deviation. How large a sample size us required?

24. If the true mean of population is not more than 1.5 than that of sample mean and population standard deviation is 15. What should be the sample size at 95% level of significance?

25. From a sample of 400 items, computed mean and standard error of mean was found to be 125 and 2.5 respectively. What sample size should be required so that computed mean and true mean will not differ more than 5 at 99% level of significance?

26. An electrical firm manufactures light bulbs that have a length of life approximately normally distributed with standard deviation of 40 hours. If sample of 30 bulbs has an average life of 780 hours, construct 96% confidence interval for the true mean of all bulbs produced by the firm. How large a sample is needed if we wish 96% confident that our sample mean will be with in 10 hours of true mean?

27. A random sample of 10 chocolate energy bars of certain brand has an average 230 calories with standard deviation of 15 calories. Construct a 99% confidence interval for the true mean calories. Content of this brand of energy bar. Assume that the distribution of the calories is approximately normal.

28. A machine produces metal pieces that are cylindrical in shape. The sample pieces having diameters 1.01, 0.97, 1.03, 1.04, 0.99, 0.98, 0.99, 1.01 and 1.01 centimeters. Construct 99% confidence interval for the mean diameter of pieces from this machine, assuming an approximate normal distribution.

29. The fraction of defective bulbs produced by two company is being analyzed. A random sample of 1000 bulbs from first company has 10 defectives, while random sample of 1200 bulbs from second company has 25 defectives. Find 99% confidence interval on the difference in fraction defectives produced by two companies.

30. An industrial engineer is interested in estimating the mean time required to assemble the printed circuit board. How large a sample is required if the engineer wishes to be 95% confident that the error in estimating the mean is 0.25 minutes? The standard deviation of assembly time is 0.45 minutes.

Answers

4. 3.58, 2.76
5. S.E. = 15.03, limits (39.77, 89.22 inches)
6. 0.4613, 0.5387 mm
7. 3.38, 3.81
8. 0.167 and (5.86, 6.54)
9. 11.95 gm, 12.13 gm
10. 0.343, 0.365
11. 0.10, 0.62
12. 0.84, 0.89
13. 0.453, 0.647
14. (0.523, 0.317)
15. (0.697, 0.503), (0.711, 0.589)
16. 3.43, 8.57 miles per gallon
17. 1.31, 2.69 cm
18. 2.71, 15.29 cm
19. 16.72, 18.78
20. –7.07, 3.73
21. 1,175
22. 207
23. 61
24. 384
25. 664
26. (765.04, 794.96), 67
27. 214.58, 245.42
28. 0.978 cm, 1.033 cm
29. – 0.0244, 0.0024
30. 13

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Unit 7

Hypothesis Testing

7.1 Introduction

In statistic generally population characteristics also known as parameters are unknown. We need to decide in many problems, whether to accept or reject a statement based on parameter. Statistical test are employed to make decision on estimated parameter whether it is valid or not. So it is important part of the inferential statistics. Statistical hypothesis is a presumption about the population parameter, whose validity are going to be tested. In another way a statement about population parameter developed for the purpose of testing is called hypothesis.

The process of testing significance or validity of the statement related to population parameter on the basis of corresponding sample statistics is called hypothesis testing. Many types of decision problems can be formulated as hypothesis testing problems. In this chapter, we will discuss about such problems.

7.2 Systematic procedure for testing of hypothesis

The following are the systematic procedure to reach in decision level for testing of hypothesis.

1. **Formulation of hypothesis:** Based on population parameter, we can formulate statistical hypothesis as positive statement and negative statement.
   a. **Null hypothesis:** Null generally means there is no difference thus a statement there is no difference between sample statistic and its expected value is known as null hypothesis. In this statement, sample static and parameter are linked by equal (=) sign. According to R.A. Fisher, "Null hypothesis is a hypothesis which is tested for the possible rejection under the assumption that it is true". Null hypothesis is denoted by H_0.

      For example, we are interested in deciding whether or not the mean life of fluorescent lamp (μ) is 3000 hours. We can express this formally as:

      $H_0 = \mu = 3000$ hrs.

      The statement $H_0 : \mu = 3000$ hrs is called null hypothesis.

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b. **Alternative hypothesis:** A complementary statement of null hypothesis is called alternative hypothesis. Alternative hypothesis is also negative or reverse statement of null hypothesis. It is denoted by H_1.

From our above example, alternative hypothesis will be:

$H_1 : \mu \neq 3000$ hrs (two tailed test)

The mean life of fluorescent lamp does not equal to 3000 hours, indicates significant difference between hypothesis value, but the direction of difference either it is less than or greater than not defined. To make it more clear,

$H_1 : \mu > 3000$ hrs (Right tailed test)

$H_1 : \mu < 3000$ hrs (Left tailed test)

2. **Selection of appropriate test statistical tool**

There are different types of statistical tools either in parametric and non parametric test. To select appropriate statistical tool, the nature of population from which sample is drawn, measurement scale is used in data collection process and sampling distribution of statistics should remember first. Then a test statistic is calculated by the following relation:

$$\text{Test statistics} = \frac{\text{Sample statistic} - \text{Parameter}}{\text{Standard error}}.$$

The commonly used test statistics are:

For large sample, i.e., $n \geq 30$

$$Z = \frac{\theta - E(\theta)}{S.E(\theta)}$$

For small sample, i.e., $n < 30$

$$t = \frac{\theta - E(\theta)}{S.E(\theta)} \text{ with } (n-1) \text{ degrees of freedom.}$$

Types of errors in testing hypothesis

The decision to accept or reject null hypothesis is based on the test statistic, which is calculated from sample data. When decision will make using the information from random sample, error will occur in decision. There are two types of error in hypothesis testing, they are type I and type II error.

Type I error

The probability of rejecting null hypothesis when it is really true is called type I error and is denoted by α.

$\alpha = P(\text{Reject } H_0, \text{ when } H_0 \text{ is true})$

The probability of type I error α is also called significance level.

Type II error

The probability of accepting null hypothesis when it is false is called type II error and it is denoted by β.

$\beta = P(\text{Accept } H_0, \text{ when } H_0 \text{ is wrong})$

Inspecting manufacturing process of quality control, α is amount to reject good objects and β is the error of accepting bad objects in the selected items. Type I and II error can not occur simultaneously in the given sample. A decrease of probability of type I error results increase in probability of type II error and vice-versa. An increase in sample size can minimize α and β. If the difference between true value and hypothesized value is greater, than β will become smaller.

Power of test

The power of the test can be computed as

$p = 1 - \beta$.

As the type II error of hypothesis testing is decreases, the power of the test will increases and becomes more powerful.

3. **Level of significance:** The level of significance is also called the level of risk. The risk of type I error which is tolerated in making a decision about null hypothesis and it is denoted by α.

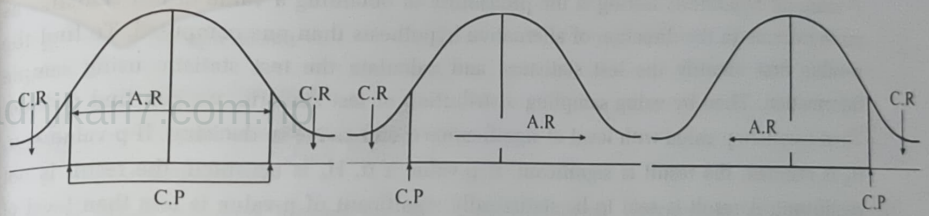
$\therefore \quad \alpha = P(\text{Type I error})$

$= P(\text{Reject } H_0, \text{ where } H_0 \text{ is really true})$

In practice for moderate precision, the level of significance are taken as 5%, i.e., $\alpha = 0.05$, for high precision it is taken as 1% ($\alpha = 0.01$). If we allow $\alpha = 0.05$, it implies that 5 samples out of 100, we are going to reject correct null hypothesis, means we are 95% confident our decision for rejecting H_0 is accurate. The level of significance always fixed before collecting the sample data for hypothesis testing and precision of the test required.

Critical region

In hypothesis testing, the test statistic follows some known sampling distributions, which own probability curve. The probability curve is divided into two mutually exclusive regions, i.e., region of acceptance and region of rejection.



A.R. = Acceptance region

C.R. = Critical region

C.P. = Critical point

The region of rejection which is the region in which null hypothesis is rejected is called critical region (C.R.). The area of the critical region is equal to the level of significance.

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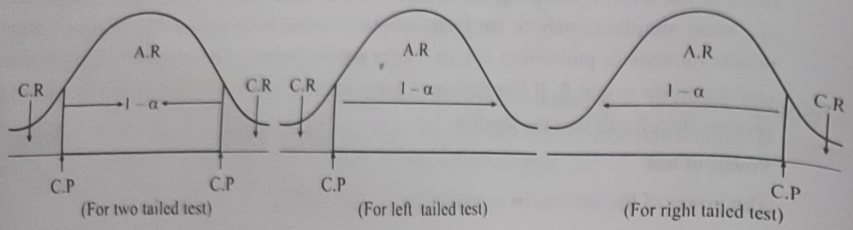
The region of acceptance of null hypothesis is called acceptance region (A.R.) and the area of acceptance region is $1 - \alpha$, which is level of confident. The critical region may lies both tail of the curve according to the alternative hypothesis.



Critical values or significant values

The value of test statistic that separates the critical region and acceptance region is called critical value or significant value. The critical value depends upon the level of significance and alternative hypothesis.

On tailed and two tailed test

If direction of difference between true value and hypothesised value is not mention then we use two tailed test. If the direction of true value and hypothesized value linked by some inequality sign $\geq$ or $<$, $\leq$ or $<$, then we use one tailed test.

$H_1: \mu \neq \theta$
$H_1: \mu_1 \neq \mu_2$ } two tailed test
$H_1: P \neq 0.7$

$H_1: \mu < \theta$
$H_1: \mu_1 < \mu_2$ } left tailed test

$H_1: \mu > \theta$
$H_1: \mu_1 > \mu_2$ } right tailed test

P-value approach in the hypothesis testing

P-value for hypothesis testing is the probability of obtaining a value of test statistics as more extreme in the direction of alternative hypothesis than one computed. To find the p-value first identify the test statistics, and calculate the test statistic using sample information. Then by using sampling distribution of test statistic, we can find p-value. Then compare p-value with level of significance α and arrive in decision. If p-value $< \alpha$, H_0 is rejected, the result is significant. If p-value $\geq \alpha$, H_0 is accepted, the result is not significant. A result is said to be statistically significant of p-value is less than level of significance α.

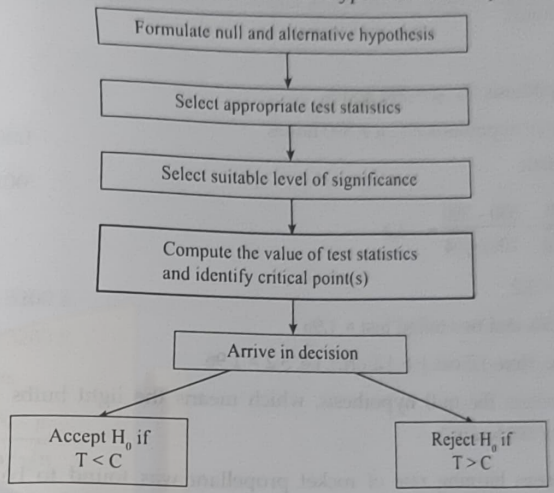
4. Take sample and analyze empirically

Collect the given sample information symbolically, then the numerical values of the information are substitute in appropriate test statistic, and calculate the value of test statistic.

5. Formulate a decision rule and arrive at decision

The critical value of test statistic should be checked from the probability table of respective test statistic. To make decision, if calculate value of test statistic lies in the acceptance region or calculated value of test statistic is less than critical value, then accept null hypothesis, otherwise it will be rejected.

In case of p-value approach, if $P \geq \alpha$, then null hypothesis is accepted, otherwise rejected.



Hypothesis Testing of single mean, for large size ($n \geq 30$)

Let the samples have been drawn from a normal populations with population mean (μ) and variance (σ^2), then to test the hypothesis we use Z-test. The test statistic is given as:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

where,

$\bar{X}$ = Sample mean

μ = Population mean

σ = Population standard derivation

n = Sample size

Z is called standard normal variate, Z follows normal distribution with mean 0, and variance 1.

Example 1. The quality control engineer at a light bulb factory needs to estimate the mean life of large shipment of light bulbs. The process standard deviation to known to be 100 hours. A random sample of 64 light bulbs indicated a sample mean life of 350 hours. Do you think that manufacture has the right to state the light bulbs last an average of 390 hours at 95% level of confidence?

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Solution

Given,

$\sigma = 100$ hours

$n = 64$

$\bar{X} = 350$ hours

$\mu = 390$ hours

$\alpha = 0.05$

Null hypothesis $H_0 : \mu = 390$ hours

Alternative hypothesis $H_1 : \mu \neq 390$ hours

Test statistic

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = \frac{350 - 390}{100/\sqrt{64}} = -3.2$$

$\therefore$ $|Z \text{ cal.}| = 3.2$

Z crit. at 5% and two tailed test = 1.96

Decision: Here $|Z \text{ cal.}| > |Z \text{ cri.}|$ i.e. $3.2 > 1.96$

Do not accept the null hypothesis, which means the light bulbs does not lasts an average of 390 hours.

Example 2. The mean burning rate of rocket propellant was found to be 40 cm/sec. with standard deviation of 2 cm/sec. sample of 36 propellant were tested, and mean burning obtained was 41.25 cm/ sec. Do you conclude that at 5% level of significance mean burning rate of propellant is more that 40 cm/ sec?

Solution

Given,

$\mu = 40$ cm/sec

$\sigma = 2$ cm/sec

$n = 36$

$\bar{X} = 41.25$ cm/sec

$Z\alpha = 1.64$ for one tailed tet.

$H_0 : \mu = 40$ cm/sec

$H_1 : \mu > 40$ cm/sec

We have,

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = \frac{41.25 - 40}{2/\sqrt{36}} = 3.68$$

$\therefore$ Z cal. = 3.68

Z cal. = 1.64

Decision:

Here Z cal. (3.68) > Z cri.(1.64). Hence reject H_0, i.e. the mean burning rate of propellant is more than 40 cm/sec.

Example 3. The mean income of the random sample of 50 engineers worked in mega project was found to be \$3000. If the population standard deviation was \$25, test whether population mean income is less that \$3200 at 10% level of significance?

Solution

Given,

$n = 50$

$\bar{X} = \$\,3000$

$\mu = \$\,3200$

$\sigma = \$\,25$

$\alpha = 0.1$

$H_0 : \mu = 3200\,\$$

$H_1 : \mu < 3200\,\$$

We have,

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

$$= \frac{3000 - 3200}{25/\sqrt{50}} = \frac{-200}{3.54} = -56.4$$

$\therefore$ $|Z| = 56.4$

Z crit. at 10% level = 1.28

Decision:

Here $|Z|$ cal. is greater than Z critical, hence do not accept H_0, i.e. mean salary of engineers is less than \$3200.

Test of significance of different between two means (large sample)

Let $\bar{X}_1$ and $\bar{X}_2$ be the sample means of two independent random samples having sizes n_1 and n_2 drawn from two different normal populations having means μ_1 and μ_2 with variances σ_1^2 and σ_2^2. Then the difference $(\bar{X}_1 - \bar{X}_2)$ also follows normal distributions. Then

Null hypothesis $H_0 : \mu_1 = \mu_2$

Alternative hypothesis $H_1 : \mu_1 \neq \mu_2$

$H_1 : \mu_1 > \mu_2$

$H_1 : \mu_1 < \mu_2$

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Test statistic:

$$Z = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

If σ_1^2 and σ_2^2 are unknown then for large sample size $\hat{\sigma}_1^2 = S_1^2$ and $\hat{\sigma}_2^2 = S_2^2$

Then

$$Z = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} \sim N(0, 1)$$

Example 4. A quality control engineer wants to know wheather there is significance difference in average life of two types of bulbs, produced by two company. The process standard deviation of company A is 120 hrs and company B is 110 hrs. Random sample of 50 bulbs from A indicates mean of 360 hours and similar that 40 bulb of B indicates 350 hours. At 5% level of significance do you find the significance difference between mean of two types of bulbs produced by companies.

Solution

Given,

$\sigma_1 = 120$ hours $\quad \sigma_2 = 110$ hours

$n_1 = 50 \quad n_2 = 40$

$\bar{X}_1 = 360$ hours $\quad \bar{X}_2 = 350$ hours

$\alpha = 5\%$

$H_0: \mu_1 = \mu_2$

$H_1: \mu_1 \neq \mu_2$

We have,

$$Z = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

$$= \frac{360 - 350}{\sqrt{\frac{120^2}{50} + \frac{110^2}{40}}}$$

$$= \frac{10}{24.34}$$

$\therefore Z_{cal} = 0.41$

Z_{crit} at 5% = 1.96

Decision: Here Z_{cal} (0.41) < Z_{crit} (1.96), hence do not reject the null hypothesis, i.e. mean life of bulbs produced by two types of company are equal.

Example 5. The average weight of 42 boys of certain school are found to be 54 kg with standard deviation of 8 kg. The weight of 36 girls of the school are 48 kg with standard deviation of 7 kg. Would you claim than mean weight of boys are more than the weight of girls at 10% level of significance?

Solution

Given,

$n_1 = 42 \quad n_2 = 36$

$S_1 = 8$ kg $\quad S_2 = 7$ kg

$\bar{X}_1 = 54$ kg $\quad \bar{X}_2 = 48$ kg

$\alpha = 0.1$

$H_0: \mu_1 = \mu_2$

$H_1: \mu_1 > \mu_2$

We have,

$$Z = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

$$= \frac{54 - 48}{\sqrt{\frac{8^2}{42} + \frac{7^2}{36}}} = 3.53$$

$\therefore Z_{cal} = 3.53$

Z_{crit} at 10% level = 1.28

Decision: Here $Z_{cal} > Z_{crit}$, hence do not accept the null hypothesis, i.e. the mean weight of boys are more than the mean weight of girls.

Example 6. In a certain factory there are two independent process manufacturing the same item. The average weight in a sample of 250 items produced by one machine is found to be 120 grams, with standard deviation of 12 gram. While the corresponding figures in a sample of 400 items from the other process are 124 and 14. Test whether two mean weight differ significantly or not at 10% level of significance?

Solution

Given,

$n_1 = 250 \quad n_2 = 400$

$S_1 = 12$ grams $\quad S_2 = 14$ grams

$\bar{X}_1 = 120$ grams $\quad \bar{X}_2 = 124$ grams

$\alpha = 0.1$

$H_0: \mu_1 = \mu_2$

$H_1: \mu_1 > \mu_2$

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We have,

$$Z = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

$$= \frac{120 - 124}{\sqrt{\frac{144}{250} + \frac{196}{400}}}$$

$$= -3.87$$

$\therefore$ $Z_{cal} = -3.87$

$|Z_{cal}| = 3.87$

$Z_{crit} = 1.64$

Decision: Here $Z_{cal} > Z_{crit}$, hence do not accept the null hypothesis, i.e. the mean weight of items produced by two process are significantly difference.

Testing hypothesis on a proportion

Testing of hypothesis concerning proportions are required in many areas. Manufacturing firm are concerned about the proportion of defective items, controller of examinations are concerned about the pass proportion of students, gambler depends on a knowledge of the proportion of outcomes that he considered favourable. In such situation population is divided into dichotomous form.

H_0: $P = P_0$, i.e. the population proportion has specified proportion P_0.

H_1: $P \neq P_0$

H_1: $P > P_0$

H_1: $P < P_0$

The test statistics will be

$$Z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} \sim N(0, 1)$$

If sampling is done from a finite population of size N, then

$$Z = \frac{p - P}{\sqrt{\frac{PQ}{n}} \cdot \sqrt{\frac{N - n}{N - 1}}}$$

Testing hypothesis of difference two population proportion

If we wish to test the null hypothesis that two proportions or binomial parameters, then .

H_0: $P_1 = P_2$

H_1: $P_1 \neq P_2$

H_1: $P_1 > P_2$

H_1: $P_1 < P_2$

Then test statistics will be

$$Z = \frac{P_1 - P_2}{\sqrt{\frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2}}} \sim N(0, 1)$$

Also we can write:

$$Z = \frac{P_1 - P_2}{\hat{P}\,\hat{Q}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$

$$\hat{P} = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}, \quad \hat{Q} = 1 - \hat{P}$$

Example 7. Random sample of 400 male and 600 female were asked whether they would line to have a airport near their residence. 200 mean and 325 women were in favour of the proposal. Test the hypothesis that proportion of men and women are in favour of proposal at 5% level of significance.

Solution

Given,

$$P_1 = \frac{200}{400} = 0.5, \; P_2 = \frac{325}{600} = 0.54$$

$\alpha = 0.05$

H_0: $P_1 = P_2$ (i.e. population proportion of male and female are favour of the proposal)

H_1: $P_1 \neq P_2$

$$Z = \frac{P_1 - P_2}{\hat{P}\,\hat{Q}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$

$$\hat{P} = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \frac{400 \times 0.5 + 600 \times 0.54}{1000} = 0.87$$

$$\hat{Q} = 1 - \hat{P} = 1 - 0.87 = 0.13$$

$$Z = \frac{0.5 - 0.54}{\sqrt{0.87 \times 0.13\left(\frac{1}{400} + \frac{1}{600}\right)}} = -.126$$

$|Z| = 1.26$

$|Z|_{tab} = 1.96$

Decision: Here $Z_{cal} < Z_{crit.}$ hence do not reject the null hypothesis, i.e. male and female proportion are in favour of proposal.

Example 8. In a factory 2% cell-phone were found to be defective in a lot of 2000 cell phone. In another factory 2.5% cell-phone were found defective in a lot of 3000 cell phone. Would you test the claim that cell phone in second factory are significantly inferior compared to the first factor at 2% level of significance?

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Solution

Given,

$P_1 = 0.02$, $P_2 = 0.025$

$n_1 = 2000$ $n_2 = 3000$

$\alpha = 0.02$

H_0: $P_1 = P_2$ (i.e. proportion of defective items in both are equal)

H_1: $P_1 < P_2$

H_1: (The proportion of defective in 1st factory is less than second) (i.e. second factory is inferior than first)

$$\hat{P} = \frac{n_1p_1 + n_2p_2}{n_1 + n_2} = \frac{2000 \times 0.02 + 3000 \times 0.025}{5000} = 0.023$$

$$\hat{Q} = 1 - \hat{P} = 1 - 0.023 = 0.977$$

$$Z = \frac{P_1 - P_2}{\hat{P}\hat{Q}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$

$$Z = \frac{0.02 - 0.025}{\sqrt{0.023 \times 0.977\left(\frac{1}{2000} + \frac{1}{3000}\right)}} = -1.15$$

$|Z| = 1.15$

Zcrit at 2% level = 2.05

Decision: Here $Z_{cal} < Z_{crit.}$ hence do not reject the null hypothesis, i.e. the defective proportion of cell phone produced by two factories are equal, no significantly difference in proportions.

Example 9. A coin is tossed 1200 times and heads occur at 520 times. Would you conclude that the coin is unbiased at 3% level of significance?

Solution

Sample of proportion of heads $(p) = \frac{520}{1200} = 0.44$

$n = 1200$

Population proportion of head $(P) = 0.5$

Population proportion of tail $(Q) = 0.5$

H_0: $P = 0.5$ (i.e. the coin is unbiased)

H_0: $P \neq 0.5$ (i.e. the coin is biased)

We have,

$$Z = \frac{p - P}{\sqrt{\frac{PQ}{n}}}$$

$$= \frac{0.44 - 0.50}{\sqrt{\frac{0.5 \times 0.5}{1200}}}$$

$$= -\frac{0.06}{0.0145}$$

$$= |-4.14|$$

$$= 4.14$$

$|Z_{cal}| = 4.14$

$Z_{crit} = 2.17$

Decision: Here $Z_{cal} < Z_{crit.}$ hence do not accept the null hypothesis, i.e. the coin is biased.

Example 10. In a sample of 200 laptop manufactured by a company the number of defective laptop was to be 15. However the owner of the company claimed that only 5% of their product is defective. Is the claim acceptable at 1% level of significance?

Solution

Sample of proportion of defective laptop $(p) = \frac{15}{200} = 0.075$

Population proportion of defective $(P) = 0.05$

Population proportion of non-defective $(Q) = 0.95$

H_0: $P = 0.05$ (i.e. defective laptop is 5%)

H_0: $P > 0.05$

We have,

$$Z = \frac{p - P}{\sqrt{\frac{PQ}{n}}}$$

$$= \frac{0.075 - 0.05}{\sqrt{\frac{0.05 \times 0.95}{200}}}$$

$$= \frac{0.025}{0.01542}$$

$$= 1.62$$

Z_{crit} at 1% level = 2.33

Decision: Here $Z_{cal} < Z_{crit.}$ hence do not reject the null hypothesis, i.e. there is no reason to reject null hypothesis. It means defective proportion is only 5%.

Example 11. A manufacture of electronic calculator takes a random sample of 1200 calculators and finds there are 8 defective units. Is there evidences to support a claim that fraction of defective units produced is 1% or less? Use $\alpha = 0.05$.

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Solution

Sample of proportion (p) $= \frac{8}{1200} = 0.0067$

n = 1200

Population proportion (P) = 0.01

Q = 0.99

H_0: P = 0.01 (i.e. defective proportion is 1%)

H_0: P < 0.01)

We have,

$$Z = \frac{p - P}{\sqrt{\frac{PQ}{n}}}$$

$$= \frac{0.0067 - 0.01}{\sqrt{\frac{0.01 \times 0.99}{1200}}}$$

$= -1.37$

$= |-1.37|$

$= 1.37$

$|Z_{cal}| = 1.37$

$Z_{crit} = 1.64$

Decision: Here $Z_{cal} < Z_{crit.}$ hence accept the null hypothesis, i.e. defective proportion is less than 1%.

Testing Hypothesis of Single Mean Small Sample Cases (n < 30)

If the sample size is small and population variances are unknown, then sample variance is not unbiased estimate of population variance. In case of small sample size Gosset developed as sampling distribution which is known as Gosset's t distribution and given as:

$t = \frac{\bar{X} - \mu}{S/\sqrt{N}}$, where S is a sample standard deviation.

Thus, we use Gosset's t-test or also known as student's t-test to estimate and test the population mean or difference of two population means.

The process of hypothesis testing in small sample case is similar as that of large sample case.

Null hypothesis, H_0: $\mu = \theta$

Alternative hypothesis, H_0: $\mu \neq \theta$, two tailed test

H_0: $\mu > \theta$, right tailed test

H_0: $\mu < \theta$, left tailed test

Test Statistic

$$t = \frac{\text{Difference of sample mean and population mean}}{\text{S.E. } \bar{X}}$$

or, $$t = \frac{\bar{X} - \mu}{S/\sqrt{N}}$$

where

$$S^2 = \frac{\Sigma(X - \bar{X})^2}{n - 1}$$

Critical value: Critical value of t-test can be checked from t-table at α% level of significance and given level of significance.

Example 12. A random sample of 16 girls of certain college showed mean weight of 52 kg with standard derivation of 4 kg. Would you claim that samples were drawn from the college having population mean weight of 50 kg at 5% level of significance?

Solution

Given,

n = 16,

$\bar{X}$ = 52 kg

S = 4 kg

μ = 50 kg

α = 0.05

Null hypothesis, H_0: $\mu = 50$ kg

Alternative hypothesis, H_0: $\mu \neq 50$ kg

We have,

$$t = \frac{\bar{X} - \mu}{S/\sqrt{n-1}} = \frac{52 - 50}{4/\sqrt{4-1}}$$

or, $t = 1.94$

t_{crit} at 5% level and (16 – 1 = 15 d.f.) = 2.13

Decision: Here $t_{cal.}$ (1.94) < $t_{critical}$ (2.13), hence do not reject null hypothesis i.e. the sample were drawn from the population having mean weight of 50 kg.

Example 13. Marks obtained by randomly selected 9 students of a class in mathematics were as follows:

45, 47, 50, 52, 48, 47, 49, 53 and 51.

Does the mean marks obtained by the students were drawn from the population mean marks having the class is 47 at 5% level of significance?

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Solution

Given,

$n = 9, \mu = 47, \alpha = 0.05$

$H_0 : \mu = 47$

$H_1 : \mu \neq 47$

We have,

$$t = \frac{\bar{X} - \mu}{S/\sqrt{n-1}} \quad \ldots(i)$$

Sample mean,

$$\bar{X} = \frac{\Sigma x}{9}$$

$$= \frac{45 + 47 + 50 + \cdots + 51}{9}$$

$$\bar{X} = 49.11$$

$$S^2 = \frac{\Sigma (X - \bar{x})^2}{n-1} = 6.8611$$

$$\therefore \quad S = 2.61$$

Using the values in (1)

$$t = \frac{49.11 - 47}{2.61/\sqrt{9-1}} = 2.28$$

$t_{critical}$ at 5% level and 9. d.f. = 2.30

Decision: Here $t_{cal} < t_{crit.}$ hence do not reject the null hypothesis i.e. samples were drawn from the population having mean 47.

Example 14. A mean breaking strength of randomly selected 12 glass tubes manufactured try a company showed a 850 pounds with standard deviation of 130 pounds, where the owner of the company claim that mean breaking strength of glass tube is more than 800 pounds. Would you support the claim of company at 10% level of significance?

Solution

Given,

$n = 12$

$\bar{X} = 850$ pound

$S = 130$ pound

$\mu = 800$ pound

$\alpha = 0.1$

Null hypothesis, H_0: $\mu = 800$ pounds

Alternative hypothesis, H_0: $\mu > 800$ pounds

We have,

$$t = \frac{\bar{X} - \mu}{S/\sqrt{n-1}}$$

$$= \frac{800 - 850}{130/\sqrt{12-1}}$$

$$t = -1.28$$

or, $|t| = 1.28$

t_{crit} at 10% level and (12 – 1 = 11 d.f.) = 1.36 (for one tail test)

Decision: Here $t_{cal.} < t_{critical}$, hence do not reject null hypothesis i.e. the mean breaking strength of the company's glass tube is 800 pounds.

Example 15. A company produces electric bulbs and claims that mean life of bulb is 30 months. A random sample of 9 such bulb gave the following mean age in months.

24, 26, 32, 28, 20, 20, 23, 27 and 24

Can you justify the company's claim at 5% level of significance?

Solution

Given,

$n = 9$

$\mu = 30$ month

$$\bar{X} = \frac{\Sigma x}{n} = \frac{234}{9} = 26$$

$$S^2 = \frac{1}{n-1}\left\{\Sigma x^2 - \frac{(\Sigma x)^2}{n}\right\}$$

$$= \frac{1}{8}\left\{6274 - \frac{(234)^2}{9}\right\}$$

$$= 23.75$$

$$\therefore \quad S = 4.87$$

Null hypothesis, H_0: $\mu = 30$ months

Alternative hypothesis, H_0: $\mu \neq 30$ months

We have,

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$$t = \frac{\bar{X} - \mu}{S/\sqrt{n-1}}$$

$$= \frac{26 - 30}{4.87/\sqrt{9-1}}$$

or, $t = |-2.31|$

$|t| = 2.31$

t_{crit} at 5% level and (9 – 1 = 8 d.f.) = 2.30

Decision: H_0 is rejected and mean does not equal to 30 months.

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Testing Hypothesis of Difference between two means Small Sample (n_1 and $n_2 < 30$)

Let $\bar{X}_1$ be the mean of the series having sizes n_1 and sample standard deviation S_1 drawn from normal population whose unknown mean is μ_1. $\bar{X}_2$, n_2 and S_2 are the resptive values drawn from second population having unknown mean μ_2. These two populations are independent to each other. The variance of these populations are not known. For small sample size to test the difference between two means of population we use t-test

Null hypothesis $H_0 : \mu_1 = \mu_2$

Alternative hypothesis $H_1 : \mu_1 \neq \mu_2$ (two tailed test)

$H_1 : \mu_1 > \mu_2$ (one tailed test)

$H_1 : \mu_1 < \mu_2$ (left tailed test)

Test statistic:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{Sp\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$Sp^2 = \frac{1}{(n_1 + n_2 - 2)}\left\{\Sigma x_1^2 - \frac{(\Sigma x_1)^2}{n_1} + \Sigma x_2^2 - \frac{(\Sigma x_2)^2}{n_2}\right\}$$

i. When sample standard S_1 and S_2 (biased estimates) are given then Sp^2 can be calculated as:

$$Sp^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}$$

ii. When unbiased estimate of population variance (i.e. s_1^2 and s_2^2) are known then Sp^2 can be calculated as:

$$Sp^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

Example 16. Two daily news papers are typed by different typist in any special day. The number of mistakes made by each of typist per paper are found as follows. Would you claim that the difference in mean number of mistakes in two papers are equal in that day at 5% level of significance?

Paper A: 8 12 13 9 3 8 10 9

Paper B: 10 8 12 15 6 11 12 12

Solution

H_0: $\mu_1 = \mu_2$ (The mean no. of mistaken in two papers are equal)

H_1: $\mu_1 \neq \mu_2$ (The mean no. of mistaken in two papers are not equal)

$\alpha = 0.05$

$$t = \frac{\bar{X}_1 - \bar{X}_2}{Sp\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \quad \ldots (i)$$

X_1	X_2	X_1^2	X_2^2
8	10	64	100
12	8	144	64
13	12	169	144
9	15	81	225
3	6	9	36
8	11	64	121
10	12	100	144
9	12	81	144
72	86	712	978

$$\bar{X}_1 = \frac{\Sigma x_1}{n_1} = \frac{72}{8} = 9$$

$$\bar{X}_2 = \frac{\Sigma x_2}{n_2} = \frac{86}{8} = 10.75$$

$$Sp^2 = \frac{1}{(n_1 + n_2 - 2)}\left\{\Sigma x_1^2 - \frac{(\Sigma x_1)^2}{n_1} + \Sigma x_2^2 - \frac{(\Sigma x_2)^2}{n_2}\right\}$$

$$= \frac{1}{14}\left\{712 - \frac{72^2}{8} + 978 - \frac{86^2}{8}\right\}$$

$\therefore$ $Sp^2 = 8.39$

Using (i)

$$t = \frac{9 - 10.75}{\sqrt{8.39\left(\frac{1}{8} + \frac{1}{8}\right)}} = -1.21$$

$|t| = 1.21$

$t_{cal.}$ at 5% level and 14 d.f. = 2.64

Decision: Here $t_{cal} < t_{crit.}$ hence accepts null hypothesis, it means average mistake in two papers are equal.

Example 17. The following random samples are measurements of heat producing capacity (in millions of calories per 100 mg.) of specimens of coal from two mine.

Mine A: 82 81 83 80 83

Mine B: 72 78 79 81 79 78

Would you claim that mean difference between heat producing capacity of two specimen are significantly different at 10% level of significance?

Solution

$$\bar{X}_1 = \frac{82 + 81 + 83 + 80 + 83}{5} = \frac{409}{5} = 81.8$$

$$\Sigma x_1^2 = 82^2 + 81^2 + 83^2 + 80^2 + 83^2 = 33463$$

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$$S_1^2 = \frac{1}{(n_1 - 1)}\left\{\Sigma x_1^2 - \frac{(\Sigma x_1)^2}{n_1}\right\}$$

$$= \frac{1}{4}(33463 - 33456.2)$$

$$= 1.7$$

$$\bar{X}_2 = \frac{\Sigma x_2}{n_2} = 77.83$$

$$\Sigma x_2^2 = 36395$$

$$S_2^2 = \frac{1}{(n_2 - 1)}\left\{\Sigma x_2^2 - \frac{(\Sigma x_2)^2}{n_2}\right\}$$

$$= 9.36$$

$$Sp^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}$$

$$= \frac{5 \times 1.7 + 6 \times 9.36}{5 + 6 - 2}$$

$$= 7.18$$

$H_0: \mu_1 = \mu_2$

$H_1: \mu_1 \neq \mu_2$

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{Sp^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

$$= \frac{81.8 - 77.83}{\sqrt{7.18\left(\frac{1}{5} + \frac{1}{6}\right)}}$$

$$= 2.43$$

$t_{cal} = 2.43$

$t_{critical}$ at 10% level and 9 d.f. = 1.83

Decision: Here $t_{cal.} > t_{crit.}$ hence do not accept null hypothesis i.e. there is significant difference between heat producing capacity of two mines.

Example 18. Two catalyst are begin analyzed to determine how they affect the mean yield of a chemical process. Following information are given for the catalyst.

Catalyst	Sample Size	Sample mean	Variance
I	8	91.75	3.89
II	8	93.75	4.02

Would you conclude that mean yield due to two catalyst are significantly different at 5% level of significance?

Solution

$H_0: \mu_1 = \mu_2$

$H_1: \mu_1 \neq \mu_2$

$\alpha = 0.05$

$$Sp^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}$$

$$= \frac{8 \times 3.89 + 8 \times 4.02}{8 + 8 - 2}$$

$$= \frac{62.24}{14}$$

$Sp^2 = 4.44$

$Sp = 2.1$

We have,

$$t = \frac{\bar{X}_1 - \bar{X}_2}{Sp\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$= \frac{91.73 - 93.75}{2.1\sqrt{\frac{1}{8} + \frac{1}{8}}}$$

$$= -\frac{2.02}{1.05}$$

$$= -1.93$$

$|t| = 1.93$

$t_{critical}$, at 5% level and 14 d.f. = 2.14

Decision: Here $t_{cal} < t_{crit}$, hence null hypothesis cannot be rejected, i.e. mean yield due to two different catalyst are equal.

Pired t-test or Dependent t-test

First elementary observations are measured and collected, then after intervening event, second type of observations measured and collected. In such situation pair of observations are created and collected from the same populations. The difference between the pairs of two types of observations must be approximately normally distributed. In these types of pair of observations we use paired t-test, which is also called dependent t-test. Dependency in observations occurred due to intervening event.

For example body temperature of some students are taken before Marathon and noted, after marathon, their body temperature again measured, here marathon is the intervening event, which creates pair of observation like before and after marathon. The hypothesis testing procedure will be:

$H_0: \mu_A = \mu_B$ [Mean effect before and after event are equal]

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$H_1: \mu_A \neq \mu_B$ [Two tailed test]

$H_1: \mu_A > \mu_B$ [Right tailed test]

$H_1: \mu_A < \mu_B$ [Left tailed test]

Level of significance: α

Test statistic

$$t = \frac{\bar{d}}{Sd/\sqrt{n}}$$

$\bar{d} = \frac{\Sigma d}{n}$, $d = X - Y$ [difference between two sets of observation]

n = Total number of observation in difference (d)

S_d^2 = Sample variance of the difference

$$= \frac{\Sigma(d - \bar{d})^2}{n - 1}$$

$$= \frac{1}{n-1}\left\{\Sigma d^2 - \frac{(\Sigma d)^2}{n}\right\}$$

Example 19. The test score of 10 candidate's priori and after training are given below. Is training effective at 5% level of significance?

Priori (X):	84	48	36	37	54	69	96	90	65
After (Y):	90	58	56	39	62	81	86	84	75

Solution

$H_0: \mu_X = \mu_Y$ (Training is not effective)

$H_1: \mu_X < \mu_Y$ (Training is effective)

$n = 10$

$\mu = 0.05$

The difference between pairs ($di = X_i - Y_i$)

d	=	- 6	- 10	- 20	- 2	- 8	- 12	10	6	- 10
Σd	=	- 52								
d^2	=	36	100	400	4	64	144	100	36	100

$\Sigma d^2 = 984$

$$\bar{d} = \frac{\Sigma d}{n} = \frac{-52}{10} = -5.2$$

$|\bar{d}| = 5.2$

$$S_d^2 = \frac{1}{n-1}\left\{\Sigma d^2 - \frac{(\Sigma d)^2}{n}\right\}$$

$$= \frac{1}{9}\left\{984 - \frac{(-5.2)^2}{10}\right\}$$

$\therefore S_d^2 = 109.04$

$\therefore S_d = 10.44$

$\therefore$ Test statistic

$$t = \frac{\bar{d}}{Sd/\sqrt{n}} = \frac{5.2}{10.44/\sqrt{10}} = \frac{5.2}{3.30}$$

$t_{cal} = 1.58$

$t_{crit.}$ at 5% level and 9 d.f. = 1.83

Decision: Here $t_{cal.} < t_{crit.}$ hence do not reject null hypothesis, i.e. the training is not effective.

Example 20. A certain stimulus administered to each of 12 patients resulted in the following increase in blood pressure (mmHg), 5, 2, 8, – 1, 3, 0, – 2, 1, 5, 0, 4 and 5. Can you conclude that stimulus will in general be accompanied by an increase in blood pressure at 10% level of significance?

Solution

$n = 12$

$\Sigma d = 30$

$\Sigma d^2 = 174$

$H_0: \mu_A = \mu_B$ [Mean blood pressure before and after stimulus are equal]

$H_1: \mu_A > \mu_B$ [Mean blood pressure before stimulus is less than after stimulus]

$\alpha = 0.1$

$$t = \frac{\bar{d}}{Sd/\sqrt{n}} \quad \ldots(i)$$

$$\bar{d} = \frac{\Sigma d}{n} = \frac{30}{12} = 2.5$$

$$S_d^2 = \frac{1}{n-1}\left\{\Sigma d^2 - \frac{(\Sigma d)^2}{n}\right\}$$

$$= \frac{1}{11}\left\{174 - \frac{(30)^2}{12}\right\}$$

$$= \frac{1}{11} \times 99$$

$$= 9$$

$\therefore S_d = 3$

$$t = \frac{2.5}{3/\sqrt{12}} = \frac{2.5}{0.87} = 2.87$$

$\therefore t_{cal} = 2.87$

$t_{crit.}$ at 10% level and 11 d.f. = 1.36

Decision: $T_{cal} > t_{crit.}$ there is no reason to accept null hypothesis.

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Example 21. An aircraft designer has theoretical evidence that painting the airplane reduces its speed at a specified power and flap settings. He tests 6 consecutive airplanes from the assembly line before and after painting. The results are:

Before painting speed mph	289	286	283	288	283	289
After painting speed mph	286	285	279	283	281	286

(Use $\alpha = 0.05$)

Solution

$d = 3 \quad 1 \quad 4 \quad 1 \quad 2 \quad 3$

$\Sigma d = 14$

$\Sigma d^2 = 42$

H_0: $\mu_A = \mu_B$ [Speed of plane before and after painting are equal]

H_1: $\mu_A > \mu_B$ [Speed of plane before painting is more than after painting]

$$\bar{d} = \frac{\Sigma d}{n} = \frac{14}{6} = 2.34$$

$$S_d^2 = \frac{1}{n-1}\left\{\Sigma d^2 - \frac{(\Sigma d)^2}{n}\right\}$$

$$= \frac{1}{5}\left\{42 - \frac{(14)^2}{6}\right\}$$

$$= \frac{1}{5} \times 9.34$$

$\therefore \quad S_d = 1.37$

$$t = \frac{\bar{d}}{Sd/\sqrt{n}} = \frac{2.34}{1.37/\sqrt{6}} = 4.17$$

$t_{cal} = 4.17$

$t_{crit.}$ at 5% level and 5 d.f. = 2.01

Decision: Here $t_{cal.} > t_{crit.}$ hence do not accept null hypothesis, i.e. speed of plane before painting is more than after painting.

Exercise 7.1

1. A quality control engineer declared the price of each set of television by a certain company is not more than $ 500 due to its durability. A businessman randomly selects 36 television set and finds mean price of $550. It is known that price of process standard deviation of television is $40. Would you justify the engineer's statement at 1% level of significance?
2. Researchers are interested in the mean marriage age of certain population, they took 50 samples from the population and the mean marriage age was found to be 27 years. It was known from past study that population s.d. was 4 years. Would you conclude that mean age of marriage is different from 30 years at 10% less of significances?

3. A manufacture of ball pens claims that mean writing life of 400 pages with s.d. of 20 pages. A purchasing agent selects a sample of 100 pens and puts them for test. The mean writing life of sample was 390 pages. Can you justify that purchasing agent reject the manufacture's claim at 5% level?
4. A researcher wishes to claim that the average age of life guard in ocean city is greater than 24 years. She selects a sample of 36 lifeguards and finds mean of sample to be 24.7 years with s.d. of 2 years. Is their evidence to support the claim at 2% level of significance?
5. A survey of 64 cement dealer revealed that the mean price charged for certain PPC cement of 50 kg pack was Rs. 840 with standard deviation of Rs. 150. Do these data provide sufficient evidence to indicate that population mean is greater than Rs. 760, at 2% level of significance?
6. A typist claims that he can type more than 60 words per minute. Test the validity of his claim on the basis of 100 trials on which he typed 55 words on a average with standard deviation of 15 words at 5% level of significance.
7. A campus chief of certain campus claims that mean marks of the students are at least 60. To test the claim 40 students of the campus are selected at random and found an average mark of 57 with variance of 225. Can you justify the claim at 10% level of significance?
8. The average breaking strength of steel rods is declared to be 175 pounds, A random sample of 100 rods is taken to test the assumption and finds the following information.

Breaking strength	150-160	160-170	170-180	180-190	190-200
No. of rods	12	20	40	20	8

Do these information justify the assumption of 5% level of significance?

9. A manufacture of sports equipment has developed new synthetic fishing line that he claims has a mean breaking strength of 8 kg with standard deviation of 1.5 kg. If a random sample of 50 lines is tested and found to have mean breaking strength of 7.6 kg. At 5% level of significance can you justify the claim?
10. Mean consumption of food grains among 400 samples middle class consumers is 380 grams per day, per person with s.d. of 120 grams. A similar sample survey of 600 working class consumers gave a mean of 410 grams with s.d. of 80 grams. Would you justify that the classes consume same quantity of food grains using 2% level of significance?
11. Random sample of 100 students from college A and random sample of 150 students from college B yield on mean length of hours to study weekly at home are 33 hours and 49 hours with variances of 900 and 1050 hours. Can you claim that the mean length of study hours of B is more than A, at 5% level of significance?
12. The average hourly wages of a sample of 150 workers in plant A was Rs. 256 with standard deviation of Rs. 108. The average wages of a sample of 200 workers in plant B was Rs. 282 with standard deviation of Rs. 128. Can you justify that hourly wages paid by plant B are more than these by plant A, at 3% level of significance?
13. The average height of 50 students who show no athletic interest was 68.2 inches with standard deviation of 2.5 inches, while another set of 60 students who showed athletic interest was average of 67.5 inches with standard deviation 2.8 inches. Test the significant difference between two types of students at 5% level of significance?

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14. A potential buyer of light bulbs bought 50 bulbs of two brands. After testing he found the brand A has a mean life of 1282 hours with standard deviation of 80 hours, where as B had a mean life of 1208 hours with standard deviation of 94 hours. Do you claim that buyer found two brands differ in quality at 5% level of significance?
15. The mean marks secured by randomly selected 100 boys of a college was 75 with standard deviation of 4, the respective values for 120 girls of same college were 70 and 5. Are the mean marks secured by boys and girls of the college are equal at 2% level of significance?
16. In a certain city 400 randomly selected labour worker found mean monthly income of $ 1600 with standard deviation of $120. After the earthquake disaster in the same city 600 random selected worker have a mean income $1800 with standard deviation of $200. Can you justify that mean salary of worker in two cares differ significantly at 5% level of significance?
17. A sample of 90 male and 110 female were taken from large population enrolled in a weight reducing program. The men weight of male was 105 pound with standard deviation 25 pound and the respective values for women was 95 pounds and 20 pounds. Would you justify that the mean weights of male and female are significantly different at 5% level of significance?
18. Two independent samples of observation were collected. For the first sample of 60 elements, the mean was 86 and S.d. of 9. The second sample of 75 elements had a mean of 82 and standard deviation of 12. At 1% level of significance test whether the two samples to have come from population of same mean.
19. The average mileage for 35 cars of brand A was 27.2 miles par gallon with s.d. of 3.8 mpg. The average mileage for 42 cars of brand B was 32.1 mpg with s.d. of 4.3 mpg. Would you justify that mean mileage of two brands of cars are significantly different at 5% level of significance?
20. Two brands of cars are compared for acceleration rate. Sample of 40 test runs are recorded for brand A car and it gives the mean elapsed time of 7.5 sec with standard deviation of 1.5 sec. For 60 test runs of brand B car gives respective values 7.1 and 1.8 sec. Would you find mean elapsed time of brand A is less than brand B at 2% level of significance?
21. Randomly selected 625 people from certain city, 300 were found to be smokers. Test the hypothesis that proportion of smokers and non smokers in the city are equal at 1% level of significance.
22. At a certain college it is estimated that 25% of the students used public bus for class. Does this seem to be a valid estimate, if a random sample of 90 students 28 used bus for class. Use α = 0.05.
23. It is claimed that tea and coffee are equally popular in certain city. If a random sample of 1200 persons 650 were regular consumers of tea. Is the claim justified at 5% level of significance?
24. In a sample of 400 cars manufactured by a company, the number of defective car was found to be 30. The company however claimed that only 5% of their product is defective. Is the claim acceptable at 1% level of significance?
25. It is claimed that both Apple and Samsung brands are equally popular in certain city. In a random sample of 500 persons 310 were in favour of Apple brand. Is the claim justified at 1% level of significance?

26. A structure engineer claims that only 3% of the residence houses made in certain city follow the building code. To test the claim 700 houses are inspected, among them only 30 follows the building code, in the phase of construction. Do you justify the claim of engineer at 5% level of significance?
27. A typist claims that only 4% of the pages typed by him have small errors. A random sample of 300 pages was inspected then found 18 errors in typing. Test the claim at 5% level of significance.
28. A coin is tossed 400 times and heads appear 176 times. Does the information support the coin is fair? (α = 0.05)
29. A manufacture of plastic bags claimed that at least 94% of the bags supplied by the factory are not defective. An inspection of 100 pieces of bags revealed that 10 were defectives. Test the claim at 5% level of significance.
30. A research is conducted to know the smoking habit of male and female of remote village. It is found that 50% of the males and 40% of the females are smokers in a random sample of each of 100. Can you conclude that proportion of smoker of male is greater than female at 5% of significance?
31. 1000 articles from publisher A are examined and found 3% have major errors. 1500 similar articles from publisher B are found 2% have major error. Would you conclude that publisher B is superior than that of A? Use α = 0.05
32. A quality control engineer inspected two brands of laptop Lenovo and Dell. Randomly selected 250 Lenevo shows 24 have some defects and 200 pieces of Dell shows 10 have some defects. Test the hypothesis that in two types of brands the proportion of defective items are equal 5% level of significance.
33. A social worker believes that fewer than 14% of the couples in a certain village used any form of contraceptive. A random sample of 120 couples was contracted, 25 of them said they use contractive for birth control. Comment on the social worker's belief at 2% level of significance?
34. A recent survey showed that in a sample of among 100 youth 30% enjoyed with Facebook page and in another 100 youth 24% enjoyed with Tiktok. Is the proportion of youth who enjoyed with Facebook page and Tiktok are significantly different at 5% level of significance?
35. Two different types of injection molding machines are used to form plastic parts. Two random samples each of size 500 are selected and 32 parts are found from machine A and 21 defective parts form machine B. Is it reasonable to conclude that both machine are produced same fraction of defective parts, at 5% level of significance.
36. From a class of civil engineering 16 students selected randomly and their mean marks obtained in mathematics was found to be 52 with standard deviation of 4. Test whether the population mean marks of the class is more than 50, at 10% level of significance?
37. A manager claims that mean thickness of the plastic bags produced by the company is 0.05 mm. A random sample of 10 bags was taken which was produced by the factory, and the mean thickness was found to be 0.053 men with s.d. of 0.003. Would you justify the claim of the manager at 5% level of significance?

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38. The breaking strength of fiber used in manufacturing cloth is required to be at least 160 Psi, past experience showed that standard deviation of breaking strength 3 Psi. A random sample of 8 specimens was tested and the average breaking was found to be 158 Psi. Should the fiber be judged acceptable at $\alpha = 0.05$?

39. the diameters of bolts are known to have standard deviation of 0.0001 inch. A random sample of 10 bolts yields an average diameter of 0.2546 inch. Test the hypothesis that the true mean diameter of the bolts equals 0.255 inch using $\alpha = 0.05$.

40. A manufacture company produces piston ring for an automobile engine. A random sample of 15 rings has a mean diameter of 74.036 mm with standard deviation of 0.001 mm. Test the hypothesis that the mean piston ring for all automobile engine is 74.035 mm. Use $\alpha = 0.01$

41. The wall thickness of randomly selected 25 glass bottles, a quality control engineer found mean was 4.05 mm with standard deviation was 0.08 mm. Test the hypothesis that population mean thickness of the bottles is more than 4.08 mm at 10% level of significance.

42. Random sample of 9 bulbs of a company found the life in month as:

    24, 26, 32, 28, 20, 20, 23, 27 and 24

    Test the hypothesis that mean life of bulbs of company is 27 months. Use $\alpha = 0.05$.

43. A random sample of 10 bags of OPC cement produced by a certain factory were weighted by the quality control engineer, he found the weight in kg as:

    50, 49, 52, 44, 45, 48, 46, 45, 49 and 45

    Test the mean weight of the cement bags produced is 50 kg at 5% level of significance.

44. The diameter of steel rods manufactured on two different machine is being investigated. Sample mean of randomly selected 15 rods produced by machine I is 8.73 mm with variance 0.30. Similarly 18 rods produced by machine II is 8.68 mm with variance 0.34. Does the mean diameter of steel rods produced by two machines are significantly different? Use $\alpha = 0.05$.

45. The methods for producing gasoline from crude oil are being investigated. The yields of both processes are assumed to be normally distributed. The following yield data have been obtained from the pilot plant.

Process	Yield in %					
1	24.2	26.6	25.7	24.8	25.9	26.5
2	21.0	22,1	21.8	20.9	22.4	22.0

Is there evidence to believe that process 1 has a greater mean yield? Use $\alpha = 0.01$.

46. The titanium content in percentage of two types of alloy is being studied in the hope of ultimately increasing the tensile strength. An analysis of six of each type chosen at random produces the following titanium contents.

Alloy I%	8	9.9	9.7	7.7	11.6	12.1
Alloy II%	8.2	9.5	9.8	8.2	10.2	11.5

Does the alloy II contains more percentage of titanium than that of alloy I. Use $\alpha = 0.05$.

47. A manufacturing company believes that the average wastage of production in a month should be 44 kg. He took the sample of 20 production units of obtained the average monthly wastage was 42 kg, with s.d. of 6 kg. Justify the manager's opinion at 5% level of significance.

48. A random sample of 8 daily wages workers in company A found to have income in average of Rs. 1234 with s.d. of Rs. 36 and 7 daily wages workers in Company B have respective values Rs. 1036 with s.d. of Rs. 40. Does the mean income of all worker in two company are significantly different?

49. Five samples of ferrous type substances are examined, if there is difference between a laboratory chemical analysis and X-ray fluorescence analysis of iron content. Each sample was split into two sub samples and two types of analysis were applied. Following coded data showing the iron content analysis.

Analysis	Iron Content Code				
X-ray	2.0	2.0	2.3	2.1	2.4
Chemical	2.2	1.9	2.5	2.3	2.4

Assuming the populations are normal, whether the two methods of analysis give on average the same results. Use $\alpha = 0.05$.

50. The following random samples are measurements of heat producing capacity (in millions of calories per 100 kg) of specimens of coal from two mines.

Mine 1	82	81	83	80	84	-
Mine 2	72	78	79	81	80	78

Test whether the mean calories of heat produced by two types of mine are significant at 1% level of significant?

51. A drug is used for 7 patients of high blood pressure. The blood pressure (in mm Hg) of patients before and after using the drug are recorded as follows.

Before drug	150	162	144	155	130	142	465
After drug	138	150	125	145	132	145	140

Is drug effective for reducing the B.P. of the patients at 5% level of significance?

52. The number of sales of Samsung television before and after advertisement on media in different 8 showroom of the company were as follows:

Before Advertisement	70	65	48	72	80	92	98	100
After Advertisement	72	70	53	75	84	95	106	104

Test whether the advertisement is effective for the company's product at 10% level of significance.

53. A principal of the college thought that memory power training is better for learning process of students, then memory power training were launched. The memory capacity of the students were noted before and after the training. The data recorded were as:

Before Training	10	16	9	3	7	12	16	17	4
After Training	12	17	8	5	6	11	18	20	3

Is training effective at 5% level of significance?

54. To increase the weight of newly born infants, a newly developed nourishing food is used for a randomly selected 10 infants in certain hospital. The difference in observed weight in gm after one month were:

    5, 2, 6, – 1, 0, 4, 3, – 2, 1, 4

    Does the nourishing food is effective for the infants at 5% level of significance?

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55. A I.Q. test was given for 6 boys to increase their ability. The score secured by the boys before and after the test was recorded as follows:

Before Training	110	120	123	132	125	130
After Training	120	118	118	136	121	135

Was the test is affective to increase the boys for 10% level of significance?

Answers

1. Z = 4.17, reject H_0
2. Z = 5.26, reject H_0
3. Z = 5, reject H_0
4. Z = 2.04, accept H_0
5. Z = 4.26, reject H_0
6. Z = – 3.33, significant
7. Z = – 1.26, not significant
8. Z = – 0.73, not significant
9. Z = – 1.81, not significant
10. Z = – 4.38, significant
11. Z = – 4, not significant
12. Zcal = 2.05, significant
13. Z = 1.4, accept H_0
14. Z = 4.24, reject H_0
15. Z = 8.24, significant
16. Z = 19.7, reject H_0
17. Z = 30.7, reject H_0
18. Z = 2.23, accept H_0
19. Z = 5.26, reject H_0
20. Z = 1.25, accept H_0
21. Z = – 1, accept H_0
23. Z_{cal} = 2.8, reject H_0
24. Z = 2.29, accept H_0
25. Z = 1.79, not significant
26. Z = 7.39, reject H_0
27. Z = 1.77, reject H_0
28. Z = – 2.4, significant
29. Z = – 1.69, significant
30. Z = – 4.5, significant
31. Z = – 1.67, reject H_0
32. Z = 1.83, accept H_0
33. Z = 2.18, accept H_0
34. Z = 0.96, accept H_0
35. Z = 2.63, reject H_0
36. t = 1.93, no significant
37. t = 3, reject H_0
38. t = 1.89, reject H_0
39. Z = – 12.65, reject H_0
40. t = – 2.14, accept H_0
41. t = 2.5, reject H_0
42. t = – 0.62, accept H_0
43. t = – 3.20, significant
44. t = 0.25, not significant
45. t = 8.49, reject H_0
46. t = 0.32, accept H_0
47. t = – 1.45, accept H_0
48. t = 9.39, significant
49. t = – 0.19, accept H_0
50. t = 0.94, not significant
51. t = 2.7, reject H_0
52. t_{cal} = 7.52, reject H_0
53. t_{cal} = 1.21, accept H_0
54. t_{cal} = 0.98, accept H_0
55. t_{cal} = 3.30, reject H_0

7.3 Chi-square (χ^2) Test

Chi-square (χ^2) is most popular non parametric test of statistical significance for bivariate tabular analysis with contingency table. This test can be applied to nominal or categorical data, but it cannot be used in ranking technique.

Condition to apply for χ^2 test

- Sample observations should be independent.
- The frequencies must be absolute not relative.
- The total frequency should be reasonably large i.e., n > 50.
- The expected frequency of any cell should not be less than 5.
- The test entirely depends on the degrees of freedom.

χ^2-test describes the magnitude of difference between the observed frequencies and expected frequencies under certain assumptions.

χ^2 Test Statistics

Let O be the observed frequency and E be the expected frequency, then

$$\chi^2 = \frac{(\text{Observed count} - \text{Expected count})^2}{\text{Expected count}}$$

or, $$\chi^2 = \frac{(O_1 - E_1)^2}{E_1} + \frac{(O_2 - E_2)^2}{E_2} + \ldots + \frac{(O_n - E_n)^2}{E_n}$$

$$\therefore \chi^2 = \sum_{i=1}^{n} \frac{(O_i - E_i)^2}{E_i}$$

or simply, $\chi^2 = \Sigma \frac{(O - E)^2}{E}$

Degrees of freedom (d.f.), in case of contingency table can be calculated as:

d.f. = (No. of rows – 1) × (No. of columns – 1)

i.e. d.f. = (r – 1) × (c – 1)

If the data are presented as a series of variate values in rows or columns, then the d.f. will be the number of items in the serious minus 1.

i.e. d.f. = n – 1

Application of χ^2- test

- Testing the goodness of fit
- Testing the independence of attributes
- Testing the homogeneity/ uniformity of attributes.

Testing the goodness of fit

χ^2-test can be used for testing the appropriateness of a particular distribution like; binomial, poison, hyper geometric, normal, etc. This test tells us whether there is significance difference between observed and expected frequencies.

Steps for test:

Formulate the null and alternative hypothesis

H_0: Given model fits the data.

H1: Given model does not fits the data.

Then, compute test statistic:

$$\chi^2 = \Sigma \frac{(O - E)^2}{E}$$

Critical value: The critical value of χ^2 in given level of significance and degrees of freedom can be checked for χ^2-table.

Decision: If $\chi^2_{cal} \leq \chi^2_{tab}$ then accept null hypothesis other wise H_0 is rejected.

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Example 22. A survey of 320 families with 5 children each revealed the following distribution.

No. of families	14	56	110	88	40	12
No. of boys	5	4	3	2	1	0
No. of girls	0	1	2	3	4	5

Does this result consistent with the hypothesis that male and female births are equally probable. Use $\alpha = 0.05$.

Solution

H_0: Probability of male birth (P) = 0.5 (or given data fits the binomial distribution)

H_1: Probability of female birth = 0.5

$\alpha = 0.05$

Here, $p = 0.5$, $q = 0.5$, $n = 5$

Probability mass function of binomial distribution is:

$P(x) = {}^nC_x\, p^x q^{n-x}$ and

Expected frequency $E(X) = N \cdot P(x)$

Let x be a r.v., which denotes the number of boys.

$P(x = 0) = {}^5C_0\,(0.5)^6\,(0.5)^{5-0} = 0.031$

$f(0) = 320 \times 0.031 = 10$

$P(X = 1) = {}^5C_1\,(0.5)^1\,(0.5)^{5-1} = 0.156$

$f(1) = 0.156 \times 320 = 50$

Similarly we can calculate:

$P(2) = 0.312$, $f(2) = 100$

$P(3) = 0.312$, $f(3) = 100$

$P(4) = 0.156$, $f(4) = 50$

$P(5) = 0.031$, $f(5) = 10$.

Then we have

0	E	(O – E)	$(O - E)^2$	$(O - E)^2 / E$
14	10	4	16	1.6
56	50	6	36	0.72
110	100	10	100	1.0
88	100	– 12	144	1.44
40	50	– 10	100	2.0
12	10	2	4	0.4
				7.16

$\chi^2 = \Sigma \dfrac{(O - E)^2}{E} = 7.16$

χ^2 tab. at 5% level and 5 d.f. = 11.07

Decision: Here $\chi^2_{cal} < \chi^2_{crit}$ at 5% level, hence null hypothesis is accepted and conclude that given data fits the binomial distribution and mean number of male and female births are equal.

χ^2-test for Independence of Attributes

Two variables can often be studied whether there is association between them or not i.e. either they are independent or dependent, such as smoking habit and cancer, weight and diabetes, marriage and divorce. The classification according to criteria of bi-variate, attributes r number of rows of attribute first and S number of column of for the second attribute, then $r \times s$ number of contingency table will be formed and we can apply χ^2-test using.

$$\chi^2 = \Sigma \frac{(O - E)^2}{E}$$

Here, degrees of freedom (d.f.) = (r – 1) (c – 1)

r = no. of rows

c = no. of columns

Example 23. In a survey of smoking habits of 100 men and 100 women were asked to classify themselves as smokers or non smokers, the results summarizes in the table below. Do these data provide any association between the smoking habits of mean and women or equally prevalent in two gender? Use $\alpha = 0.05$.

Smoking	Men	Women	Total
Yes	54	32	86
No.	46	68	114
Total	**100**	**100**	**200**

Solution

H_0: The smoking habits of mean and women are independent. (or the prevalence of smoking is equal in two genders).

H_1: The smoking habits of men and women are dependent.

To determine expected frequency of each cell use:

$$E = \frac{(\text{row total}) \times (\text{Column total})}{\text{Grand total}}$$

i.e. $$E = \frac{(\text{r.t.} \times \text{c.t.})}{\text{G.t.}}$$

when 0 = 54

$E(54) = 86 \times 100/200 = 43$

$E(32) = 86 \times 100/200 = 43$

$E(46) = 114 \times 100/200 = 57$

$E(68) = 114 \times 100/200 = 57$

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Then

0	E	(O - E)	$(O - E)^2$	$(O - E)^2 / E$
54	43	11	121	2.81
32	43	- 11	121	2.81
46	57	- 11	121	2.12
68	57	11	121	2.12
				9.87

$$\chi^2 = \Sigma \frac{(O - E)^2}{E} = 9.87$$

Decision: Here χ^2_{tab} at 5% level and (2 - 1) (2 - 1) = 1 d.f. = 6.31

Since χ^2 calculated value > χ^2 critical value, hence do not accept the null hypothesis i.e. smoking habit of men and women are dependent.

Example 24. A random sample of 1000 registered voters from Kathmandu valley are classified as to whether they are in low, medium or high income level and whether or not they favour a new tax reform. The observed frequencies are formed as:

Tax \ Income Level	Low	Medium	High	Total
For	182	213	203	598
Against	1554	138	110	402
Total	**336**	**351**	**313**	**1000**

Does the level of tax and income level are independent at 5% level of significance?

Solution

H_0: There is no association between tax level and income level.

H_1: There is association between tax level and income level.

Then, to calculate the expected frequency.

$$E = \frac{R.T. \times C.T.}{G.T.}$$

$$E(182) = \frac{598 \times 336}{1000} = 201$$

$$E(213) = \frac{598 \times 351}{1000} = 210$$

$$E(203) = \frac{598 \times 313}{1000} = 187$$

$$E(154) = \frac{402 \times 336}{1000} = 135$$

$$E(138) = \frac{402 \times 351}{1000} = 141$$

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$$E(203) = \frac{402 \times 313}{1000} = 126$$

To find χ^2-value

0	E	(O - E)	$(O - E)^2$	$(O - E)^2 / E$
182	201	- 19	361	1.79
213	210	3	9	1.04
203	187	16	256	1.37
1554	135	19	361	2.67
138	141	- 3	9	0.06
110	126	- 16	256	2.03
				7.96

$$\chi^2 = \Sigma \frac{(O - E)^2}{E} = 7.96$$

Here χ^2_{crit} at 5% level and (3 - 1) (2 - 1) = 2 d.f. = 5.99

Decision: Here χ^2cal (7.96) > χ^2 crit (5.99), hence do not accept H_0, i.e. there is association between tax level and income level.

χ^2-test for homogeneity/ Uniformity of Attributes

To test the uniformity or homogeneity between variables χ^2-test is used. The homogeneity test is concerned with the answer are the samples drawn from two populations are homogenous with respect toe some criteria of classification.

H_0: The two samples are homogeneous

H_1: The two samples are not homogeneous

Example 25. In an air pollution study, a random sample of 200 households are selected from each of two communities. A responds in each households are asked whether or not anyone in the household bothered by air pollution. The responses are given below. Can you conclude that the two communities are homogenous with respect to the variables of interest at 5% level of significance?

Community	Any member bothered by air pollution		
	Yes	No	Total
I	43	157	200
II	81	119	200
Total	124	276	400

Does the level of tax and income level are independent at 5% level of significance?

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Solution

H_0: The two communities are homogenous with respect to the air pollution.

H_1: The two communities are not homogenous with respect to the air pollution.

The expected frequencies are calculated by using

$$E = \frac{R.T. \times C.T.}{G.T.}$$

Then,

0	E	(O – E)	$(O - E)^2$	$(O - E)^2 / E$
43	62	– 19	361	5.82
81	62	19	361	5.82
167	138	19	361	2.61
119	138	– 19	361	2.61
				16.86

$$\chi^2 = \Sigma \frac{(O - E)^2}{E} = 16.86$$

Here χ^2_{crit} at 5% level and (2 – 1) (2 – 1) = 1 d.f. = 3.84

Decision: Here χ^2cal (7.96) > χ^2 crit (5.99) at 5%, hence reject null hypothesis and the two communities are not homogenous with respect to air pollution.

Yate's Correction

In case of 2 × 2 contingency table the number of degrees of freedom (2 – 1) (2 – 1) equals 1, and χ^2 value is

a	b	a + b
c	d	c + d
a + c	b + d	N

$$\chi^2 = \frac{N(ad - bc)^2}{(a + c)(b + d)(a + b)(c + d)}$$

If any of the cell frequency is less than 5, then use of pooling method of χ^2-test, result is meaningless. Since 1 d.f. is loss due to pooling χ^2-test with zero d.f. is meaningless. In this situation χ^2-test is used.

According to Yate's correction adding 0.5 to the cell frequency which is less than 5 and adjusting the cell frequencies accordingly. Then finally χ^2 value will be.

$$\chi^2 = \frac{N\left\{|ad - bc| - \frac{N}{2}\right\}^2}{(a + c)(b + d)(a + b)(c + d)}$$

Exercise 7.2

1. A group of 266 healthy men and women were grouped according to their number of relations such as parent, friend, neighbour, etc. They are the exposed to virus that causes colds. A adoption of results is shown in table.

Cold	No. of relations			Total
	2 or fewer	3 to 5	6 or more	
Yes	49	43	34	126
No	31	37	62	140
Total	**80**	**90**	**96**	**266**

Are they are association between number of relationship and common cold at 5% level of significance?

2. The number of accidents per week in a certain area were as follows.
Observed frequencies: 12, 8, 20, 14, 10, 15, 6, 9, 4, 2
Are the frequencies in agreement with the belief that accidents were uniformly distributed per the 10 period of the week. Use $\alpha = 0.1$.

3. In an experiment to study the dependence of hypertension smoking habits, the following data were taken on 180 individuals.

Cases	Non smokers	Moderate smokers	Heavy smokers
Hypertension	21	36	30
No Hypertension	48	26	19

Test the hypothesis that presence or absence of hypertension is independent of smoking habits. Use $\alpha = 0.05$.

4. A random sample of 200 married men, all retired were classified according to education and number of children.

Education	Number of children		
	0 - 1	2 - 3	Over - 3
Elementary	14	37	32
Secondary	19	42	17
College	12	17	10

Test the hypothesis that, at 5% level of significance, the size of a family is independent of the level of education attained by the father.

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5. 64 boys and girls appeared in the examination of mathematics yielded the results as given below.

Gender	Fail	Pass	Total
Boys	2	42	44
Girls	6	14	20
Total	**8**	**56**	**64**

Does the data provide there exists relationship between the result and gender at 10% relationship?

6. A random sample of 90 adults are classified according to gender and the number of hours they watch television during a week.

Hours	Male	Female
Over 25 years	15	29
Under 25 years	27	19

Test the hypothesis that time spent on watching television is independent of whether viewer is male or female at 5% level of significance?

Answers

1. $\chi_{cal} = 14.5$, reject H_0
2. $\chi_{cal} = 26.6$, reject H_0
3. $\chi_{cal} = 15.12$, reject H_0
4. $\chi_{cal} = 5.27$, accept H_0
5. $\chi_{cal} = 5.56$, reject H_0
6. $\chi_{cal} = 4.47$, reject H_0

❐

Unit 8

Correlation and Regression Analysis

8.1 Simple correlation

8.1.1 Introduction

If we are trying to find the relationship between two variable like height and weight of children, temperature and volume of the metals, change in income and expenditure, change in price and sales, etc. we need to find the degree of association between such variables. The two or more variables are said to be correlated if the change in value of one variable affects the change in value of other variables. Karl Pearson was able to suggest a method for measuring the association between two variables.

A statistical measure used to determine the degree of linear relationship between two or more variables is called correlation. Correlation describes the relationship or association between two or more numeric variables.

Types of correlation

According to nature and number of associated variables the correlation can be categorised as follows:

1. **Positive correlation:** If the relationship between two variables varies in the same direction i.e. values of one variable increases the values of other variable also increases and values of one variable decreases. The values of other also decreases i.e. they tend to vary in the same direction, such variables are positively correlated. For example, temperature increases volume of the metal also increases and vice versa.
2. **Negative correlation:** If the change in value of one variable affects the change in value of other variable in opposite direction, such correlation is said to be negative correlation. In this correlation values tend to deviated in opposite direction. For example values of acceleration due to gravity increases, when the radius of earth decreases and vice-versa.
3. **Linear and non linear correlation:** If the change in value of one variable to the change in other variable are proportional to each other i.e. variables bears a constant ratio then the variables are linearly correlated and we can say there is linear correlation. If the variables are correlated and does non bears a constant ratio then there is non linear correlation.

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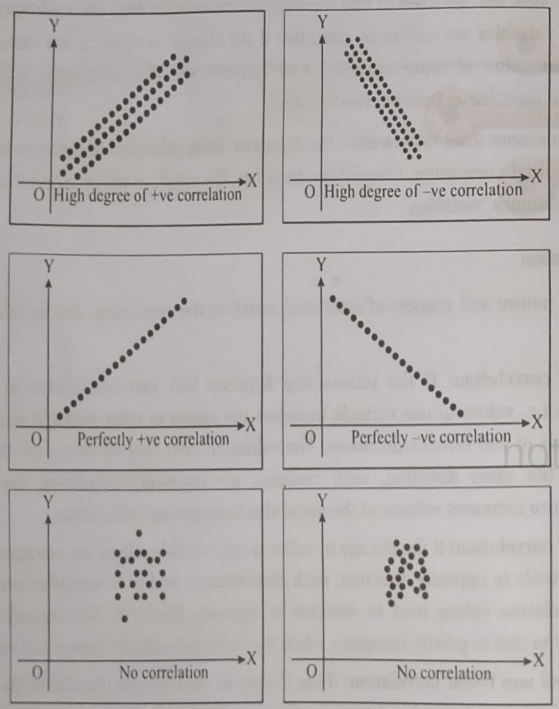
4. **Multiple correlation:** If the association between one dependent variable and two or more independent variables are measured simultaneously, then it is said to the multiple correlation. For example yield of the Cow (is milk) as dependent variable studided simultaneously independent variables like caste of cow, nutrition, habitat, etc. gives multiple correlation values.
5. **Partial correlation:** If there are two or more than two independent variable, and the relationship of one dependent variable is studied by taking the effects of only independent variable, keeping the effects of other variables fixed or constant is called partial correlation.

Methods of Studying Correlations

1. Scatter diagram

If two variables X and Y are plotted on graph keeping the value of independent variable is X-axis and values of dependent variable in Y-axis, the diagram of dots so obtained is called scatter diagram. It is the simplest method of studying correlation. The plotted dot show some trend of upward or downward, they we can say the two variables are correlated.



8.1.2 Karl Pearson's correlation coefficient

Karl Pearson's correlation coefficient measures the degree of association between two linear variables. Let X and Y be two correlated variables, then Karl Pearson's correlation coefficient denoted by r_{XY} or simply is defined as:

$$r = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)} \times \sqrt{\text{Var}(Y)}} \quad \ldots (i)$$

where,

$$\text{Cov}(X, Y) = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$$

or, $$\text{Cov}(X, Y) = \frac{1}{n}\sum_{i=1}^{n}(x_i y_i - x_i\bar{y} - \bar{x}y_i + \bar{x}\bar{y})$$

$$= \frac{1}{n}\Sigma x_i y_i - \frac{1}{n}x_i\bar{y} - \bar{x}\frac{1}{n}y_i + \bar{x}\bar{y}$$

$$= \frac{1}{n}\Sigma x_i y_i - \bar{x}\bar{y} - \bar{x}\bar{y} + \bar{x}\bar{y}$$

$\therefore$ $$\text{Cov}(X, Y) = \frac{1}{n}\Sigma x_i y_i - \bar{x}\bar{y}$$

$$= \frac{1}{n}\Sigma xy - \bar{x}\bar{y}$$

$$\sigma_X = \sqrt{\frac{1}{n}\Sigma(x - \bar{x})^2}$$

$$\sigma_Y = \sqrt{\frac{1}{n}\Sigma(y - \bar{y})^2}$$

We can write equation (i)

$$r = \frac{\Sigma(X - \bar{X})(Y - \bar{Y})}{\sqrt{\Sigma(X - \bar{X})^2}\sqrt{\Sigma(Y - \bar{Y})^2}} \quad \ldots (ii)$$

On simplification we can write

$$r = \frac{n\Sigma XY - \Sigma X\,\Sigma Y}{\sqrt{n\Sigma X^2 - (\Sigma X)^2}\sqrt{n\Sigma Y^2 - (\Sigma Y)^2}} \quad \ldots (iii)$$

Formula (iii) is more continent to use when calculating correlation.

Properties of Karl Pearson's Correlation Coefficient

i. Correlation coefficient lies between – 1 and + 1.

i.e. $-1 \le r \le 1$.

ii. Correlation coefficient r is the geometric mean between two regression coefficients.

$$r = \sqrt{b_{YX} \cdot b_{XY}}$$

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b_{YX} = regression coefficient of Y on X

b_{YX} = regression coefficient of X on Y

ii. For two variables X and Y, correlation coefficient is symmetric

i.e. $r_{XY} = r_{YX}$

iv. Correlation coefficients are not addition.

v. Correlation coefficients r is independent of change of origin and scale.

Let $U = \frac{X-a}{h}$ and $V = \frac{Y-b}{k}$

Then

$$r = \frac{n\Sigma UV - \Sigma U\,\Sigma V}{\sqrt{n\Sigma U^2 - (\Sigma U)^2}\sqrt{n\Sigma V^2 - (\Sigma V)^2}}$$

i.e. $r_{XY} = r_{UV}$

Interpretation of r:

i. If r = 0 to 0.5, there is low degree of correlation.

ii. If r = 0.51 to 0.699, there is moderate degree of correlation.

iii. If r = 0.7 to 0.999, there is high degree of correlation.

iv. If r = 1, there is perfectly positive correlation.

v. If r = – 1, there is perfectly negative correlation.

vi. If r = 0, there is no correlation, the two variables are independent.

Example 1. Calculate the coefficient of correlation from the following data;

X	4	3	7	9	6
Y	5	2	8	12	3

Solution

X	Y	X^2	Y^2	XY
4	5	16	25	20
3	2	9	4	6
7	8	49	64	56
9	12	81	144	108
6	3	36	9	18
29	30	191	246	208

We have

$$r = \frac{n\Sigma XY - \Sigma X\,\Sigma Y}{\sqrt{n\Sigma X^2 - (\Sigma X)^2}\sqrt{n\Sigma Y^2 - (\Sigma Y)^2}}$$

$$= \frac{5 \times 208 - 29 \times 30}{\sqrt{5 \times 191 - 29^2}\sqrt{5 \times 246 - 30^2}}$$

$$= \frac{170}{\sqrt{114}\sqrt{330}}$$

$\therefore$ r = 0.88, shows that there is almost perfect positive correlation between X and Y.

Example 2. The height in inches (X) and weight (in kg) of 7 students are given below.

X	65	66	67	68	69	70	71
Y	67	68	66	69	72	72	69

Calculate the coefficient of correlation between height and weight.

Solution

We can find

$\Sigma X = 476$

$\Sigma X^2 = 32.396$

$\Sigma XY = 32864$

$\Sigma Y = 483$

$\Sigma Y^2 = 33,359$

We have

$$r = \frac{n\Sigma XY - \Sigma X\,\Sigma Y}{\sqrt{n\Sigma X^2 - (\Sigma X)^2}\sqrt{n\Sigma Y^2 - (\Sigma Y)^2}}$$

$$= \frac{7 \times 32864 - 476 \times 483}{\sqrt{7 \times 32396 - 476^2}\sqrt{7 \times 33359 - 483^2}}$$

$\therefore$ r = 0.67

There is moderate degree of correlation between X and Y.

The Coefficient of Determination

The square of correlation coefficient is also called the coefficient of determination and it is denoted by r^2. The correlation coefficient measure the linear association between variable, whereas the coefficient of determination measures the proportion of variation in one variable which can be explained by the variation in the second variable.

Suppose r = 0.8, the r^2 = 0.64, it means 64% of the variability between two variables have been noted for and still 36% of the variability unaccounted. To measure the percentage variation in the dependent variable as explained by the independent variable r^2 is also used in regression analysis.

Probable Error

To test the reliability of correlation coefficient, probable error is used. The probable error P.E. (r) of r is

$$\text{P.E. (r)} = 0.675 \times \text{S.E. (r)} = 0.6745 \times \frac{1-r^2}{\sqrt{n}}$$

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where standard error of r i.e. S.E. (r) $= \frac{1-r^2}{\sqrt{n}}$

If r > 6 P.E. (r) the values of r is highly significant.

If r > 6 P.E. (r) the values of r is not significant.

8.1.3 Spearman's rank correlation coefficients

Qualitative characteristics such as beauty, honesty, intelligence, taste, etc. cannot measured quantitatively. So they can be measured by assigning ranks. In such cases, we use Spearman's rank correlation coefficient. Which is denoted by r_s to find the relation.

Spearman's rank correlation (r_s) is given as:

$$r_s = 1 - \frac{6\Sigma d^2}{n^3 - n}$$

where, $d = R_X - R_Y$ R_X = Rank of X-series

R_Y = Rank of Y-series n = Number of observation.

When rank is repeated

For repeated ranks the Spearman's rank correlation coefficient is given as:

$$r_s = 1 - \frac{6\left\{\Sigma d^2 - \frac{m_1(m_1^2-1)}{12} + \frac{m_2(m_2^2-1)}{12}\right\} + \ldots}{n^3 - n}$$

Where, $d = R_X - R_Y$

m_1 = Number of times the first value is repeated

m_2 = Number of times the second value is repeated

m_3 = Number of times the third value is repeated and so on.

Example 3. Calculate the rank correlation coefficient from the following data.

X:	80	78	75	75	68	67	68	59
Y:	12	13	14	14	14	16	15	17

Solution

X	R_X	Y	R_Y	$d = R_X - R_Y$	d^2
80	1	12	8	-7	49
78	2	13	7	-5	25
75	3.5	14	5	-1.5	2.25
75	3.5	14	5	-1.5	2.25
68	5.5	14	5	0.5	0.25
67	7	16	2	5	25
68	5.5	15	3	2.5	6.25
59	8	17	1	7	49
					159

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$$\therefore \quad r_s = 1 - \frac{6\left\{\Sigma d^2 - \frac{m_1(m_1^2-1)}{12} + \frac{m_2(m_2^2-1)}{12} + \frac{m_3(m_3^2-1)}{12}\right\}}{n^3 - n}$$

$$= 1 - \frac{6\left\{159 - \frac{2\times 3}{12} + \frac{2\times 3}{12} + \frac{3\times 8}{12}\right\}}{8^3 - 8}$$

$$= -0.1929$$

$\therefore$ There is perfect negative correlation.

8.2 Regression analysis

The term regression means 'stepping back towards the average'. Sir Francis Galton showed that, the height of children tends to move back or regress towards the mean height of population. Now a days regression analysis becomes a most important branch of statistical theory that is widely used in all the scientific discipline.

Definition: Regression analysis is a mathematical measures of the average relationship between two or more variables in terms of the original units of data". We can also say the determination or prediction of values of one variable with the help of known values of another variable(s), also known as regression.

In regression analysis there are two types of variables. The variable whose value are known and used for prediction that it is called independent or explanatory variable and whose values are being predicted are known as dependent or explained variable.

8.3 Regression equation of Y on X

The regression equation of Y on X is given by

$$Y = a + bX \qquad \ldots(1)$$

where,

Y is the dependent variable

X is independent variable

a is y-intercept of the line

$b = b_{yx}$ is the regression coefficient of Y on X, it measures the rate of relationship between Y and X, also called slope of the line.

The value of a and b in (1) can be determined by solving following two normal equations.

$$\Sigma Y = na + b\Sigma X \qquad \ldots(2)$$

$$\Sigma XY = a\Sigma X + b\Sigma X^2 \qquad \ldots(3)$$

Solving above two normal equations, we get

$$b = byx = \frac{n\Sigma xy - \Sigma x \cdot \Sigma y}{n\Sigma x^2 - (\Sigma x)^2}$$

$$a = \bar{Y} - b\bar{X}$$

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If we substitute the values of a and b in (1), we get fitted regression of Y on X.

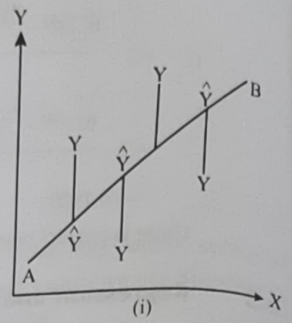
$$\hat{Y} = a + bX.$$

The line AB in figure (1) will be best fit when:

$$\Sigma(Y - \hat{Y}) = 0$$

and $\Sigma(Y - \hat{Y})^2$ is minimum.

Here Y is the actual observations and $\hat{Y}$ is the corresponding estimated values based up on the method of least square.



8.4 Regression equation of X on Y

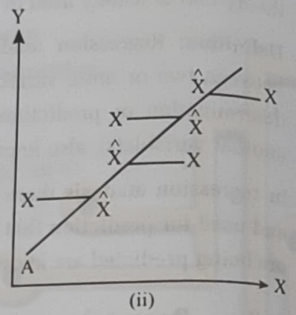
Regression equation of X on Y is:

$$X = a_1 + b_1Y \qquad \ldots(4)$$

The line AB in figure (ii) will be best fit when:

$$\Sigma(X - \hat{X}) = 0$$

and $\Sigma(Y - \hat{Y})^2$ is minimum from the method of principle of least square.



The normal equations for estimating a_1 and b_1 in (4) will be

$$\Sigma X = na_1 + b_1\Sigma Y \qquad \ldots(5)$$

$$\Sigma XY = a_1\Sigma Y + b_1\Sigma Y^2 \qquad \ldots(6)$$

Solving (5) and (6), we get

$$b_1 = b_{xy} = \frac{n\Sigma xy - \Sigma x\,\Sigma y}{n\Sigma y^2 - (\Sigma y)^2}$$

$$a_1 = \bar{X} - b_1\bar{Y}$$

Then fitted regression of X on Y will be:

$$\hat{X} = a_1 + b_1Y$$

8.5 Measure of variation

As predicting the values of dependent variable, by using the values of independent variable(s), several measures of variation requires to develop.

Let the regression equation of Y on X be:

$$Y = a + bX \qquad \ldots(A)$$

The variation in Y measured in terms of mean $\bar{Y}$ is $(Y - \bar{Y})$.

The sum of square of deviation from mean $\Sigma(Y - \bar{Y})^2$ is called total sum of square (TSS) or total variation in Y.

The variation in Y can be decomposed into two components, i.e.,

$$(Y - \bar{Y}) = (Y - \hat{Y}) + (\hat{Y} - \bar{Y}) \qquad \ldots(B)$$

The terms $(Y - \hat{Y})$ presents the variation about regression line, it can not be explained by regression itself. The sum of square of this variation is called error sum of square.

Taking square both side and sum both sides of (B), we get

$$\Sigma(Y - \bar{Y})^2 = \Sigma(Y - \hat{Y})^2 + \Sigma(\hat{Y} - \bar{Y})^2 \quad [\because \text{the product term} = 0]$$

or, TSS = ESS + RSS

i.e., Total sum of square = Error sum of square + Regression sum of square

RSS may be considered as a measure of variation in Y, associated with regression line.

8.6 Coefficient of determination

The coefficient of determination measures the strength of association between dependent variable Y and independent variable X. It is denoted by r^2 and measures the proportion of variation in Y due to X.

$$r^2 = \frac{RSS}{TSS}$$

$$= \frac{\Sigma(\hat{Y} - \bar{Y})^2}{\Sigma(Y - \bar{Y})^2}$$

$$= \frac{a\Sigma y + b\Sigma XY - n\bar{Y}^2}{\Sigma Y^2 - n\bar{Y}^2}$$

8.7 Standard error of estimate

It measures the average variation or scatteredness of the data point around the regression line. The standard error of estimate, denoted by (Se) = 0 means there is no variation. If Se is large the fitted regression line is very weak for predicting the values of dependent variable. If Se is small, there is less variation, and fitted line will be better for prediction.

The standard error of estimate is given as:

$$Se = \sqrt{\frac{SSE}{n - 2}}$$

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$$Se = \sqrt{\frac{\Sigma(Y - \hat{Y})^2}{n - 2}}$$

$$Se = \sqrt{\frac{\Sigma y^2 - a\Sigma y - b\Sigma xy}{n - 2}}$$

Example 4. The following table presents the height in inches of father (X) and heights in inches of their sons (Y).

X:	65	66	67	67	68	69	70	72
Y:	67	68	65	68	72	72	69	71

Fit the regression equation of Y on X, find Y, when X = 74 inch. Also calculate the standard error of estimate.

Solution

	X	Y	XY	X^2	Y^2
	65	67	4355	4225	4489
	66	68	4488	4356	4624
	67	65	4355	4489	4355
	67	68	4556	4489	4624
	68	72	4896	4624	5784
	69	72	4968	4761	5184
	70	69	4830	4900	4760
	72	71	5112	5184	5040
Total	**544**	**552**	**37560**	**37028**	**38262**

The regression of Y on X is:

$Y = a + bX$ ...(1)

$\bar{X} = \frac{544}{8} = 68$ and $\bar{Y} = 69$

$b = b_{YX}$

$$= \frac{n\Sigma XY - \Sigma X \Sigma Y}{n\Sigma X^2 - (\Sigma X)^2}$$

$$= \frac{8 \times 37560 - 544 \times 552}{8 \times 37028 - (544)^2}$$

or, $b = 2.85$

$a = \bar{Y} - b\bar{X}$

$= 69 - 2.85 \times 68$

$= -124.8$

∴ The fitted regression equation will be:

$\hat{Y} = -124.8 + 2.85X$

When X = 74 inch, then

$\hat{Y} = -124.8 + 2.85 \times 74$

$= 86.9$ inch

The standard error of estimate

$$Se = \sqrt{\frac{\Sigma Y^2 - a\Sigma Y - b\Sigma xy}{n - 2}}$$

$$= \sqrt{\frac{38262 - (-124.8) \times 552 - 2.85 \times 37560}{8 - 2}}$$

∴ $Se = 17.6$

Example 5. Fit the regression equation of Y on X from the following data and predict the value of Y When X = 5, also find Se.

X:	2	4	6	8
Y:	10	20	25	30

Solution

Regression equation of Y on X is $Y = a + bX$ ...(1)

	X	Y	X^2	Y^2	XY
	2	10	4	100	20
	4	20	16	400	80
	6	25	36	625	150
	8	30	64	900	240
Total	**20**	**85**	**120**	**2025**	**490**

$$\bar{X} = \frac{\Sigma X}{n}$$

$$= \frac{20}{4}$$

$= 5$

$$Y = \frac{\Sigma Y}{n}$$

$$= \frac{85}{4}$$

$= 21.25$

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$$b = b_{YX} = \frac{n\Sigma XY - \Sigma X\,\Sigma Y}{n\,\Sigma X^2 - (\Sigma X)^2}$$

$$= \frac{4 \times 490 - 20 \times 85}{4 \times 120 - (20)^2}$$

$$= \frac{260}{80}$$

$\therefore$ $b = 3.25$

$a = \bar{Y} - b\bar{X}$

$= 21.25 - 3.25 \times 5$

$= 5$

Fitted regression equation will be:

$\hat{Y} = 5 + 3.25\,X$

When X = 5, then

$\hat{Y} = 5 + 3.25 \times 5$

$\therefore$ $\hat{Y} = 22.5$

Standard error of estimate

$$Se = \sqrt{\frac{\Sigma Y^2 - a\Sigma Y - b\Sigma XY}{n-2}}$$

$$= \sqrt{\frac{2025 - 5 \times 85 - 3.25 \times 490}{2}}$$

$$= \sqrt{\frac{7.5}{2}}$$

$\therefore$ $Se = 1.94$

Example 6. The age in years of patients and blood pressure in mmHg are given in the table below. (a) Compute the blood pressure of the patients at the age of 50. (b) Calculate the standard error of estimate and coefficient of determination.

Age (Years) (X)	56	42	72	36	63	47
Y: BP (mm HG) (Y)	147	125	160	118	149	129

Solution

Age in years = X, B.P. mmHg = Y

The regression equation of Y on X is

$Y = a + bX$ ...(1)

$\Sigma X = 316$, $\Sigma Y = 827$, $\Sigma X^2 = 17{,}558$

$\Sigma Y^2 = 1{,}15{,}343$, $\Sigma XY = 44653$

$$\bar{X} = \frac{316}{6} = 52.67$$

$$\bar{Y} = \frac{827}{6} = 137.83$$

$$b = b_{YX} = \frac{n\Sigma XY - \Sigma X\,\Sigma Y}{n\,\Sigma X^2 - (\Sigma X)^2}$$

$$= \frac{6 \times 44653 - 316 \times 827}{6 \times 17558 - (316)^2}$$

$$= \frac{6586}{5492}$$

$b = 1.199$

$a = \bar{Y} - b\bar{X}$

$= 137.83 - 1.199 \times 52.67$

$a = 74.67$

$\therefore$ Fitted regression equation of Y on X is:

$\hat{Y} = 74.67 + 1.199X$

a. When X = 50 years

$\hat{Y} = 74.67 + 1.199 \times 50$

$= 134.62$

Hence at the age of 50, blood pressure = 134.62 mmHg

b. Standard error of estimate

$$Se = S_{YX} = \sqrt{\frac{\Sigma Y^2 - a\Sigma Y - b\Sigma XY}{n-2}}$$

$$= \sqrt{\frac{1{,}15{,}343 - 74.67 \times 827 - 1.199 \times 44653}{4}}$$

$$= \sqrt{\frac{51.963}{4}}$$

$= 3.6$

$\therefore$ $Se = 3.6$

Coefficient of determination

$$r^2 = \frac{a\Sigma Y - b\Sigma XY - n\bar{Y}^2}{\Sigma Y^2 - n\bar{Y}^2}$$

$$= \frac{74.67 \times 827 - 1.199 \times 44653 - 6 \times (137.83)^2}{115343 - 6(137.83)^2}$$

$\therefore$ $r^2 = 0.96$

It means there is around 96% variation when predicting B.P. by using the ages of patients.

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Example 7. A data on family expenditure on '000 Rs. and their family size are as follows. Estimate the family expenditure having 6 family members and also standard error of estimate.

Family expenditure '000 Rs. (Y)	5	7	8	9	11	12
Family size (X):	3	2	4	5	1	7

Solution

We have to estimate the family expenditure, so let Y = family expenditure, X = family size. Then,

	X	Y	X^2	Y^2	XY
	3	5	9	25	35
	2	7	4	49	14
	4		16	64	32
	5	9	25	81	45
	1	11	1	121	11
	7	12	49	144	84
Total	**22**	**55**	**104**	**484**	**201**

Regression equation of Y on X is:

$$Y = a + bX \qquad \ldots(1)$$

$$\bar{X} = \frac{22}{6}$$

$$= 3.67$$

$$\bar{Y} = \frac{52}{6}$$

$$= 8.67$$

$$b = \frac{n\Sigma XY - \Sigma X\,\Sigma Y}{n\Sigma X^2 - (\Sigma X)^2}$$

$$= \frac{6 \times 201 - 22 \times 52}{6 \times 104 - (22)^2}$$

or, $b = 0.44$

$a = \bar{Y} - b\bar{X}$

$= 8.67 - 3.67 \times 0.44$

$= 7.05$

∴ Fitted regression is

$$\hat{Y} = 7.05 + 0.44X$$

When X = 6

$$\hat{Y} = 7.05 + 0.44 \times 6$$

$$= 9.69 \text{ thousand}$$

∴ The family having 6 members, the family expenditure

$Y = 9.69$ thousand Rs.

The standard error of estimate

$$Se = \sqrt{\frac{\Sigma Y^2 - a\Sigma Y - b\Sigma XY}{n-2}}$$

$$= \sqrt{\frac{484 - 7.05 \times 52 - 0.44 \times 201}{6-2}}$$

or, $Se = 2.69$

∴ Standard error of estimate (Se) = 2.69.

8.8 Properties of regression coefficient

Let X and Y be two variables, then regression equation of Y on X is:

$$Y = a + bX \qquad \ldots(1)$$

Where, b is the regression coefficients of Y on X and also denoted by b_{yx}.

$$b = b_{yx} = \frac{Cov(x, y)}{Var(x)} = \frac{r\sigma_x\sigma_y}{\sigma_x^2} = \frac{r\sigma_y}{\sigma_x}.$$

Similarly for regression equation of X on Y

$$X = a + bY \qquad \ldots(2)$$

Here, b is the regression coefficients of X on Y.

$$b = b_{xy} = \frac{Cov(x, y)}{Var(y)} = \frac{r\sigma_x\sigma_y}{\sigma_y^2} = \frac{r\sigma_x}{\sigma_y}.$$

1. Correlation coefficient is the geometric mean of regression coefficients.

   i.e., $r = \sqrt{b_{yx} \cdot b_{xy}}$.

   We have,

   $$b_{yx} \cdot b_{xy} = \frac{r\sigma_y}{\sigma_x} \cdot \frac{r\sigma_x}{\sigma_y} = r^2$$

   $$\therefore r = \sqrt{b_{yx} \cdot b_{xy}}.$$

2. Product of two regression coefficient lies between 0 and 1.

   i.e., $(b_{yx})\,(b_{xy}) \le 1$ $\qquad [\because 0 \le r^2 \le 1]$

3. If one of the regression coefficient is greater than 1, the other must be less than 1.

   If $b_{yx} > 1$, then $b_{xy} < 1$.

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4. $b_{yx} = r\frac{\sigma_Y}{\sigma_X}$ and $b_{xy} = r\frac{\sigma_X}{\sigma_Y}$

Let $y = a + bX$ ...(1)

The normal equations for solving a and b are

$\Sigma Y = na + b\Sigma X$ ...(2)

$\Sigma XY = a\Sigma X + b\Sigma X^2$ ...(3)

Multiplying (2) by ΣX and (3) by n and subtracting (3) from (2), we get

$n\,\Sigma XY = na\,\Sigma X + hb\Sigma X^2$

$\Sigma X \cdot \Sigma Y = na\,\Sigma X + b(\Sigma X)^2$

– – –

$\therefore b = b_{YX} = \frac{n\,\Sigma XY - \Sigma X \cdot \Sigma Y}{n\,\Sigma X^2 - (\Sigma X)^2}$

$\therefore \quad b_{YX} = b = \frac{Cov\,(X, Y)}{\sigma_X^2}$ ...(4)

We have,

$Cov(XY) = \frac{1}{n}(X - \bar{X})(Y - \bar{Y})$

$= \frac{1}{n}\Sigma XY - \frac{\Sigma X}{n} \cdot \bar{Y} - \frac{\Sigma Y}{n} \cdot \bar{X} + \bar{X}\,\bar{Y}$

$= \frac{1}{n}\Sigma XY - \bar{X}\,\bar{Y} - \bar{X}\,\bar{Y} + \bar{X}\,\bar{Y}$

$\therefore \quad Cov(XY) = \frac{1}{n}\Sigma XY - \bar{X}\,\bar{Y}$

or, $\frac{1}{n}\Sigma XY = Cov(XY) + \bar{X}\,\bar{Y}$ ...(5)

Also, we have

$\sigma_X^2 = \frac{1}{n}\Sigma X^2 - \bar{X}^2$ and $\sigma_Y^2 = \frac{1}{n}\Sigma Y^2 - \bar{Y}^2$ ...(6)

Dividing both sides of equation (3) by n, we get

$\frac{\Sigma XY}{n} = a\frac{\Sigma X}{n} + b\frac{\Sigma X^2}{n}$

Using (5) and (6) in above equation, we get

$Cov(X, Y) + \bar{X}\,\bar{Y} = a\bar{X} + b(\sigma_X^2 + \bar{X}^2)$ ...(7)

Now, $\bar{Y} = a + b\bar{X}$

or, $\bar{X}\,\bar{Y} = a\bar{X} + b\bar{X}^2$ ...(8)

Subtracting (8) from (7), we get

$b = \frac{Cov\,(X, Y)}{\sigma_X^2}$

$\therefore \quad b_{YX} = \frac{r\,\sigma Y}{\sigma X}$

Similarly, $b_{xy} = r\frac{\sigma_X}{\sigma_Y}$.

The line of regression passes through $(\bar{X}, \bar{Y})$

$\therefore \quad Y - \bar{Y} = b(X - \bar{X})$

or, $Y - \bar{Y} = r\frac{\sigma_Y}{\sigma_X}(X - \bar{X})$

and $X - \bar{X} = r\frac{\sigma_X}{\sigma_Y}(Y - \bar{Y})$

8.8.1 Difference between correlation and regression analysis

The differences between correlation and regression analysis are as follows:

S.N.	Correlation	Regression
1.	It describes the degree to which variables are linearly related.	It describes the average relationship between the variables are linearly or nonlinear relationship.
2.	It is a measure of degree and direction of linear relationship between X and Y.	It establishes the functional relationship between dependent and independent variables.
3.	It is symmetric, i.e., $r_{XY} = r_{YX}$.	The coefficients are not symmetric, i.e., $b_{YX} \neq b_{xy}$.
4.	It does not imply cause and effect relationship between variables,.	It indicates cause and effect relationship between dependent and independent variables.
5.	It has limited applications.	It has wide applications and used to estimate dependent variables by the help of independent variables.
6.	r_{XY} or ρ have no any units.	Regression coefficients have units, i.e., attached from measurement.

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Example 8. The two random variables have the regression lines with equations $3x + 2y = 26$ and $6x + y = 31$. Find the mean values and correlation coefficient between X and Y.

Solution

$3x + 2y = 26$

$6x + y = 31$

The regression passes through their means $(\bar{x}, \bar{y})$

$\therefore \quad 3\bar{x} + 2\bar{y} = 26$

$6\bar{x} + \bar{y} = 31$

Solving these two we get $\bar{x} = 4$ and $\bar{y} = 7$.

To calculate regression coefficients

$3x + 2y = 26$

or, $y = \dfrac{26 - 3x}{2}$

or, $y = -1.5x + 13$

or, $b_{yx} = -1.5$

$6x = -y + 31$

$x = -0.167y + 5.167$

$\therefore \quad b_{xy} = -0.167$

$r = \sqrt{b_{yx} \cdot b_{xy}}$

$= \sqrt{(-1.5)(-0.167)}$

$\therefore \quad r = 0.2505$ and $\bar{x} = 4, \bar{y} = 7$.

Example 9. For 10 observations on price (X) and supply (Y), the following data were obtained.

$\Sigma X = 130, \Sigma Y = 220, \Sigma X^2 = 2288, \Sigma Y^2 = 5506, \Sigma XY = 3467$

a. Obtain two lines of regression and estimate the supply when the price is 16 units.

b. Find the correlation coefficients and interpret the result.

Solution

$$\bar{X} = \frac{\Sigma X}{n}$$

$$= \frac{130}{10} = 13$$

$$\bar{Y} = \frac{\Sigma Y}{n}$$

$$= \frac{220}{10} = 22$$

$$b_{YX} = \frac{n\Sigma XY - \Sigma X \cdot \Sigma Y}{n \cdot \Sigma X^2 - (\Sigma X)^2}$$

$$= \frac{10 \times 3467 - 130 \times 220}{10 \times 2288 - (130)^2} = 1.02$$

$$b_{XY} = \frac{n\Sigma XY - \Sigma X \cdot \Sigma Y}{n \cdot \Sigma Y^2 - (\Sigma Y)^2}$$

$$= \frac{10 \times 3467 - 130 \times 220}{10 \times 5506 - (220)^2}$$

$\therefore \quad b_{XY} = 0.91$

Regression equation of Y on X is:

$(Y - \bar{Y}) = b_{YX}(X - \bar{X})$

or, $Y - 22 = 1.02(X - 13)$

or, $Y = 1.02x + 22 - 13.26$

$\therefore \quad Y = 1.02X + 8.74$, when $X = 6$ units

Again, $Y = 1.02 \times 16 + 8.74 = 256.06$

$(X - \bar{X}) = b_{XY}(Y - \bar{Y})$

or, $X - 13 = 0.91(Y - 22)$

$\therefore \quad X = 0.914 - 7.02$

Correlation coefficient

$r = \sqrt{b_{YX} \cdot b_{XY}} = \sqrt{1.02 \times 0.91}$

$\therefore \quad r = 0.963$

Here X and Y are highly correlated.

Coefficient of determinant

$r^2 = b_{YX} \cdot b_{xy}$

$= 1.02 \times 0.91$

$= 0.9282$

$\therefore$ One variable accounts the total verification. Only 7% is unaccounted.

Exercise 8.1

1. A researcher wants to find out if there is relationship between the heights of sons and heights of their fathers. Calculate the correlation between two heights.

Father's height (in inch) X:	63	65	66	67	64	68
Son's height (in inch) Y:	66	68	65	67	69	70

2. Find correlation coefficient from the marks obtained by boys and girls.

Boys marks:	23	27	28	29	30	31	33	35
Girls marks:	21	22	23	24	25	26	28	29

3. The following table gives the age and blood pressure of 10 women:

Age (X):	56	42	36	47	49	42	60	72	63	55
Girls marks:	147	125	118	128	145	140	155	160	149	150

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a. Find the correlation coefficient between age and B.P.

b. Fit the regression equation and estimate the B.P. of the women at the age of 65.

4. The following table gives the distribution of the total population and those who are totally or partially blind among them. Find if there is any relation between age and blindness.

Age in years:	0-10	10-20	20-30	30-40	40-50	50-60	60-70
No. of persons:	100	60	40	36	24	11	6
Number of blind:	55	40	40	28	36	22	18

5. Find the correlation coefficient between X and Y. Also find the two regression lines.

X:	1	2	3	5	6	7	8	9	10
Y:	10	12	16	28	25	36	41	40	50

6. Find the Karl Pearson's correlation coefficient from the following data:

Advertisement cost (in '000 Rs.)	39	65	62	90	82	75	25	36	40
Sales (in lakh Rs.)	47	53	58	86	62	68	60	51	20

7. The following data gives the monthly income and expenditure in food, in thousand Rs. of 10 families.

Income:	120	90	80	150	130	140	105	92	70	98
Expenditure:	40	36	40	45	40	42	46	38	50	32

a. Calculate the correlation coefficient between income and expenditure.

b. Also fit the regression equation of expenditure on income and find expenditure when the income of the family is Rs. 150 thousands.

8. In order to find the correlation coefficients of 12 pairs observations of X and Y, the following calculation were made.

$\Sigma X = 30, \Sigma Y = 25, \Sigma X^2 = 670, \Sigma Y^2 = 280, \Sigma XY = 334$

After checking, it was found that, the pair (X = 11, Y = 14) was copied thoroughly instead of (X = 10, Y = 24). Find the correct value of correlation coefficient.

9. The grade of mathematics and physics of 7 students of the classes are:

Mathematics:	6	4	8	5	3.5	5.4	6.5
Physics:	6.5	4.5	7	5	4	6	8

Determine the regression lines and calculate the expected grade in physics for a student who has a 7.5 in mathematics.

10. Two random variables have the regression line with equations $7x + 5y = 35$ and $4x + y = 25$, find the mean values and correlation coefficient between X and Y.

11. 10 articles were ranked by two professor's A and B by becoming the judges. The rank given by them were as follows:

Prof. A:	2	1	4	6	5	8	9	10	7	3
Y:	4	3	2	5	1	6	8	9	10	7

Does the ranks by pair of judges are correlated?

12. A part of waste removal project, a new compression machine for processing sewage sludge is being studied. Engineers collected the data as moisture control of compressed pellets (in percentage) Y and machine filterate rate (in kg - DS/m/hr) X; which are given below:

Machine filterate rate (X):	125.2	98.5	160.1	78.9	159.5	145.8	156.6	110.0
Moisture pellets (Y)	77.9	96.8	80.1	82.2	79.4	78.2	80.3	95.1

Fit the regression equation of Y on X and estimate the moisture control compressed pellets. When machine filterate rate is 120 in kg-DS/m/hr. Also calculate the standard error of estimate.

13. The data from investigating the relationship between roadway surface temperature (X) and permanent deflection (Y). The summary quantities were $n = 20, \Sigma X^2 = 4, 43, 215, \Sigma X = 1478, \Sigma Y = 1275, \Sigma Y^2 = 8860, \Sigma XY = 1083.67$.

Fit the regression equation, what is the mean permanent deflection when the surface temperature is 90°F.

14. An article in a certain journal presented data in green liquor concentration (in gm per liter) X and paper machine production (in tons per day) Y are as shown as:

X:	825	830	890	895	890	910
Y:	40	42	49	46	44	48

Find the fitted value of Y corresponding to X = 850 and associated standard error of estimate.

15. An article in certain journal described a study investigating the relationship between noise exposure and hypertension. The following data are representative of those reported in article.

X:	60	63	65	70	70	72	80	90	80
Y:	1	3	4	5	3	6	4	2	6

a. Draw the scatter diagram of (Y) blood pressure rise in mmHg versus X (sound pressure level in decibels.)

b. Fit the linear regression equation and predict the mean rise in blood level associated with sound pressure level of 85 decibels.

16. An article presents data on fretting wear of mild steel Y (wear volume 10^{-4} mm^3) and viscosity (X) as follows:

X:	240	181	190	155	172
Y:	1.6	9.4	15.5	20	22

Fit the regression of Y on X and predict the value of Y when X = 12.5, also find coefficient of determination.

17. A metal rod was gradually heated and its length (in cm) increased was measured at various temperature, the data recorded were as:

Temperature °C	15	20	25	30	35	40
Length (cm)	90	93.8	96.8	100.2	103.4	107.1

Find the correlation coefficient between temperature and length of rod. Fit the regression equation and estimate the length of rod at 50°C.

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18. An experiment was carried out to investigate variation of solubility of chemical X in water. The quantities in kg that dissolved at 1 liter at various temperature are shown.

Tem°C (y):	15	20	25	30	35	50	70
Mass of X, (x):	2.1	2.6	2.9	3.3	4	5.1	7

Fit the regression line and find what quantity of mass X dissolved at 42°C?

19. The following performance were recorded in high jump and long jump of 5 competition in 2 days. All distances are measured in metres as follows:

Competitor:	A	B	C	D	E
High jump (X):	1.9	1.8	1.9	1.8	1.9
Long jump (Y):	6.2	6.3	6.5	6.0	6.1

What performance might have in expected when competitor have high jump value (X) = 2.9 m, then long jump. Also calculate correlation coefficient between X and Y.

20. Two regression lines of random variables X and Y giving the regression lines of Y on X and X on Y are $7x - 16y + 9 = 0$ and $5y - 4x - 3 = 0$. Then find (a) the mean of X and Y. (b) The regression coefficient of Y on X and X on Y. (c) Also calculate the coefficient of determination.

21. Obtain the regression equation of Y on X and estimate Y when X = 55 from the following:

X:	40	50	38	60	65	50	35
Y:	38	60	55	70	60	48	30

Also calculate standard error of estimate.

22. Find the equation of regression line of Y on X, if the observations (X, Y) are: (1, 4), (2, 8), (3, 2), (4, 12), (5, 10), (6, 14), (7, 16), (8, 6) and (9, 18).

23. The two regression lines were found to $4x - 5y + 33 = 0$ for Y on X and for X on Y is $20x - 9y - 107 = 0$. Find the mean values and correlation coefficient between X and Y.

Answers

1. 0.32 **2.** 697 **3. a.** 0.89 **b.** $\hat{Y} = 83.75 + 1.11x$, Y = 155.9 mmHg
4. −0.92, negatively correlated **5.** r = 0.98, $\hat{Y} = 3.90 + 4.37X$, $\hat{X} = -0.92 + 0.23Y$
6. 0.64 **7. a.** 0.052 **b.** 42.8 **8.** 0.58 **9.** $\hat{Y} = 5.58 + 0.051X$, 5.96
10. 4.06, 8.75 and r = 0.59 **11.** 0, no correlation **12.** $\hat{Y} = 80.9$, 3.2
13. $\hat{Y} = 59.25$ **14.** $\hat{Y} = 42.88$, $S_{e=1.77}$ **15.** $\hat{Y} = 4.29$ **16.** When X = 125, $\hat{Y} = 28.05$, $r^2 = 0.26$
17. 0.86, $\hat{Y} = 148.31$ **18.** 3.6 kg **19.** $\hat{Y} = 7.43$, 0.33 **20.** (7.25, 5.2), (0.44, 1.25) and 0.55
21. $\hat{Y} = 57.88$ and Se = 9.71 **22.** $\hat{Y} = 3.71 + 1.29x$ **23.** [(6.98, 3.62) and r = 0.6]

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Appendix

Table 1: Table of Random Numbers

36518	36777	89116	05542	29705	83775	21564	81639	27973	62413	85652	62817	57881
46132	81380	75635	19428	88048	08747	20092	12615	31046	67753	69630	10883	13683
31814	77367	47091	97402	27569	90148	02338	39318	54936	34641	95525	86316	87384
84108	93793	64953	51472	65358	23701	75230	47200	78176	85248	90589	74567	22633
78435	37586	07015	98729	76703	16224	97661	79907	06611	26501	93389	92725	68158
41859	94198	37182	61345	88857	53204	86721	59613	67494	17292	94457	89520	77771
13019	07274	51068	93129	40386	57731	44254	66685	72835	01270	42523	45323	63481
82448	72430	29041	59208	95266	33978	70958	60017	39723	00606	17956	19024	15879
25432	96593	83112	96997	55340	80312	78839	09815	16887	22228	06206	54272	83516
69226	38655	03811	08342	47863	02743	11547	38250	58140	98470	24364	99797	73498
245837	68821	66426	20496	84843	18360	91252	99134	48931	99538	21160	09411	44659
38914	82707	24769	72026	56813	49336	71767	04474	32909	74162	50404	68562	14088

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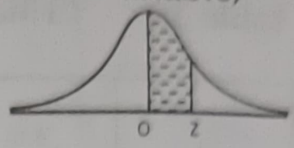
Table 2: Area Under Normal Curve (Table of Standard Normal Variable)

Entry represents area under the standard normal distribution from the mean to Z



Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0199	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2517	0.2549
0.7	0.2580	0.2612	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2967	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3531	0.3554	0.3577	0.3599	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974
2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.4979	0.4980	0.4981
2.9	0.4981	0.4982	0.4982	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986
3.0	0.49865	0.49869	0.49874	0.49878	0.49882	0.49886	0.49889	0.49893	0.49897	0.49900
3.1	0.49903	0.49906	0.49910	0.49913	0.49916	0.49918	0.49921	0.49924	0.49926	0.49929
3.2	0.49931	0.49934	0.49936	0.49938	0.49940	0.49942	0.49944	0.49946	0.49948	0.49950
3.3	0.49952	0.49953	0.49955	0.49957	0.49958	0.49960	0.49961	0.49962	0.49964	0.49965
3.4	0.49966	0.49968	0.49969	0.49970	0.49971	0.49972	0.49973	0.49974	0.49975	0.49976
3.5	0.49977	0.49978	0.49978	0.49979	0.49980	0.49981	0.49981	0.49982	0.49983	0.49983
3.6	0.49984	0.49985	0.49985	0.49986	0.49986	0.49987	0.49987	0.49988	0.49988	0.49989
3.7	0.49989	0.49990	0.49990	0.49990	0.49991	0.49991	0.49992	0.49992	0.49992	0.49992
3.8	0.49993	0.49993	0.49993	0.49994	0.49994	0.49994	0.49995	0.49995	0.49995	0.49995
3.9	0.49995	0.49995	0.49996	0.49996	0.49996	0.49996	0.49996	0.49996	0.49997	0.49997

Short Cut Key for Critical Values of Z

Critical Values (Z_α)	Level of significance (α)		
	1%	5%	10%
Two tailed test	$\lvert Z_{\alpha} \rvert = 2.575$	$\lvert Z_{\alpha} \rvert = 1.96$	$\lvert Z_{\alpha} \rvert = 1.645$
Right tailed test	$Z_{\alpha} = 2.33$	$Z_{\alpha} = 1.645$	$Z_{\alpha} = 1.28$
Left tailed test	$Z_{\alpha} = -2.33$	$Z_{\alpha} = -1.645$	$Z_{\alpha} = -1.28$

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Table 3: Critical Values of Student's t-distribution

d.f.	Level of significance for one-tailed test					
	0.10	0.05	0.025	0.01	0.005	0.0005
	Level of significance for two-tailed test					
	0.20	0.10	0.05	0.02	0.01	0.001
1	3.078	6.314	12.706	31.821	63.657	636.619
2	1.886	2.920	4.303	6.965	9.925	31.598
3	1.638	2.353	3.182	4.541	5.841	12.941
4	1.533	2.132	2.776	3.747	4.604	8.610
5	1.476	2.015	2.571	3.365	4.032	6.859
6	1.440	1.943	2.447	3.143	3.707	5.959
7	1.415	1.895	2.365	2.998	3.499	5.405
8	1.397	1.860	2.306	2.896	3.355	5.041
9	1.383	1.833	2.262	2.821	3.250	4.781
10	1.372	1.812	2.228	2.764	3.169	4.587
11	1.363	1.796	2.201	2.718	3.106	4.437
12	1.356	1.782	2.179	2.681	3.055	4.318
13	1.350	1.771	2.160	2.650	3.012	4.221
14	1.345	1.761	2.145	2.624	2.977	4.140
15	1.341	1.753	2.131	2.602	2.947	4.073
16	1.337	1.746	2.120	2.583	2.921	4.015
17	1.333	1.740	2.110	2.567	2.898	3.965
18	1.330	1.734	2.101	2.552	2.878	3.922
19	1.328	1.729	2.093	2.539	2.861	3.883
20	1.325	1.725	2.086	2.528	2.845	3.850
21	1.323	1.721	2.080	2.518	2.831	3.819
22	1.321	1.717	2.074	2.508	2.819	3.792
23	1.319	1.714	2.069	2.500	2.807	3.767
24	1.318	1.711	2.064	2.492	2.797	3.745
25	1.316	1.708	2.060	2.485	2.787	3.725
26	1.315	1.706	2.056	2.479	2.779	3.707
27	1.314	1.703	2.052	2.473	2.771	3.690
28	1.313	1.701	2.048	2.467	2.763	3.674
29	1.311	1.699	2.045	2.462	2.756	3.659
30	1.310	1.697	2.042	2.457	2.750	3.646
40	1.303	1.684	2.021	2.423	2.704	3.551
60	1.296	1.671	2.000	2.390	2.660	3.460
120	1.289	1.658	1.980	2.358	2.617	3.373
∞	1.282	1.645	1.960	2.326	2.576	3.291

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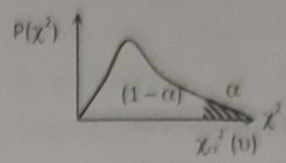
Table 20: Critical Values of chi-square



This table contains the values of χ^2 that correspond to a specific right-tail area and specific number of degrees of freedom df.

Degree of Freedom	Right-tail Area			
	0.10	0.05	0.02	0.01
1	2.706	3.841	5.412	6.635
2	4.605	5.991	7.824	9.210
3	6.251	7.815	9.837	11.345
4	7.779	9.488	11.668	13.277
5	9.236	11.070	13.388	15.086
6	10.645	12.592	15.033	16.812
7	12.017	14.067	16.622	18.475
8	13.362	15.507	18.168	20.090
9	14.684	16.919	19.679	21.666
10	15.987	18.307	21.161	23.209
11	17.275	19.675	22.618	24.725
12	18.549	21.026	24.054	26.217
13	19.812	22.362	25.472	27.688
14	21.064	23.685	26.873	29.141
15	22.307	24.996	28.259	30.578
16	23.542	26.296	29.633	32.000
17	24.769	27.587	30.995	33.409
18	25.989	28.869	32.346	34.805
19	27.204	30.144	33.687	36.191
20	28.412	31.410	35.020	37.566
21	29.615	32.671	36.343	38.932
22	30.813	33.924	37.659	40.289
23	32.007	35.172	38.968	41.638
24	33.196	36.415	40.270	42.980
25	34.382	37.652	41.566	44.314
26	35.563	38.885	42.856	45.642
27	36.741	40.113	44.140	46.963
28	37.916	41.337	45.419	48.278
29	39.087	42.557	46.693	49.588
30	40.256	43.773	47.962	50.892

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Exam Question - 2023

POKHARA UNIVERSITY

Level: Bachelor | **Full Marks:** 100

Programme: BE | **Pass Marks:** 45

Course: Probability and Statistics (New) | **Time:** 3 hrs

Attempt all the questions.

1. a) Cell phones have made our lives easier, they also have negative effect on us. Daily use of cell phones for a long time may lead to phobia of phones may be cause of depression. The randomly selected some teenage boys and girls of certain college daily spends their time in hours in cell phone are given below. [8]

Boys (in hrs):	3.0	4.0	2.5	5.0	4.5	1.5	2.0	3.5
Girls (in hrs):	4.5	1.5	5.5	3.0	2.5	4.0	5.0	4.5

i) In boys and girls which one may lead near to phone phobia and why?

ii) Which one has more uniform in phone using time?

b) A Dell company has two plants to manufacture computer chips. Out of 100 chips, plant A produces 80% and plant B produces 20% of the chips. Whereas production from plan A and plant B chips rated 85% and 65% as standard quality, a chip is selected at random from the whole production of the two plants and found to be standard quality; i) what is the probability that it was produced by plant A? ii) What is the probability that the selected chips being non standard quality? [7]

2. a) The number of hardware failure of a computer system in a week of operation has the following probability mass function

No. of failures:	0	1	2	3	4	5	6
Probability:	0.18	0.28	0.25	0.16	0.04	0.06	0.03

Find the variance of the number of failures in a week. [7]

b) The phone lines to an airline reservation system are occupied 40% of the time. Assume that the events the lines are occupied on successive calls are independent. Assume that 10 calls are placed to the airline then what is the probability that i) for exactly 3 calls the lines are occupied? ii) For at least one call the call lines are not occupied? [8]

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3 a) What is standard normal variate? From past experience it is known that time to download on certain university webpage is normally distributed with mean of 9 seconds with standard deviation of 2 seconds. What is the probability that the download will be completed i) under 3.5 seconds ii) more than 3 seconds? [8]

OR

A set of final examination grades in an introductory statistics course was found to be normally distributed with mean of 73 and standard deviation of 8. i) What is the probability of getting a grade no higher than 91 on this exam? ii) Only 5% of the students taking the test scored than what grade? [8]

b) The amount of time that a surveillance camera will run without having to be reset is a random variable having exponential distribution with parameter $\lambda = 1/50$ day. Find the probability that such camera will have to reset in less than 20 days. [7]

OR

Out of 3000 families 4 children each, assume boys and girls are equally probable. How many families would you expect to have i) at most 2 girls ii) at least one boy? [7]

4. a) The joint probability distribution of X and Y are:

$f(x, y) = 2 - x - y, 0 < x < 1, 0 < y < 1$

$= 0$, otherwise

Find marginal probability density function of x and y, are X and Y independent? [7]

b) A population consists of the values 7, 6, 8, 4 and 10. Prove that sample mean of size 3 is unbiased estimate of population mean. [8]

5 a) The following are the Brinell hardness values obtained for eight samples of magnesium alloys:

Hardness of alloys: 107, 148, 123, 165, 102, 119, 115, 125

Find 95% confidence interval of the mean Brinell hardness of the magnesium alloys. [8]

OR

A population consists of the value 7, 6, 8, 4, 10. Show that sample mean of size 3 is unbiased estimator of the population mean. [8]

b) In random sample 1000 houses in a certain city it was found that 318 have internet facilities. In the neighboring city it was found that 450 houses out of 1200 houses have internet facility. Test whether there is significance difference between houses of two cities having internet facilities at 5% level of significance? [7]

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OR

A quality control engineer at a light bulb factory needs to estimate the average life of a large shipment of light bulbs. The process standard deviation is known to be 100 hours. A random sample 64 light bulbs indicated a sample mean of 350 hours. Would you claim that the manufacture has the right to state that the light bulbs last an average of 400 hours at 5% level of significance? [7]

6. a) The score of 8 candidates priori and after training are given below.

Priori:	82	48	36	54	72	84	90	65
After:	92	42	45	81	84	75	86	70

Was training effective at 5% level of significance? [7]

b) The following data presents the marks obtained by students in mathematics and physics.

Marks in mathematics:	85	60	73	40	82	90
Marks in physics:	92	72	75	65	50	80

Fit the regression equation of marks of physics on marks of mathematics and predict the marks of mathematics whose marks of physics are 52. [8]

7. Write short note on any two: [2×=10]

a) Poisson distribution

b) Interval estimation

c) Karl Pearson's correlation coefficient

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